

H/W Version: A
S/W Version: 1.0

RADIO NAVIGATION AIDS
300 DME
310 / 320 DME
TECHNICAL MANUAL



RADIO NAVIGATION AIDS

300 DISTANCE MEASURING EQUIPMENT

310 DME / 320 DME

Rev. C/2022-12-19

H/W Version: A
S/W Version: 1.0

COMMERCIAL IN CONFIDENCE

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SAFETY INFORMATION

Please take a moment to review these safety precautions. They are provided for your protection and to prevent damage to the equipment. This safety information applies to all operators and service personnel.

GENERAL SAFETY MEASURES

Observe all **CAUTION** and **DANGER** signs mounted on the equipment and all **WARNINGS**, **CAUTIONS**, and **NOTES** printed in this manual.

Safety Labeling and Signal Words



DANGER

indicates immediate hazard which will result in severe personal injury or death.



WARNING

indicates a situation where a potentially hazardous condition exists that, if not avoided, could result in serious injury or loss of life.



CAUTION

indicates a situation where a non-immediate or potentially hazardous condition exists that, if not avoided, could result in minor or moderate injury, and/or damage to the equipment or other property.



HIGH VOLTAGE

indicates hazardous high voltage. It calls your attention to items or operations that could be dangerous to you and other persons operation this equipment.



NOTE

is used to notify users of important information or suggestions which will result in enhanced installation, operation, maintenance or reliability, but not hazard-related.

SPECIFIC PRECAUTIONS

- Observe all of these precautions to ensure your personal safety and to prevent damage to either the equipment or ancillary equipment connected to it.
- DO NOT wear conductive articles of jewelry or clothing (such as watchbands, bracelets, rings, key chains, necklaces, metalized aprons, cloth with conductive thread, metal headgear, or metal frame glasses) while working on the equipment. They may present an electrical contact hazard with exposed live electrical parts.

- DO NOT work on the equipment unless there is another person nearby who is familiar with the operation and hazards of the equipment and who is competent at administering first aid.
- If personnel have been exposed to electrical shock, perform the required first aid procedures and contact local medical service immediately.
- DO NOT, under any circumstances, attempt to defeat any safety feature built into the equipment by interfering with interlocks, using jumpers, modifying software, or by any other action that disables the safety features of the system.
- Review proper lifting techniques when lifting heavy objects. Use sufficient personnel when lifting or moving the equipment.
- DO NOT come into contact with high voltage or high current connections when operating or maintaining the equipment. Keep fingers and conductive tools away from high-voltage and high-current areas.
- Remove all tools from the equipment and replace all covers and panels removed from the equipment after completing the replacement or repair.
- DO NOT use or store flammable sprays, solvents, or other combustible materials close to the system equipment. Keep proper fire extinguishers rated for use on electrical equipment available and in working order. Know how to use an extinguisher properly.



WARNING

SAFETY FIRST! Whenever it is likely that the protection has been impaired, the equipment must be made inoperative and be secured against any unintended operation.



WARNING

RADIO FREQUENCY: This equipment is capable of delivering harmful levels of radio frequency power. Ensure all times during operation that the RF output is properly terminated with an adequately rated termination or a suitable antenna and that the cables and connectors attached to the equipment are in good condition. Never operate the equipment without proper termination or with defective cables or connectors.



WARNING

FUSES: Make sure that only fuses with required rated current and of the specified type (normal blow time delay, etc.) are used for replacement. The use of repaired fuses and short-circuiting of fuse holders must be avoided.



WARNING

RF EXPOSURE: This product is not intended to be operated close to the body. To meet EU Maximum Permissible Exposure (MPE) conditions, ensure that at least 5.7 meters (for 310 DME) / 17.8 meters (for 320 DME) separation distance is maintained between the transmitter antenna and the body of the user or animals.



CAUTION

GROUNDING: Any interruption of the protective ground connection inside or outside the equipment is likely to cause damage to the equipment. To avoid damage, this equipment and all the powered devices connected to it must be connected to the same earth ground.



CAUTION

SEMI-REGID COAX: DO NOT bend the cables more than necessary while working with and around the semi-rigid coaxial cables in the equipment.



CAUTION

RF CONNECTORS: DO NOT apply excessive torque to the RF connectors. Use a proper torque wrench whenever available.



CAUTION:

ELECTROSTATIC DISCHARGE: Carefully review and follow the electrostatic discharge information.

ELECTROSTATIC DISCHARGE INFORMATION

Certain components used in the equipment are sensitive to static electricity and can be damaged by its discharge. Proper packaging and grounding techniques are required to prevent damage.



Boards, assemblies and sub-assemblies sensitive to Electrostatic Discharge (ESD) are marked with this symbol.

Follow these guidelines while handling Electrostatic Discharge (ESD)-sensitive equipment:

- Assume all components and assemblies are ESD sensitive.
- Ensure that your body and all system equipment are at a ground potential to avoid damage from ESD.
- Wear a current-limited, grounded static strap on your wrist to ensure that any accumulated charge is discharged from your body to ground.
- Handle devices and circuit board assemblies only by the edges when possible. Avoid touching the components, conductors or connector pins.
- Avoid direct contact between ESD-sensitive device and any plastic materials with no anti-static treatment.
- Keep all static-sensitive devices in their protective packaging until needed. Storing or transporting devices in conventional plastic containers could be destructive.
- Handle ESD-sensitive material only when properly grounded and at an ESD-safe workstation.
- Place removed devices, circuit boards and sub-assemblies on an antistatic mat.
- Do not place an unprotected device/board on top of any unprotected component or on any surface that is not dissipative or antistatic.
- Avoid activities such as rubbing, which cause ESD.
- Keep materials such as plastic, Styrofoam, cellophane tap, nylon, wool, food, and beverage out of work area.
- Keep unprotected personnel out of work area.
- Handle defective devices using the same precautions as with good material.

*310 DME
CE Marking
Declaration of Conformity*

*320 DME
CE Marking
Declaration of Conformity*

*310 DME
CE Marking
NB Opinion*

*320 DME
CE Marking
NB Opinion*

Change / Correction Request Form

Rev. C/2022-12-19

INSTRUCTIONS:

1. Use this form to report deficiencies or problems, or to make recommendations to this manual.
2. Complete form and submit to: MOPIENS, Inc.

MOPIENS, Inc.

2F, Segi Bldg., 111 Ogeum-ro

Songpa-gu, Seoul 05548

Republic of Korea

E-mail : support@mopiens.com or Fax : +82-2-421-8388

Name / Title	
Company / Organization	
Mailing Address	
Phone / Fax / E-mail	

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STANDARD EQUIPMENT WARRANTY

MOPIENS, Inc. (“MOPIENS”) warrants that all MOPIENS manufactured equipment will be free from defects in material and workmanship for the period of (1) year. Warranty begins from the date of shipment from a MOPIENS facility. The warranty is extended to customers and applies to all MOPIENS manufactured equipment purchased, installed, and used for the purpose for which such equipment was originally designed. The above warranties cover only defects arising under normal use and do not include malfunctions or failures resulting from misuse, abuse, neglect, alteration, problems with electrical power, usage not in accordance with product instructions, acts of nature, or improper installations or repairs made by anyone other than MOPIENS or a MOPIENS authorized representative. MOPIENS reserves the right to substitute functionally equivalent new or serviceable refurbished parts.

WARRANTY CLAIMS AND PROCEDURES

1. During the applicable Standard Equipment Warranty Period outlined above, customer’s sole and exclusive remedy for any breach of the Standard Equipment Warranty will be, at MOPIENS’s sole discretion and option, the repair or replacement of the defective product. Components that customer claims to be defective must be available to MOPIENS for inspection and evaluation. To be entitled to rights under the Standard Equipment Warranty, the customer must notify MOPIENS in writing within thirty (30) days after discovering a suspected defect in any product, but in any event prior to the expiration of the applicable Standard Equipment Warranty Period. Notice to a MOPIENS dealer, systems integrator, sales representative or other third party is not notice to MOPIENS. Following its receipt of any such customer notice, MOPIENS will determine whether the reported problem is covered by this Standard Equipment Warranty. If MOPIENS determines that the problem is covered, MOPIENS will authorize repair or replacement of the defective product, as deemed appropriate by MOPIENS in its sole discretion.

2. Before shipping any product to MOPIENS, the customer must obtain a written return authorization from MOPIENS, and provide any proof of warranty eligibility requested by MOPIENS. Any product received by MOPIENS without a return authorization may, at MOPIENS’s option, be returned to the customer collect. If a warranty replacement part is required, customer shall provide Purchase Order to MOPIENS prior to shipment of the replacement, to guarantee the return of the rejected unit. Purchase Order is valid until suspected part is received and warranty is confirmed by assessment. Once a return authorization is obtained, the customer is responsible for packing and shipping the product/component to which its warranty claim relates to a service facility designated by MOPIENS, within thirty (30) days after receipt of the return authorization. Upon receipt of replacement equipment (or part thereof), customer has thirty (30) days to tender the defective equipment (or part thereof) to the return carrier for shipment to the service center designated by MOPIENS. If customer does not timely return the defective equipment (or part thereof), MOPIENS shall invoice customer for the list price of such equipment (or part thereof), plus applicable shipping. Such failure to return the equipment (or part thereof) may, in MOPIENS’s discretion, be grounds for termination of the warranty and/or suspension of any future advance exchange privileges until such outstanding defective equipment has been returned.

3. MOPIENS will provide customer with new, rebuilt, refurbished or alternate equipment (or part thereof) of equal or improved quality, as exchange equipment (or part thereof) to replace eligible defective equipment (or part thereof). Any alternate equipment (or part thereof) will meet or exceed the specifications of the replaced equipment (or part thereof). Rebuilt or refurbished equipment may bear cosmetic blemishes that do not affect performance. Unless otherwise specified by MOPIENS in writing, repaired or replaced equipment (or parts thereof) are covered only for the remainder of the term of the applicable Standard Equipment Warranty. All defective equipment (or parts thereof) replaced by MOPIENS become the property of MOPIENS. MOPIENS has no obligation to (i) service, exchange or otherwise replace any equipment (or part thereof) that has been damaged, modified, abused, misused or over-used as determined by MOPIENS or has been used with non-MOPIENS supplies or products that have caused damage or malfunction; (ii) paint, refinish, refurbish, restore or exchange any equipment (or part thereof) with cosmetic blemishes; (iii) service, exchange or otherwise replace any equipment (or part thereof) if the same would interfere with, impede or be redundant with normal or scheduled maintenance of such equipment (or part thereof); (iv) service, exchange or otherwise replace any equipment (or part thereof) that is within sixty (60) days of the end of its production life; or (v) provide any 3rd party application software support or service involving application hardware or replace any accessories. If MOPIENS elects to

perform any such services at customer's request, then such services will be deemed a service call and all labor, parts and materials used for the service call will be charged at MOPIENS's then - prevailing rates.

EQUIPMENT WARRANTY EXCLUSIONS

MOPIENS does not warrant or guarantee, and is not responsible for:

1. Defects, failures, damages or performance limitations caused in whole or in part by (A) power failures, surges, fires, floods, snow, ice, lightning, excessive heat or cold, highly corrosive environments, accidents, actions of third parties, or other events outside of MOPIENS's control, or (B) customer's abuse, mishandling, misuse, negligence, improper storage, servicing or operation, or unauthorized attempts to repair or alter the equipment in any way. Customer must provide qualified technical personnel to maintain and repair the equipment.
2. Alterations and/or Modifications to any part of MOPIENS's product, without MOPIENS's written authorization unconditionally VOIDS the MOPIENS Standard Equipment Warranty.
3. Equipment built to customer's specifications that are later found not to meet customer's needs or expectations.
4. The performance of the equipment when used in combination with equipment not purchased, specified, or approved by MOPIENS.
5. Batteries and other consumable goods.
6. Wearable items, such as tooling, cables, part harnesses, contacts etc.

ADDITIONAL WARRANTY NOTES

1. OEM or third-party equipment that is incorporated into MOPIENS equipment is covered under the applicable MOPIENS Standard Equipment Warranty unless the OEM or Third-Party equipment carries its own limited warranty, in which event the OEM or third-party warranty will apply to such equipment incorporated into MOPIENS equipment. For example and not limitation, PCs, monitor displays, printers, modems and UPSs are OEM products that have a limited 1 year manufacturer's warranty.
2. Items Sold As Resale. Items sold as resale are such items that are not manufactured by MOPIENS but may be utilized in conjunction with or independently of MOPIENS manufactured equipment (such as computers, printers and network adapters) and shall be covered only by the specific warranty terms of the supplier or original equipment manufacturer of those items.
3. The MOPIENS Warranty applies to the original purchaser, and is not transferrable. Used Equipment. IF THE EQUIPMENT SPECIFIED IN AN ORDER IS DESCRIBED AS USED, UNLESS OTHERWISE AGREED IN WRITING BY THE PARTIES, IT IS SOLD "AS IS" AND WITH NO WARRANTY.

SERVICES WARRANTY

MOPIENS warrants that the services will be performed in a professional manner (the "Services Warranty"). Notice of a breach of the Services Warranty must (i) specify in reasonable detail, the nature of the claim, and (ii) be received within ninety (90) days from the last day of performance of the services. Upon notice of a breach of the services warranty and MOPIENS's determination of the validity of such breach of the Services Warranty, MOPIENS will re-perform the applicable services at MOPIENS's expense. If after reasonable opportunity MOPIENS is unable to re-perform such services to the reasonable satisfaction of customer, customer may, as its exclusive remedy, obtain a refund of the fees paid to MOPIENS under the applicable order for such services.

DISCLAIMER OF WARRANTY

EXCEPT AS EXPRESSLY PROVIDED IN THIS STANDARD WARRANTY POLICY STATEMENT, MOPIENS HEREBY EXPRESSLY DISCLAIMS ALL REPRESENTATIONS, CONDITIONS AND WARRANTIES, WHETHER EXPRESS OR IMPLIED, INCLUDING BY WAY OF EXAMPLE AND NOT LIMITATION, THE IMPLIED WARRANTIES OF TITLE, MERCHANTABILITY, NONINFRINGEMENT AND FITNESS FOR A PARTICULAR PURPOSE. LIMITATION ON LIABILITY NOTWITHSTANDING ANYTHING HEREIN TO THE CONTRARY, IN NO EVENT WILL MOPIENS BE LIABLE FOR ANY

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TECHNICAL SUPPORT

To get technical support, please contact our Customer Support at support@mopiens.com.

HOW TO ORDER REPLACEMENT PARTS

To order replacement parts, please contact our customer service by e-mail, Fax, phone or your local authorized representative. To help us assist you better, please be sure to provide as much information about the equipment and part as possible including the description, manufacturer, any information stamped on the part, and all the information from the label on the equipment when you place an order:

INFORMATION NEEDED:

1. Model number, name, and serial number of principal equipment.
2. MOPIENS's Part number and description.
3. Unit subassembly number and/or Reference designation (where applicable).
4. Manufacturer's name and manufacturer's part number (where applicable).
5. Quantity of each replacement part required.

ADDRESS:	MOPIENS, Inc.
	Customer Support
	2F, Segi Bldg., 111 Ogeum-ro
	Songpa-gu, SEOUL 05548
	Republic of Korea
E-mail:	support@mopiens.com
Phone:	+82-2-3433-8388
Fax:	+82-2-414-8388
	or
	your local Authorized Representative

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TERMS AND ABBREVIATIONS

AC	<i>Alternating Current</i>
A/C	<i>Aircraft</i>
AC/DC	<i>AC-to-DC Converter</i>
ADC	<i>Analog-to-Digital Converter</i>
ADN	<i>Antenna Distribution Network</i>
AGC	<i>Automatic Gain Control</i>
AM	<i>Amplitude Modulation</i>
BITE	<i>Built-In Test Equipment</i>
CAN	<i>Controller Area Network</i>
CFB	<i>Cartesian Feedback</i>
CPLD	<i>Complex Programmable Logic Device</i>
CSP	<i>Control Status Panel</i>
DAC	<i>Digital-to-Analog Converter</i>
DC	<i>Direct Current</i>
DC/DC	<i>DC-to-DC Converter</i>
DDS	<i>Direct Digital Synthesis</i>
DME	<i>Distance Measuring Equipment</i>
DPX	<i>Duplexer</i>
DPX-MSC	<i>Duplexer - Mute Switch, Combiner</i>
DSP	<i>Digital Signal Processing</i>
EMC	<i>Electro Magnetic Compatibility</i>
EMI	<i>Electro Magnetic Interference</i>
EMU	<i>Environmental Monitor Unit</i>
ESD	<i>Electro Static Discharge</i>
EUROCAE	<i>European Organisation for Civil Aviation Equipment</i>
FAA	<i>Federal Aviation Administration</i>
FPGA	<i>Field Programmable Gate Array</i>
GP	<i>Glide Path</i>
GUI	<i>Graphic User Interface</i>
HPA	<i>High Power Amplifier</i>
I/Q	<i>In-phase / Quadrature</i>
ICAO	<i>International Civil Aviation Organization</i>
IDENT	<i>Identification</i>
IFB	<i>Interface Board</i>
ILS	<i>Instrument Landing System</i>
IMC	<i>International Morse Code</i>
ISP	<i>In-System Programming</i>

LAN	<i>Local Area Network</i>
LCD	<i>Liquid Crystal Display</i>
LDSE	<i>Long Distance Echo Suppression</i>
LED	<i>Light Emitting Diode</i>
LMI	<i>Local Maintenance Interface</i>
LOC	<i>Localizer</i>
LRU	<i>Line Replaceable Unit</i>
MCN	<i>Monitor Combiner Network</i>
MCU	<i>Micro Controller Unit</i>
MKR	<i>Marker</i>
MON	<i>Monitor</i>
MSC	<i>Mute Switch, Combiner</i>
NM	<i>Nautical Mile</i>
PLL	<i>Phase Locked Loop</i>
PMDT	<i>Portable Maintenance Data Terminal</i>
PMU	<i>Power Management Unit</i>
RCS	<i>Remote Control System</i>
RCU	<i>Remote Control Unit</i>
RF	<i>Radio Frequency</i>
RFG	<i>Radio Frequency Generator</i>
RSU	<i>Remote Status Unit</i>
RXU	<i>Receiver Unit</i>
SCU	<i>System Controller Unit</i>
SDES	<i>Short Distance Echo Suppression</i>
SMPS	<i>Switched Mode Power Supply</i>
SRU	<i>Shop Replaceable Unit</i>
STDBY	<i>Standby</i>
SYN	<i>Synthesizer</i>
TCP/IP	<i>Transmission Control Protocol/Internet Protocol</i>
TCU	<i>Transponder Control Unit</i>
TCXO	<i>Temperature-Controlled Crystal Oscillator</i>
TFT	<i>Thin Film Transistor</i>
THD	<i>Total Harmonic Distortion</i>
TMA	<i>Terminal Maneuvering Area</i>
T/P	<i>Test Point</i>
T/R	<i>Transmit/Receive</i>
TX	<i>Transmitter</i>
TXU	<i>Transmitter Unit</i>
UART	<i>Universal Asynchronous Receiver/Transmitter</i>

UDP	<i>User Datagram Protocol</i>
USB	<i>Universal Serial Bus</i>
UTP	<i>Unshielded Twisted Pair</i>
VGA	<i>Variable Gain Amplifier</i>
VOR	<i>Very High Frequency Omnidirectional Radio Range</i>
VSWR	<i>Voltage Standing Wave Ratio</i>
WPM	<i>Word Per Minute</i>

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SECTION 1

GENERAL INFORMATION

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1. GENERAL INFORMATION

1.1. INTRODUCTION

This technical manual describes the 300 DME manufactured by MOPIENS, Inc. The purpose of this document is to provide information required to install, operate and maintain the MOPIENS 300 DME equipment when used together with the relevant Antenna System technical manual.

Included are a general description and specifications, block diagram theory, detailed theory of operation, operating procedures, standards and tolerances, periodic and corrective maintenance procedures, parts list, installation procedures, schematics, and other diagrams.

1.2. DME PRINCIPLES

1.2.1. DME OVERVIEW

The DME system provides for continuous and accurate indication in the cockpit of the slant range distance of an equipped aircraft from an equipped ground reference point.

The system comprises two basic components, one fitted in the aircraft, and the other installed on the ground. The aircraft component is referred to as the interrogator and the ground component as the transponder.

In operation, interrogators interrogate transponders which, in turn, transmit to the interrogator replies synchronized with the interrogations, thus providing means for accurate measurement of distance. The airborne interrogator uses a random timing algorithm in transmitting interrogation pulse pairs in series to recognize the transponder's replies synchronized to its own interrogations.

1.2.2. PRINCIPLES OF OPERATION

DME operates on principle of secondary radar system, and is used to determine the slant range distance of an aircraft (interrogator) to a ground station (transponder).

The airborne interrogator transmits a series of pulse pairs to the ground based transponder. The transponder evaluates the interrogation signals by measuring the time spacing of each pulse pair and, if the signals are determined to be valid; i.e. 12 μ s for X channel and 36 μ s for Y channel, transmits a series of reply pulse pairs on a different frequency. The reply frequency will be either above or below the interrogation frequency by 63 MHz depending on the channel number. The interrogator measures the total elapsed time for interrogation and reply, and it is converted into a distance in nautical miles for cockpit indication.

A measure of the time for a radio signal to travel a distance of one nautical mile and return is called "Radar Mile". A nautical mile is defined as 1852 meters, and therefore, a radar mile is derived as follows:

$$1 \text{ Radar Mile} = \frac{1852 \text{ m} \times 2}{c} = \frac{3704 \text{ m}}{299,792,458 \text{ m/s}} \approx 12.36 \mu\text{s}$$

Where, c is the speed of light (299,792,458 m/sec).

Note:

A radar mile may also be derived from the substitution of feet for meter in the above calculation. With the knowledge that a radio signal travels at approximately 984 feet per microsecond and a

nautical mile is approximately 6,080 feet, we can figure the approximate time required for the radio signal to travel back-and forth a nautical mile:

$$1 \text{ Radar Mile} = \frac{6080 \text{ ft} \times 2}{984 \text{ ft}/\mu\text{s}} \approx 12.36 \mu\text{s}$$

DME transponder imposes a 50 μs delay (for X channel; 56 μs for Y channel) from the receipt of an interrogation to the transmission of a reply. Therefore, the total elapsed time measured in the airborne interrogator is the sum of the two-way travel time to and from the ground station and the delay time. Accordingly the distance will be determined by subtracting 50 μs (delay time) from T (total elapsed time), and dividing the remaining two-way propagation time (T-50 μs) by the radar mile as given below:

$$D = \frac{(T - 50 \mu\text{s})}{12.36 \mu\text{s}} \text{ NM}$$

Where, D is the distance in nautical miles, and T is the total elapsed time in microseconds.

The time delay, also referred to as reply delay or system delay, must be the same for all ground stations in order to have an accurate measurement of distance. The channel code, pulse pair spacing and time delay established by ICAO Annex 10 are given on the following table.

Table 1-1. Pulse Spacing and Time Delay for X and Y Channel

Channel	Pulse pair spacing (μs)		Time delay (μs)	
	Interrogation	Reply	1st pulse timing	2nd pulse timing
X	12	12	50	50*
Y	36	30	56	50*

The DME signal is encoded by transmitting two pulses. Because of such simple coding, the DME receiver cannot completely eliminate false triggering due to noise but encoding does provide some improvement.

A DME interrogation consists of two Gaussian pulses separated by 12 μs for X channel and 36 μs for Y channel. The reply pulse pair is separated by 12 μs for an X channel and 30 μs Y channel. The master reference point for making all timing measurements is the 50% amplitude point of the transmitter envelope as shown in Figure 1-1

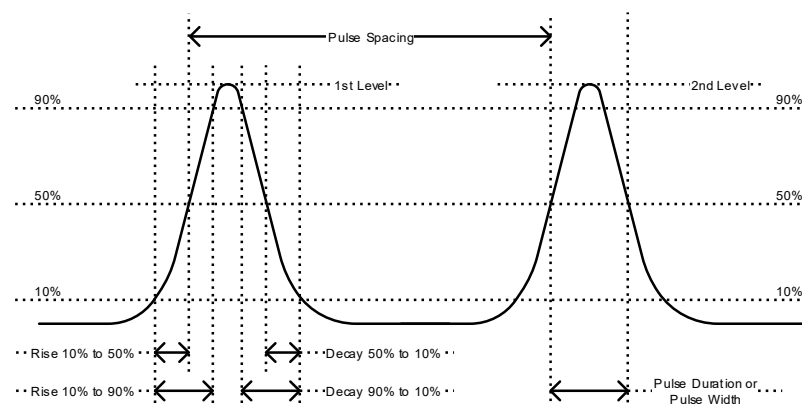


Figure 1-1. Pulse Spacing and Timing Definition

The interrogation is made on an interrogation frequency and received by the DME transponder at the ground station. After a specified time period of delay (50 μs X channel and 56 μs Y channel for first pulse timing), the ground transponder transmits a reply with a pulse pair with exactly the

same pulse width and pulse separation. (Refer to Figure 1-2 and Figure 1-3). The time delay shall be the interval from the half voltage point on the leading edge of the first pulse of the interrogation pulse pair and the half voltage point on the leading edge of the first pulse of the reply pulse pair.

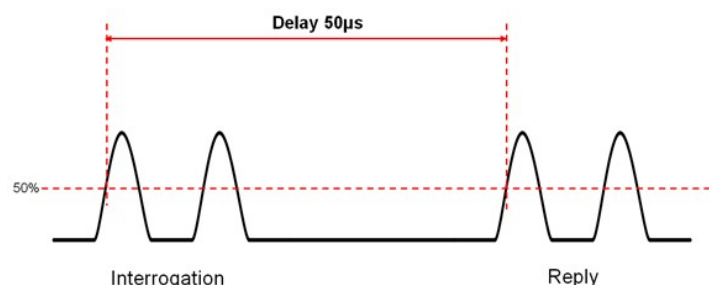


Figure 1-2. X channel Time Delay

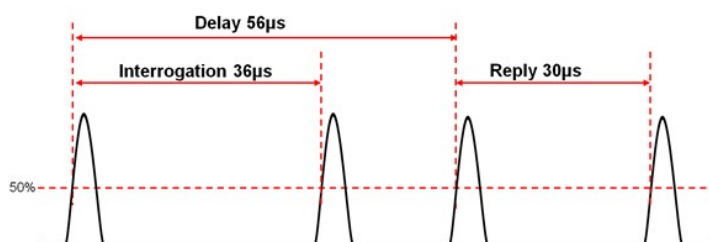


Figure 1-3. Y channel Time Delay

Both X-channel and Y-channel share the same interrogation frequencies but operate on different reply frequencies. . (Refer to Table 1-2).

Table 1-1. Interrogation and Reply Frequency Assignment for X and Y channel

962 to 1024 MHz	1025 to 1087 MHz	1088 to 1150 MHz	1151 to 1213 MHz
Reply CH 1-63X	Interrogation CH 1-63X	Interrogation CH 64-126X	Reply CH 64-126X
N/A	Interrogation CH 1-63Y	Reply CH 1-63Y	N/A
	Reply CH 64-126Y	Interrogation CH 64-126Y	

The DME channels may be paired with its collocated VHF localizer and UHF glide path so that the pilot can dial one control and simultaneously tune a runway's ILS equipment. It is also channel paired for VOR/DME combination. (Refer to Table 1-2).

Table 1-2. Channel Pairing with VOR and ILS

DME Channel Number	Interrogation Frequency MHz	Reply Frequency MHz	VHF Frequency MHz	Glide Path Frequency MHz
1X	1 025	962	-	-
1Y	1 025	1 088	-	-
2X	1 026	963	-	-
2Y	1 026	1 089	-	-
3X	1 027	964	-	-
3Y	1 027	1 090	-	-
4X	1 028	965	-	-
4Y	1 028	1 091	-	-
5X	1 029	966	-	-
5Y	1 029	1 092	-	-
6X	1 030	967	-	-
6Y	1 030	1 093	-	-
7X	1 031	968	-	-
7Y	1 031	1 094	-	-
8X	1 032	969	-	-
8Y	1 032	1 095	-	-
9X	1 033	970	-	-
9Y	1 033	1 096	-	-
10X	1 034	971	-	-
10Y	1 034	1 097	-	-
11X	1 035	972	-	-
11Y	1 035	1 098	-	-
12X	1 036	973	-	-
12Y	1 036	1 099	-	-
13X	1 037	974	-	-
13Y	1 037	1 100	-	-
14X	1 038	975	-	-
14Y	1 038	1 101	-	-
15X	1 039	976	-	-
15Y	1 039	1 102	-	-
16X	1 040	977	-	-
16Y	1 040	1 103	-	-
17X	1 041	978	108.00	-
17Y	1 041	1 104	108.05	-
18X	1 042	979	108.10	334.7
18Y	1 042	1 105	108.15	334.55
19X	1 043	980	108.20	-
19Y	1 043	1 106	108.25	-
20X	1 044	981	108.30	334.1
20Y	1 044	1 107	108.35	333.95
21X	1 045	982	108.40	-
21Y	1 045	1 108	108.45	-
22X	1 046	983	108.50	329.9
22Y	1 046	1 109	108.55	329.75
23X	1 047	984	108.60	-
23Y	1 047	1 110	108.65	-
24X	1 048	985	108.70	330.5
24Y	1 048	1 111	108.75	330.35
25X	1 049	986	108.80	-
25Y	1 049	1 112	108.85	-
26X	1 050	987	108.90	329.3
26Y	1 050	1 113	108.95	329.15
27X	1 051	988	109.00	-

27Y	1 051	1 114	109.05	-
28X	1 052	989	109.10	331.4
28Y	1 052	1 115	109.15	331.25
29X	1 053	990	109.20	-
29Y	1 053	1 116	109.25	-
30X	1 054	991	109.30	332.0
30Y	1 054	1 117	109.35	331.85
31X	1 055	992	109.40	-
31Y	1 055	1 118	109.45	-
32X	1 056	993	109.50	332.6
32Y	1 056	1 119	109.55	332.45
33X	1 057	994	109.60	-
33Y	1 057	1 120	109.65	-
34X	1 058	995	109.70	333.2
34Y	1 058	1 121	109.75	333.05
35X	1 059	996	109.80	-
35Y	1 059	1 122	109.85	-
36X	1 060	997	109.90	333.8
36Y	1 060	1 123	109.95	333.65
37X	1 061	998	110.00	-
37Y	1 061	1 124	110.05	-
38X	1 062	999	110.10	334.4
38Y	1 062	1 125	110.15	334.25
39X	1 063	1 000	110.20	-
39Y	1 063	1 126	110.25	-
40X	1 064	1 001	110.30	335.0
40Y	1 064	1 127	110.35	334.85
41X	1 065	1 002	110.40	-
41Y	1 065	1 128	110.45	-
42X	1 066	1 003	110.50	329.6
42Y	1 066	1 129	110.55	329.45
43X	1 067	1 004	110.60	-
43Y	1 067	1 130	110.65	-
44X	1 068	1 005	110.70	330.2
44Y	1 068	1 131	110.75	330.05
45X	1 069	1 006	110.80	-
45Y	1 069	1 132	110.85	-
46X	1 070	1 007	110.90	330.8
46Y	1 070	1 133	110.95	330.65
47X	1 071	1 008	111.00	-
47Y	1 071	1 134	111.05	-
48X	1 072	1 009	111.10	331.7
48Y	1 072	1 135	111.15	331.55
49X	1 073	1 010	111.20	-
49Y	1 073	1 136	111.25	-
50X	1 074	1 011	111.30	332.3
50Y	1 074	1 137	111.35	332.15
51X	1 075	1 012	111.40	-
51Y	1 075	1 138	111.45	-
52X	1 076	1 013	111.50	332.9
52Y	1 076	1 139	111.55	332.75
53X	1 077	1 014	111.60	-
53Y	1 077	1 140	111.65	-
54X	1 078	1 015	111.70	333.5
54Y	1 078	1 141	111.75	333.35
55X	1 079	1 016	111.80	-
55Y	1 079	1 142	111.85	-

56X	1 080	1 017	111.90	331.1
56Y	1 080	1 143	111.95	330.95
57X	1 081	1 018	112.00	
57Y	1 081	1 144	112.05	
58X	1 082	1 019	112.10	
58Y	1 082	1 145	112.15	
59X	1 083	1 020	112.20	
59Y	1 083	1 146	112.25	
60X	1 084	1 021	-	
60Y	1 084	1 147	-	
61X	1 085	1 022	-	
61Y	1 085	1 148	-	
62X	1 086	1 023	-	
62Y	1 086	1 149	-	
63X	1 087	1 024	-	
63Y	1 087	1 150	-	
64X	1 088	1 151	-	
64Y	1 088	1 025	-	
65X	1 089	1 152	-	
65Y	1 089	1 026	-	
66X	1 090	1 153	-	
66Y	1 090	1 027	-	
67X	1 091	1 154	-	
67Y	1 091	1 028	-	
68X	1 092	1 155	-	
68Y	1 092	1 029	-	
69X	1 093	1 156	-	
69Y	1 093	1 030	-	
70X	1 094	1 157	112.30	
70Y	1 094	1 031	112.35	
71X	1 095	1 158	112.40	
71Y	1 095	1 032	112.45	
72X	1 096	1 159	112.50	
72Y	1 096	1 033	112.55	
73X	1 097	1 160	112.60	
73Y	1 097	1 034	112.65	
74X	1 098	1 161	112.70	
74Y	1 098	1 035	112.75	
75X	1 099	1 162	112.80	
75Y	1 099	1 036	112.85	
76X	1 100	1 163	112.90	
76Y	1 100	1 037	112.95	
77X	1 101	1 164	113.00	
77Y	1 101	1 038	113.05	
78X	1 102	1 165	113.10	
78Y	1 102	1 039	113.15	
79X	1 103	1 166	113.20	
79Y	1 103	1 040	113.25	
80X	1 104	1 167	113.30	
80Y	1 104	1 041	113.35	
81X	1 105	1 168	113.40	
81Y	1 105	1 042	113.45	
82X	1 106	1 169	113.50	
82Y	1 106	1 043	113.55	
83X	1 107	1 170	113.60	
83Y	1 107	1 044	113.65	
84X	1 108	1 171	113.70	

84Y	1 108	1 045	113.75
85X	1 109	1 172	113.80
85Y	1 109	1 046	113.85
86X	1 110	1 173	113.90
86Y	1 110	1 047	113.95
87X	1 111	1 174	114.00
87Y	1 111	1 048	114.05
88X	1 112	1 175	114.10
88Y	1 112	1 049	114.15
89X	1 113	1 176	114.20
89Y	1 113	1 050	114.25
90X	1 114	1 177	114.30
90Y	1 114	1 051	114.35
91X	1 115	1 178	114.40
91Y	1 115	1 052	114.45
92X	1 116	1 179	114.50
92Y	1 116	1 053	114.55
93X	1 117	1 180	114.60
93Y	1 117	1 054	114.65
94X	1 118	1 181	114.70
94Y	1 118	1 055	114.75
95X	1 119	1 182	114.80
95Y	1 119	1 056	114.85
96X	1 120	1 183	114.90
96Y	1 120	1 057	114.95
97X	1 121	1 184	115.00
97Y	1 121	1 058	115.05
98X	1 122	1 185	115.10
98Y	1 122	1 059	115.15
99X	1 123	1 186	115.20
99Y	1 123	1 060	115.25
100X	1 124	1 187	115.30
100Y	1 124	1 061	115.35
101X	1 125	1 188	115.40
101Y	1 125	1 062	115.45
102X	1 126	1 189	115.50
102Y	1 126	1 063	115.55
103X	1 127	1 190	115.60
103Y	1 127	1 064	115.65
104X	1 128	1 191	115.70
104Y	1 128	1 065	115.75
105X	1 129	1 192	115.80
105Y	1 129	1 066	115.85
106X	1 130	1 193	115.90
106Y	1 130	1 067	115.95
107X	1 131	1 194	116.00
107Y	1 131	1 068	116.05
108X	1 132	1 195	116.10
108Y	1 132	1 069	116.15
109X	1 133	1 196	116.20
109Y	1 133	1 070	116.25
110X	1 134	1 197	116.30
110Y	1 134	1 071	116.35
111X	1 135	1 198	116.40
111Y	1 135	1 072	116.45
112X	1 136	1 199	116.50
112Y	1 136	1 073	116.55

113X	1 137	1 200	116.60
113Y	1 137	1 074	116.65
114X	1 138	1 201	116.70
114Y	1 138	1 075	116.75
115X	1 139	1 202	116.80
115Y	1 139	1 076	116.85
116X	1 140	1 203	116.90
116Y	1 140	1 077	116.95
117X	1 141	1 204	117.00
117Y	1 141	1 078	117.05
118X	1 142	1 205	117.10
118Y	1 142	1 079	117.15
119X	1 143	1 206	117.20
119Y	1 143	1 080	117.25
120X	1 144	1 207	117.30
120Y	1 144	1 081	117.35
121X	1 145	1 208	117.40
121Y	1 145	1 082	117.45
122X	1 146	1 209	117.50
122Y	1 146	1 083	117.55
123X	1 147	1 210	117.60
123Y	1 147	1 084	117.65
124X	1 148	1 211	117.70
124Y	1 148	1 085	117.75
125X	1 149	1 212	117.80
125Y	1 149	1 086	117.85
126X	1 150	1 213	117.90
126Y	1 150	1 087	117.95

DME transponder transmits IMC coded identification signal which consists of a series of paired pulses transmitted for an appropriate period of time at a repetition rate of 1 350 Hz, which means a 740 μ s separation between consecutive ident pulse pairs. The transmission of identification repeats approximately every 30 seconds so that the pilots can identify the selected ground DME station.

Since the identification of ground-based DME station is very important, the highest priority in the emission of DME signals is given to identification. The second priority is given to the reply signals, and the third priority to squitter pulses. Consequently the transmission of identification signal temporarily replaces all reply pulses that would normally occur during that key-down time. These identification pulses have similar characteristics to the other pulses of the reply signals.

1.3. EQUIPMENT DESCRIPTION

MOPIENS 300 DME is type “N” distance measuring equipment, primarily serving operational needs of en-route or TMA navigation, where the “N” stands for narrow spectrum characteristics.

The DME system provides for continuous and accurate indication in the cockpit of the slant range distance of an equipped aircraft from an equipped ground reference point.

MOPIENS 300 DME is fully compliant with ICAO Annex 10 and EUROCAE ED-57 minimum performance specification.

MOPIENS 300 DME is designed to be fully solid state design with state-of-the-art surface mount technology. All materials, workmanship and tests are in accordance with the highest standards for this class of equipment.

Figure 1-4 shows the equipment cabinet of MOPIENS 300 DME



Figure 1-4. Equipment Cabinet

1.3.1. OVERVIEW

Key features of MOPIENS 300 DME are as follows:

Compact Design

Full dual high-power transponder, dual monitor and dual backup batteries are included in an industry standard 19" rack cabinet for easy accessibility and maintenance. The equipment is fully solid-state design with no mechanically moving parts.

Modular Construction

Construction with plug-in or easily replaceable sub-assemblies (i.e., no special tools required) is used throughout the equipment in order to provide the specified level of maintainability. Modularization is based on logical functional block concepts. Design is such as to minimize the cost and number of different types of modules required for equipment support. Modules are

printed circuit boards whenever practical from the standpoint of component size weight and consistent with circuit performance requirements.

Hot-swappable Plug-in Units

Most of the system hardware components are line replaceable units, which are designed to be hot-swappable plug-in modules. Therefore operators can replace a live unit without powering off, and this makes it easier to carry out routine preventive maintenance service.

Modernized microprocessor-based digital control

All the system functions are monitored and controlled by high performance 16/32-bit microcontrollers. Most of the pulse waveforms are electronically synthesized using state-of-the-art direct digital synthesis technology.

Non-volatile memory is used for storing all the system configuration, adjustments, and calibration data that any system data will not be damaged or lost after any accidental power outage or scheduled maintenance.

Long design life & durability

- Minimum 15 years of design life through normal and regular maintenance
- Designed to operate on 24 hours a day and 365 days per year basis
- Protection against lightning and switching transients
- Protection from EMI radiation, high voltage, etc.

Built-in Test Equipment

The Built-In Test Equipment (BITE) is included to check the integrity of system operation. The Built-In Self-Test function is independent from the monitoring operation

Full Local and Remote Control/Monitor

By using high performance microprocessor based control/monitor circuitry, full local and remote monitoring and maintenance as well as fault finding from remote are possible with detailed system status including each LRU status. Also up to two slave indicators (RSU) can be connected for providing status indicator.

Collocation with any VOR or ILS

By using software programmable PLL frequency synthesizer and flexible IDENT keying interface, a collocation with any VOR or ILS equipment on any operational frequency/channel pair can be easily achieved. Both coaxial collocation and offset allocation can be implemented by means of programmable delay offset adjustment.

Wideband Operation

The equipment is completely wideband that it can operate on any DME channel without any module change or modification. The channel or the identification assigned to the station can be easily changed on site by software operation and no tuning or adjustment is required.

1.3.2. GENERAL DESIGN

The dual equipment, MOPIENS 300 DME system comprises two DME/N transponders, two independent monitors and two AC/DC power supplies with backup batteries.

The transponders can be configured as dual system capable of working as main and standby. Any of the system can be designated as main. The other automatically works as standby. The

changeover between the main and standby can be either initiated by an operator command or automatically by the monitor system when an executive alarm condition is detected.

The monitors can be configured such that both monitors are monitoring the main transponder simultaneously or such that one monitor is monitoring the main transponder and the other is monitoring the standby transponder. When two monitors are monitoring the main transponder, they can be configured either in 'AND' mode or in 'OR' mode for a changeover or a complete shutdown in the event of failure.

The two power supplies work in parallel sharing load currents in normal condition. When one of the power supplies fails, the other power supply continues to provide enough power for the dual system with "no break" operation.

The power supplies also charge the backup batteries. When the mains power is interrupted, the backup batteries supply the power for the system with "no-break" operation.

1.4. EQUIPMENT IDENTIFICATION

The 300 DME is part of a family of comprehensive and latest-generation radio navigation equipment. The versions share many features, modules and principles of operation, and differ only slightly in frequency and purpose. All of the versions are available in a single or dual equipment configuration and can be used with a set of antenna configurations also. A single equipment version is denoted with a suffix -S, preceded by the type number, i.e. 310S DME refers to a low power, single equipment version DME.

- 310S DME: a low-power (100 W), single equipment DME
- 320S DME: a high-power (1 kW), single equipment DME
- 310 DME: a low-power (100 W), dual equipment DME
- 320 DME: a high-power (1 kW), dual equipment DME

1.5. COMPOSITION AND IDENTIFICATION

The Table 1-1 shows the various units and the related reference labels of the 300 DME for single and dual equipment configurations. The features and function of the units are common to all configurations with only difference in the quantity of some specific units.

Table 1-3. Reference Labels

Unit	Label	Low Power		High Power	
		310S	310	320S	320
AC to DC Converter Unit p/n: 3151-0016	AC/DC	1	2	1	2
Control Status Panel p/n: 0330-7560	CSP	1	1	1	1
DC to DC Converter Unit p/n: 0330-7710	DC/DC	1	2	1	2
DC to DC Converter Unit-A p/n: 0130-7720	DC/DC-A	1	2	1	2
Duplexer - Mute Switch, Combiner p/n: 0330-7120	DPX-MSC	1	2	1	2
Environment Monitoring Unit p/n: 0130-7540	EMU	(1)	(1)	(1)	(1)
Interface Board p/n: 0130-3410	IFB	1	1	1	1

Local Maintenance Interface p/n: 0130-7530	LMI	1	1	1	1
Receiver Unit p/n: 0330-7180	RXU	1	2	1	2
Transponder Control Unit p/n: 0330-7570	TCU	1	2	1	2
Transmitter Unit p/n: 0330-7140	TXU	1	2	1	2
High Power Amplifier p/n: 0332-7150	HPA	0	0	1	2
Modem p/n: 0130-7420	Modem	(1)	(2)	(1)	(2)
Monitor Unit p/n: 0330-7550	MON	1	2	1	2
Radio Frequency Generator p/n: 0330-7190	RFG	1	2	1	2
Power Management Unit p/n: 0130-7730	PMU	1	2	1	2
System Controller Unit p/n: 0130-7520	SCU	1	2	1	2

*The number in () is optional.

1.6. PHYSICAL DESCRIPTION

The 300 DME subsystems are housed in an enclosed, floor-mounted type cabinets.

The cabinet material is durable and non-corrodible. Major assemblies or LRUs are designed to be completely removable from their enclosures without disassembly. The cabinet has two doors. The front door provides access to the front panel, LRUs and all the other assemblies of the subsystem. The rear door provides access to the back of the backplane, cables and connectors of the assemblies. With the front door open, all the maintenance operations can be performed from the front of the equipment. The locking, hinged door protects the front of the cabinet.

The front control panel is on the front door of the cabinet. On the front-door, the LMI is mounted to provide a quick access to the system maintenance as well as the basic system control and status information.

Figure 1-5 shows the equipment cabinet front and rear view.

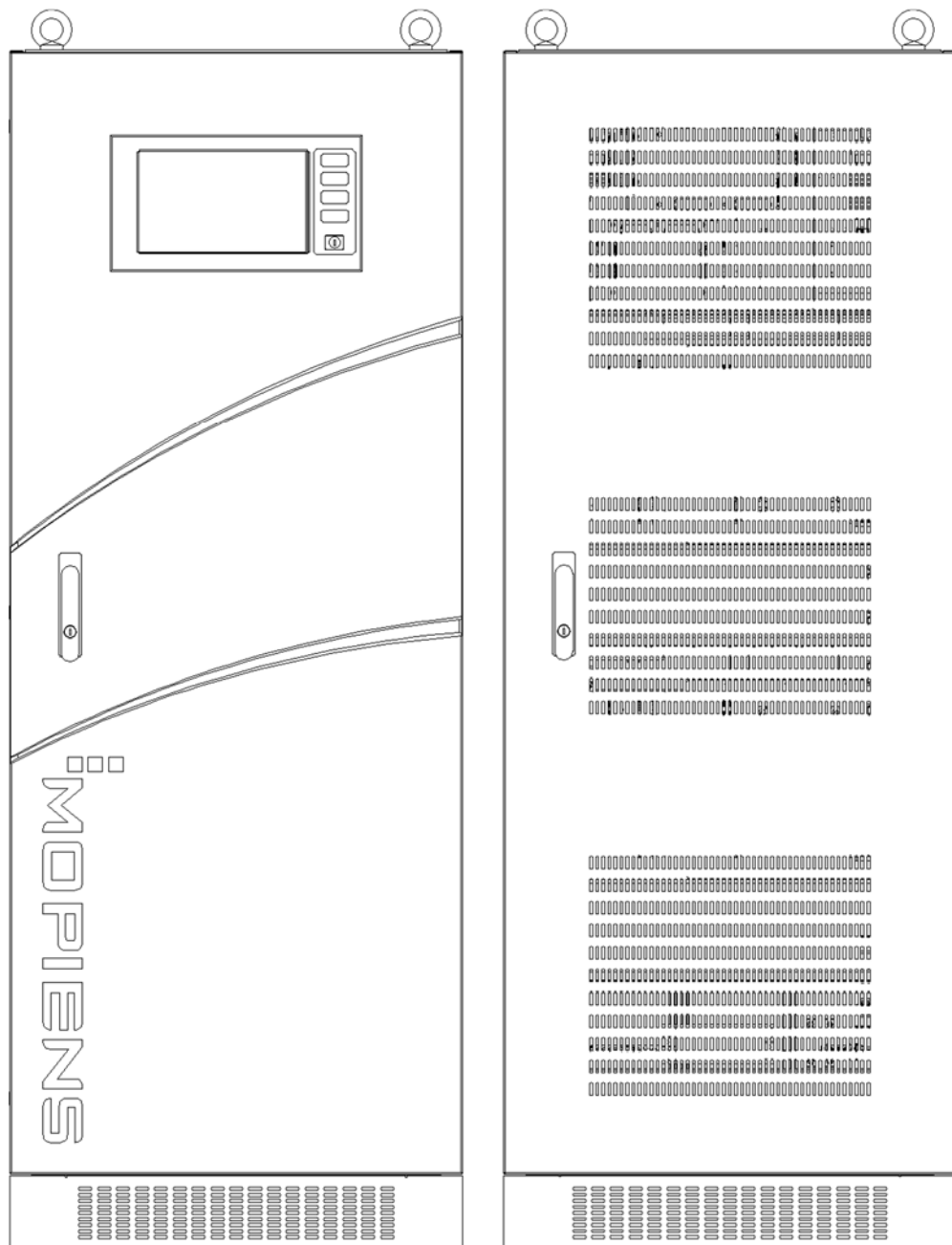


Figure 1-5. Equipment Cabinet, front and rear view

The cabinet is self-ventilated. It has got perforated metal plate on top and bottom so no forced ventilation is required. The top of the cabinet has four threaded holes for screwing in the eyebolts that are used to lift the equipment.

The standard configuration of the 300 DME is composed of two transmitters, a dual monitor, a dual power supply and control and monitoring unit. The equipment assemblies are housed in a single 19-inch standard cabinet. The steel cabinet can accommodate three assembled sub-racks and plug-in 6-U or 3-U Eurocard printed circuit boards with depth of 200 mm or 6-U size modules with metal shield case of 300 mm depth.

Each module or unit is designed to be plugged-in to each designated slot. If it is inserted into a wrong place, it does not fit as a special keying-mechanism prevents it for protection. The modules

and units with printed circuit boards has hot swap and hot plug-in circuits and mechanism so that it can be plugged and unplugged even when the equipment is powered on.

The RF output connector to the antenna distribution network and the monitor combining network are located on top of the cabinet. The RF components of the modules are enclosed in metal housings.

The RF components of the modules are enclosed in metal housings for prevention of electronic crosstalk and electro-magnetic interferences (EMI) as well as for protection of human body. Thanks to such design, it does not cause interference to the equipment and systems nearby airport and has no harmful effect on human body and equipment.

Figure 1-6 and Figure 1-8 show the interior layout of 300 DME equipment cabinets respectively, showing the location of the major LRUs and modules.

Each sub-rack can be unscrewed and pulled out. All the LRUs are plug-in types with either ejectors or captive screws. Each LRU slides and can be easily pulled out by its handles or ejectors.

1.7. MECHANICAL DESCRIPTION

The 300 DME equipment comprises of a 19" standard rack cabinet. The Figure 1-6 and Figure 1-8 show the location of the LRUs and modules in 300 DME. The figures refer to dual equipment configuration which equips two transponders for redundancy.

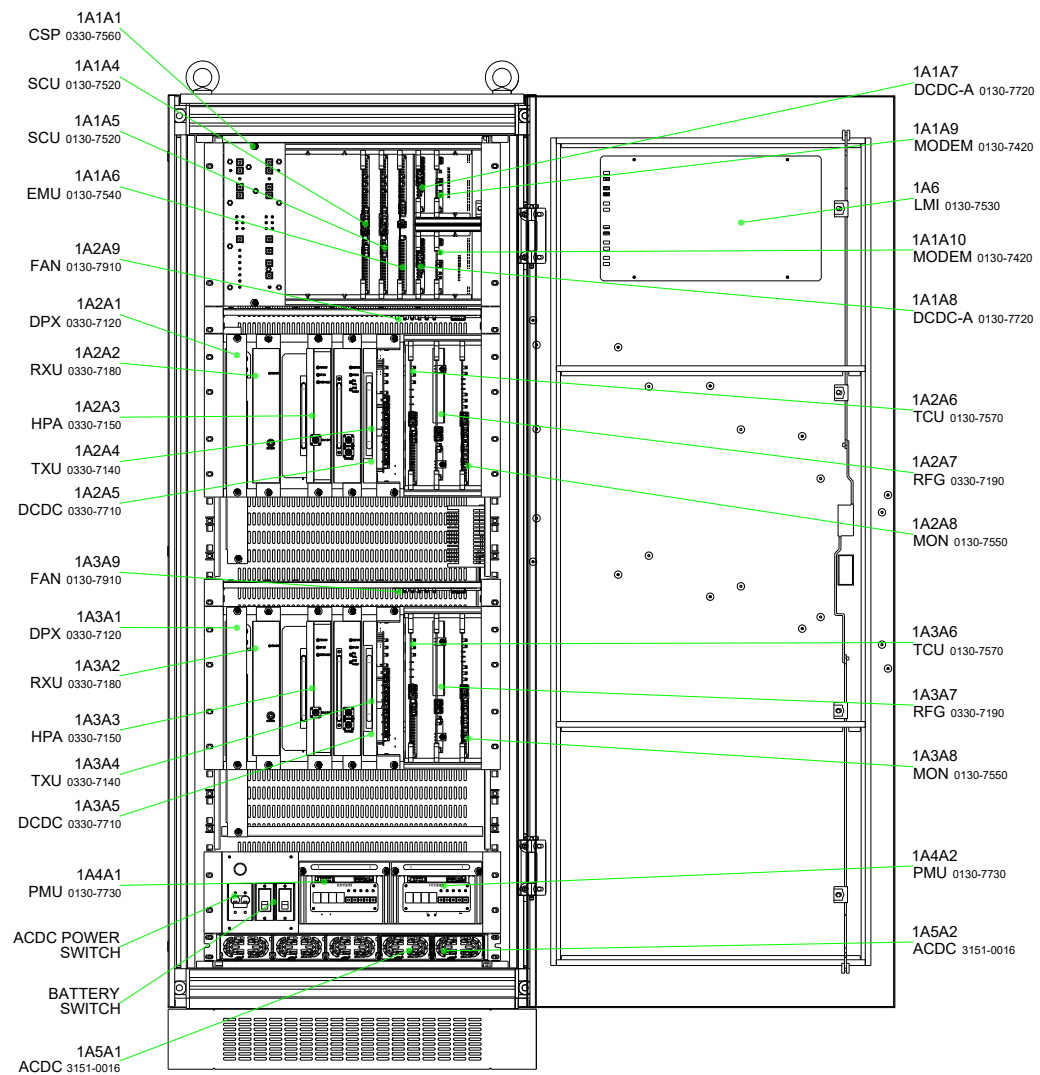


Figure 1-6. Equipment Cabinet, front view; door open

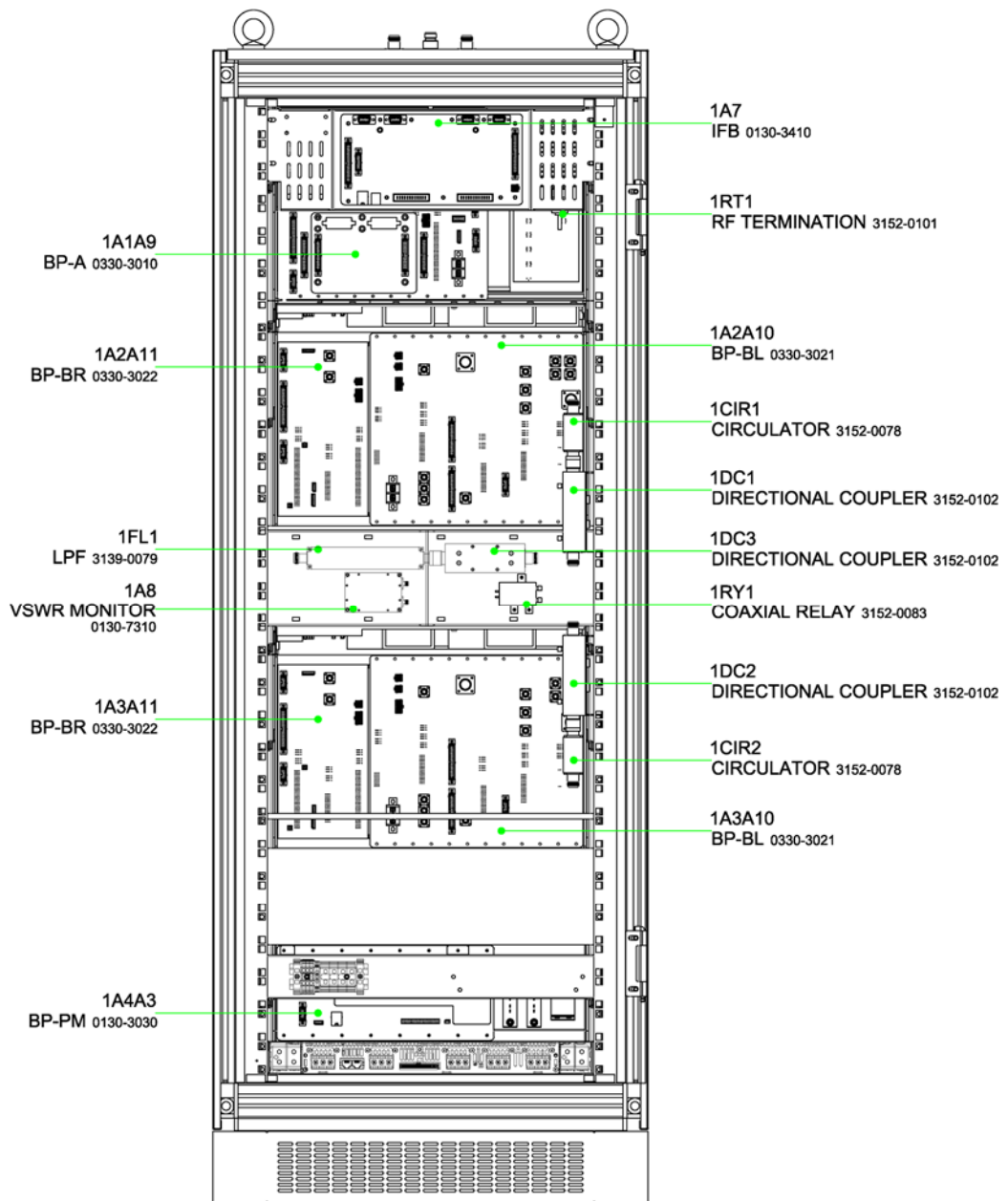


Figure 1-7. Equipment Cabinet, Rev. 0, rear view; door detached

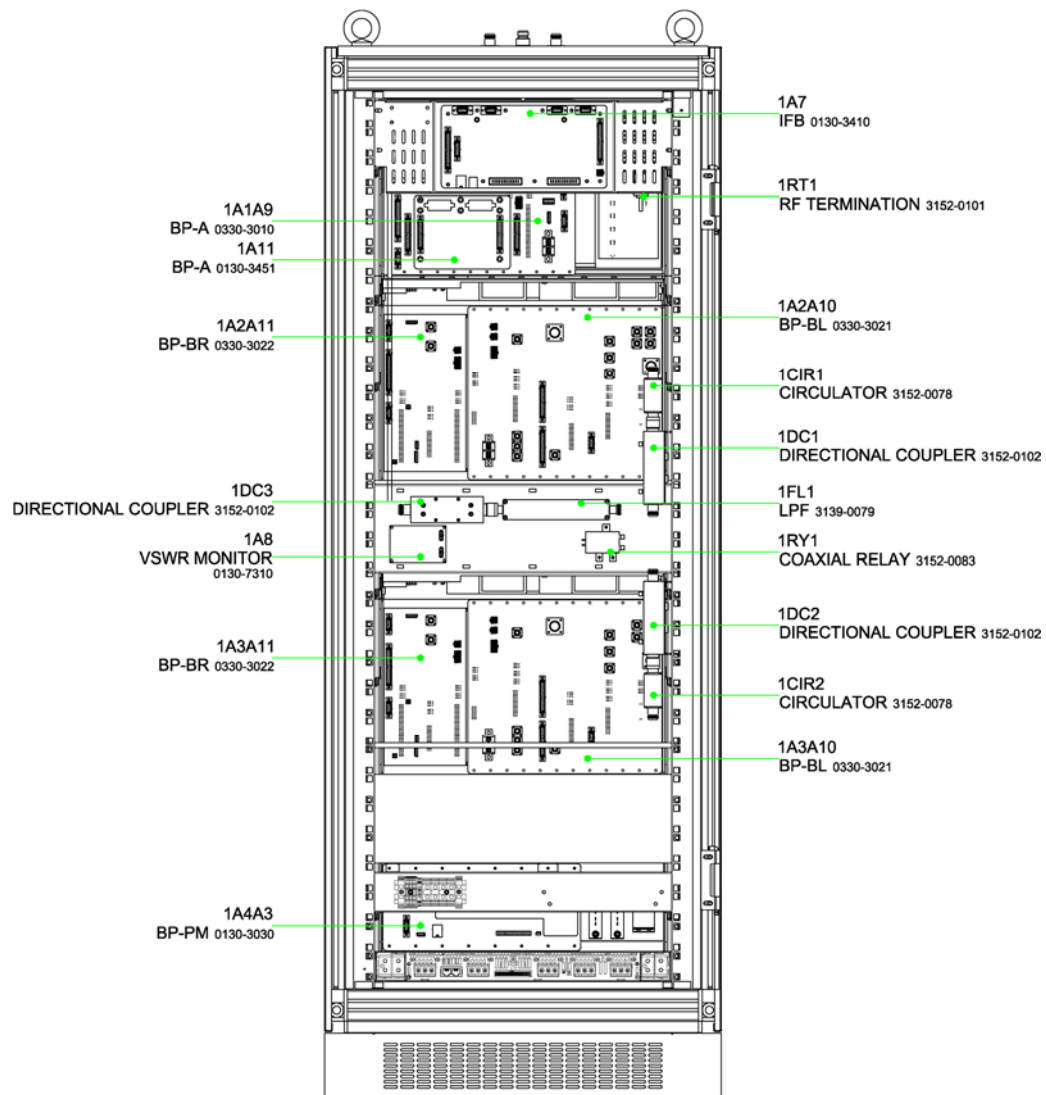


Figure 1-8. Equipment Cabinet, Rev. A, rear view; door detached

1.8. FUNCTIONAL DESCRIPTION

1.8.1. OVERVIEW

The 300 DME equipment consists of dual transponders and monitors whereby each monitor performs monitoring for both transponders. The hardware is based on RF and digital assemblies and the software that controls them.

1.8.2. TRANSPONDER

The transponder receives and validates the interrogation signals and transmits reply pulse pair if the integration signal is valid. The random squitter pulses are transmitted in order to maintain the minimum pulse rate. Identity pulse pairs are transmitted for identification of ground DME stations.

The transponder can be either single or dual redundant equipment. In case of dual redundant equipment, one transponder is designated as the main transponder and the other transponder designated as the standby transponder. While main transponder is radiating through the antenna, the standby transponder is either off or on the dummy load. If the monitor senses a monitor action

condition, the control circuit causes the main transponder turned off and the standby transmitter takes over.

The transponder consists of RXU, TCU, TXU, HPA (for 320 DME only), DC/DC, FAN, T/R DPX-MSC, Changeover relays and etc.

1.8.3. MONITOR

The Monitor can be either a single or dual redundant system. Two monitors watch for the active main transponder and the standby transponder simultaneously.

It generates alarms in case of transmitter or antenna system failure. The monitor analyzes the DME parameters such as time delay, pulse spacing, reply efficiency and voltage, current, temperature of subrack and compares with operator programmed threshold values. If any parameter is beyond the tolerance values, an alarm is generated.

This system provides fault detection by injecting test interrogation signals and monitoring the replies from the transponder. It also initiates changeover to standby transponder or shutdown of transponder in the event of an out-of-tolerance condition.

If dual monitor system is used, the voting logic determines the action to take. The 'AND' configuration requires both monitors sense an alarm condition in order to take the monitor action. In the 'OR' configuration, only one of the either monitor requires to report an alarm condition to cause the monitor action.

The monitor consists of the MON unit, envelope detector, and RFG unit. The baseband pulse processing is handled by the MON unit and the RF signal processing is done in the envelope detector and the RFG. The test interrogation carrier signal is generated and pulse-modulated by the RFG.

1.8.4. POWER SUPPLY

The Power Supply system consists of:

- *AC/DC converter*, which takes 110V/220V AC mains power input and converts it into 28V DC. The 28V power source is fed to the multi-voltage DC/DC converters and also used to charge external backup batteries.
- *DC/DC converter*, which receives +28V DC input from either AC/DC or Battery backup source and converts to the output voltages of +5V, +8V, +15V, -15V, +28V and +50V which is used for VOR/DME only.
- *DC/DC-A converter* to provide power to the auxiliary electronics subrack by converting +28 V DC to +5V, +15V and -15V output voltages.
- *Power Management Unit (PMU)*, which monitors the output voltages, current, temperature and battery status.
- *Backup batteries*, which provide DC 24V and are connected in parallel with the AC/DC converter outputs to provide uninterrupted power supply.

The AC/DC power supply units are placed at the bottom of the cabinet. The primary power source of the equipment is the commercial AC line power (mains). For uninterrupted continuous operation, a secondary power source by backup batteries is supported. The primary power and secondary power run in parallel without switching so that "no-break" operation is possible.

1.8.5. CONTROL AND MONITORING

The Control and Monitoring subsystem of the 300 DME provides system status information and allows monitoring functions to be performed locally and remotely. The major building blocks of Control and Monitoring subsystem are the SCU, CSP, LMI, IFB and EMU.

The System Controller Unit (SCU) provides control and monitoring of all system units. The SCU monitors the alarm/fault signal lines from all internal units. Also it exchanges data with some units such as TCU, MON, PMU, EMU and external components such as RCU and PMDT. The Control Area Network (CAN) bus is used for communication with internal units. The RCU and PMDT exchange data with SCU via RS 232C, modem or Ethernet interface.

When the SCU senses alarm signals from MON units, it initiates a transfer of the equipment or shutdown. In case of a dual equipment system, the voting logic in the SCU determines the monitor action to take based on the preconfigured option 'AND' or 'OR'. In case of 'AND' configuration, the transfer or shutdown action takes place only when both monitors signal alarm status at the same time, while either one of the MON units triggers the action in case of 'OR' configuration. For increased reliability, emergency terminator logic is included in the SCU, which shuts down the system if the voting logic is malfunctioning.

The LMI provides the system status and control using a combination of LCD and touchscreen panel. Touchscreen-based, graphic user interface software running on the LMI provides simple, easy-to-use and intuitive access to the system status and equipment control. The LMI also provides a message routing service for remote control and maintenance devices via a USB port and an Ethernet LAN port which supports TCP/IP protocol. With these interfaces, the operator can connect a PC running the PMDT software to view equipment status and alarm history, set up system parameters, and perform troubleshooting either at local or at remote.

The CSP consists of push buttons and light emitting diode (LED) indicators using which the system status and configuration can be displayed and modified.

The IFB provides surge protection (transient voltage suppression) and physical interfaces to various units such as LMI, EMU, remote voice, RCU and PMDT via RS-232, USB, modem (telephone line) and Ethernet.

The equipment is protected from any unexpected incoming high voltages by means of IFB. Digital input and output signals are photo coupled for galvanic insulation from outside. Also in case both TX1 and TX2 shutdown, the SCU is directed to block the +28V DC power being supplied to DUCU.

The EMU provides interface to various sensors such as smoke, temperature and intrusion sensor using which it monitors the environmental conditions that may affect continuous operation of the equipment. Also user-definable, analog and digital expansion input/output is provided. The EMU sends the information on the environmental conditions and the expansion I/O data to SCU via CAN bus.

1.9. PMDT

1.9.1. GENERAL

The Portable Maintenance Data Terminal (PMDT) is an input/output device, which enables the user to communicate with the transponder and monitor of DME equipment.

The PMDT consists of the PMDT software with a laptop computer (or a desktop PC with a monitor display as a substitute for laptop computer), a printer (optional), and etc. For convenient

handling of laptop computer, an external mouse may be used as an option. With the PMDT, the maintenance personnel are able to not only easily monitor and control the equipment but also diagnose, troubleshoot and calibrate the equipment. Each PMDT is designed to individually connect to the equipment to work on.

The PMDT software is based on IBM PC compatible computers and Microsoft Windows operating system. It provides an intuitive and easy-to-use graphic user interface (GUI) for the maintenance of the navigation aids equipment.

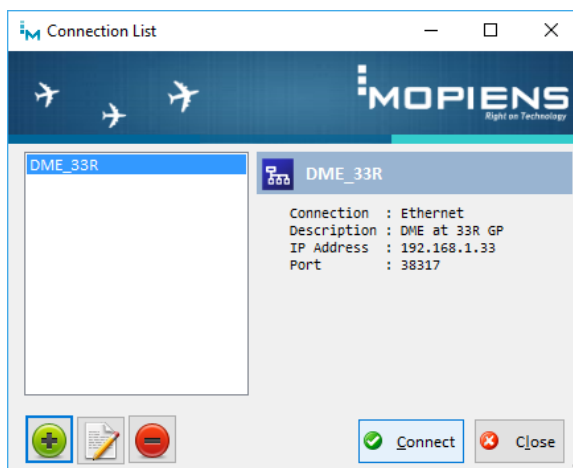


Figure 1-9. PMDT screenshot showing connection setup

The PMDT can be connected to the equipment either locally or remotely over a suitable data link. The functions listed in 1.9.2 below are equally available to both locally connected PMDT and remotely connected PMDT.

The PMDT provides a log-on security to protect the system from an unauthorized access. By the access control, it is ensured that only the authorized personnel can get access to the equipment.

1.9.2. FUNCTIONS

Functions that the PMDT provides include, but not limited to, the following:

- Selection of Main/Standby transmitter
- Transmitter On/Off
- Changeover between Main/Standby transmitters
- Bypass of Monitors for maintenance
- Control and adjustments of transmitter and power supply
- Control and adjustment of monitors
- Setup and configuration of system parameters
- Access to the Built-In Test Equipment (BITE) and identifying faulty units or antennas
- Readout and printout of system parameters at present
- Printout of transmitter and monitor status
- Data logging of alarms, warnings, system parameter values, etc.
- Flight check/inspection and calibration support

With the PMDT, it is possible to control or adjust the standby transmitter, monitor or power supply while the main transmitter is in normal operation. The results of the maintenance works are recorded and saved into a file on the computer storage. The data logging is performed on real time basis. The size of the data logging is only limited by the capacity of the computer storage.

1.9.3. SECURITY

Access to the system is controlled by security levels assigned to the users: i.e. level 1 for read-only, level 2 for read-only plus limited control of non-critical parameters, and level 3 for complete access. Only level-3 user is allowed to create and delete the user ID account.

The built-in security system enforces the following policies by default:

- The user ID consists of up to 16 alpha-numeric characters. The user ID is case insensitive.
- The password consists of up to 16 alpha-numeric or special characters. The password is case-sensitive.
- The access control list (ACL) which contains the IDs and passwords are kept in a protected storage area and updated on real-time basis when the administrator makes changes on it.
- Failed log-on attempts are recorded with its timestamp and relevant information and can be notified to the administrator.

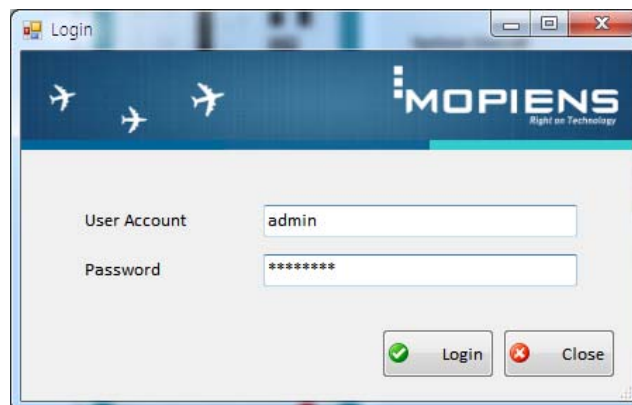


Figure 1-10. PMDT screenshot showing login

1.10. ANTENNA SYSTEM

1.10.1. GENERAL

The interrogation signals from airborne interrogators are received and the reply pulse pairs from the transponder are transmitted by the DME antenna. The radiation from DME antenna is vertically polarized.

Typically the DME antenna has one or two monitor pick-up probes to sample the radiated signal which is to be used for monitoring the transmitted signal.

Typically omnidirectional antenna is used when the DME is used in stand-alone or collocated with a VOR. Optionally directional sector antenna can be used when the DME is used in association with an ILS system. For the stand-alone DME, a tower is provided for installation next to the equipment shelter. If the DME is to be collocated with VOR system, appropriate

mounting adapter kits may be furnished for a coaxial installation above the VOR antenna.

MOPIENS-supplied DME antenna system is housed in a weatherproofing fiberglass radome to withstand adverse environmental conditions like extreme temperatures, fog, rain, snow, ice, salinity and etc. An obstruction light assembly with two incandescent lamps is furnished for installation above the DME antenna radome.

1.11. REMOTE CONTROL AND STATUS MONITORING

1.11.1. GENERAL

The 300 DME system can provide indications at designated remote control points of the operational status of ground system components. For this purpose, the 600 Remote Control System is installed at the designated remote control points.

The 600 RCS provides visual indications, when systems are operating normally, provides audible and visual alerts when alarm conditions exist, and when communication faults exist. The aural alert can be stopped by pressing the SILENCE button. The visual indicators and aural alert can be fully activated by pressing LAMP TEST button to verify its integrity.

The RCS is capable of displays and remote control of an ILS, VOR, DME, FFM, or a combination of all.

The 600 RCS can contain one or a combination of the following optional kits; the DME (single or dual) interface kit, Localizer / Glides Path (single or dual) interface kit, Marker Beacon (single or dual) interface kit, VOR (single or dual) interface kit, and FFM interface kit for facility performance category III localizer.

The 600 RCS consists of two distinct units: 610 Remote Control Unit (RCU) and 620 Remote Status Unit (RSU).

1.11.2. REMOTE CONTROL UNIT

The remote control unit provides the following capabilities:

- to transfer to standby equipment, if system is a dual equipment system;
- to turn the system ON and OFF;
- to designate one transponder as main and the other as standby;
- to manually transfer the equipment from the main transponder to standby transponder;
- to indicate the status of environmental sensors such as fire, intrusion or temperature;
- to indicate whether AC power or backup battery is in operation;
- to reset alarm status and put the transmitter into initial state

The 610 RCU contains an equipment cabinet assembly, a backplane, and a power supply.

The RCU is generally located in the remote maintenance equipment room for monitoring and control of the navigational aids equipment.

The RCU has a built-in dedicated line modem and a generic RS-232C serial interface. Also it supports various types of remote access including UTP copper cable, fiber-optic and radio-link via TCP/IP Ethernet LAN port.



Figure 1-11. MOPIENS 610 Remote Control Unit, standard version



Figure 1-12. MOPIENS 610 Recmote Control Unit, mini version

1.11.3. REMOTE STATUS UNIT

The 620 RSU processes information which is transformed into a visual and audible indication of the operating status of the systems being monitored. The equipment status is indicated as normal, warning, or alarm.

The 620 RSU contains an equipment cabinet assembly, a backplane, and a power supply.

The RSU is generally located in the control tower for monitoring of the ILS/VOR/DME and connected to the RCU via RS-485 link.



Figure 1-13. MOPIENS 620 Remote Status Unit, standard version



Figure 1-14. MOPIENS 620 Remote Status Unit, mini version

SECTION 2

TECHNICAL SPECIFICATIONS

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2. TECHNICAL SPECIFICATIONS

2.1. APPLICABLE REFERENCE DOCUMENTS

Applicable reference documents for the technical specifications of MOPIENS 310/320 DME are listed below.

- International Civil Aviation Organization (ICAO), International Standards and Recommended Practices, Aeronautical Telecommunications, Annex 10 to the Convention on International Civil Aviation, Radio Navigational Aids, Volume I, Seventh edition, July 2018
- International Civil Aviation Organization (ICAO), *Manual on Testing of Radio Navigation Aids, Testing of Ground-Based Radio Navigation Systems*, Doc. 8071, Volume 1, Fifth Edition, 2018
- The European Organisation for Civil Aviation Electronics (EUROCAE), Minimum Operational Performance Specification for Distance Measurement Equipment (DME/N and DME/P) (GROUND EQUIPMENT), ED-57, Edition 2, October 26, 1992.
- United States Department of Transportation, Federal Aviation Administration (FAA), *Performance Specification Distance Measuring Equipment (DME)*, FAA-E-2996, April 1, 2008

2.2. ENVIRONMENTAL SERVICE CONDITIONS

The 310/320 DME environmental service conditions are listed below.

For Indoor Equipment

- Temperature -10°C to +55°C
- Relative Humidity up to 95% for temperature range -10°C to +35°C
up to 60% for temperature range +35°C to +55°C
- Atmospheric Pressure 840 hectopascal

For Outdoor Equipment

- Temperature -40°C to +70°C
- Relative Humidity up to 95% for temperature range up to +35°C
up to 60% for temperature range up to +70°C
- Wind Velocity up to 45 m/sec, average,
up to 62 m/sec, instantaneous maximum
- Icing encased in 1.27 cm thick clear ice

2.3. PHYSICAL CHARACTERISTICS

- Housing 19" Standard rack cabinet
- Dimension 600 mm (w) × 600 mm (d) × 1600 mm (h)
- Weight < 120 kg
- Transponder Output Impedance 50 Ω nominal
- Monitor Input Impedance 50 Ω nominal

2.3.1. EQUIPMENT VERSIONS

There are four versions of DME available in the MOPIENS 300 DME.

- 310 DME – Low-power (100 W), dual-equipment version
- 320 DME – High-power (1000 W), dual-equipment version
- 310S DME – Low-power (100 W), single-equipment version
- 320S DME – High-power (1000 W), single-equipment version

Single equipment versions and dual equipment versions differ as listed below.-

- Single Equipment: a standard single transponder, single monitor, single or dual AC/DC power supply
- Dual Equipment: fully-redundant dual transponder with automatic changeover function, dual monitors, dual power supply configuration

2.4. SYSTEM FEATURES

2.4.1. GENERAL

The DME system provides for continuous and accurate indication in the cockpit of the slant range distance of an equipped aircraft from an equipped ground reference point.

The system comprises two basic components, one fitted in the aircraft, the other installed on the ground. The aircraft component is referred to as the interrogator and the ground component as the transponder.

In operation, interrogators interrogate transponders which, in turn, transmit to the interrogator replies synchronized with the interrogations, thus providing means for accurate measurement of distance.

The MOPIENS 310/320 DME is a DME/N transponder system, which is the ground component, primarily serving operational needs of en-route or TMA navigation, where the “N” stands for narrow spectrum characteristics.

2.4.2. COVERAGE

With a combination of suitable antenna system properly installed at normal site, the coverage limit is up to 200 NM for the 1 kW power output. The actual coverage is dependent upon site location, terrain, and aircraft altitude. The peak pulse power density within the service range and level is at least -83 dBW/m².

The DME system is capable of providing enough peak equivalent isotopically radiated power that required to ensure peak pulse power density of -89 dBW/m² under all operational weather conditions at any point within the coverage.

2.4.3. ACCURACY

The total system limits include errors from all causes such as those from airborne equipment, ground equipment, propagation, and random pulse interference effects. As the 310/320 DME is the ground equipment, only the transponder contribution is concerned and specified in §2.7.3 on a 95% probability basis.

2.4.4. RADIO FREQUENCIES AND POLARIZATION

The system operates with vertical polarization in the frequency band 960 MHz to 1,215 MHz. The interrogation and reply frequencies are assigned with 1 MHz spacing between channels.

A total of 126 channels is available for each X and Y channel.

2.4.5. CHANNELING

DME operating channels are formed by pairing interrogation and reply frequencies and by pulse coding on the paired frequencies.

2.4.6. AIRCRAFT HANDLING CAPACITY

The aircraft handling capacity (also referred to as traffic handling capacity or simply traffic capacity) of the 310/320 DME transponder is adequate for peak traffic of 200 interrogators with following assumptions:

95% of the interrogators in track mode where an average interrogation rate is 30 pulse pairs per second or lower

5% of the interrogators in search mode where an average interrogation rate is 150 pulse pairs per second or lower

Average transponder reply efficiency of 70%

The 310/320 DME provides slant range adequate for peak traffic load of 5,400 transmissions per second including the identification and squitter pulses.

2.4.7. TRANSPONDER IDENTIFICATION

MOPIENS 310/320 DME transponder transmits identification signal coded (dotted and dashed) with International Morse Codes (IMC) approximately every 30 seconds so that the pilots can identify the selected ground DME station. For each dot and dash period, a series of paired pulses are transmitted at a repetition rate of 1,350 Hz.

The identification of ground-based DME station is so important that the highest priority is given to identification in the emission of DME signals. The second priority is given to the reply signals, and the third priority to squitter pulses.

Consequently the transmission of identification signal temporarily replaces all reply pulses that would normally occur during that key-down time. This is accomplished in the way that the identification code inhibits the replies and the squitter pulses as long as the dots and dashes continue. And the decoded pulses inhibit the squitter pulses at the encoder input until the reply is transmitted.

Reply pulses are transmitted between code letters, and also between dots and/or dashes, i.e. between key down times.

The DME can transmit an identification signal in one of the following forms:

- An “independent” identification consisting of coded International Morse Code identity pulses which can be used with all transponders.
- An “associated” signal which can be used for transponders specifically associated with a VHF navigation or an MLS angle guidance facility which itself transmits an identification signal.

Optionally, an equalizing pair of pulses, having the same characteristics as the identification pulse pairs, can be transmitted $100 \pm 10 \mu\text{s}$ after each identity pair, if it is desired to preserve a constant duty cycle.

The characteristics of the “independent” identification signal are as follows:

- The identity signal consists of the transmission of the beacon code in the form of dots and dashes (International Morse Code) of identity pulses at least once every 40 seconds, at a rate of at least 6 words per minutes; and
- The identification code characteristic and letter rate for the DME transponder conforms to the following to ensure that the maximum total key down time does not exceed 5 seconds per identification group.
 - The time duration of a dot is 0.1 second to 0.160 second.
 - The time duration of a dash is three times the dot.
 - The time duration between dots and/or dashes is equal to that of one dot plus or

minus 10 per cent.

- The time duration between letters or numerals is not less than three dots.
- The total period of transmission of an identification code group does not exceed 10 seconds.

The characteristics of the “associated” signal are as follows:

- When associated with a VHF or an MLS angle facility, the identification is transmitted in the form of dots and dashes (International Morse Code) and synchronized with the VHF facility identification code;
- Each 40-second interval is divided into four or more equal periods, with the transponder identification transmitted during one period only and the associated VHF and MLS angle facility identification, where these are provided, transmitted during the remaining periods;

Normally, the “independent” identification code is employed wherever a transponder is not specifically associated with a VHF navigational facility or an MLS facility. Wherever a transponder is specifically associated with a VHF navigational facility or an MLS facility, identification is provided by the “associated” code. When voice communications are being radiated on an associated VHF navigational facility, an “associated” signal from the transponder is not suppressed.

2.5. TRANSMITTER

2.5.1. FREQUENCY

The transponder transmits on the reply frequency appropriate to the assigned DME channel. The radio frequency of operation does not vary more than $\pm 0.0005\%$ from the assigned frequency.

The transmitter frequency band is between 960 MHz and 1215 MHz. A total of 126 channels is available for each X and Y channel.

The frequency is controlled by a PLL frequency synthesizer driven by a precision temperature-controlled crystal oscillator (TCXO) clock source with ± 2.5 ppm stability characteristics.

2.5.2. PULSE SHAPE

Pulse rise time does not exceed 3 μs .

Pulse duration is 3.5 $\mu\text{s} \pm 0.5 \mu\text{s}$.

Pulse decay time is normally 2.5 μs but does not exceed 3.5 μs .

The instantaneous amplitude of the pulse does not, at any instant between the point of the leading edge which is 95% of maximum amplitude and the point of the trailing edge which is 95% of the maximum amplitude, fall below a value which is 95% of the maximum voltage amplitude of the pulse.

2.5.3. PULSE SPECTRUM

The spectrum of the pulse modulated signal is such that during the pulse the effective radiated power contained in a 0.5 MHz band centered on frequencies 0.8 MHz above and 0.8 MHz below the nominal channel frequency in each case does not exceed 200 mW, and the effective radiated power contained in a 0.5 MHz band centered on frequencies 2 MHz above and 2 MHz below the nominal channel frequency in each case does not exceed 2 mW. The effective radiated power contained within any 0.5 MHz band decreases monotonically as the band center frequency moves away from the nominal channel frequency.

The instantaneous magnitude of any pulse turn-on transients which occur in time prior to the virtual origin is less than 1% of the peak amplitude. This is to ensure proper operation of the thresholding techniques. Initiation of the turn-on process does not commence sooner than 1 μs prior to the virtual origin.

2.5.4. PULSE SPACING

The spacing of the constituent pulses of transmitted pulse pairs is as follows:

Channel suffix	Interrogation pulse spacing	Reply pulse spacing
X	12 μs	12 μs
Y	36 μs	30 μs

The tolerance on the pulse spacing is $\pm 0.10 \mu\text{s}$.

2.5.5. PEAK POWER OUTPUT

The nominal peak power output at the connector on the equipment cabinet is 100 W for the 310 DME and 1,000 W for the 320 DME. With a combination of suitable antenna system, this gives sufficient peak equivalent isotopically radiated power to ensure a peak pulse power density of approximately - 83 dBW/m² at the maximum specified service range and level. The peak power of the constituent pulses of any pair of pulses does not differ by more than 1 dB.

The reply capability of the transmitter is such that the transponder is capable of continuous operation at a transmission rate of $5,400 \pm 180$ pulse pairs per second.

2.5.6. SQUITTER PULSES

The transmitter operates at a transmission rate, including randomly distributed pulse pairs and distance reply pulse pairs, of not less than 700 pulse pairs per second except during identity. The minimum transmission rate is 700 pulse pairs per second. Even in the absence of interrogation signals from aircraft, the squitter pulse pairs are generated and transmitted for the purpose of stabilizing AGC circuits of the aircraft interrogator.

The time spacing between squitter pulse pairs is random. The spacing is not less than 200 μ s but not in the range of 730 to 750 μ s which is typical spacing for identification code repetition.

2.5.7. SPURIOUS RADIATION

During the intervals between transmission of individual pulses, the spurious power received and measured in a receiver having the same characteristics as transponder receiver, but tuned to any DME interrogation or reply frequency, is more than 50 dB below the peak pulse power received and measured in the same receiver tuned to the reply frequency in use during the transmission of the required pulses. This specification refers to all spurious transmissions, including modulator and electrical interference.

The spurious power level specified herein is more than 80 dB below the peak pulse power level. (Ref: ICAO Annex 10 ¶3.5.4.1.6.1)

Out-of-band Spurious Radiation

At all frequencies from 10 to 1800 MHz, but excluding the band of frequencies from 960 to 1215 MHz, the spurious output of the DME transponder transmitter does not exceed -40 dBm in any one kHz of receiver bandwidth.

2.6. RECEIVER

2.6.1. FREQUENCY

The receiver center frequency is the interrogation frequency appropriate to the assigned DME operating channel.

The center frequency of the receiver does not vary more than $\pm 0.001\%$ from the assigned frequency.

The receiver frequency band is between 1025 MHz and 1150 MHz. A total of 126 channels is available for each X and Y channel.

The frequency is controlled by a PLL frequency synthesizer driven by a precision temperature-controlled crystal oscillator (TCXO) clock source.

2.6.2. TRANSPONDER SENSITIVITY

In the absence of all interrogation pulse pairs, with the exception of those necessary to perform the sensitivity measurement, interrogation pulse pairs with the correct spacing and nominal frequency trigger the transponder if the peak power level of the interrogation signal at the RF connector on the equipment cabinet is at least -91 dBm and this cause the transponder to reply with an efficiency of at least 70%.

When a properly installed, suitable antenna system is used, this signal level is equivalent to the peak power density of approximately -103 dBW/m² at the transponder antenna.

2.6.3. DYNAMIC RANGE

The performance of the transponder properly installed in combination with suitable antenna system can be maintained when the power density of the interrogation signal at the transponder antenna has any value between -103 dBW/m² up to a maximum of -22 dBW/m² when installed with ILS or MLS and -35 dBW/m² when installed for other applications.

The transponder sensitivity level does not vary by more than 1 dB for transponder loadings between 0% and 90% of its maximum transmission rate.

When the spacing of an interrogator pulse pair varies from the nominal value by up to $\pm 1 \mu\text{s}$, the receiver sensitivity is not reduced by more than 1 dB.

2.6.4. LOAD LIMITING

When transponder loading exceeds 90% of the maximum transmission rate, the receiver sensitivity is automatically reduced in order to limit the transponder replies, so as to ensure that the maximum permissible transmission rate is not exceeded.

The available range of sensitivity reduction is more than 50 dB.

2.6.5. NOISE-GENERATED PULSE PAIRS

When the receiver is interrogated at the power densities of -103 dBW/m² to produce a transmission rate equal to 90% of the maximum, the noise-generated pulse pairs does not exceed 5% of the maximum transmission rate.

2.6.6. BANDWIDTH

The bandwidth of the receiver is such that the transponder sensitivity level does not deteriorate by more than 3 dB when the total receiver drift is added to an incoming interrogation frequency drift of ± 100 kHz.

The receiver bandwidth is sufficient to allow compliance with the accuracy specifications when the pulse shape and spectrum of the input signals are those specified in 2.5.2 and 2.5.3 above.

2.6.7. OFF-CHANNEL AND SPURIOUS RESPONSE

Signals greater than 900 kHz removed from the desired channel nominal frequency and having power densities up to the value -22 dBW/m² (when installed with ILS) or -35 dBW/m² (when installed for other applications) do not trigger the transponder.

Signals arriving at the intermediate frequency (IF) is suppressed at least 80 dB. All other spurious response or signals within the 960 MHz to 1215 MHz band and image frequencies are suppressed at least 75 dB.

2.6.8. RECOVERY TIME

Within 8 μ s of the reception of a signal between 0 dB and 60 dB above minimum sensitivity level, the minimum sensitivity level of the transponder to a desired signal is within 3 dB of the value obtained in the absence of signals.

This specification is met with echo suppression circuits rendered inoperative. The 8 μ s are to be measured between the half voltage points on the leading edges of the two signals, both of which conform in shape, with the specifications in 2.5.2 above.

2.6.9. CW AND ECHO SUPPRESSION

- Short Distance Echo Suppression (SDES):

If enabled, echo pulses that occur between the pulses of a valid interrogation pair do not affect the reply timing by more than 0.15 μ s. The presence of such synchronous pulse signals does not cause the reply efficiency to be reduced by more than 10% from the measured in the absence of the pulse signals.

- Long Distance Echo Suppression (LDES):

If enabled, echo pulses that fall after the dead time (60 μ s) interval are suppressed. The duration for LDES can be set between 0 μ s and 300 μ s.

The presence of CW interference signal on the assigned channel frequency or elsewhere within the receiver pass-band does not reduce the on-channel threshold sensitivity by more than 2 dB when the level of CW is 10 dB and more below the on-channel sensitivity level in the absence of CW interference.

Additionally, within the range of the receiver desensitization provided by automatic gain reduction the reply efficiency to a single aircraft interrogation is not reduced by more than 10% when the level of the interrogation signal is 6 dB and more above the level of the interfering CW signal.

2.7. TRANSPONDER

2.7.1. DECODING

The transponder includes a decoding circuit such that the transponder can be triggered only by pairs of received pulses having pulse duration and pulse spacings appropriate to interrogator signals.

The decoding circuit performance is not affected by signals arriving before, between, or after, the constituent pulses of a pair of the correct spacing.

An interrogation pulse pair with a spacing of $\pm 2 \mu\text{s}$, or more, from the nominal value and with any signal level up to the value of -22 dBW/m^2 is rejected such that the transmission rate does not exceed the value obtained when interrogations are absent.

2.7.2. TIME DELAY

The time delay is the interval from the half voltage point on the leading edge of the first constituent pulse of the interrogation pair and the half voltage point on the leading edge of the first constituent pulse of the reply transmission. This delay is consistent with the following table:

Channel suffix	Pulse pair spacing		Time delay	
	Interrogation	Reply	1st pulse timing	2nd pulse timing
X	12 μs	12 μs	50 μs	50 μs
Y	36 μs	30 μs	56 μs	50 μs

The time delay is digitally controlled and adjustable from 35 μs to 75 μs by software.

The stability of time delay is within $\pm 0.5 \mu\text{s}$ for the interrogation signal level -10 dBm to -81 dBm and within $\pm 0.8 \mu\text{s}$ for the interrogation signal level -81 dBm to -91 dBm .

2.7.3. ACCURACY

When the system is installed and adjusted properly, the transponder does not contribute more than $\pm 0.5 \mu\text{s}$ (75 m) to the overall system error if associated with instrument landing system, and not more than $\pm 1 \mu\text{s}$ (150 m) for other application.

The contribution of the ground component to the overall system error may include the transponder errors, transponder location coordinate errors, propagation effects and random pulse effects. The combination of those errors can be managed not to contribute more than $\pm 185 \text{ m}$ (0.1 NM) to the overall system error.

2.7.4. REPLY EFFICIENCY

The transponder reply efficiency is at least 70% at all values of the transponder loading up to 200 interrogators and at the minimum sensitivity level of -103 dBW/m^2 .

2.7.5. TRANSPONDER DEAD TIME

The transponder is rendered inoperative for a period normally not to exceed 60 μs after a valid interrogation decode has occurred. In extreme cases when the geographical site of the transponder is such as to produce undesirable reflection problems, the dead time may be increased but only by the minimum amount necessary to allow the suppression of long distance echoes.

2.8. MONITORING AND CONTROL

2.8.1. ALARM CONDITIONS

The conditions that cause the system to initiate monitor action vary depending on the installation and configuration setup. The following basic conditions are specified in ICAO Annex 10:

- The transponder delay differs from the assigned value by 1 μ s (150 m) or more;
- In the case where the DME is associated with a landing aid, the transponder delay differs from the assigned value by 0.5 μ s (75 m) or more.
- The spacing between the first and second pulse of the transponder pulse pair differs from the nominal value by 1 μ s or more.

These conditions are fully implemented with the 310/320 DME. The actual alarm conditions that trigger monitoring actions can be programmed by software in detail. Monitored parameters are categorized as primary or secondary depending on the severity of the effect that occurs if outside specification. Primary parameters are those parameters which, if outside specification, may be the cause of false guidance or a degradation of accuracy. Other monitored parameters are defined as secondary. The conditions listed above are primary parameters.

Alarms can be triggered in the 310/320 DME if the following conditions are detected:

- The time delay deviates outside the preset tolerance range. The tolerance range can be set 30 μ s to 80 μ s in 0.01 μ s resolution. This is a primary parameter.
- Time (spacing) between the constituent pulses of a pulse pair deviates outside the preset tolerance range. The tolerance range can be set 5 μ s to 40 μ s in 0.01 μ s resolution. This is a primary parameter.
- The reply efficiency falls below preset limit. The limit can be set from 0% to 100% in 1% resolution. This is a secondary parameter but can be escalated to primary by software.
- There is a reduction in the transmitter power output to below preset value. The limit can be set from 20 W to 150 W in 0.1 W resolution for 310 DME and 200 W to 1500 W in 1W resolution for 320 DME. This is a secondary parameter but can be escalated to primary by software.
- The pulse pair transmission rate falls below a preset lower limit value or exceeds a preset upper limit value. The limit can be set from 0 to 6000 pp/s. This is a secondary parameter but can be escalated to primary by software.
- Identification code group lasts more than a preset period of time or there is no identification in any period of a preset value. The time value can be set up to 30 seconds in 0.1 seconds resolution. This is a secondary parameter but can be escalated to primary by software.
- The effective radiated power (ERP) drops by more than a preset limit. The threshold can be programmed by software. The ERP alarm threshold is typically 3 dB.

2.8.2. MONITORING ACTIONS

According to the ICAO Annex 10, in the event that any of the conditions specified in 2.8.1 above, which is categorized as primary parameters, the monitor should cause the following action to take place:

- A suitable indication is given at a control point;
- The operating transponder is automatically switched off; and

- The standby transponder, if configured, is automatically placed in operation.

By default, the alarms by both primary and secondary parameters give both aural and visual alert locally on the CSP, LMI and PMDT if connected as well as RSU, RCU and PMDT if connected at remote site.

The primary parameter alarms cause the operating transponder automatically switched off and the standby transponder, if configured, is automatically placed in operation. The secondary parameters by default do not trigger the transfer or shutdown action. However the secondary parameters can be escalated to primary by software so that the transfer or shutdown action can be initiated.

2.8.3. ALERT CONDITIONS

According to ICAO Annex 10, the monitor should cause a suitable indication to be given at a control point if any of the following conditions arise:

- A fall of 3 dB or more in transponder transmitted power output;
- A fall of 6 dB or more in the minimum transponder receiver sensitivity (provided that this is not due to the action of the receiver automatic gain reduction circuits);
- The spacing between the first and second pulse of the transponder reply pulse pair differs from the normal value by 1 μ s or more;
- Variation of the transponder receiver and transmitter frequencies beyond the control range of the reference circuits.

Except for the spacing, these conditions are secondary parameter by default but can be escalated to primary.

(추가: pulse duration, decay time, rise time, transmission rate, frequency, VSWR as secondary by default)

2.8.4. ALARM DELAY

A configurable delay timer is provided so that any of the condition and malfunctioning enumerated above which are monitored can persist for a certain period before the monitor takes action. This period is software adjustable.

2.8.5. MONITOR INTERROGATION

The transponder is not triggered more than 120 times per second or preset value for either monitoring or automatic frequency control purposes, or both.

Baseband monitor interrogation pulse pairs for testing the transponder are generated by the monitor unit and sent to the RF generator unit (RFG) which generates a suitable RF carrier signal, performs pulse modulation and signal conditioning.

The characteristics of the monitor interrogation pulse pairs are:

- | | |
|-------------------------|---|
| • Output Frequency | Adjustable 960 MHz to 1215 MHz
in 100 kHz step |
| • Output Level | Adjustable -95 dBm to -10 dBm in 0.1 dB step |
| • Pulse Spacing | Adjustable 10 μ s to 49 μ s in 0.1 μ s step |
| • Pulse Repetition Rate | Adjustable 0 pp/s to 5400 pp/s |

2.8.6. MONITOR FAILSAFE

Failure of any part of the monitor itself automatically produces the same results as the malfunctioning of the element being monitored.

2.9. POWER SUPPLY

2.9.1. POWER INPUT

- AC Input Voltage 85 to 265 V; 47 Hz to 63 Hz
- DC Input Voltage 24 V nominal, 18 V to 36 V

2.9.2. POWER CONSUMPTION

- 310 DME 190 W (cold-standby mode)
230 W (hot-standby mode) or less
- 320 DME 450 W (cold-standby mode)
700 W (hot-standby mode) or less
- excluding battery charging current.

2.9.3. BACKUP BATTERY

- Action on power interruption “No-break” – continuous operation
- Backup capacity 4 hours of full hot-standby operation
with the standard 65 Ah battery bank.
Longer backup option is available.

2.10. RELIABILITY AND MAINTAINABILITY

2.10.1. RELIABILITY

The DME system in single equipment configuration has a mean time between failures (MTBF) of not less than 8760 hours as a serial reliability model, as per MIL-STD-217F.

The mean time between outage (MTBO) of the dual equipment DME system is longer than 20,000 hours.

2.10.2. MAINTAINABILITY

The mean time to isolate and replace a faulty Line Replaceable Unit (LRU) does not exceed 30 minutes. In no case the time required to isolate and replace a faulty LRU exceeds 120 minutes with the assumptions that a single maintenance technician performs the repair; the technician has had adequate equipment-specific training; and, the repair work is performed under normal conditions in the equipment shelter.

SECTION 3

TECHNICAL DESCRIPTION

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3. TECHNICAL DESCRIPTION

3.1. SYSTEM DESCRIPTION

3.1.1. OVERVIEW

Figure 3-1 shows a simplified system block diagram of the 300 DME with dual transponder and dual monitor configuration.

- Transponder
- Monitor
- Control and Monitoring
- Power Supply

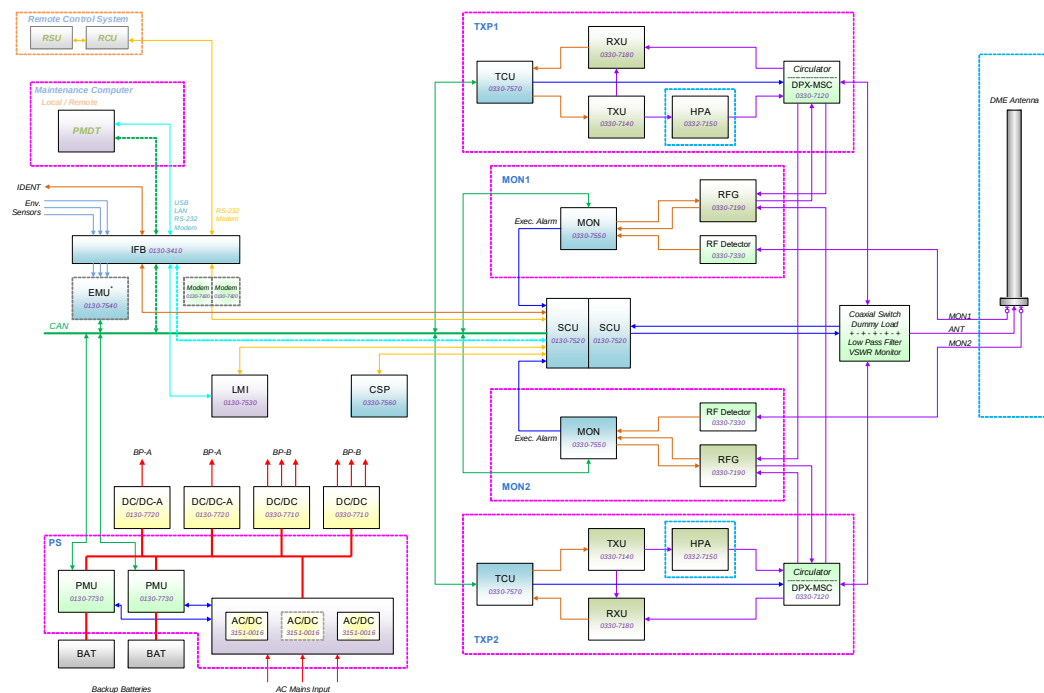


Figure 3-1. System Block Diagram

3.1.2. AUXILIARY ELECTRONICS SUBRACK 1A1

The Auxiliary electronics subrack, 1A1, contains the following LRUs:

- A1 – Control Status Panel, CSP
- A2, A3 – System Controller Unit, SCU
- A4 – Environmental Monitor Unit, EMU
- A5, A6 – DC-to-DC Converter for Aux, DC/DC-A
- A7, A8 – Modem
- A9 – Backplane A, BP-A

3.1.3. TRANSPONDER/MONITOR SUBRACK 1A2, 1A3

The Transponder/Monitor subrack consists of the following LRU's.

Low Power-DME:

- A1 – Duplexer – Mute Switch / Coupler, DPX-MS
- A2 – Receiver Unit, RXU
- A4 – Transmitter Unit, TXU
- A5 – DC-to-DC converter, DC/DC
- A6 – Transponder Controller Unit, TCU
- A7 – Backplane B Left, BP-BL
- A8 – Backplane B Right, BP-BR
- A9 – Cooling fan, FAN

High Power DME:

- A1 – Duplexer – Mute Switch / Coupler, DPX-MS
- A2 – Receiver Unit, RXU
- A3 – High Power Amplifier, HPA
- A4 – Transmitter Unit, TXU
- A5 – DC-to-DC converter, DC/DC
- A6 – Transponder Controller Unit
- A7 – Backplane B Left, BP-BL
- A8 – Backplane B Right, BP-BR
- A9 – Cooling fan, FAN

3.1.4. POWER MANAGEMENT UNIT SUBRACK 1A4

The Power Management Unit subrack consists of PMU LRU:

- A1, A2 – Power Management Unit, PMU
- A3 – Backplane, BP-PM

3.1.5. AC/DC CONVERTER SUBRACK 1A5

Up to 5 AC to DC converter units which provide 28 VDC, 1600 W each, can be plugged in although only two units are plugged, which still provide more than enough capacity. The AC/DC units used for ILS equipment are readily available Commercial-off-the-shelf (COTS) units placed at the bottom shelf of the equipment below the power On/Off switch and PMUs.

- A1, A2, A3, A4, A5 – AC/DC Converter, AC/DC
- A6 – Rack, HFE1600-S1U-TB



Figure 3-3. DPX-MSC

The signal attenuation of the transmit signal is obtained by shorting out the receiver input while the transmitter is sending out pulse pairs. For this, single pole single throw RF switches (mute switch) using PIN diodes are employed. The PIN diodes are turned on or off by the control signal from the TCU. When the PIN diodes are forward-biased, it forms a low impedance signal path between the receiver input and the signal ground. When the PIN diodes are reverse-biased, it turns into high impedance circuit and the received signal flows through the mute switch and gets to the receiver input.

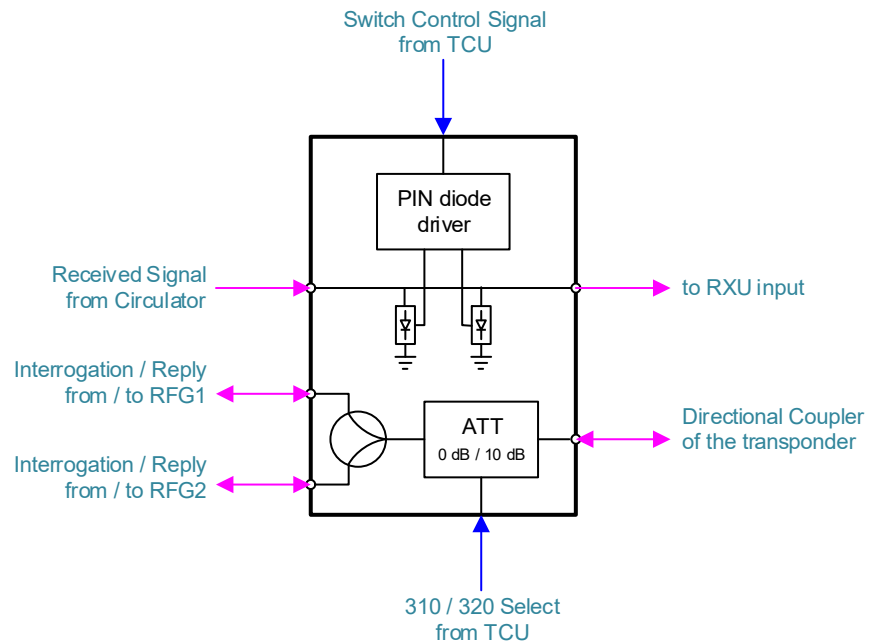


Figure 3-4. DPX-MSC Block Diagram

Besides the mute switch, DPX-MSC contains a two-way splitter and switchable attenuator.

The two-way splitter is to feed the test interrogation signals from RFG1 and RFG2 to the receiver and to distribute the reply signals from the transponder back to the detectors in RFG1 and RFG2. The actual signal injection is taken place at the directional coupler of the transponder.

The switchable 10 dB attenuator is activated by the select signal from the TCU only for the 320 DME, 1 kW version. It is disabled for the 310 DME, 100 W version, in which case the signal is pass-through without any attenuation. By this way, the signal level at the detectors on the RFGs remain relatively constant regardless of whether 310 DME or 320 DME.

3.2.3. RXU *1A2A2, 1A3A2*

OVERVIEW

The RXU provides the following functions:

- Detection and recovery of incoming interrogation pulse pair
- Discrimination of ON channel and OFF channel
- Adjusting receiver sensitivity



Figure 3-5. RXU

DESCRIPTION

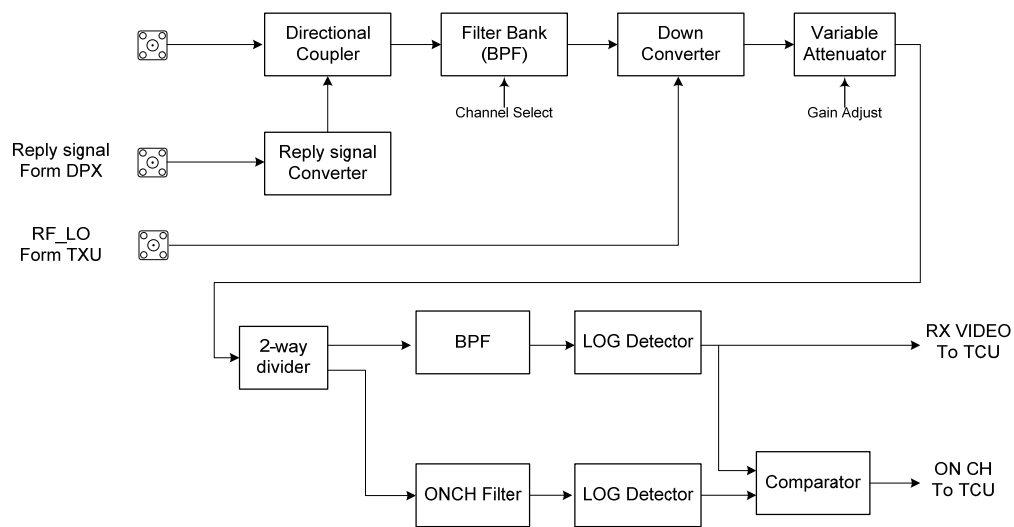


Figure 3-6. RXU Simplified Block Diagram

The received signal which was separated from transmit signal at DPX-MSC is delivered to the RXU via the backplane.

Inside the RXU, the signal first passes through a directional coupler and goes into the band pass filter bank.

The reply signal converter is a frequency conversion mixer to translate the transmit signal, which is always 63 MHz higher or lower than the receive frequency, into the receive frequency. A crystal controlled, precision 63 MHz oscillator is used as a local oscillator. By mixing with this local oscillator signal, the transmit signal sample is frequency converted to the receiver frequency and then injected to the receiver via the directional coupler. As a result, the transmit signal waveform will be appeared along with the interrogation signal waveform at the RX VIDEO test point. This is useful for the purpose of test and maintenance. This feature is can be on/off controlled by TCU.

The band pass filter bank is a programmable pre-selector, which cuts away the undesired image frequency response inherent to a super heterodyne receiver. The filter is built around four groups of dielectric resonator filters which have 30 MHz passband but with four different mid frequencies. Each group of the band pass filters consists of a front filter F2, F4, F6 or F8, a low noise amplifier U1, U2, U5 or U7 and a tail filter FL1, FL3, FL5 or FL7. Programmable, four-way (“SP4T”) RF switches U3 and U4 are incorporated to select a particular pair of the filters that are best fit for the station frequency. The filter bank suppresses the image response by more than 70 dB.

The RF mixer U16 converts the preselected RF signal, which is in the frequency range of between 1025 MHz and 1150 MHz, down to 63 MHz intermediate frequency. A high dynamic range double balanced type mixer is used for this purpose. Following the frequency converter, LC low pass filter removes unwanted high side frequency components. The local oscillator signal is a CW signal with the frequency same as that of transmitter and supplied from the TXU.

RF variable attenuators U18, U19 attenuate the received signal to desensitize the receiver in case the load level is over a threshold point set by TCU. The attenuation range is from 0 dB to 63 dB.

The signal from the attenuators is fed to the first IF filter FL9 which suppress out-of-channel response, specifically outside 1 MHz bandwidth from the mid frequency. The signal is then split into two paths: one is the main signal path to recover the baseband pulse waveform known as

“RX VIDEO” and the other is a sub signal path to discriminate on-channel from off-channel.

The received signal can be on the valid interrogation frequency assigned to the station, i.e. ON channel or it might be off the station channel frequency i.e. “OFF channel” which is on an adjacent channel frequency and possibly strong enough to pass through the IF filters. The RXU provides “ON CH” signal to the TCU to indicate that the signal on “RX VIDEO” at given moment is valid on-channel input and it needs to reply.

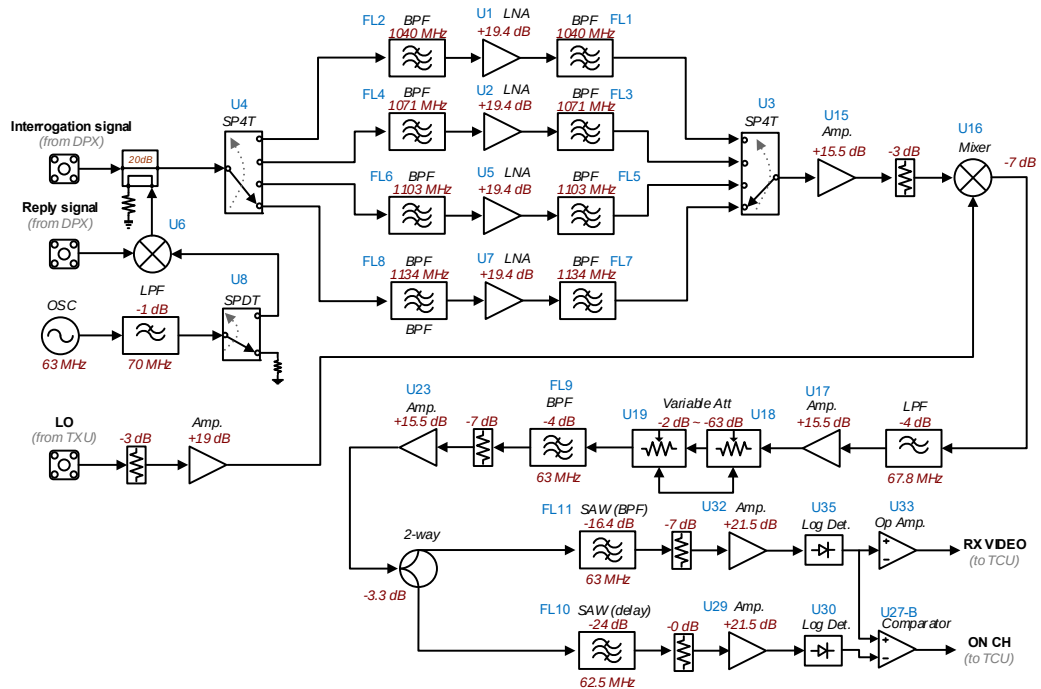


Figure 3-7. RXU Block Diagram

The main path IF signal passes through the second IF filter FL11 which is a SAW band pass filter with passband of 0.9 MHz (-3 dB bandwidth). Most of the selectivity is obtained by FL11. The signal is then amplified and demodulated by the logarithmic amplifier/detector U35. The demodulated signal is logarithmically compressed video envelope of the interrogation pulse pair, aka. “RX LOG VIDEO” and finally sent to the TCU for decode and reply processing.

The sub path IF signal is processed to discriminate “ON channel” signal from “OFF channel” signal. The signal passes through a SAW filter FL10 which has the same 63 MHz mid frequency but much larger passband than the main path. It is to equalize the sub path delay same as to the main path delay. Then the signal is amplified and demodulated by the logarithmic amplifier/detector U30. Same the main path, the demodulated is also logarithmically compressed video envelope of the interrogation pulse pair but sub path video come out of wider bandwidth filter than the main path. Therefore the off-channel signals pass through the sub path without attenuation, but they are attenuated by the filter FL11 when passing through the main path. However the on channel signals pass through both the main path and the sub path without any attenuation. By comparing two outputs from the main path and sub path, it is possible to discriminate the on-channel signals from off-channel signals.

The BNC connectors on the front panel are test points that can be used to measure the interrogation signal coming into the DME.

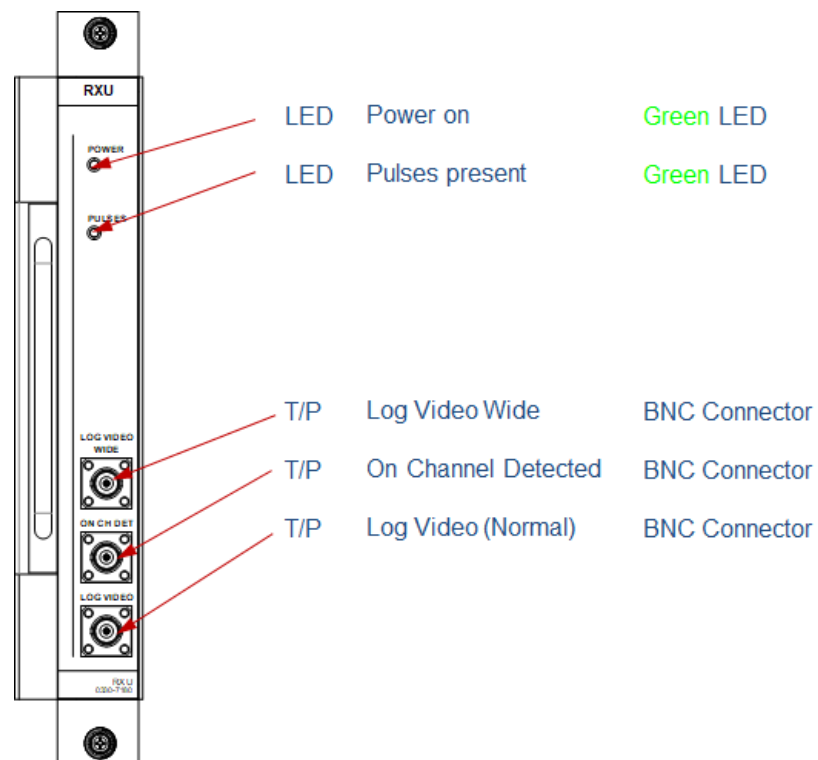


Figure 3-8. Indicators and Test points on RXU

3.2.4. HPA 1A2A3, 1A3A3

OVERVIEW

HPA is equipped only in MOPIENS 320 DME, which is 1 kW output version. The HPA provides the following functions:

- High-power amplification of DME response signal
- Additional Gaussian shape pulse modulation
- Adjustment of power output
- Monitoring of inside temperature
- Detection of RF output power



Figure 3-9. HPA

DESCRIPTION

The HPA takes input of peak power 44.5 dBm from the output of TXU and amplifies it with maximum gain of up to 17.5 dB. To maintain stable constant amplitude, a sample of output power is detected and fed back to TCU for digital control of output amplitude.

The HPA comprises two stages of amplification. The first stage is driven by a bipolar transistor operating as class C. Besides the modulation in TXU, additional pulse modulation is applied to the first stage of HPA. The final stage is driven by four transistors combined in parallel operating as class C.

Like TXU, the final output signal is fed to an isolator for protection against possible mismatching and the resulting reflected power.

Supervisory circuits similar to that of TXU are provided including a digital temperature sensor

for monitoring inside temperature, a directional coupler, an envelope detector circuits for detecting RF output power.

A hot-swap controller controls the power supply line for safe unit swapping.

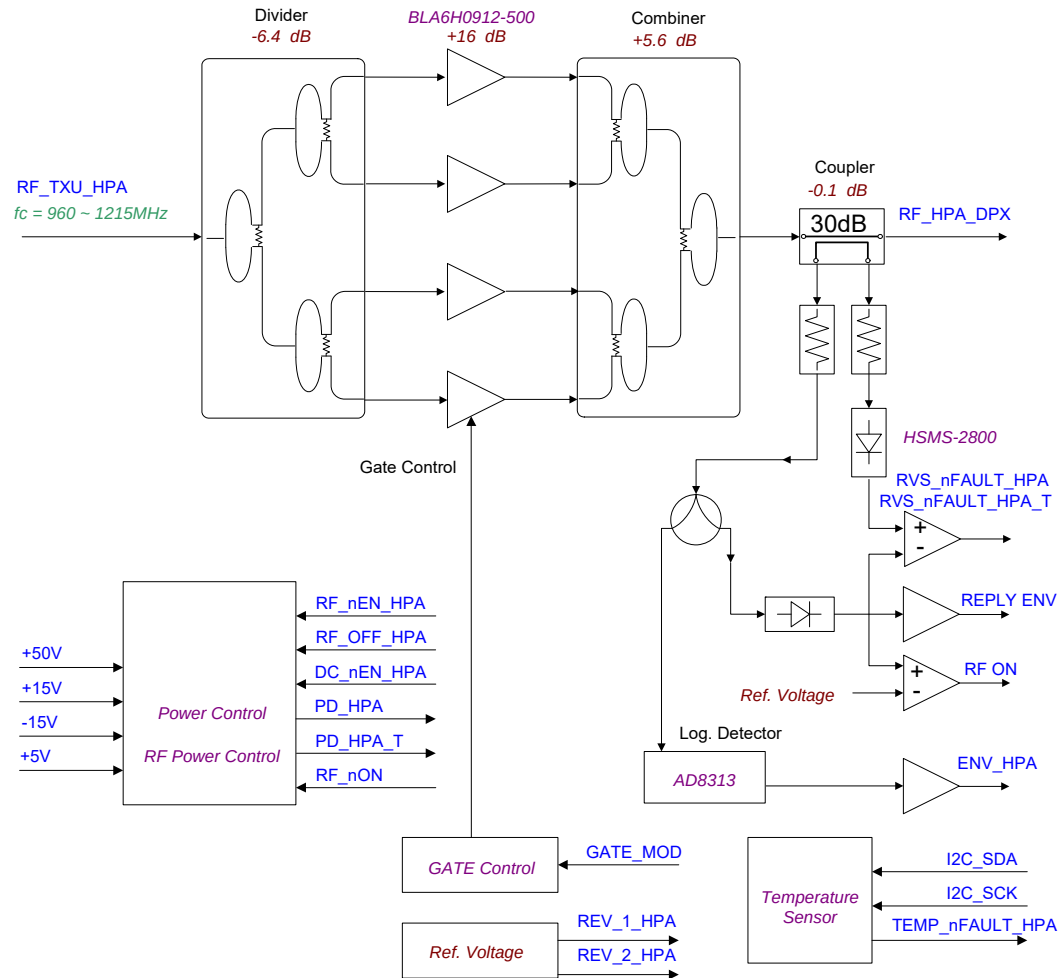


Figure 3-10. HPA Block Diagram

The HPA amplifies the transmitted signal of the DME received from the TXU at a high power of 1.6 kW or more. The amplified transmission signal is transmitted to the DPX-MSC.

The HPA consists of a high output amplifier, a power and output detector, a temperature monitor, and a gate voltage control circuit. The structure is shown in the block diagram below.

HPA receives +46.2 dBm signal from TXU and amplifies +14.8 dB minimum to give +61 dBm output to DPX-MSC. To do this, the HPA combines four amplifiers in parallel and maintains at least 1.6 kW output at all frequencies in DME frequency band.

On the output signal line, the output is sampled through a forward coupler and input to a logarithmic detector U1. And transmits the detected signal “ENV_HPA” to the TCU. There is another forward coupler on the output signal line, which is connected to the diode detector D2 and the detected signal can be measured by the BNC connector J7 of the HPA front panel. The signal detected by the diode detector D2 is input to the comparator to turn on the LED2 labeled as “RF ON” on the HPA front panel.

When the HPA output is open or shorted, amplifier damage can occur due to the reverse power signal reflected back to the amplifier. The protection circuit is operated so that the reverse power signal is detected through the output-side reverse coupler and pulse modulation is not performed.

When a reverse signal is detected, the protection circuit operation sends an alarm signal RVS_nFAULT_HPA_T to the TXU to interrupt the pulse modulation for a predetermined time 200 ~ 300 ms to block the RF signal to the HPA.

It also sends a reverse signal alarm signal RVS_nFAULT_HPA to the TCU.

The temperature sensor is inside the HPA and sends the HPA temperature information to the TCU in I2C format. When the reference temperature is set at the temperature sensor, the RF output is cut off when the temperature is higher than the set temperature, and at the same time, a shutoff state signal TEMP_nFAULT_HPA is sent to the TCU.

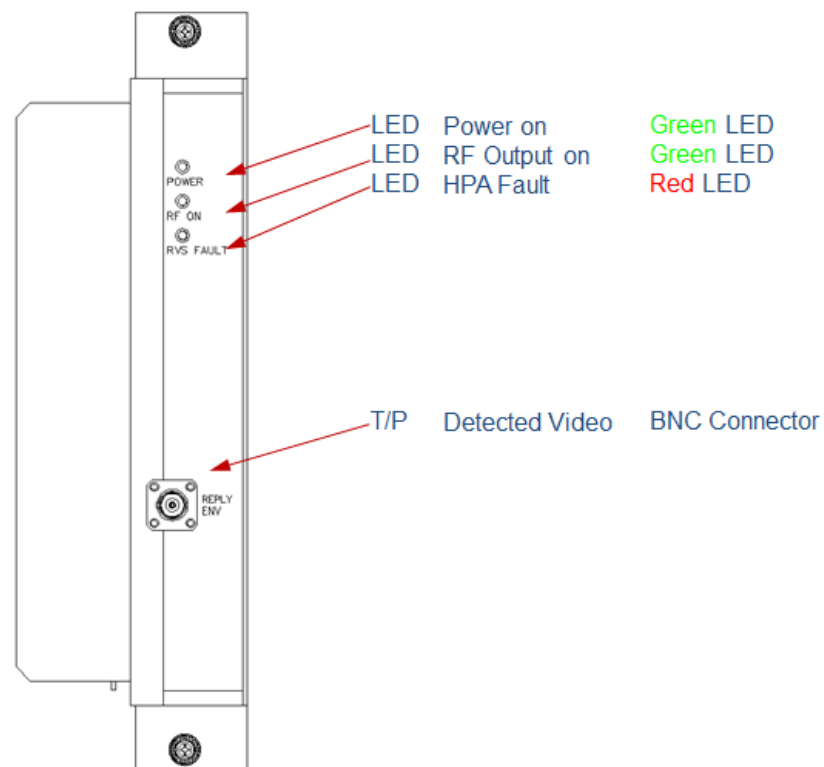


Figure 3-11. Indicators and Test points on HPA

3.2.5. TXU *1A2A4, 1A3A4*

OVERVIEW

TXU provides the following functions:

- Generation of transmit carrier reference signal
- Pre-scaling the reference signal for use with frequency monitoring
- Binary modulation of transmit signal
- Providing the local oscillator signal for the RXU
- Providing automatic gain control to maintain constant output power level



Figure 3-12. TXU

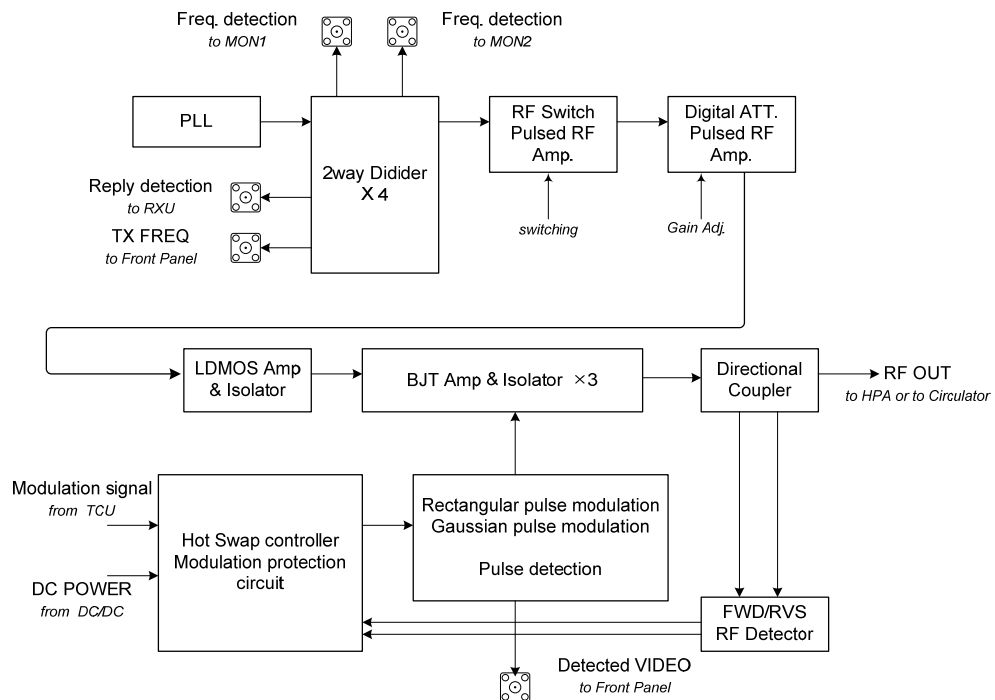


Figure 3-13. TXU Simplified Block Diagram

TCXO

generates a stable reference frequency for the PLL frequency synthesizer

PLL frequency synthesizer

synthesizes CW signal with a station frequency from the reference signal from the TCXO.

Driver amplifier

The output signal from the PLL frequency synthesizer is amplified to -20 dBm, a level to drive the final amplifier

AGC attenuator

The transmit carrier signal should be regulated by an AGC circuit to maintain stable amplitude. The attenuator is used to control amplitude of the transmit signal. The AGC control signal is generated by converting detected sample of output signal into a dc signal. A two-way power splitter, RF Schottky diodes and OP amp circuitry are used for this purpose.

Frequency detector

To monitor integrity of the station frequency, a sample of generated transmit signal is pre-scaled by 1/80 and fed to a TCU, which measures the station frequency by counting the number of pulses per unit time.

RF Switch

RF switch is used to modulate the CW input signal with rectangular pulse shape. This is done by switching on/off the input signal in sync with the modulation signal.

Final amplifier

The final amplifier amplifies the input signal from the driver amplifier to a 100W of peak power.

Coupler & Detector

The signal from the final amplifier is supplied to either DPX or HPA through a directional coupler. The directional coupler samples a small amount of RF energy and RF Schottky diode detects the RF signal envelope.

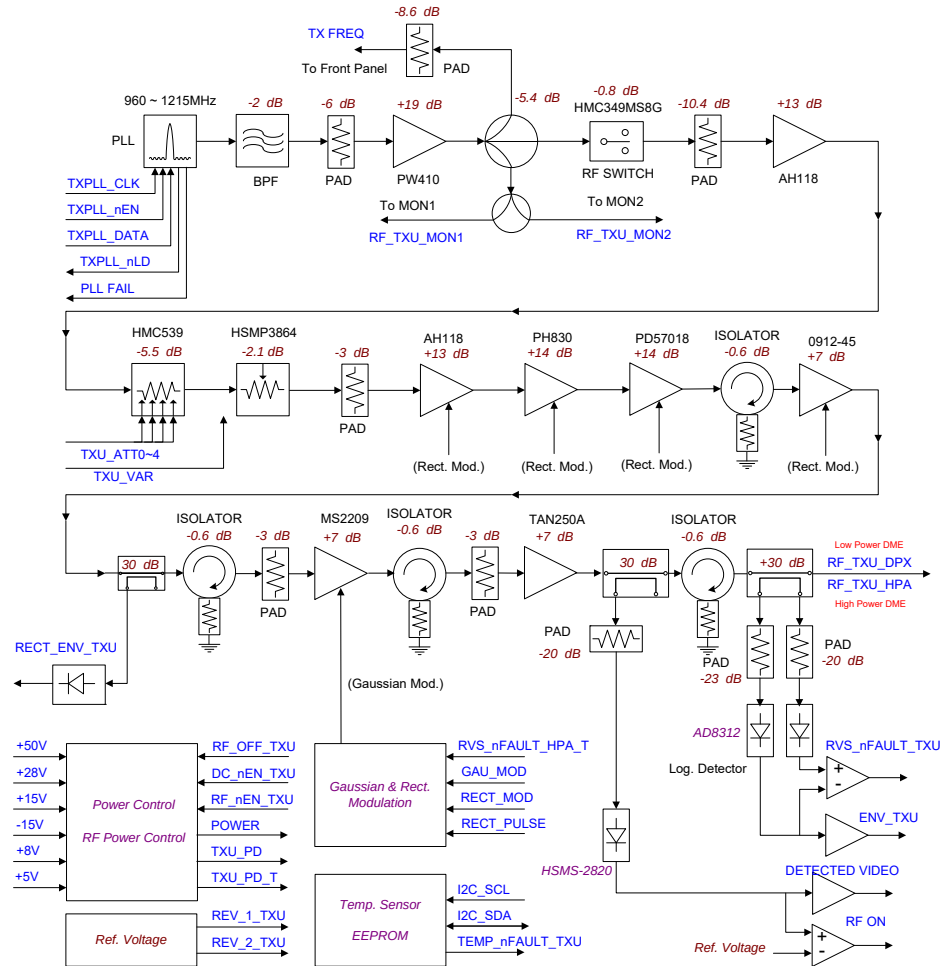


Figure 3-14. TXU Simplified Block Diagram

The TXU generates the transmission frequency of the DME, amplifies the transmission signal with high output power, and performs square-wave pulse modulation and Gaussian pulse modulation. The modulated transmit signal is sent to the DPX in the low power DME and to the HPA (High Power Amplifier) in the high power DME.

The TXU performs the following functions.

- Generate the transmit frequency.
- High output amplification of DME transmission signal.
- Rectangular and Gaussian pulse modulation
- Power adjustment and gain control of the transmitted signal
- Detect RF output signal.
- Detect reflected power.

- TXU temperature measurement and protection

In TXU, the frequency required for DME transmission is generated by using PLL module.

The PLL with built-in TCXO receives the control signals (TXPLL_DATA, TXPLL_CLK, TXPLL_nEN) from the TCU and generates the transmission signal by fixing the PLL frequency.

The output frequency range of the PLL is 960MHz to 1215MHz and the output level is + 3dBm. The output frequency can be set in 0.1MHz increments. The distributor distributes the signal and provides the T / P so that the front panel can measure the transmission frequency.

It also distributes the signal to MON1 and MON2 so that it can analyze the frequency in MON.

In the TXU, the transmission signal of the DME is amplified using a power amplifier.

Low power DME receives +3 dBm signal from PLL and amplifies by 48 dB to send +51 dBm output to DPX, high power DME amplifies 43.2dB and sends +46.2 dBm output to HPA.

The transmission signal from the PLL is amplified with high power using five amplifiers. The overall gain is 48 dB in 100 Watt DME and has an error within ± 1 dB. Signals above + 52 dBm peak power must be available. The first stage amplifier used two MMIC amplifier PW410s in series, followed by another MMIC amplifier PH465.

The LDMOS FET MMRF1304GNRI and MMRF1008H are used as drive amplifiers. A gain attenuator and an isolator were used for each amplifier and the terminal amplifier to obtain gain control and output side VSWR.

The TXU receives the Gaussian waveform mixed with the square wave pulse modulation and the pedestal shape from the TCU and performs pulse modulation. Receives the Gaussian pulse signal GAU_MOD of the TCU and inputs it as the IF signal of the mixer U35 via the buffer U26. The transmission frequency signal CW of the PLL is input to the LO signal roll of the mixer U35 and the RF signal modulated with the Gaussian pulse waveform is transmitted to the RF switch U34.

Receives the pulse modulation signal RECT_PULSE_TXU from the TCU and turns on / off the RF switches U35, U30 to block the low CW signal from leaking to the output side when there is no transmission signal.

In the case of the 320 DME, a reverse power is detected after the output of the HPA, and a protection circuit that cuts off the output of the TXU is operated to protect the HPA. The operation cuts out the spherical pulse modulation to prevent the pulse modulation of the TXU by the RVS_nFAULT_HPA_T signal of the HPA, and cuts off the RF switch

The TCU adjusts the modulation signal GAU_MOD used as the modulating signal of the IF of the mixer, so that the output power has a variable range between -6 dB and +2 dB.

The output detection signal RECT_ENV_TXU of the TXU is constantly monitored by the TCU to adjust the modulation signal GAU_MOD to maintain the constant output.

At the frequency change, the digital attenuator for adjusting the gain of the TXU receives the control signal TXU_ATT0 through TXU_ATT4 from the TCU and the gain of the transmission signal can be adjusted from 0 dB to -7.25 dB

The output signal sampled by the forward coupler of the AMP circuit is input to the logarithmic detector U1 of the detection circuit on the back of the TXU. It transmits the detected signal ENV_TXU to the TCU. The output signal sampled by another forward coupler of the AMP circuit section is input to the Schottky diode and envelope detection is performed. The detected envelope signal is inputted to the comparator so that the LED of the front panel operates at a certain level

or higher. This detected signal can be measured on the front panel BNC connector.

When the TXU output is open or shorted, the signal reflected back to the amplifier is detected through a reverse coupler. When the reflected power signal is detected, it shuts off the transmission signal for a certain period of time 200 - 300 ms, during which the alarm signal RVS_nFAULT_TXU is sent to the TCU.

The temperature sensor measures the TXU internal temperature and sends the temperature information to the TCU. When the temperature reaches a temperature higher than the TUC, the temperature sensor generates a FAULT signal to cut off the RF output and at the same time send a shutdown status signal TEMP_nFAULT_TXU to the TCU.

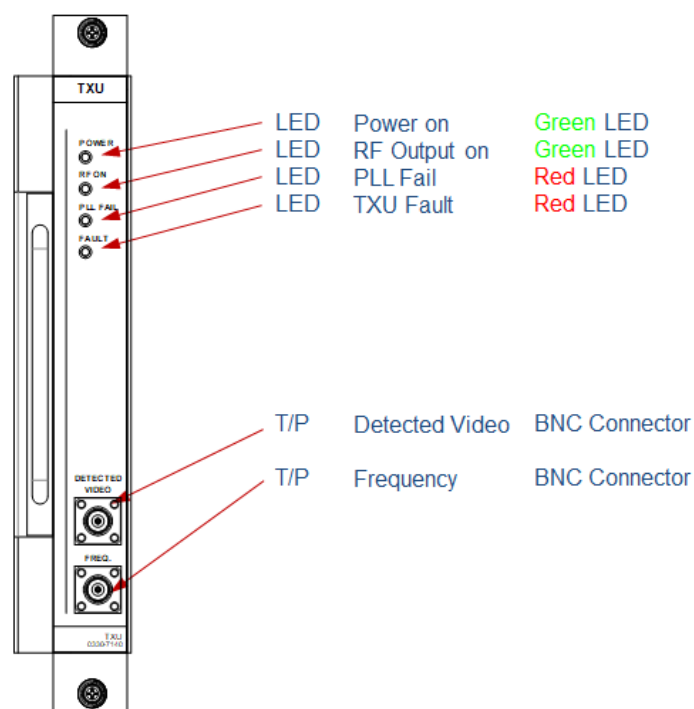


Figure 3-15. Indicators and Test points on TXU

3.2.6. DC/DC 1A2A5, 1A3A5

OVERVIEW

The DC/DC converter provides the multi-voltage dc power needed for the transmitter / monitor subrack. The DC/DC unit receives +28V DC input from either AC/DC converter or battery backup source via PMU subrack and converts it into +5V, +8V, +15V, -15V, +28V output voltages and +50V.



Figure 3-16. DC/DC

DESCRIPTION

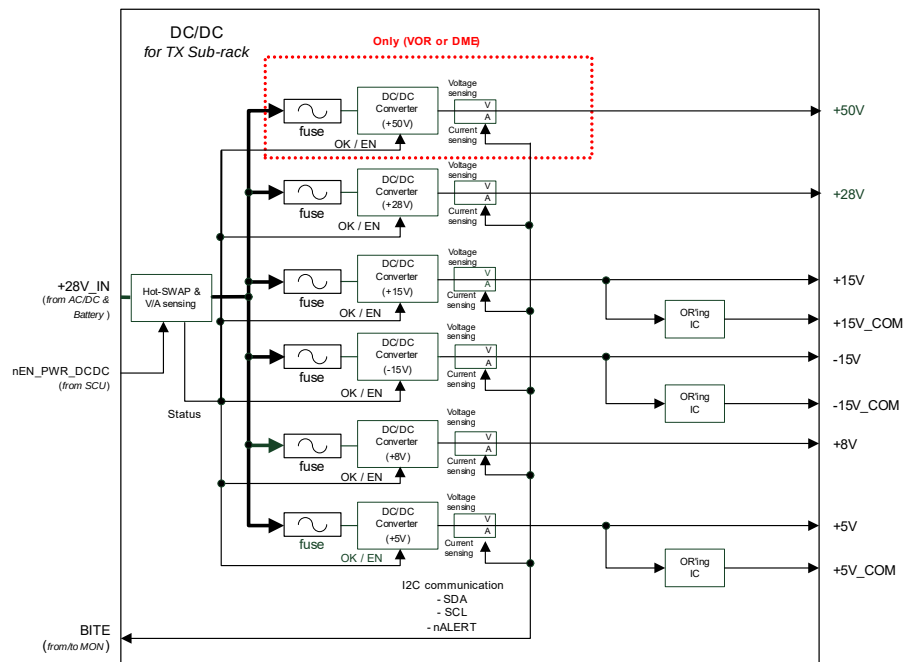


Figure 3-17. DC/DC Block Diagram

Figure 3-18 illustrates front view of the DC/DC.

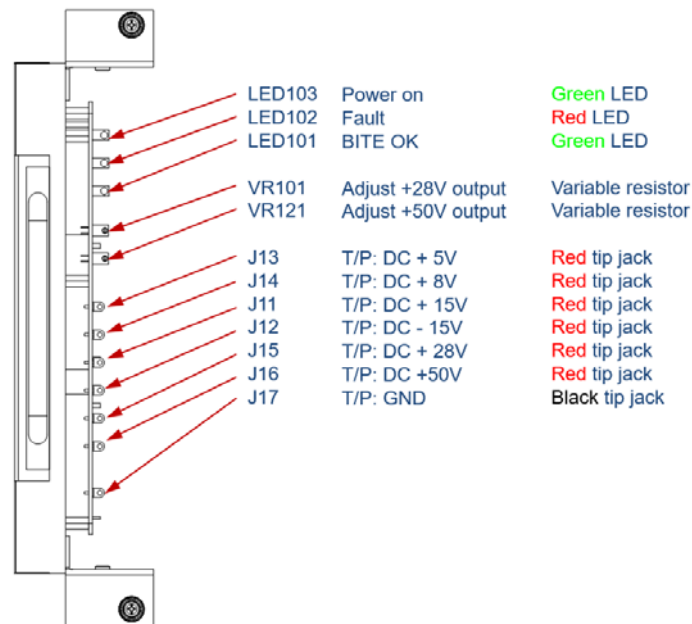


Figure 3-18. Indicators and Test points on DC/DC

The Table 3-2 describes the control and indicators on the DC/DC.

Table 3-1. Indicators on DC/DC

No	Name	Type	Color	Description
1	POWER	LED	Green	Power on status indicator
2	FAULT	LED	Red	Fault indicators
3	BITE OK	LED	Green	Indicates BITE result is OK

Table 3-2. Controls on DC/DC

No	Name	Type	Description
1	ADJ1	Potentiometer	Adjustment of DC + 28V output voltage
2	ADJ2	Potentiometer	Adjustment of DC +50 V output voltage (VOR/DME only)

Table 3-3. Test points on DC/DC

No	Name	Type	Color	Description
1	+5V_TP	2 mm tip jack	Red	Test point for DC +5 V rail
2	+8V_TP	2 mm tip jack	Red	Test point for DC +8 V rail
3	+15V_TP	2 mm tip jack	Red	Test point for DC +15 V rail
4	-15V_TP	2 mm tip jack	Red	Test point for DC -15 V rail
5	+28V_TP	2 mm tip jack	Red	Test point for DC +28 V rail
6	+50V_TP	2 mm tip jack	Red	Test point for DC +50 V rail
7	GND	2 mm tip jack	Black	Test point for common (GND)

3.2.7. TCU 1A2A6, 1A3A6

OVERVIEW

TCU provides the following functions:

- Sampling and quantization of received signal
- Decoding of received pulse pair
- Generation of system delay
- Echo suppression
- Generation of pulse modulation signal
- Measurement and adjustment of reply pulse pairs
- Generation of squitter pulse pairs and IDENT pulse pairs
- Load limiting by controlling receiver sensitivity
- Control of station frequency for both transmitter and receiver



Figure 3-19. TCU

DESCRIPTION

In the heart of the TCU, two large scale field programmable gate arrays (FPGAs) and a high-performance 16/32-bit microcontroller are comprised.

The log video signal of received pulse pair is fed via buffer to a high speed analog-to-digital converter, which digitize each video sample of input pulse pair. The stream of digitized video signals are queued onto a FIFO and numerically processed to evaluate the pulse pair coding. All the processing logic is built into a FPGA 1. When valid interrogation pulse pair is detected, a trigger pulse is generated at the point of 50% amplitude.

The trigger pulse is fed to the reply processing block of the FPGA and causes it to initiate a reply pulse generation after fixed time delay. The reply pulse is generated by using combination and

sequential digital logic circuitry.

The pulse modulation signal is a Gaussian shape pulse signal. It is numerically synthesized and converted to an analog pulse through a digital-to-analog converter.

The echo suppression is provided under control of the second FPGA.

Measurements of reply pulse pair is done by sampling the transmit signal from the TXU or HPA. The sampled pulse signal is digitized by an analog-to-digital converter. The stream of digitized data samples are processed in similar way as the MON does and measured parameters are used to complement output pulse shape.

The squitter signal is a pulse pair with random interval, transmitted even in no interrogation signal is present. The squitter signal is generated in the FPGA 1 and its pulse repetition is programmable via the microcontroller on the TCU.

The IDENT signal is also generated in the FPGA 1 and its Morse code keying is controlled by TCU software.

The TCU monitors the load level of received interrogation pulse pairs. If the load level is increased beyond a preset threshold level, the TCU lowers the control voltage supplied to the attenuator in the DPX, so as to desensitize the receiver responding range and eventually limits load level.

The TCU provides the serial setup data for the phase locked loop (PLL) frequency synthesizers of both transmitter and receiver with the station frequency.

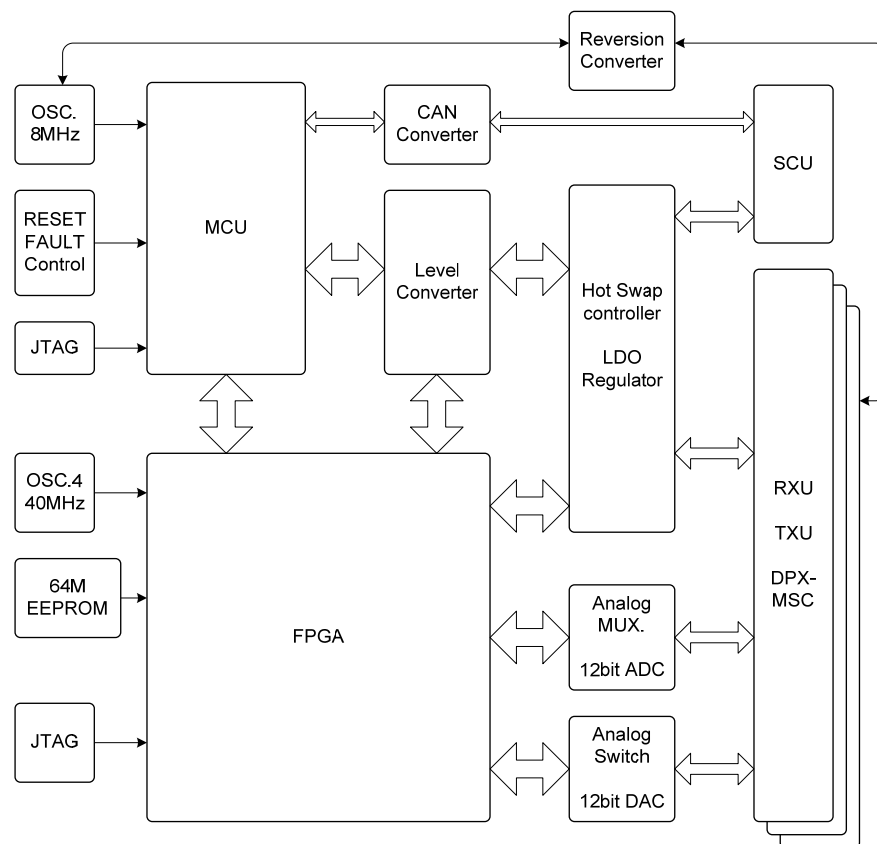


Figure 3-20. TCU Block Diagram

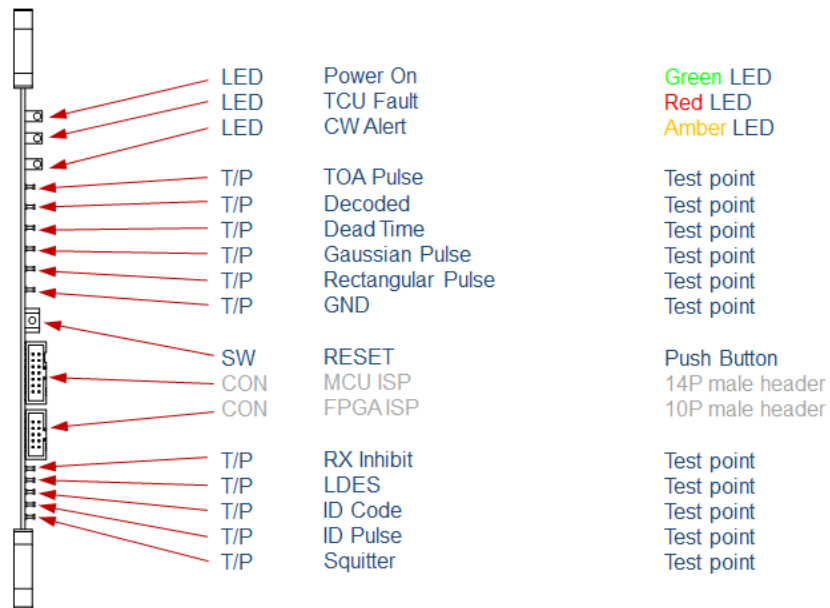


Figure 3-21. Control, Indicators and Test points on TCU

3.2.8. FAN 1A2A9, 1A3A9

The fan provides air flow to cool down the transponder subrack.



Figure 3-22. Cooling Fan

Figure 3-23 illustrates front panel of the FAN.

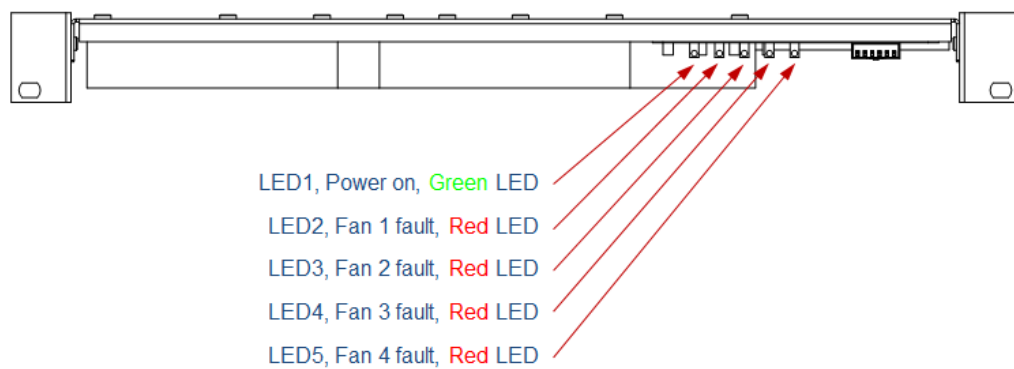


Figure 3-23. Indicators on FAN Unit

The Table 3-4 describes the control and indicators on the FAN.

Table 3-4. Control and Indicators on FAN

No	Name	Type	Color	Description
1	POWER	LED	Green	Power on status indicator
2	FAN1 Fail	LED	Red	FAN1 fail indicator
3	FAN2 Fail	LED	Red	FAN2 fail indicator
4	FAN3 Fail	LED	Red	FAN3 fail indicator
5	FAN4 Fail	LED	Red	FAN4 fail indicator

3.3. MONITOR

3.3.1. OVERVIEW

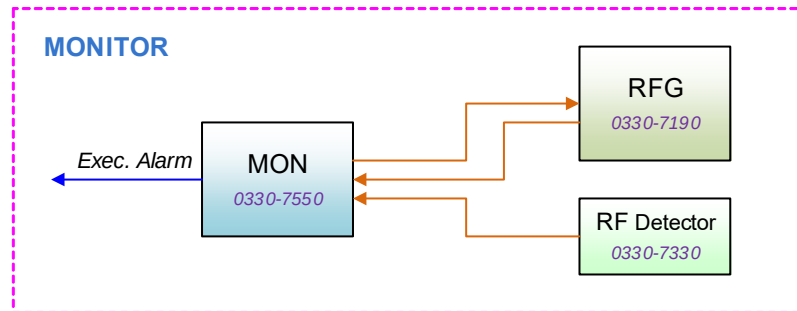


Figure 3-24. MONITOR Block Diagram

3.3.2. MON **1A2A6, 1A3A8**

OVERVIEW



Figure 3-25. MON

DESCRIPTION

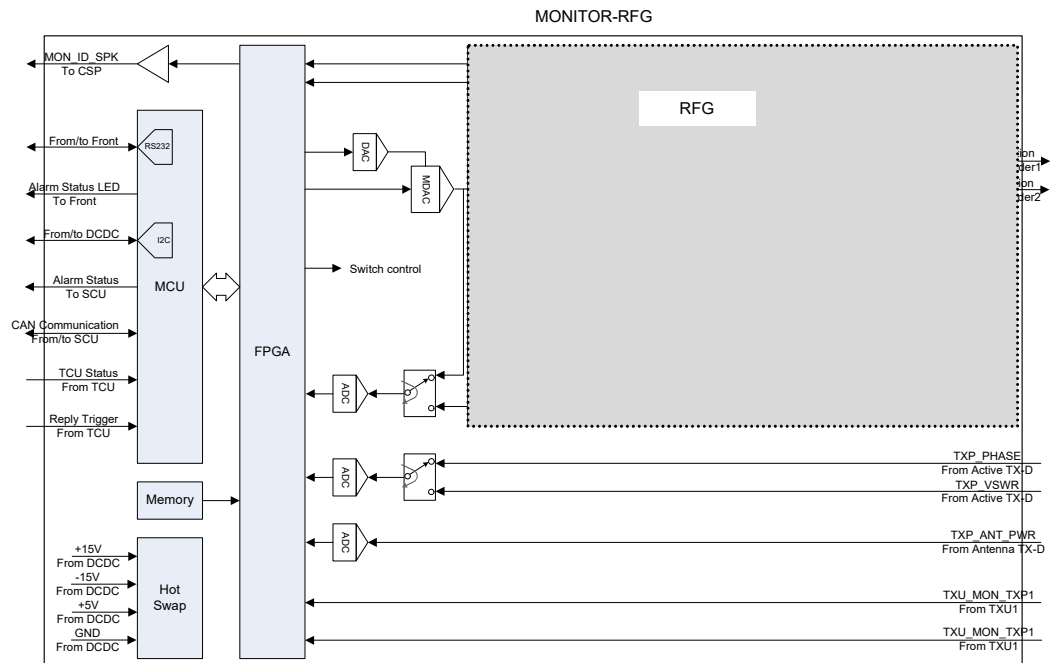


Figure 3-26. MON Unit Simplified Block Diagram

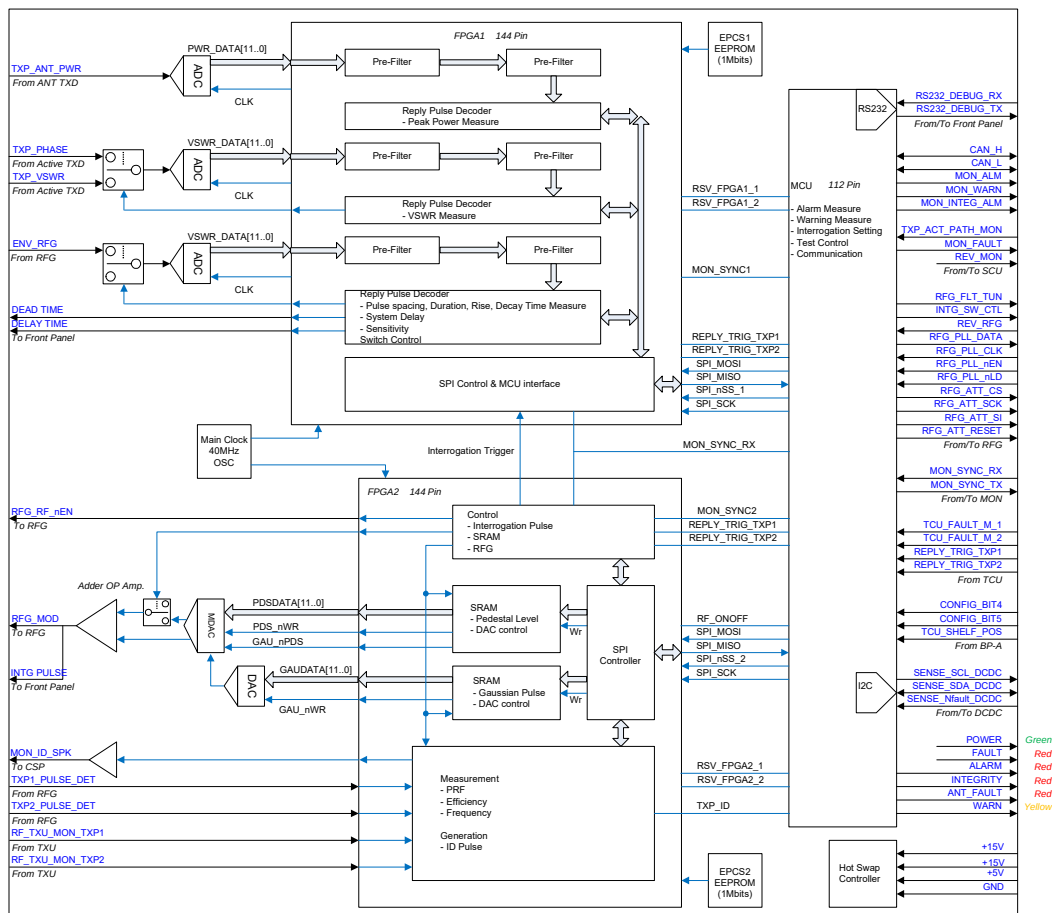


Figure 3-27. MON Unit Block Diagram

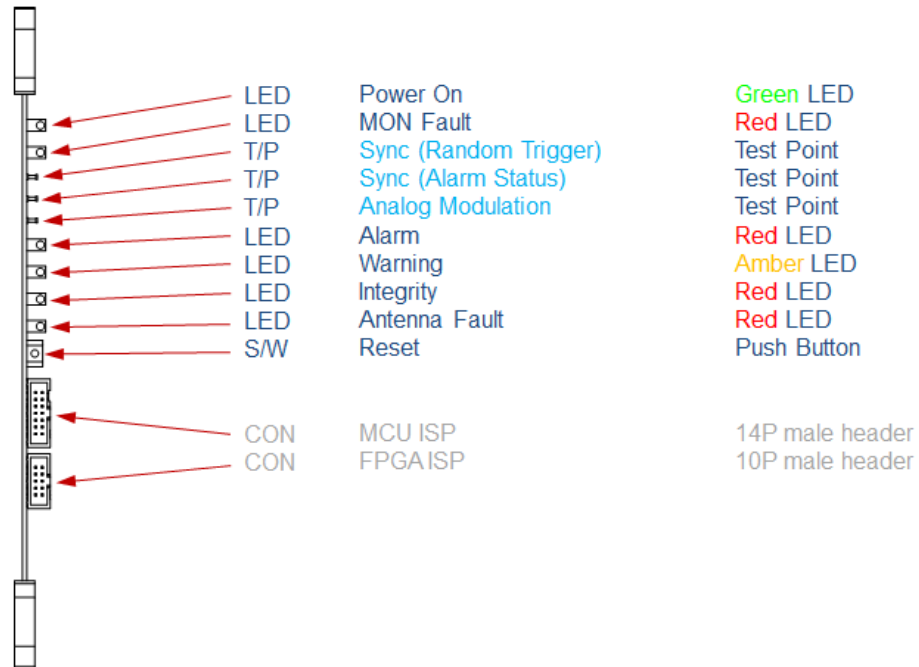


Figure 3-28. Control, Indicators and Test points on MON (Rev. C or prior)

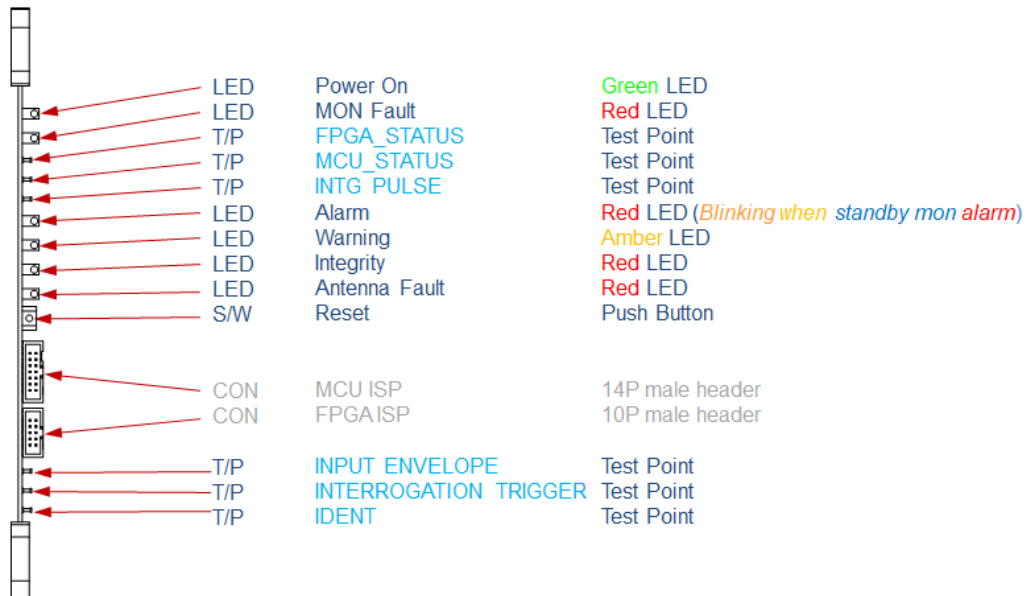


Figure 3-29. Control, Indicators and Test points on MON (Rev. D or later)

3.3.3. RFG 1A2A7, 1A3A7

OVERVIEW



Figure 3-30. RFG

DESCRIPTION

The RFG generates the question frequency of the DME, amplifies the question signal at high power, and performs square-wave pulse modulation and Gaussian pulse modulation. The modulated signal of the interrogation signal is received from MON and generates a question signal.

The block diagram of the RFG is shown in Figure 3-23.

The TCXO (X1) signal of 63MHz is amplified by 60dB or more with MMIC AMP. The amplified signal is frequency-synthesized with the PLL signal and frequency-converted to a signal spaced 63 MHz from the response frequency of the DME transponder. The signal is amplified by more than +38dB at the question frequency, and the signal of + 23dBm maximum is input to the DME RXU through the 30dB coupler (TXP1: 1DC1, TXP2: 1DC2).

The PLL with built-in TCXO receives the control signal from MON and fixes the frequency of the PLL to generate the RFG question signal.

The output frequency range of the PLL is 960MHz to 1215MHz and the output level is + 3dBm. The output frequency can be set in 0.1MHz increments. It distributes the signal to the distributor and provides T / P (J4, RFG FREQ.) So that the front panel can measure the frequency of the PLL.

In RFG, DME question signals are amplified using power amplifiers (MMIC AMP: U27, U6, U35, U16, U15).

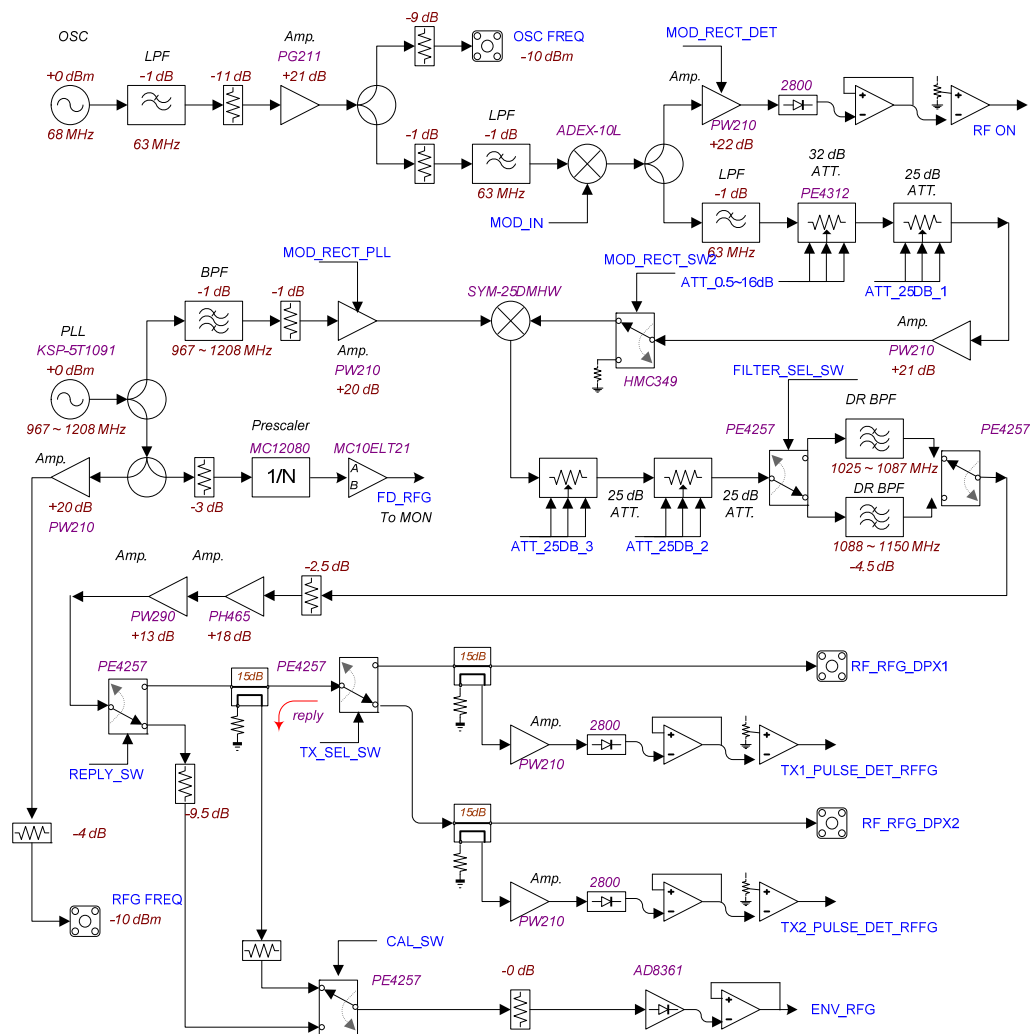


Figure 3-31. RFG Block Diagram

RFG receives the Gaussian waveform mixed with square wave pulse modulation and pedestal shape from MON and performs pulse modulation. Modulation is performed by inputting 63MHz signal of TCXO (1X) and Gaussian modulation signal into LO and IF of Mixer respectively, and RF signal which is the output signal of Mixer (U2) is input to MMIC AMP and amplified.

Square wave pulse modulation is performed using the RF switch (U31).

Since RFG's output variable range is -83dBm to + 23dBm, the RF attenuator is placed in the middle of the amplification circuit to control the output in 0.5dB steps.

At the IF frequency of 68MHZ, it has a variable range of 57dB and controls the output at 0.5dB intervals, and after the mixer (U32) it has a variable range of 50dB and controls the output at 25dB intervals.

The attenuator of RFG can be adjusted more than 100dB by receiving D0 (0 ~ 8) _ATT_RFG signal of MON

RFG sends a detection signal of question and response signals to MON by using one linear detector.

The Q signal is sent to the detector (LTC5535) through the RF switches (U9, U10) by the REPLY_SW signal and CAL_SW.

The response signals to TXP1 and TXP2 respectively enter the RFG output port and are coupled to the 15dB directional coupler via three RF switches (U11, U7, U5) and sent to the detector (LTC5535: U28) for detection. The control of each switch is switched by a control signal of MON.

The detector for detecting the ID of MON is composed of TXP1 and TXP2, respectively, and the detected pulse signal is converted from a comparator to a square wave to generate a TTL signal.

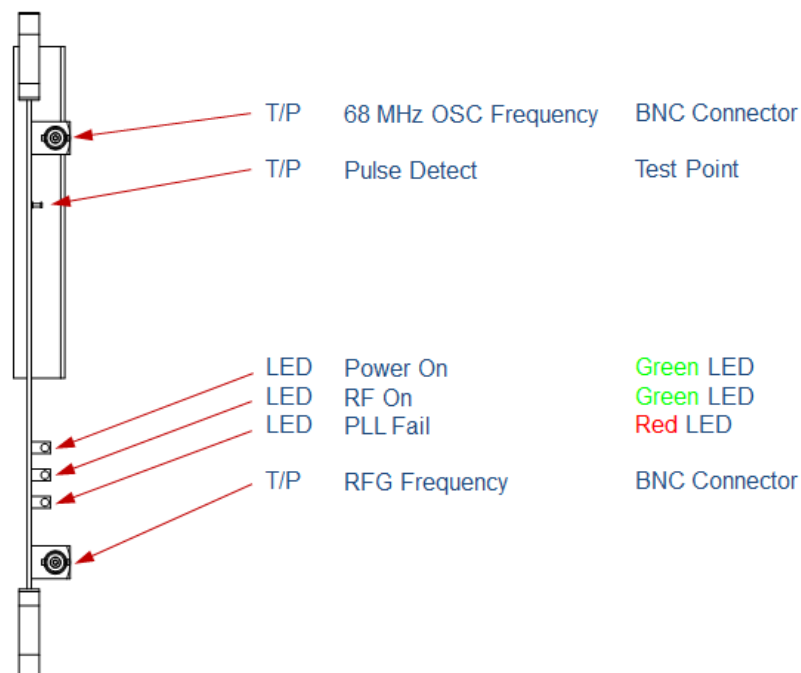


Figure 3-32. Control, Indicators and Test points on RFG

3.4. CONTROL AND MONITORING

3.4.1. OVERVIEW

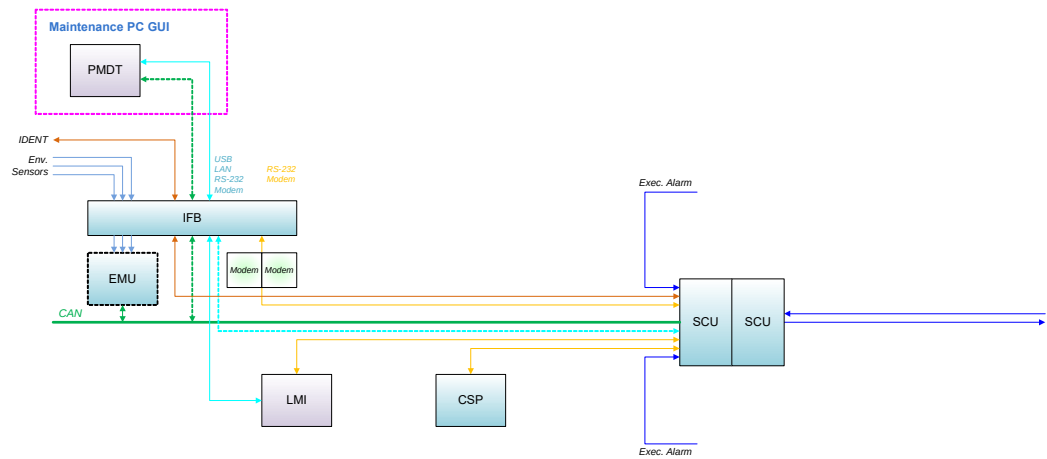


Figure 3-33. Control and Monitoring

3.4.2. LMI 1A6

OVERVIEW



Figure 3-34. LMI

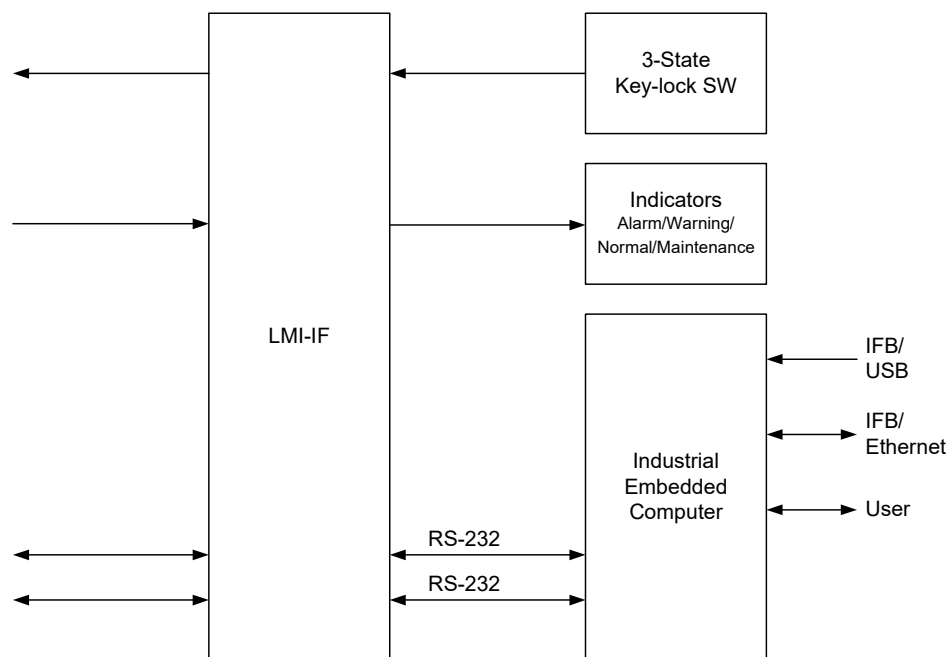


Figure 3-35. LMI Simplified Block Diagram

DESCRIPTION

The Local Maintenance Interface (LMI) consists of a 10.2" color TFT LCD with touchscreen, light emitting diode (LED) indicators. With the LCD display in combination with the touchscreen, the system status and configuration can be displayed and modified.

LMI has the following functions:

- Equipment status and change settings

LMI displays the system status, power supply status and shelter status. It also displays the TX configuration settings and monitor settings.

LMI receives commands from the operator via GUI and exchanges it with SCU to carry out changes in the system setting such as TX setting, monitor setting, any setting needed to be changed in sensors inside the shelter and settings of the individual units.

- Log history and query

The status of the equipment and individual units are saved in a event log file. The queries made by operator are also recorded in a log file. These information can be used at a later time using LMI to check history of commands and events.

- Unit Firmware Upgrade

It is possible to upgrade firmware for all units SCU, PMM, EMU, TCU, MON with the microcontroller MCU.

- Communication Interface with an embedded computer

LMI consists of an operator interface GUI in the form of a touch screen display with embedded computer. It is installed on the top of rack door.

It provides USB, Ethernet and RS 232 communication interface with PMDT, and RS-232, RS - 485 communication interface with SCU.

3.4.3. CSP 1A1A1

OVERVIEW

The Control and Status Panel (CSP) consists of push buttons and light emitting diode (LED) indicators for control and display of the system status and configuration. Figure 3-36 shows the CSP Front Panel.

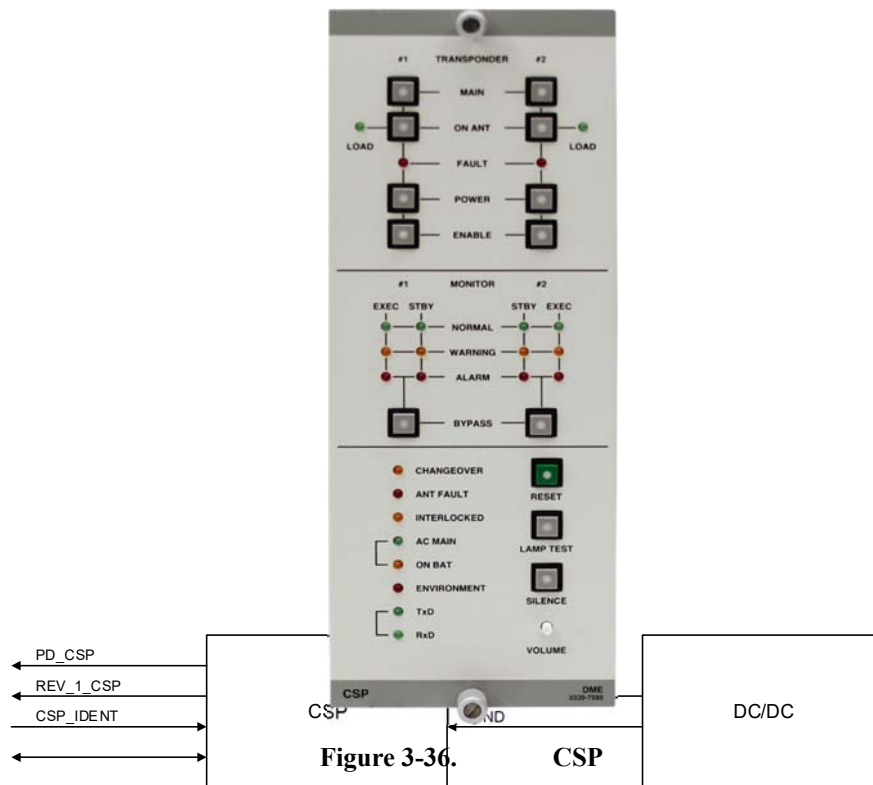


Figure 3-36.

There are 5 push button controls and 2 LED indicators for each transmitter- TX1 and TX2.

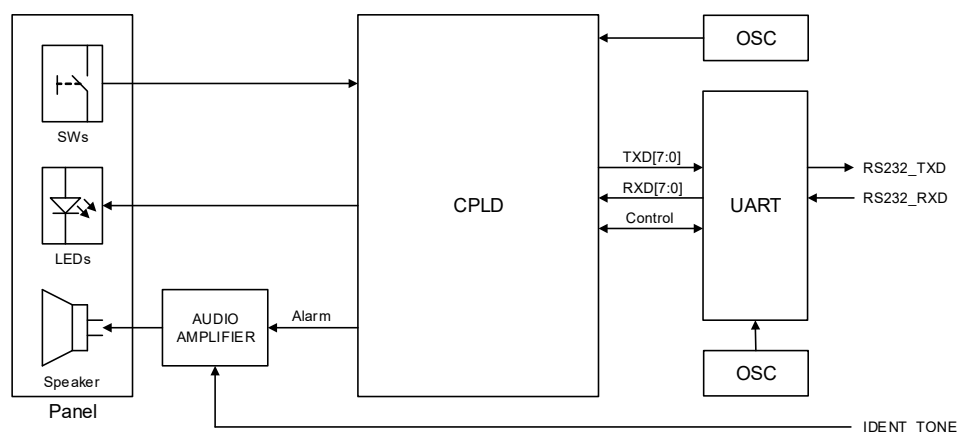


Figure 3-37. CSP Simplified Block Diagram

DESCRIPTION

The CSP indications for the monitors provide control and display of EXEC (executive) monitor. Each monitor reports the alarm status to CSP and consists of 4 push buttons and 18 LEDs (3 for

each monitor). The LEDs for each monitor indicate the normal, warning and alarm states depending on the parameter values measured and tolerance limits set.

There are LEDs to indicate CHANGEOVER in case there has been changeover of transmitter, ANT FAULT for indicating fault in the antenna, and INTERLOCKED to indicate an active interlock in case of mutual exclusive operation on different ILS systems.

The active power source is indicated by either of the 2 LEDs – AC MAIN or ON BAT.

The TxD and RxD LEDs indicate transmission and reception of signals respectively.

There are push buttons for RESET to reset the active transmitter and SILENCE to mute the system. The LAMP TEST switch causes all installed front panel indicator lamps, including those normally inactive in a single-equipment system, to illuminate, so as to avoid confusion with “dark” lamps when performing this test.

A potentiometer knob VOLUME, below the silence button provides volume adjustment.

The Block diagram of CSP is shown in Figure 3-38. The front end panel consists of push-button switches and LEDs and a speaker to hear IDENT tone. The microprocessor logic based on a large-scale complex programmable logic device (CPLD) processes the input commands via switches and indicates the status information via the LEDs.

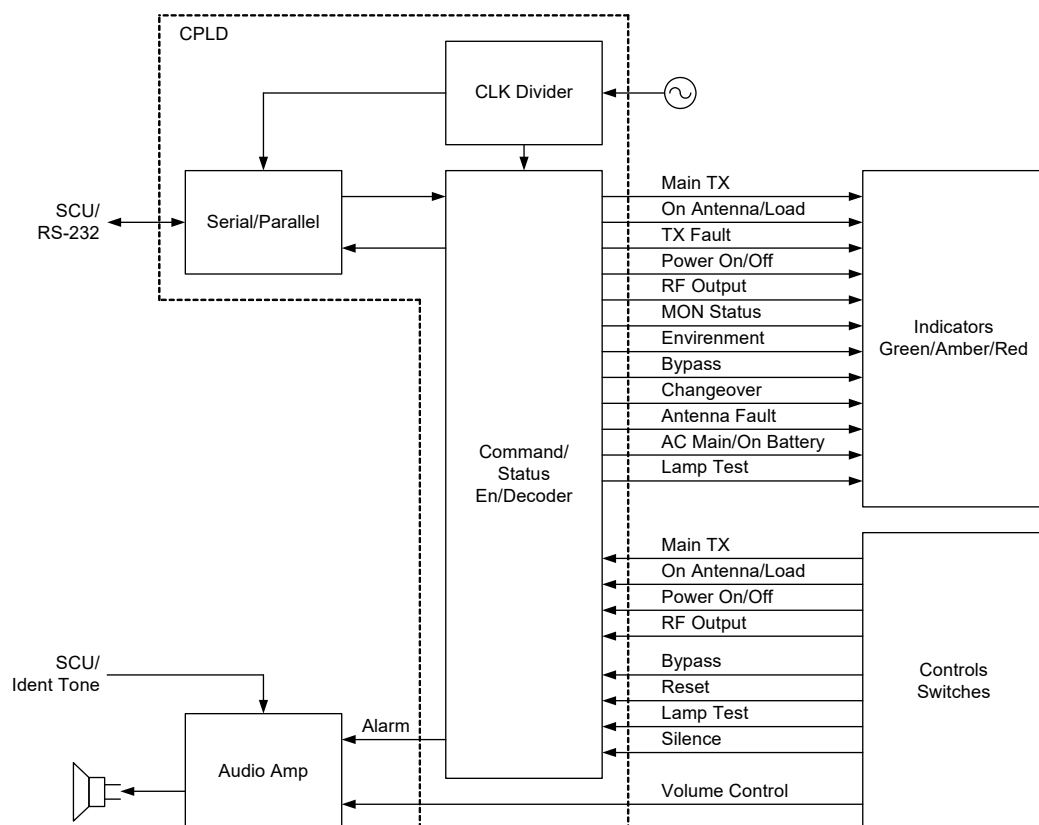


Figure 3-38. CSP Block Diagram

3.4.4. SCU 1A1A2, 1A1A3

OVERVIEW

The System Controller Unit, SCU is responsible to control the switching of the transmitters (changeover). It also serves as the communication interface between the control units (CSP, LMI) and remote control unit (RCU, PMDT) and provides the status of the equipment.

The block diagram of SCU is shown in Figure 3 10.

The main functional units of SCU are: Microcontroller and Peripheral Logic and SCU Software.



Figure 3-39. SCU

DESCRIPTION

Microcontroller and Peripheral Logic

The microcontroller and peripheral logic are almost identical to those of the TCU and the MON.

The SCU communicates with the TCU and the MON via internal CAN bus, issuing and receiving control commands and responses. For communicating with user both locally and remotely, the SCU is equipped with two RS-232C ports on the rear panel of the system cabinet. Various other communication links including direct null-modem cable, dedicated-line modem, dial-up modem, twisted-pair balanced-line are also supported.

The SCU provides the following communication interfaces

- CAN : MON, TCU, PMM, EMU
- RS-485 : LMI
- Modem : PMDT
- RS-232C : PMDT, Debug

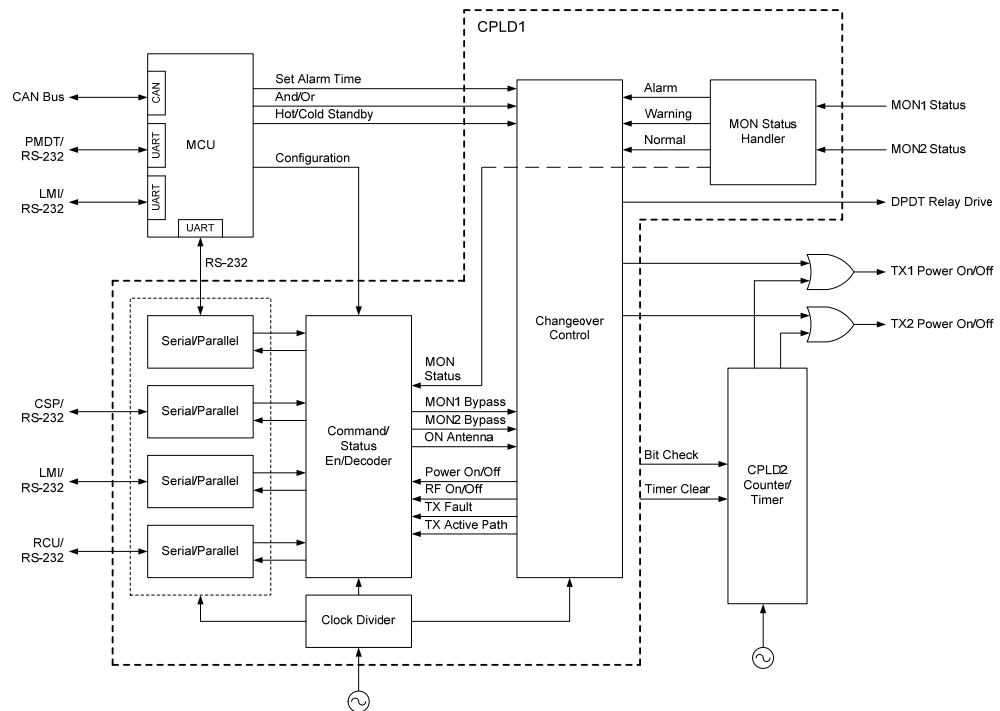


Figure 3-40. SCU Block Diagram

Figure 3-41 illustrates front view of the SCU.

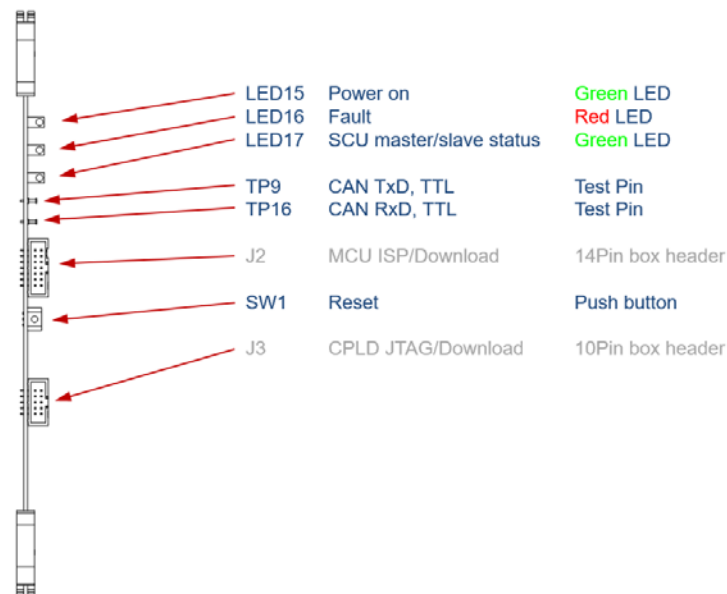


Figure 3-41. Control, Indicators and Test points on SCU

The Table 3-5 and Table 3-6 show the controls and indicators on the SCU.

Table 3-5. Controls on SCU

No	Name	Type	Description
1	RESET	Push button	Reset and initialize the SCU

Table 3-6. Indicators on SCU

No	Name	Type	Color	Description
1	POWER	LED	Green	Power on status indicator
2	FAULT	LED	Red	Fault indicators
3	MASTER	LED	Green	Indicates that the SCU is selected as master

Table 3-7. Test points on SCU

No	Name	Type	Description
1	TP9	Turret terminal	Test point for DC +5 V rail
2	TP16	Turret terminal	

Table 3-8. ISP/Debug Connectors on SCU

1	MCU ISP	10-pin Header	MCU In system programming / debug
2	CPLD ISP	14 pin Header	CPLD In system programming

3.4.5. EMU 1A1A4



Figure 3-42. EMU

Environmental Monitoring Unit (EMU) consists of sensors using which it monitors the temperature and other environment conditions. Its function is common to CVOR, DVOR, DME, LOC, GP and the MKR units. It performs the following functions.

- Fire Detector Monitoring using a smoke sensor
- Temperature Measurement and Monitoring using a temperature sensor
- Shelter Door On/Off Status Monitoring using an intrusion sensor
- I/O Pin Interface Control using 16 general-purpose TTL-level digital input/output ports or 8 general-purpose galvanically-isolated digital input/output ports
- Communication with SCU via the CAN bus. The EMU receives the configuration data and setup information of sensors for monitoring from the SCU and exchanges the measurement information and status information of the EMU with the SCU.
- Firmware upgrade

3.4.6. DC/DC-A 1A1A5, 1A1A6

The DC/DC-A converter provides the multi-voltage dc power which is needed for the auxiliary electronics subrack. Two DC/DC-A converters are connected in parallel to provide redundancy and current boost. The main functions of DC/DC-A are as follows:

- **DC/DC conversion**
The DC/DC-A unit receives +28V DC input from either AC/DC or battery backup source via PMU subrack and converts it into +5V, +15V, -15V output voltages.
- **Monitor the output current and voltage**
A monitor IC monitors the status of the input voltage and output current and the measured values are conveyed to the SCU
- **BITE and Alarm signal**
A BITE signal is generated to check any abnormal state in the SCU and an alarm is generated and Fault LED (Red) is ON, if any value is outside the tolerance limit.
- **Presence Detector (PD)**
It is possible to determine if any unit has been detached by the signal associated with revision check.



Figure 3-43. DC/DC-A

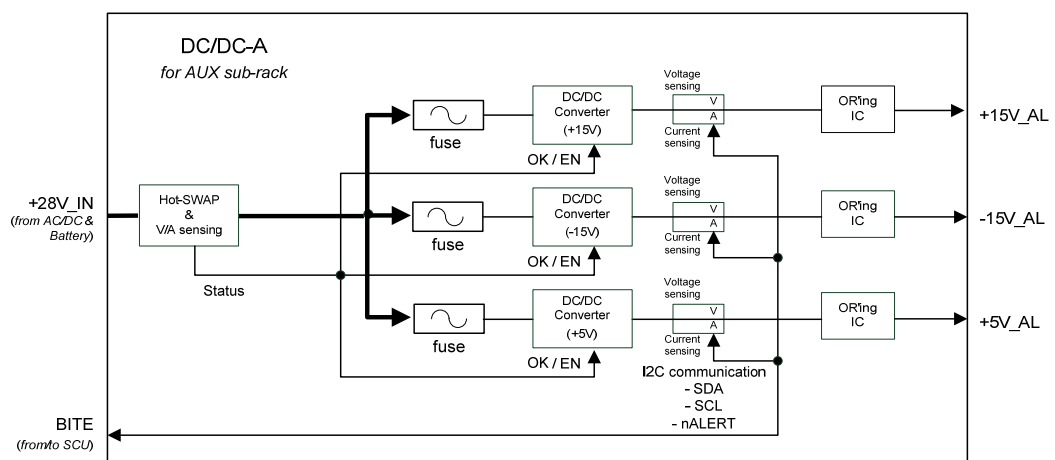


Figure 3-44. DC/DC-A Block Diagram

Figure 3-45 illustrates front panel of the DC/DC-A.

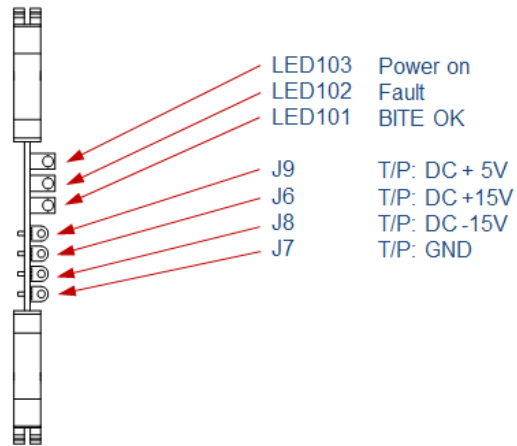


Figure 3-45. Indicators and Test points on DC/DC-A

The Table 3-9 describes the control and indicators on the DC/DC-A

Table 3-9. Indicators on DC/DC-A

No	Name	Type	Color	Description
1	POWER	LED	Green	Input Power Status Power ON = LED ON
2	FAULT	LED	Red	Indicates fault if the input voltage is below the set limit Alarm = LED ON
3	OUTPUT	LED	Green	Indicates that the DC output voltage is at least the minimum set value Output =LED ON

Table 3-10. Test Points on DC/DC-A

No	Name	Type	Color	
1	+5V_TP	2mm Tip jack	Red	Test point for DC +5 V rail
2	+15V_TP	2mm Tip jack	Red	Test point for DC +15 V rail
3	-15V_TP	2mm Tip jack	Red	Test point for DC -15 V rail
4	GND	2mm Tip jack	Black	Test point for common (GND)

3.4.7. MODEM 1A1A7, 1A1A8

OVERVIEW

The Modems provide a communication medium to connect to maintenance PC PMDT. There are two types of modems used for ILS equipment.

- High-Speed (up to 33.6kbps) leased-line/dialup Modem
- Low-speed (1200 bps) FSK leased-line modem



Figure 3-46. Modem

DESCRIPTION

Figure 3-47 shows the block diagram of the Modem.

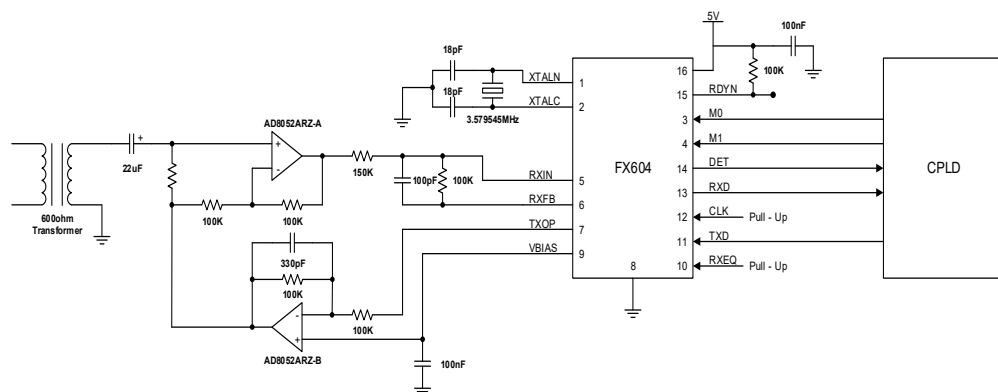


Figure 3-47. Modem Block Diagram

3.5. POWER SUPPLY

3.5.1. OVERVIEW

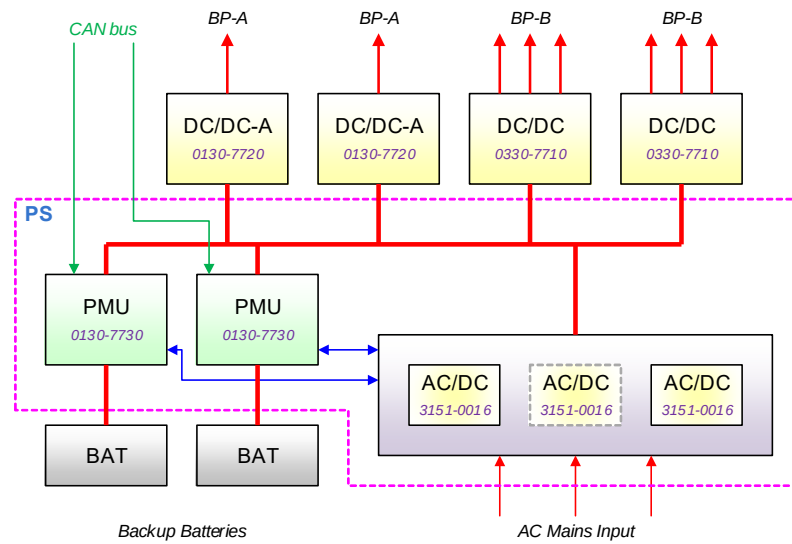


Figure 3-48. Power Supply Block Diagram

3.5.2. PMU 1A4A1, 1A4A2

OVERVIEW

The Power Monitoring Unit (PMU) manages the power supplied to the equipment. Two PMU devices are used in order to improve the reliability of the system and either of the two can be used to monitor the AC/DC output. With the help of 7-segment display battery temperature, battery current, battery voltage, AC/DC current, AC/DC voltage, etc. can be displayed. There are selection buttons available in PMU to allow the operator to select the measured value to be displayed on the seven-segment display.

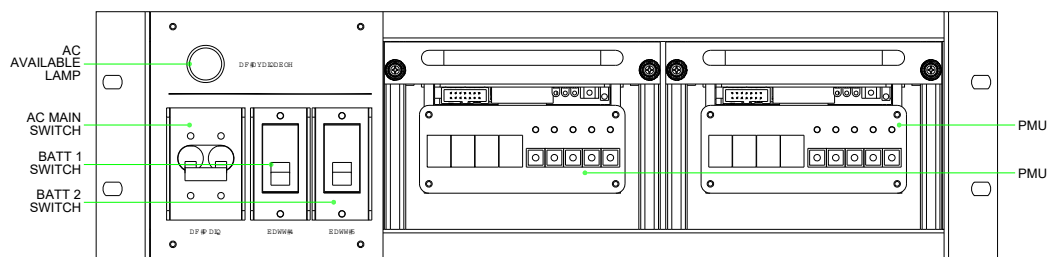


Figure 3-49. PMU Subrack, Front View



Figure 3-50. PMU

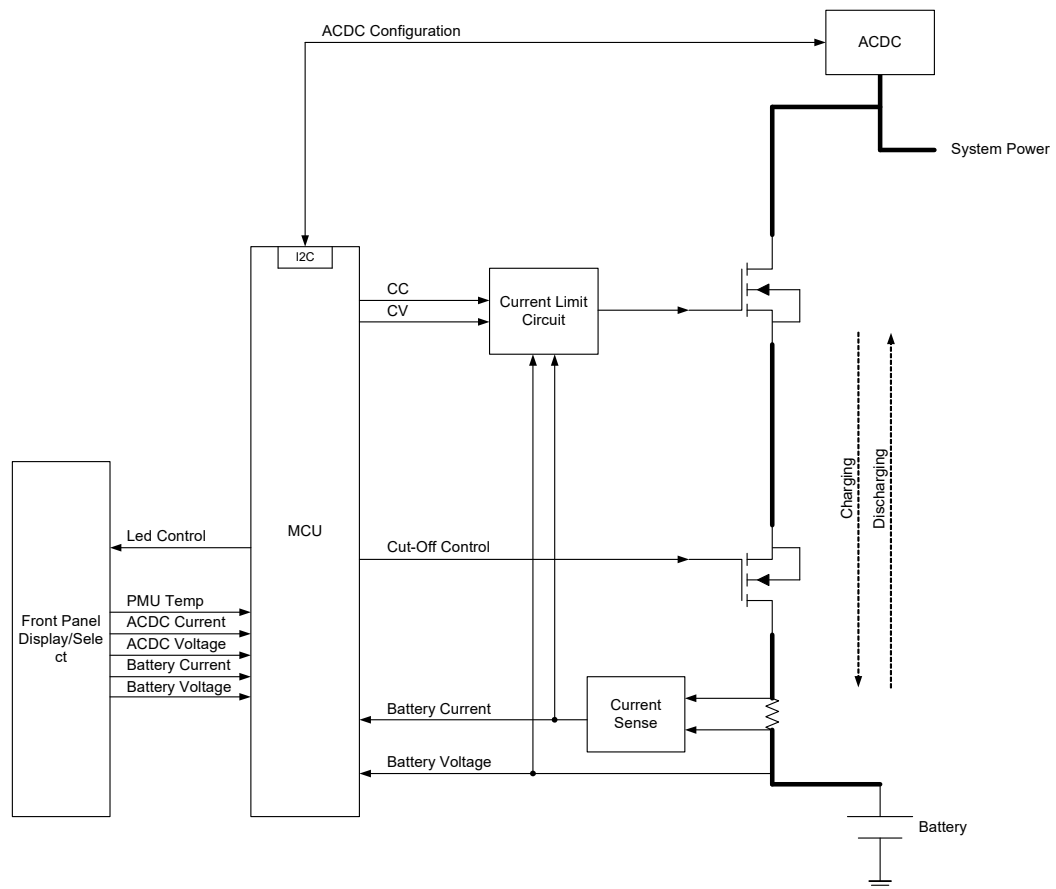


Figure 3-51. PMU Block Diagram

DESCRIPTION

The PMU performs the following functions.

1) PM Bus Monitoring

The maximum permitted voltage and the output voltage of the AC/DC is set and can be read at the display along with current value and temperature.

An AC/DC fault signal is generated if the input/output voltages falls above or below the set value or the equipment gets overheated due to fan failure. The SCU stores the status register value from PMU and generates an alarm in case of fault.

If the AC/DC unit has malfunctioned, the PMU switches the power to battery power source. However, the DC/DC is supplied power by the PMU by OR-ing the output of the two PMU in order to improve the reliability. Either of the two PMUs work as master or slave depending on their state and are used to monitor the AC/DC.

2) Battery Charging and Monitoring

The PMU provides a current path to float charge the backup batteries. The backup batteries are connected to the main +28V DC rail inside the PMU. Thanks to this float charge connection, the batteries can provide the backup power to the system instantly without any switching, when the mains power is interrupted. When the mains power comes back, the batteries are continuously float charged via the PMU.

The PMU monitors the charge and discharge state of the battery by measuring its output voltage

and current.

The PMU transmits alarm information to SCU in case the output voltage of battery decreases below a preset voltage level, typically 22V, and cuts off the battery when it reaches the preset cutoff voltage, typically set to 20V.

3) Reporting Monitoring results to SCU

The PMU monitors and reports the values of output voltage of AC/DC, current, temperature, fan status, to SCU via CAN bus. PMU also communicates any fault occurred to the SCU.

PMU monitors the output and the current status of the battery and reports it to SCU via CAN bus. Depending on the voltage values, PMU communicates with SCU via CAN bus whether to use AC/DC or battery as the power source.

Any fault in PMU functioning is also reported to the SCU.

Each time hardware in a unit is changed, a revision status signal is generated for the value changed and sent to SCU.

SCU receives a control for setting the various parameters of the PMU through the monitor results sent to SCU via CAN bus.

4) Control and status indicators on the front panel

PMU provides LED display for any fault occurrence, power of AC/DC, charging of battery and PMU status.

PMU provides a 7-segment display for the voltage and current measurements.

5) Acts as an Interface

PMU uses PM Bus (I2C interface) for communicating with AC/DC and communicates via CAN with other units outside the power supply unit.

Figure 3-52 illustrates front panel of the PMU.

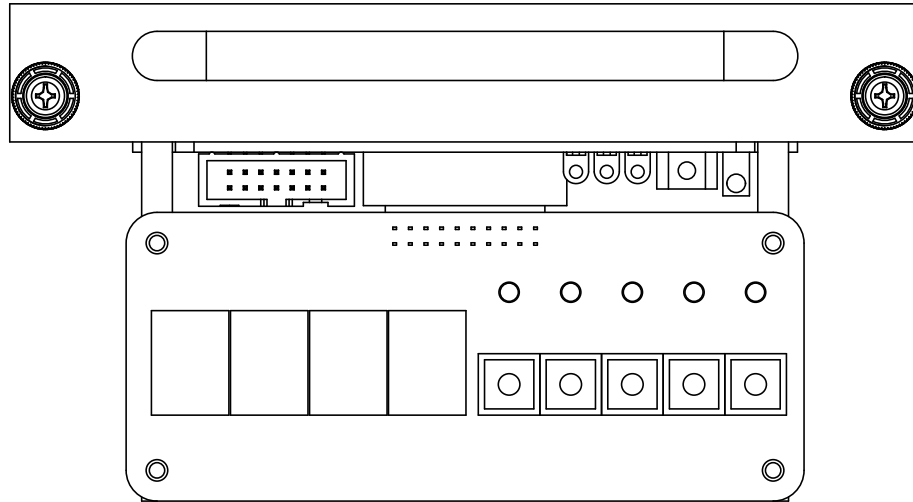


Figure 3-52. PMU Front View

The Table 3-12 describes the control and indicators on the PMU.

Table 3-11. Indicators on PMU

No	Name	Type	Color	Description
1	Numeric Display	7 segment	Green	4-digit multi-function display Contents selected by the push button
2	AC FAIL	LED	Red	AC fail indicator
3	ON BATT	LED	Amber	Battery in use indicator
4	CHARGING	LED	Amber	Battery charging indicator

Table 3-12. Controls on PMU

No	Name	Type	Description
1	TEMP	Push button with LED	Select PMU temperature display
2	DC V	Push button with LED	Select AC/DC output voltage display
3	DC I	Push button with LED	Select AC/DC output current display
4	BATT V	Push button with LED	Select battery voltage display
5	BATT I	Push button with LED	Select battery current display

Table 3-13. Test Points on PMU

No	Name	Type	Color	Description
1	+5V_TP	2mm Tip jack	Red	Test point for DC +5 V rail
2	+15V_TP	2mm Tip jack	Red	Test point for DC +15 V rail
3	-15V_TP	2mm Tip jack	Red	Test point for DC -15 V rail
4	GND	2mm Tip jack	Black	Test point for common (GND)

3.5.3. AC/DC **1A5A1..1A5A5**

OVERVIEW

The 300 DME system uses readily accessible Commercial-Off-The-Shelf AC/DC units to ensure recent standards. The AC/DC is a switched-mode power supply (SMPS), which takes 110V/220V AC mains power input and converts it into 28V DC. The 28V power source is fed to the multi-voltage DC/DC converters and also used to charge external backup batteries.

For safety, both over-voltage protection and over-current protection circuits are included.



Figure 3-53. AC/DC Unit

3.6. EQUIPMENT CABINET

3.6.1. OVERVIEW

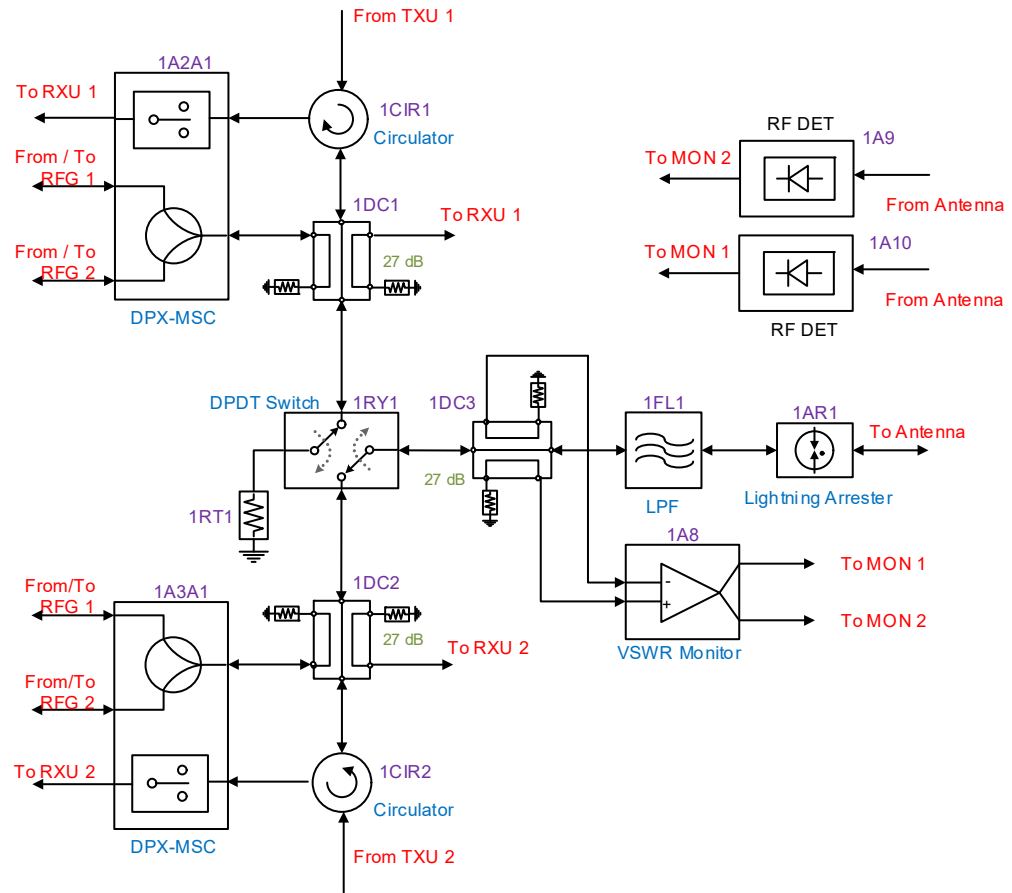


Figure 3-54. Modules on Equipment Cabinet

3.6.2. IFB 1A7

OVERVIEW

The Interface Board (IFB) acts as an interface between devices inside and outside the equipment.

It has the following functions.

- Interface board between devices

IFB acts as an interface between the various units inside the equipment such as SCU, LMI, EMU and TCU. It also acts as an interface for communicating with external units such as Sensors, collocation equipment, PMDT (maintenance PC, RCU (remote control and monitoring), remote voice, FFM (Far Field Monitor), and any other external devices.

IFB provides various interface signals via RS-232, RS-485, USB, modem (telephone line), Ethernet, CAN, It also provides additional 8-channel analog input/output and 16-channel digital input/output ports which are reserved for future use.

- Surge Protection

IFB uses surge protection to protect the equipment from any unexpected incoming high voltages.

- Photo Coupling

Digital input and output signals are photo coupled for insulation from outside.

- ASU power supply interruption

In case both TX1 and TX2 shutdown, the SCU is directed to block the +28V DC power being supplied to ASU.

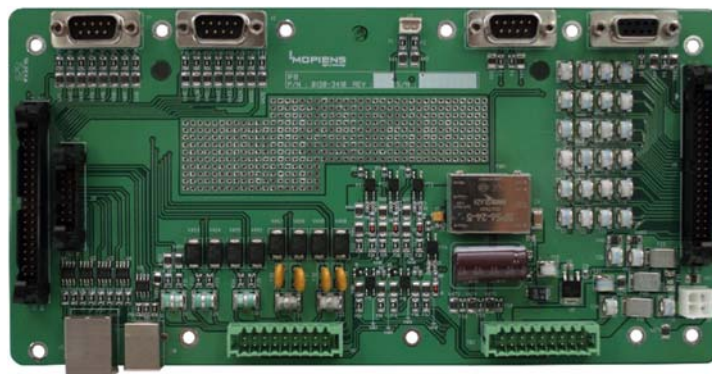


Figure 3-55. IFB

DESCRIPTION

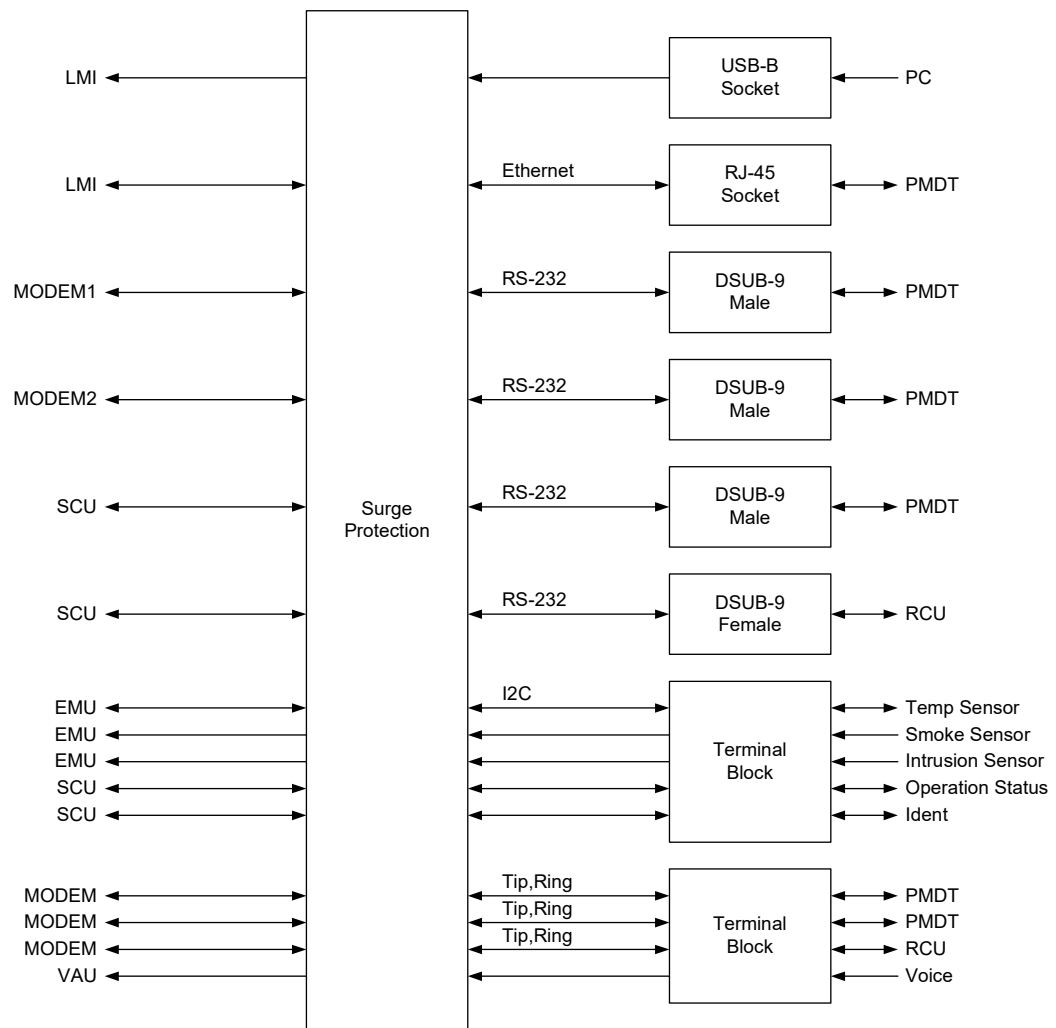


Figure 3-56. IFB Block Diagram

3.6.3. COAXIAL RELAY **1RY1**

The coaxial relay 1RY1 performs a switching function so that two transponders can be connected to one antenna.

J4 of Coaxial Relay 1RY1 is the On Antenna path, which connects the DME transmission / reception signal with the antenna. J3 is connected to 1RT1J1, so that the transmission output of the spare transponder is consumed by the dummy load.

J2 is connected to Transponder 1 and J1 is connected to Transponder 2.

3.6.4. DUMMY LOAD **1RT1**

The dummy load 1RT1 is a high-power RF resistor to terminate the standby transmitter output properly. The dummy load has a heat sink to dissipate the heat converted from the RF energy from the standby transmitter and is completely shielded to prevent any RF leakage.

3.6.5. CIRCULATOR **1CIR1, 1CIR2**

The circulators 1CIR1 and 1CIR2 are used for combining and routing DME transmit / receive signals.

The circulators separate the interrogation signal and the transmission signal from the transponder. The separated interrogation signal is sent to the receiver but the transmission signal is prevented to reach the receiver.

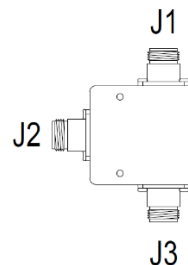


Figure 3-57. Circulator

J1 is output port of the received signal and connected to RXU input. J2 is input port of the transmit signal and connected to TXU or HPA output. J3 is duplexed port and connected to antenna via directional coupler 1DC1 or 1DC2.

3.6.6. DIRECTIONAL COUPLER **1DC1, 1DC2, 1DC3**

The directional coupler 1DC1 and 1DC2 of each transponder to inject the interrogation signal generated by RFG 1A2A7 or 1A3A7 into the receiver of the transponder.

Other functions include sampling of the transmitted signal for delay adjustment of the transponder. The transmitted signal is coupled at J5 and input to J9 of RXU and demodulated at the receiver. The TCU analyzes the demodulated signal to correct the delay error.

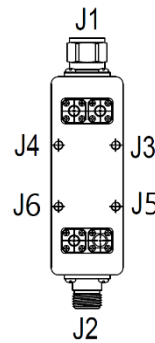


Figure 3-58. Directional Coupler

The directional coupler 1DC3 samples the forward and reverse signals of the transmitted signal and sends them to VSWR monitor module 1A8.

J3 is the forward couple port and connected to J1 of VSWR monitor. J6 is the reverse couple port and connected to the J2 of the VSWR monitor.

3.6.7. VSWR MONITOR 1A8

The VSWR monitor 1A8 measures VSWR of the antenna connected to the output port of the DME and presents a dc voltage corresponding to the measured value to the system monitor 1A2A8 and 1A3A8.

It takes two inputs from the directional coupler 1DC3: one for forward power and the other for reflected power. Since the input power level is normally different between 310 DME and 320 DME by 10 dB, the input signals are attenuated by 10dB at the attenuator U4 and U3 in the case of 320 DME so that the appropriate signal enters the log detector but the attenuators are bypassed in case of 310 DME.



Figure 3-59 VSWR monitor

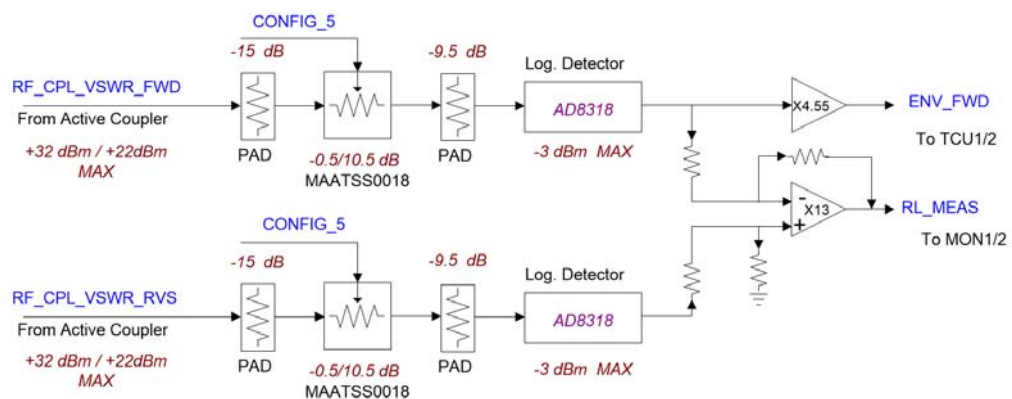


Figure 3-60. Block diagram of VSWR monitor

The two detected signals are connected to the input signal of the OP Amplifier U2. The OP amplifier compares the difference between the two signals and sends the result to MON.

3.6.8. RF DETECTOR 1A9, 1A10

Usually DME antennas have monitor ports which take samples of the transmitted signal with typical couple ratio of 25 dB or so. The sampled signal is then provided to the RF detector 1A9 and 1A10, which detects the pulse power of the transmitted signal and sends it to the system monitor 1A2A8 and 1A3A8 respectively.

The signal samples from the coupled port of the antenna is attenuated by 10dB at the attenuator U4 and U3 in the case of 320 DME, but in the case of 310 DME, no attenuation takes place so that the same level is applied to the detector regardless of the DME type 310 or 320.

The signal is then detected by the U3, amplified by OP Amplifier U1, and sent to the system monitors.

3.6.9. LPF 1FL1

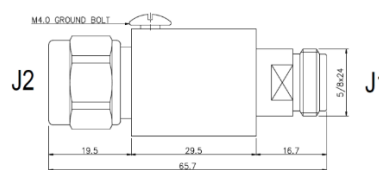
The LPF 1FL1 is used to suppress the second or higher order harmonic components of the transmission signal of the DME.



Figure 3-61. LPF

3.6.10. LIGHTNING ARRESTER 1AR1

The lightning arrester 1AR1 protects the DME equipment from lightning. Peak surge current that can be handled is 10 kA for duration of 50 μ s.



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SECTION 4

OPERATION

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4. OPERATION

4.1. INTRODUCTION

This section explains how to control and operate the 300 DME equipment.

Typically, the PMDT is the main operator interface. PMDT provides most comprehensive, easy-to-use graphic user interface. In order to use the PMDT, the PC must be connected to the equipment via suitable data link. This section shows how to install, setup and use the PMDT.

Local maintenance interface (LMI) is another user interface permanently built-into the equipment.

4.2. CONTROLS AND INDICATORS

4.2.1. LMI

The Local Maintenance Interface (LMI) consists of a 10.2" color TFT LCD with touchscreen and light emitting diode (LED) indicators. With the LCD touchscreen display, the system status and configuration can be displayed and modified.

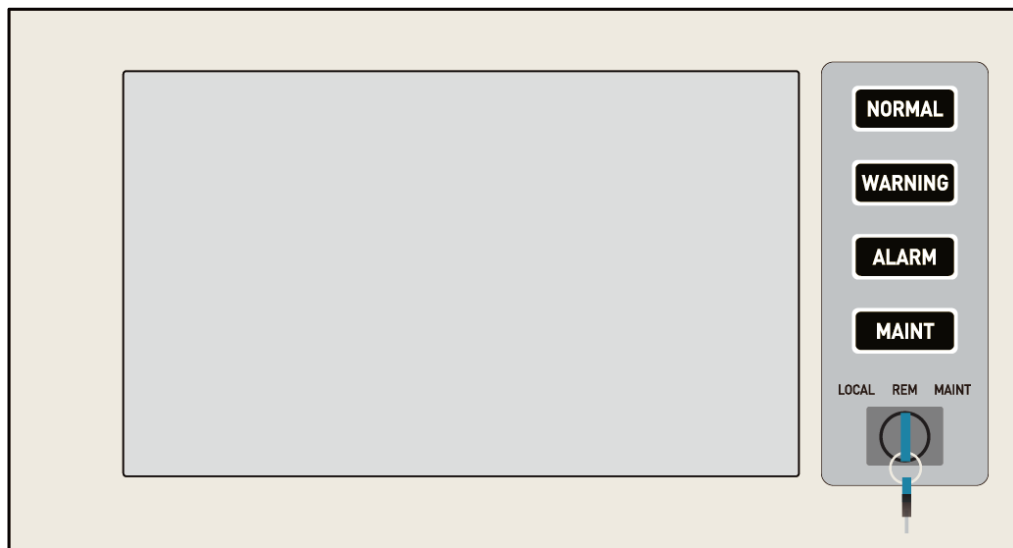


Figure 4-1. LMI faceplate

Figure 4-1 shows the LMI faceplate. Table 4-1 describes the control and indicators on LMI. The LMI located on the front door of cabinet consists of the following controls and indications.

- A 10.2" color TFT LCD with touchscreen for indication of status of transmitters, monitors, measured parameter values, etc.
- Four indication lamps for the main system- **NORMAL**, **WARNING**, **ALARM** and **MAINT**.
- A keylock switch to select three operational modes of the station- **LOCAL**, **REM** and **MAINT** for operation from local site, remote site and in maintenance mode respectively.

Table 4-1. LMI Indicators

Indicator	Color	Description
NORMAL	<i>Green</i>	- The transmitters and monitors are switched ON and serviceable for normal operation - There is no Warning or Alarm
WARNING	<i>Amber</i>	- MON 1 and MON2 are in Warning state. - The equipment is operating on battery supply - BITE warning
ALARM	<i>Red</i>	- MON1 and MON2 are in Alarm state. - CSP fault MON 1 and MON2 are in bypass/maintenance mode
MAINT	<i>Blue</i>	- The equipment is under maintenance mode. - Remote control is disabled – Local control only. - All monitors are automatically bypassed. - If interlock is enabled, it is overridden automatically.

Table 4-2. Keylock Switch

Position	Description
LOCAL	The equipment is enabled for Local operation.
REM	The equipment is enabled for Remote control.
MAINT	- The equipment is in maintenance mode. - Remote control is disabled – Local control only. - All monitors are automatically bypassed. - If interlock is enabled, it is automatically overridden.

4.2.2. CSP

The Control and Status Panel has the following controls and indications as described below.

- Main TXP Select control / indications
- TXP On Antenna Select control / indication
- TXP Power On/Off control / indication
- RF Output Enable/Disable control / indication
- Monitor Bypass control / indication
- Changeover, Antenna Fault, Interlock indication
- AC Main / Battery indication
- System Reset control
- Lamp Test control
- Data exchange between CSP and SCU indication (TxD/RxD)

Figure 4-2 illustrates front faceplate of the CSP.

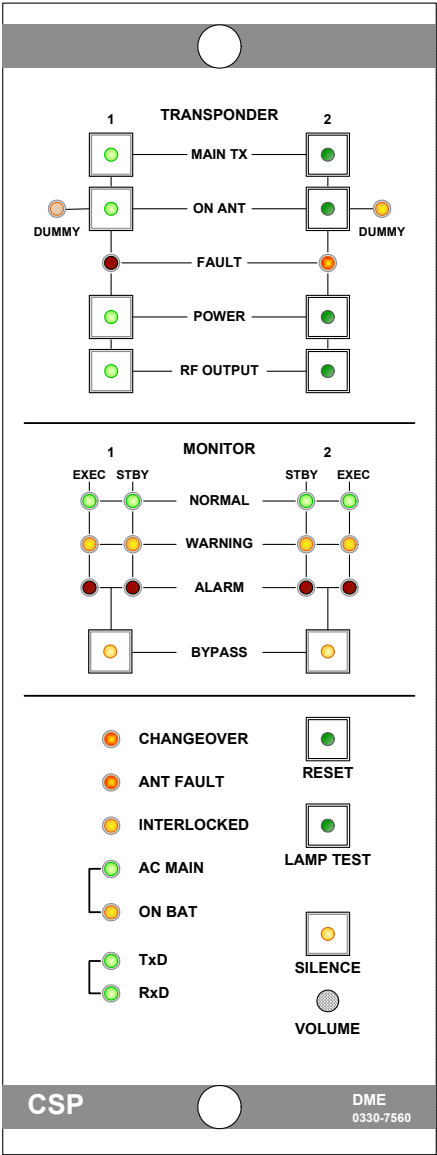


Figure 4-2. CSP Faceplate

4.3. BASIC OPERATION PROCEDURES

4.3.1. TURN ON PROCEDURE

This section provides the procedure to Turn ON the DME equipment and to verify a normal operating state.

The following conditions should be verified before turning ON the equipment.

- Make sure proper antenna or suitable dummy load is connected to the RF output of the equipment. Also two monitor probes are also connected properly.
- It is not mandatory but it is recommend having the remote control system connected to the equipment and turned ON and also the PMDT connected and ready to use.
- The **AC AVAILABLE** lamp on the front panel of the equipment cabinet is illuminated.
- The Backup Battery circuit breaker switch **BATT 1** and **BATT 2** are in OFF position, i.e. the lever downward.
- The main power supply circuit breaker switch **AC MAIN** is in OFF position, i.e. the lever downward.
- The keylock switch on the LMI is in **LOCAL** position, i.e. the key is turned to left.

To turn ON the DME equipment:

- a) Turn ON the **AC MAIN** circuit breaker switch, i.e. the lever upward.
- b) Turn ON the **BAT1** and **BAT 2** circuit breaker switches, i.e. the lever upward.

4.3.2. INDICATIONS DURING NORMAL OPERATION

During normal operation, in TRANSPONDER section on the CSP:

- **MAIN** indicator: either #1 or #2 is illuminated.
- **ON ANT** indicator: either #1 or #2 is illuminated.
- **FAULT** indicator: neither of #1 nor #2 is illuminated.
- **POWER** indicator: both #1 and #2 are illuminated in hot-standby mode. Either #1 or #2 is illuminated in cold-standby mode.
- **ENABLE** indicator: both #1 and #2 are illuminated in hot-standby mode. Either #1 or #2 is illuminated in cold-standby mode.

In MONITOR section on the CSP:

- **NORMAL** indicators: both #1 and #2 are illuminated for both **EXEC** and **STDBY** monitors.
- **WARNING** indicators: neither #1 nor #2 is illuminated for both **EXEC** and **STDBY** monitors.
- **ALARM** indicators: neither #1 nor #2 is illuminated for both **EXEC** and **STDBY** monitors.
- **BYPASS** indicators: both #1 and #2 may be illuminated if monitors are bypassed. However for normal unattended operation, both monitors should not be in bypassed state and the **BYPASS** indicators should not be illuminated.

In the lower section on the CSP:

- The **CHANGEOVER** indicator is OFF.
- The **ANT FAULT** indicator is OFF.
- The **INTERLOCKED** indicator may or may not be illuminated if interlock function is enabled. If interlock function is not used, the **INTERLOCKED** indicator should not be illuminated.
- The **AC MAIN** indicator is illuminated unless the AC line is not interrupted.
- The **ON BAT** indicator is illuminated only when the AC line is interrupted.
- The **ENVIRONMENT** indicator is not illuminated.
- The **TxD** is blinking once in a while
- The **RxD** is blinking constantly.
- The **RESET**, **LAMP TEST**, **SILENCE** indicators are not illuminated.

4.3.3. SHUTDOWN PROCEDURE / TURN OFF PROCEDURE

If there is any unsaved changes that needs to be preserved, save it to the equipment. Optionally, it is recommended to make a backup copy of the current configuration by executing **File → Save Configuration** on PMDT.

Turn the RF outputs of both transponder #1 and transponder #2 OFF by using CSP, LMI or PMDT.

Turn the power of both transponder #1 and transponder #2 OFF by using CSP, LMI or PMDT.

Log out of the PMDT if anyone is logged on.

Turn the backup battery circuit breaker switch **BATT 1** and **BATT 2** OFF, i.e. the lever downward.

Turn the main power supply line circuit breaker switch **AC MAIN** OFF, i.e. the lever downward.

4.4. PMDT

4.4.1. INTRODUCTION

Portable Maintenance Data Terminal (PMDT) is software program used for setup, operation and maintenance of the MOPIENS Radio Navigation Aids equipment. It is provided as a single unified version of user interface for LOC, GP, MKR and FFM and DME. This section describes the functions and screens relating to DME only.

4.4.1.1. Requirements

- IBM® PC compatible computer
- Microsoft Window 7 or later Operating System
- Microsoft .NET Framework version 3.5 or later

Recommendation

- Close all other applications before continuing with the installation.
- Make sure that at least 100 MB of free space available on the drive where the PMDT is installed on.

4.4.2. INSTALLATION AND SETUP

4.4.2.1. Installation Procedure

- a. Run the PMDT Setup file on the installation media and the screen as shown in Figure 4-3 will be briefly displayed.



Figure 4-3. PMDT installation start screen

- b. Then the screen of Figure 4-4 will be displayed. Verify the version number is right and click on the **Next** button.

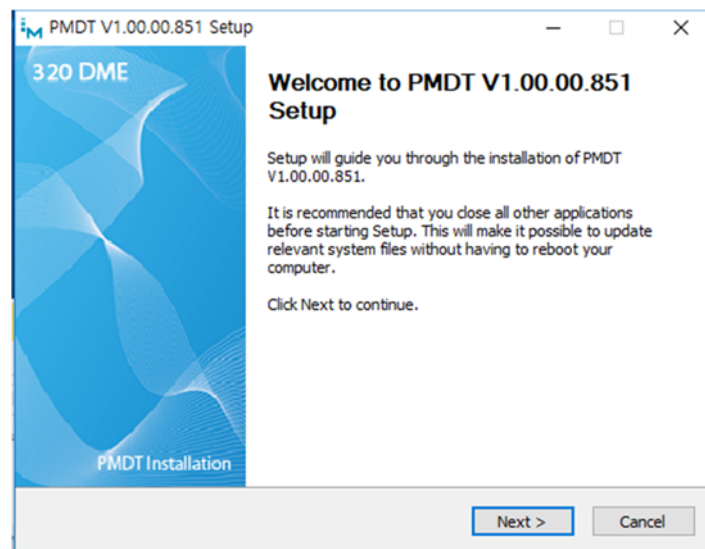


Figure 4-4. PMDT installation welcome screen

- c. Then the screen as in Figure 4-5 will appear. Select appropriate type of install or select the optional components you wish to install and click on **Next** button.

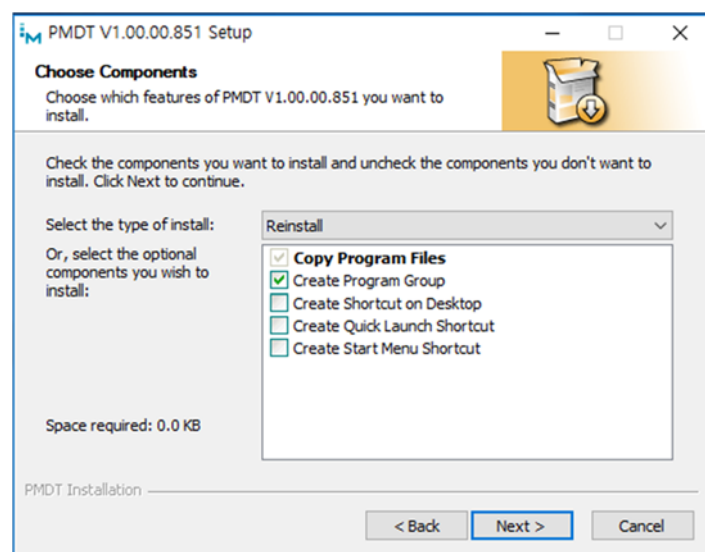


Figure 4-5. Select type of install or select optional components

- d. Then the screen as shown in Figure 4-6 will appear. The default installation folder is "C:\Program Files (x86)\MOPIENS\PMDT". If you want to install in a different folder, click **Browse** button and select another folder. Click **Install** to start the installation.

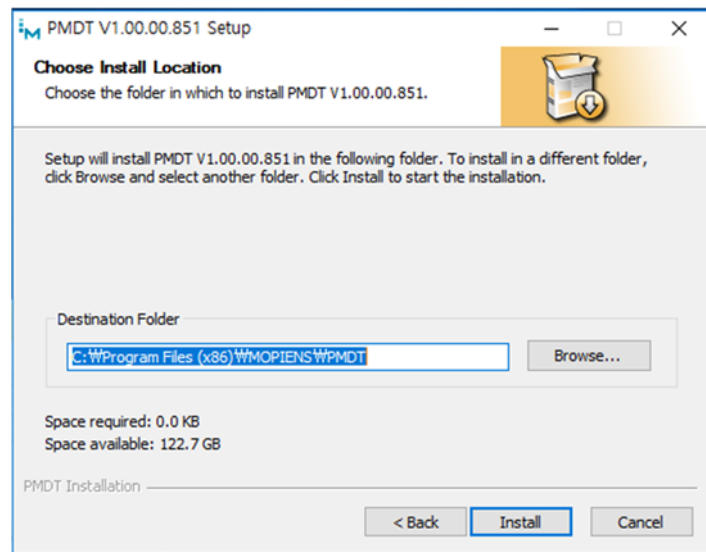


Figure 4-6. Select destination folder

- e. Wait while the PMDT program is being installed.

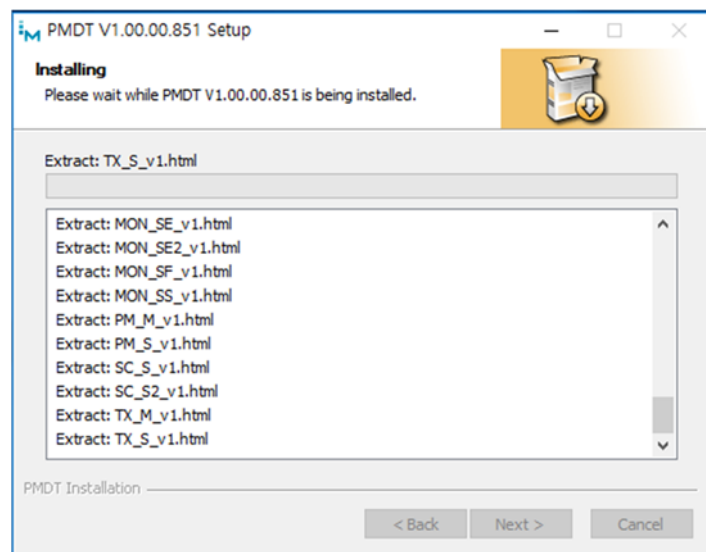


Figure 4-7. Installing screen

- f. When the installation is complete, the screen as shown in Figure 4-8 will be displayed. Click **Finish** button to close setup. If you do not want to run the PMDT after closing setup, unset the **Run PMDT** check box.

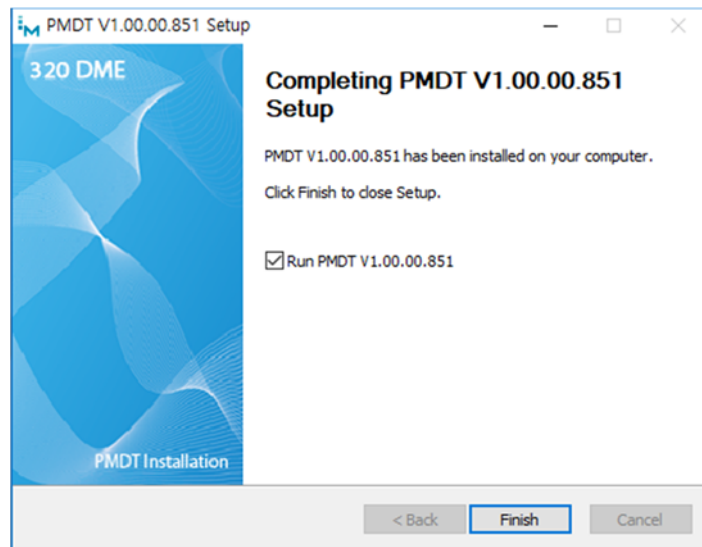


Figure 4-8. Finish screen

4.4.2.2. Uninstallation Procedure

- a. Using windows file explorer and open the folder where PMDT is installed in and run the uninstallation program, *uninstall.exe*.

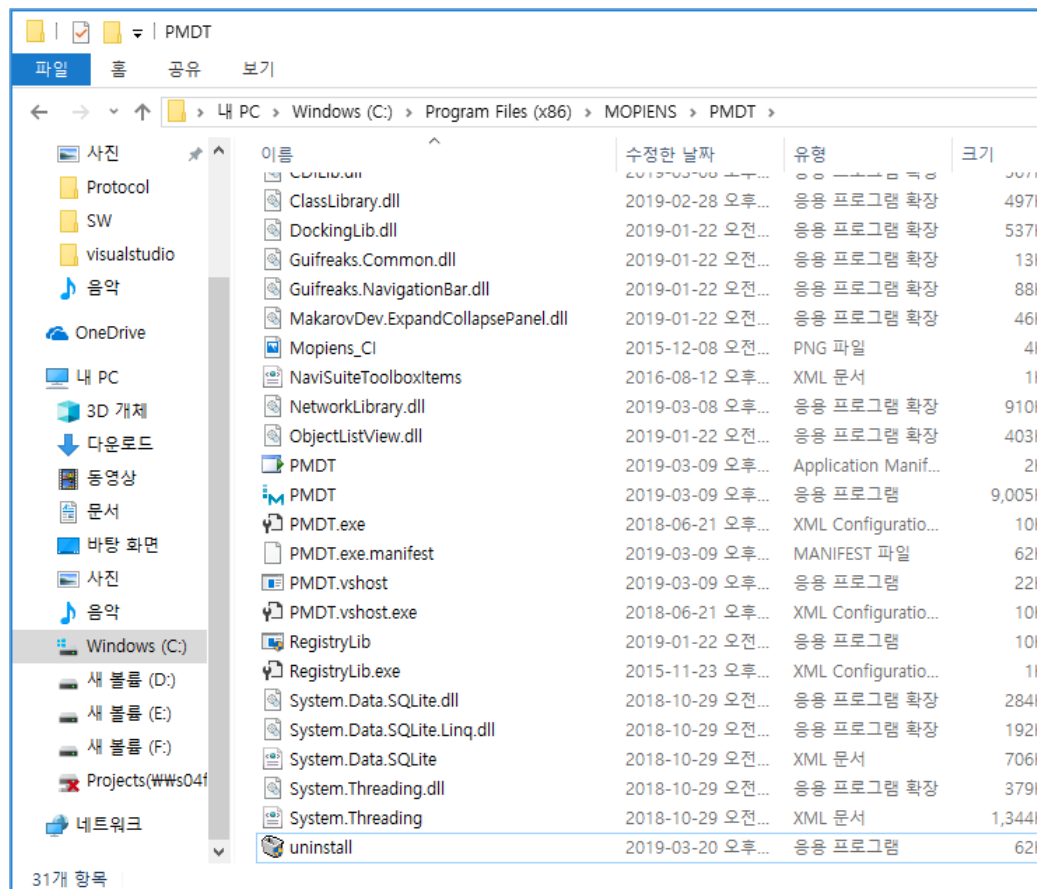


Figure 4-9. uninstall.exe in PMDT folder

- b. Then the screen as shown in Figure 4-10 will appear. Click on **Uninstall** button to continue with the uninstallation or press **Cancel** button to exit without uninstallation.

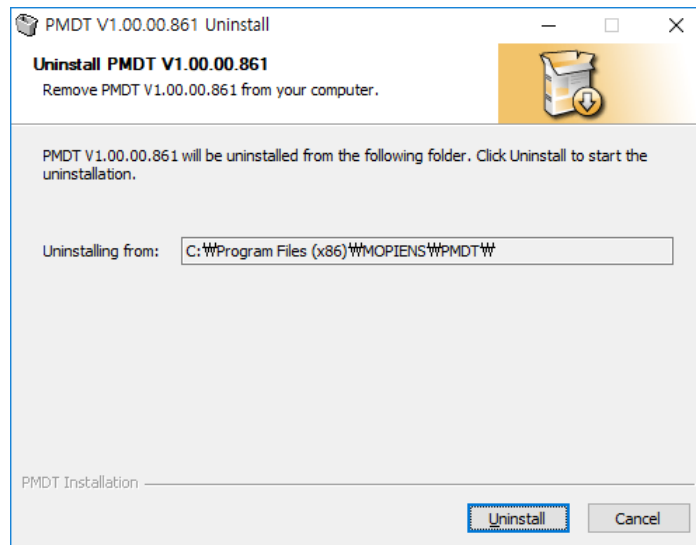


Figure 4-10. Uninstallation start screen

- c. When the uninstallation is complete, the screen as shown in Figure 4-11 will appear. Click on the **Close** button to close the uninstallation program.

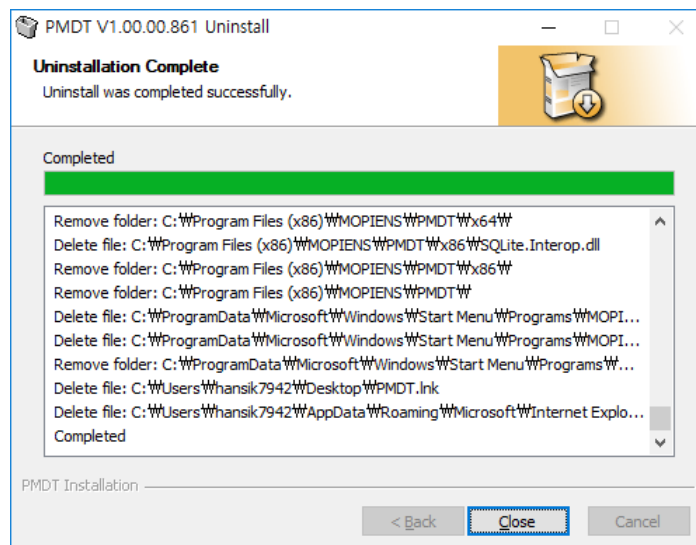


Figure 4-11. Uninstallation complete screen

4.4.2.3. Maintenance

PMDT Application Data Folder

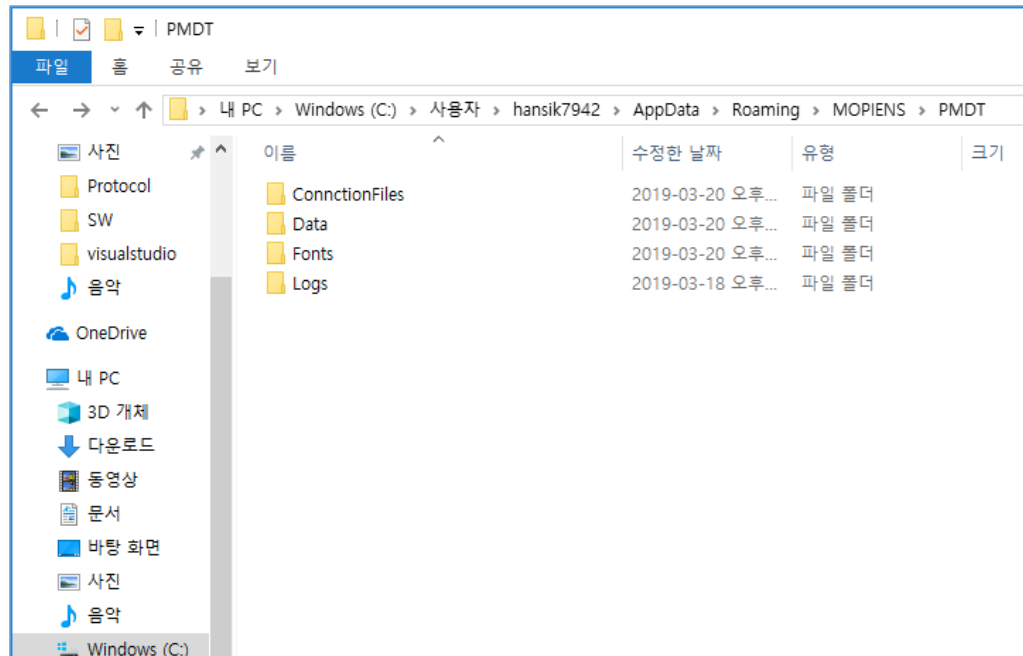


Figure 4-12. PMDT Application Data Folder

PMDT application specific data is stored in the folder: %AppData%\MOPIENS\PMDT.

- “ConnectionFile”: Connection specific data or profiles is stored.
- “Data”: PMDT event log data is stored.
- “Logs”: Log files for unexpected disconnection and other exception data are stored.

Command Line Options

PMDT takes the following command line arguments. The command line arguments can be inserted into the PMDT application shortcut as shown in Figure 4-13.

- “- loc_demo”: Starts PMDT in offline demo mode for LOC.
- “- gp_demo”: Starts PMDT in offline demo mode for GP.
- “- dme_demo”: Starts PMDT in offline demo mode for DME.
- “- ffm_demo”: Starts PMDT in offline demo mode for FFM.
- “- connectionName”: Starts PMDT with the connection profile specified by connectionName without the prompt dialog box “connection list” which is to select the destination station to connect.



Figure 4-13. Adding command line option in PMDT shortcut

4.4.3. MAIN MENUS

4.4.3.1. File Menu

4.4.3.1.1. New Connection

Selecting the menu item **New Connection** will bring a popup dialog box as shown in Figure 4-15 to start a new session to preconfigured equipment or to create, delete or modify connections.

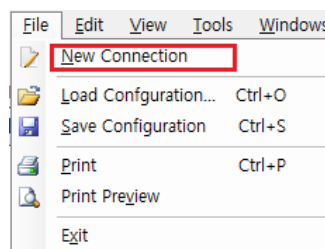


Figure 4-14. New Connection

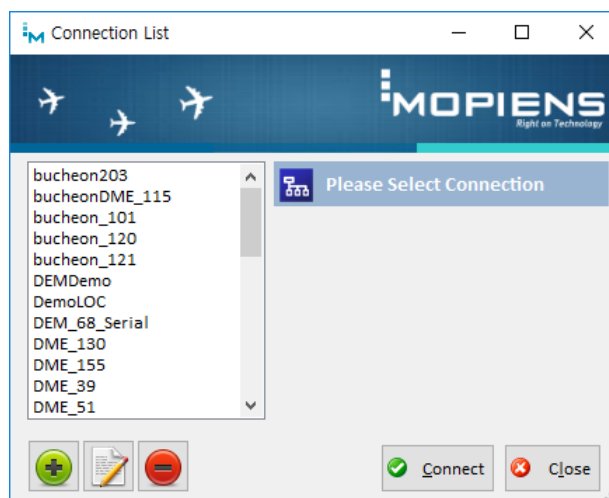


Figure 4-15. Connection List

-  This button is to create a new connection.
-  This button is to modify selected connection.
-  This button is to delete selected connection.

To connect to the equipment station, select the name of the station from the list on the left and click **Connect** button.

4.4.3.1.2. Connection Configuration

USING ETHERNET INTERFACE:

Figure 4-16. Ethernet Connection Configuration

- **Session Name:** Enter a name for the connection.
- **Session Description:** Enter descriptive text for the connection.
- **Connection Timeout:** Enter the timeout for automatic retry attempts.
- **IP Address:** Enter IP address of the station to connect to.
- **Ethernet Port:** Enter the UDP port number to use (default: 38317).
- **Automatic Login:** If checked, it will try to automatically log-in using the configured User Name and Password automatically.

USING RS-232 INTERFACE:

The screenshot shows the 'Connection Configuration' window for the RS-232 interface. The window is titled 'Connection Configuration' and features the MOPIENS logo and tagline 'Right on Technology' at the top. The configuration fields are as follows:

- Interface:** RS-232 (dropdown menu)
- Session Name:** (text field, note: Blank is not Allowed.)
- Session Description:** (text field)
- Connection Timeout:** 10 sec (Minimum 10seconds) (spin box)
- Serial Port:** COM1 (dropdown menu) with a 'Refresh' button
- Baudrate:** 230400 (dropdown menu)
- Flow Control:** None (dropdown menu)
- Parity:** None (dropdown menu)
- Data Bit:** 8 (dropdown menu)
- Stop Bit:** 1 (dropdown menu)
- Automatic Login:**
 - ☐ Enable
 - User Name:** (text field)
 - Password:** (text field)

At the bottom right, there are 'OK' and 'Cancel' buttons.

Figure 4-17. Connection Configuration RS-232

- **Session Name:** Enter a name for the connection.
- **Session Description:** Enter descriptive text for the connection.
- **Connection Timeout:** Enter the timeout for automatic retry attempts.
- **Serial Port:** Select the serial COM port to use.
- **Baudrate:** Select the baud rate to use.
- **Flow Control:** Choose flow control option.
- **Parity:** Choose parity bit option.
- **Data Bit:** Choose data bit option.
- **Stop Bit:** Choose stop bit option.
- **Automatic Login:** If checked, it will try to automatically log-in using the configured User Name and Password automatically.

USING LEASED LINE MODEM:

The screenshot shows the 'Connection Configuration' window for a leased line modem. The window is titled 'Connection Configuration' and features the MOPIENS logo at the top. The interface is organized into several sections: 'Interface' with a dropdown set to 'Leased Line'; 'Session Name' and 'Session Description' text boxes; 'Connection Timeout' set to 10 seconds; a 'Serial Port' dropdown set to 'COM1' with a 'Refresh' button; and a series of dropdowns for 'Baudrate', 'Flow Control' (None), 'Parity' (None), 'Data Bit' (8), and 'Stop Bit' (1). Below these is an 'AT Command' text box. An 'Automatic Login' section contains an 'Enable' checkbox and fields for 'User Name' and 'Password'. At the bottom right are 'OK' and 'Cancel' buttons.

Figure 4-18. Connection Configuration Leased Line

- **Session Name:** Enter a name for the connection.
- **Session Description:** Enter descriptive text for the connection.
- **Connection Timeout:** Enter the timeout for automatic retry attempts.
- **Baudrate:** Select the baud rate to use.
- **Flow Control:** Choose flow control option.
- **Parity:** Choose parity bit option.
- **Data Bit:** Choose data bit option.
- **Stop Bit:** Choose stop bit option.
- **AT Command:** Enter AT command string for initializing the modem.
- **Automatic Login:** If checked, it will try to automatically log-in using the configured User Name and Password automatically.

USING DIAL-UP MODEM:

The screenshot shows the 'Connection Configuration' window for a dial-up modem. The window is titled 'Connection Configuration' and features the MOPIENS logo and tagline 'Right on Technology' at the top. The interface is organized into several sections:

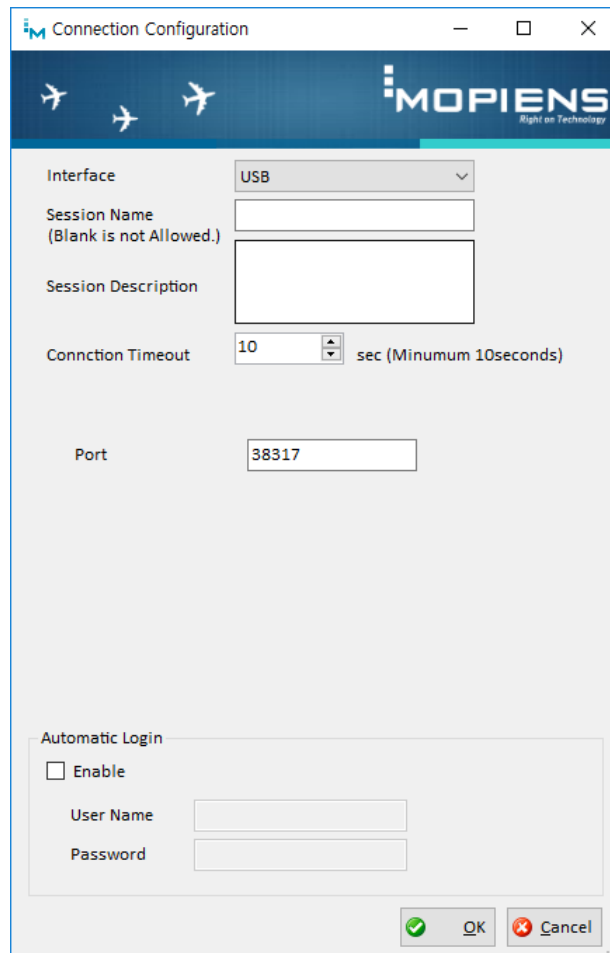
- Interface:** A dropdown menu set to 'Dial-up'.
- Session Name:** A text field with the note '(Blank is not Allowed.)'.
- Session Description:** A larger text area for descriptive text.
- Connection Timeout:** A numeric field set to '10' with a unit of 'sec (Minimum 10seconds)'.
- Serial Port:** A dropdown menu set to 'COM1', accompanied by a 'Refresh' button.
- Baudrate:** A dropdown menu.
- Flow Control:** A dropdown menu set to 'None'.
- Parity:** A dropdown menu set to 'None'.
- Data Bit:** A dropdown menu set to '8'.
- Stop Bit:** A dropdown menu set to '1'.
- Telephone No.:** A text field for the phone number to dial.
- AT Command:** A text field for the AT command string.
- Automatic Login:** A section containing an 'Enable' checkbox and fields for 'User Name' and 'Password'.

At the bottom right, there are 'OK' and 'Cancel' buttons.

Figure 4-19. Connection Configuration Dial-Up

- **Session Name:** Enter a name for the connection.
- **Session Description:** Enter descriptive text for the connection.
- **Connection Timeout:** Enter the timeout for automatic retry attempts.
- **Baudrate:** Select the baud rate to use.
- **Flow Control:** Choose flow control option.
- **Parity:** Choose parity bit option.
- **Data Bit:** Choose data bit option.
- **Stop Bit:** Choose stop bit option.
- **Telephone No.:** Enter the phone number to dial.
- **AT Command:** Enter AT command string for initializing the modem.
- **Automatic Login:** If checked, it will try to automatically log-in using the configured User Name and Password automatically.

USING USB INTERFACE:



The screenshot shows a window titled "Connection Configuration" with the MOPIENS logo and tagline "Right on Technology" in the header. The window contains the following fields and controls:

- Interface:** A dropdown menu set to "USB".
- Session Name:** A text input field with the note "(Blank is not Allowed.)" below it.
- Session Description:** A larger text input field.
- Connection Timeout:** A spinner box set to "10" with the unit "sec (Minumum 10seconds)".
- Port:** A text input field containing the value "38317".
- Automatic Login:** A section containing an "Enable" checkbox (which is unchecked), a "User Name" text input field, and a "Password" text input field.
- Buttons:** "OK" and "Cancel" buttons at the bottom right.

Figure 4-20. Connection Configuration USB

- **Session Name:** Enter a name for the connection.
- **Session Description:** Enter descriptive text for the connection.
- **Connection Timeout:** Enter the timeout for automatic retry attempts.
- **Port:** Enter the UDP port number to use (default: 38317).
- **Automatic Login:** If checked, it will try to automatically log-in using the configured User Name and Password automatically.

USING OFFLINE DEMO MODE:

Figure 4-21. Connection Configuration Demo

- **Session Name:** Enter a name for the connection.
- **Session Description:** Enter descriptive text for the connection.
- **Connection Timeout:** Enter the timeout for automatic retry attempts.
- **Equipment:** Choose the demo equipment type to run.
- **Automatic Login:** If checked, it will try to automatically log-in using the configured User Name and Password automatically.

4.4.3.1.3. Load Configuration

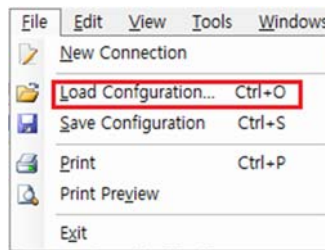


Figure 4-22. Load Configuration

- Loads the system configuration file in XML format into the equipment.
- The settings will be applied on execution of the command but it will not be automatically preserved after the power is turned off, unless the profile setting is saved to the non-volatile flash memory on the equipment.

4.4.3.1.4. Save Configuration

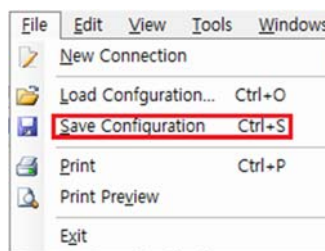


Figure 4-23. Save Configuration

- Saves the system configuration into a computer file in XML format on the PC.
- All the set items currently in effect are saved as an XML file, which can be loaded later back to the equipment using **Load Configuration** command. The file also can be open and edited using a suitable text editor or xml editor.

4.4.3.1.5. Print

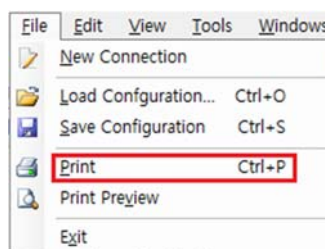


Figure 4-24. Print

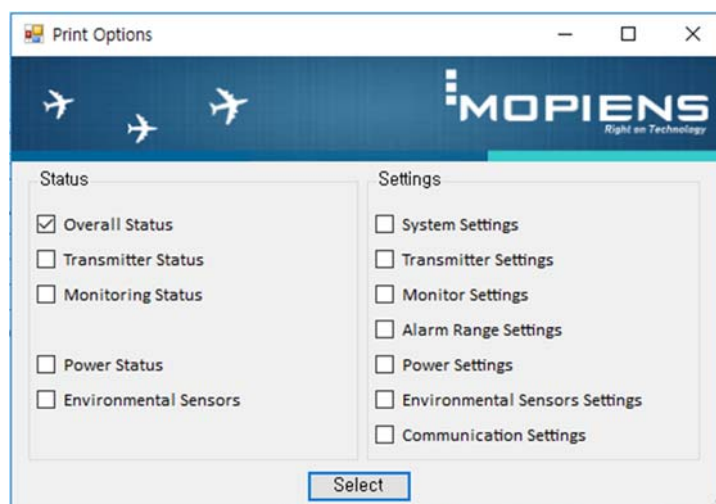


Figure 4-25. Print Option

- Prints output the current status and setting information of the equipment.
- Only checked items are printed out.

4.4.3.1.6. Print Preview

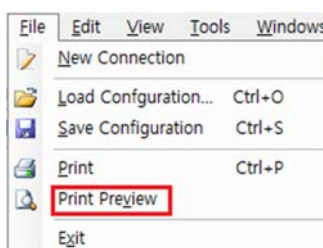


Figure 4-26. Print Preview

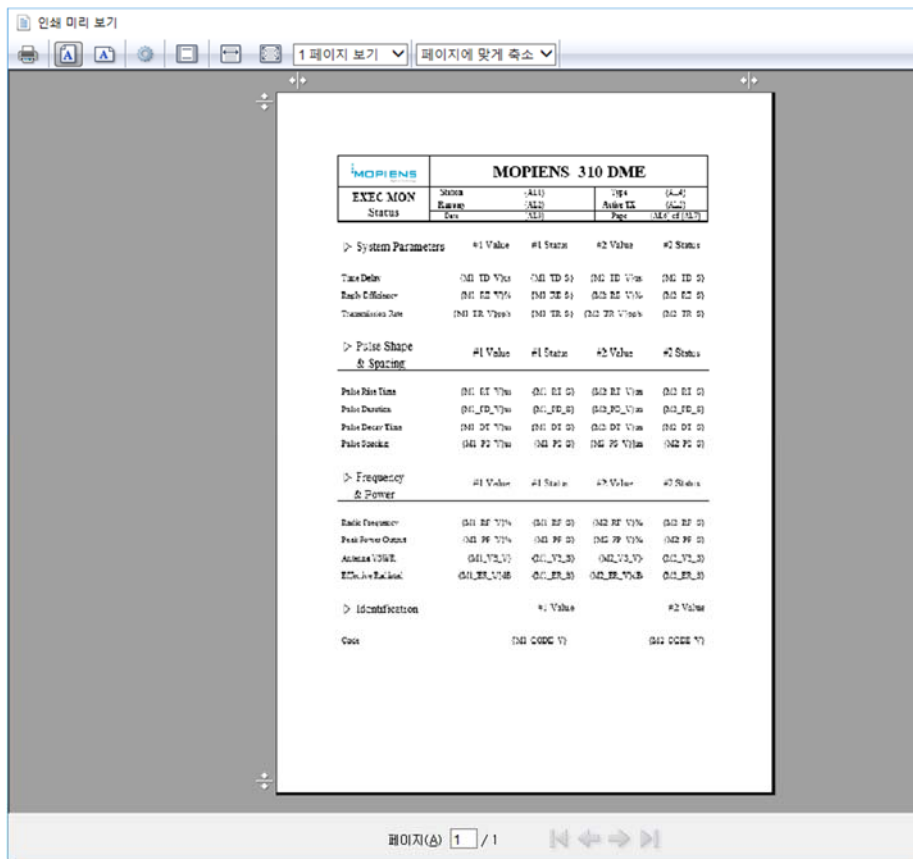


Figure 4-27. Preview

- The print preview function allows you to check what you want to print.
- Same as the Print menu, only checked item will appear on the preview.

4.4.3.1.7. Exit

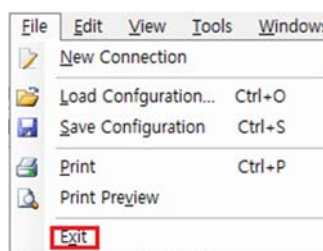


Figure 4-28. Exit

Exit terminates PMDT and exit.

4.4.3.2. Edit Menu

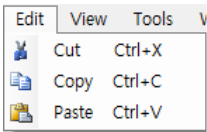


Figure 4-29. Menu Edit

- **Cut:** cuts the selection and moves into the clipboard
- **Copy:** copies the selection and moves into the clipboard.
- **Paste:** pastes the text in the clipboard at current cursor location

4.4.3.3. View Menu

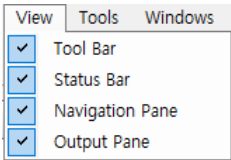


Figure 4-30. Menu View



Figure 4-31. Tool Bar



Figure 4-32. Status Bar

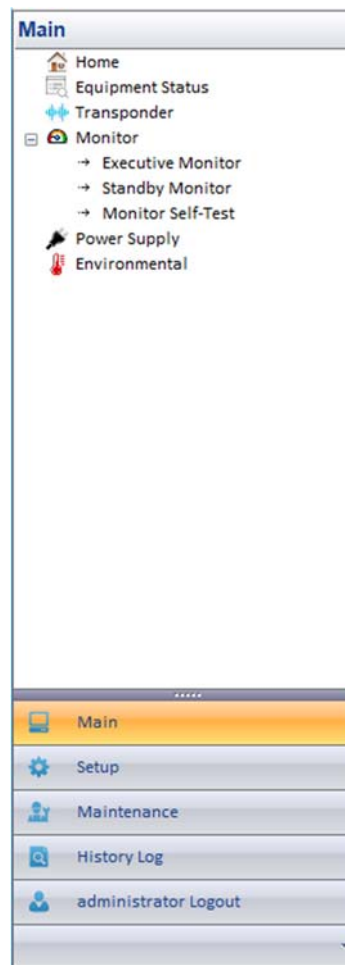


Figure 4-33. Navigation Pane

Control			
Time	Detail	UserID	
2018-12-14 오후 2:42:38.4	User Logged Out : administrator	administrator	
2018-12-14 오후 12:50:57.8	User Logged Out : administrator	administrator	
2018-12-14 오후 12:27:02.8	User Logged Out : administrator	administrator	
2018-12-14 오후 2:36:25.9	User Logged Out :		
2018-12-14 오후 2:59:56.7	User Logged In : administrator(Level3)	administrator	

Figure 4-34. Output Pane

- **Tool Bar** shows a group of icons which contains frequently used commands.
- **Status Bar** shows concise status of the connected equipment.
- **Navigation Pane** shows menu groups that can be navigated through.
- **Output Pane** shows alarms, control or event history.

4.4.3.4. Tools Menu

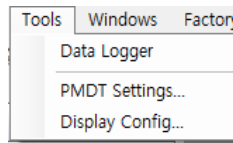


Figure 4-35. Menu Tools

4.4.3.4.1. Data Logger

The 310/320 DME includes a Data Logger to record equipment status variables including monitor parameters. This function is enabled only when the user sets it as an option. When you set the Data Logger, the log of the measured value is saved as a CSV file at the set location.

Data Logger function setting is available through the Tool - Data Logger Options menu on the PMDT top menu.

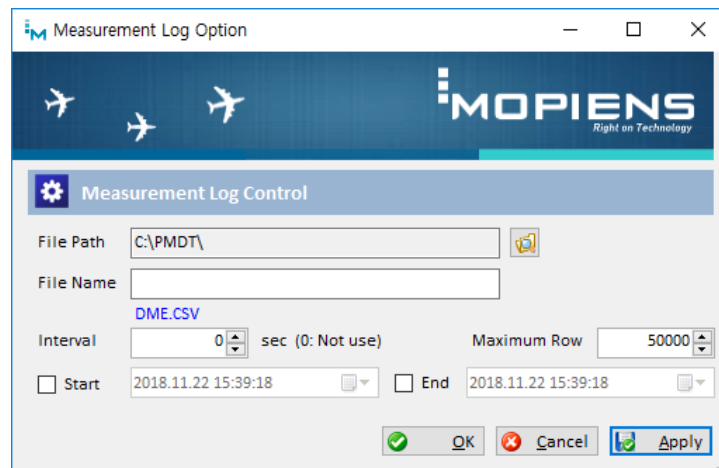


Figure 4-36. Data Logger Option

Figure 4-36 shows the Data Logger Option screen.

Items that can be set are as follows

- **File Path:** Measurement Log Storage Location
- **File Name:** The name of the Measurement Log file
 - If multiple files are created because the size of one file is large, the file name is followed by (1), (2), ... is added.
- **Interval:** Cycle of storing measured value, 0 is disabled, can be set in 1 second
- **Maximum Row:** The maximum number of lines a file has. If it exceeds, separate it into another file.
- **Start:** Start date of the Data Logger
- **End:** Data Logger end date, do not exit unless selected

The measurement log file saved by Data Logger can be open and examined by external application programs such as Microsoft Excel.

4.4.3.4.2. Menu PMDT Settings

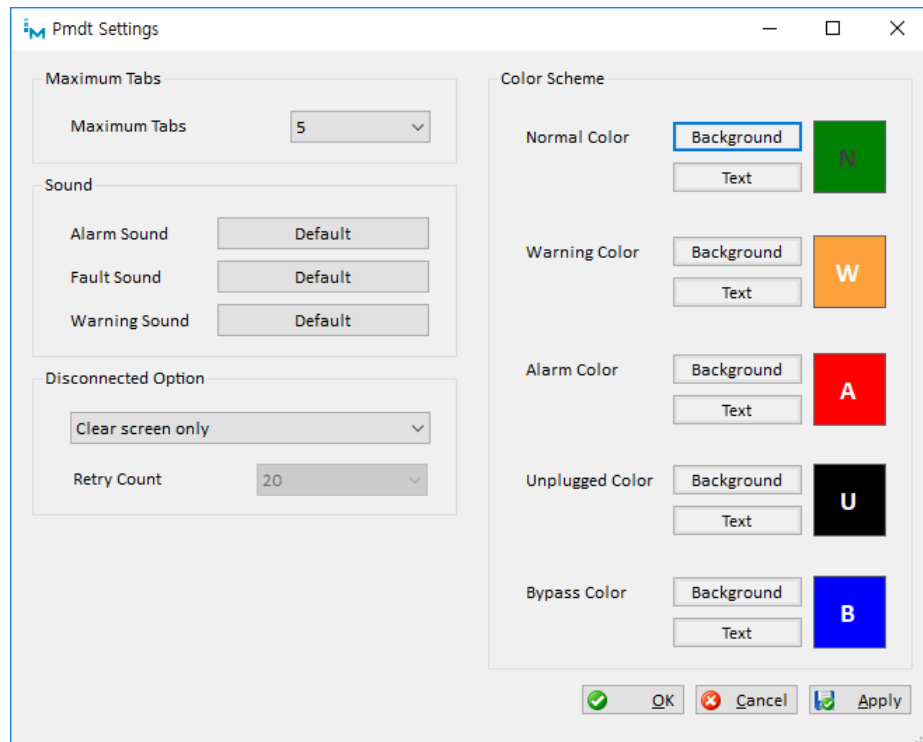


Figure 4-37. PMDT Settings

Category	Item	Description
Maximum Tabs	Maximum Tabs	Set the number of PMDT tabs. At least 3, up to 10.
Sound	Alarm Sound	Alarm Sound can be set by the user by calling Sound File.
	Fault Sound	Fault Sound can be set by the user by calling Sound File.
	Warning Sound	Warning Sound can be set by the user by calling Sound File.
Disconnected option	option	Set the action when PMDT connection is released. • Clear screen only: Initializes the PMDT screen. • Exit Program: Exit PMDT. • Clear Screen and Automatic Retry: Initializes the screen and retries the connection.
	Retry Count	Set the number of connection retries. Minimum: 1, Maximum: Continuous
Far Field Monitor	IP Address	Far Field Monitor is available. Enter the IP Address to connect.

Color Scheme	Normal Color	Set the Normal Color. Sets the text in the normal state.
	Warning Color	Sets the Color of the Warning state. Sets the Text of the Warning state.
	Alarm Color	Set the alarm status color. Sets the text of the alarm status.
	Unplugged Color	Sets the color of the unplugged state. Sets Text in Unplugged state.
	Bypass Color	Sets the Bypass color. Sets the Bypass text.

Table 4-3. PMDT Settings

4.4.3.4.3. Display Configuration

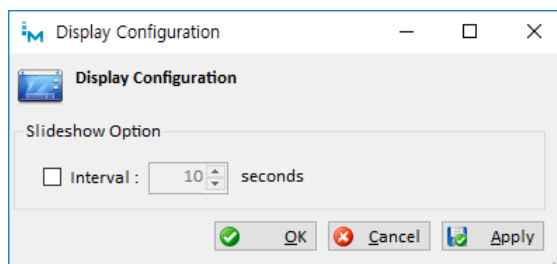


Figure 4-38. Display Configuration

Category	Item	Description
Slideshow Option	Interval	Set the Slideshow Interval on the Home screen.

Table 4-4. Display Configuration

4.4.3.5. Windows Menu

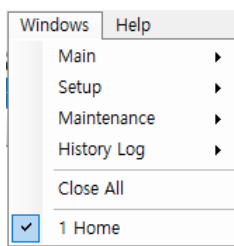


Figure 4-39. Window

- **Close All:** Closes all open windows on the current PMDT screen.
- Displays the currently active screen and displays the list of open screens.
- Move to each screen of PMDT.

First Depth	Second Depth	Third Depth
Main	Home	
	Equipment Status	

	Transponder	
	Monitor	Executive Monitor
		Standby Monitor
	Power Supply	
Setup	Environmental	
	Station Setup	
	Transmitter	Transmitter Setup
		Overheat Prevent Setup
	Monitor	Executive Monitor Setup
		Monitor Limit Setup
	System Setup	
	Environmental Setup	
Maintenance	Communication Setup	
	Miscellaneous Setup	
	Calibration	
	User Management	
	Time Sync	
History Log	Version	
	PMDT History Log	
	LMI History Log	

Table 4-5. Window

4.4.3.6. Help

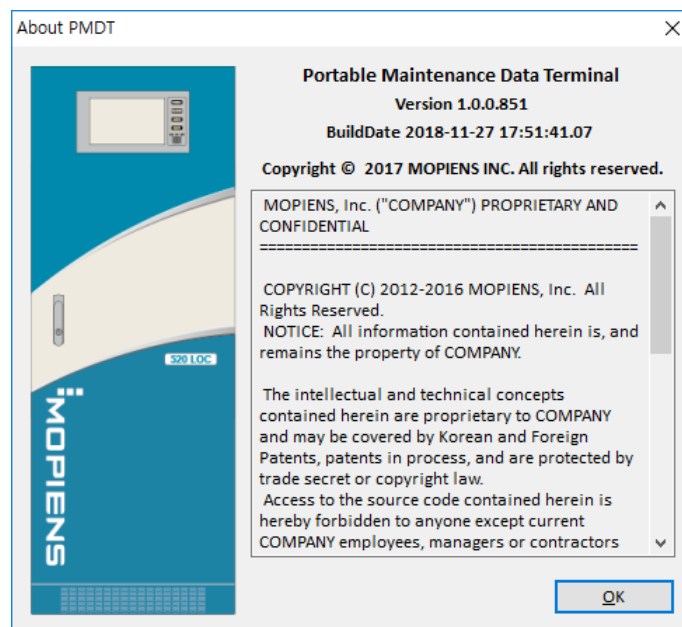


Table 4-6. About PMDT

The **About** box shows information and explanation about PMDT.

4.4.4. SCREEN LAYOUT

4.4.4.1. Title Bar



Figure 4-40. Title Bar

- Display the Session Name.
- Display session information.

4.4.4.2. Menu Status



Figure 4-41. Menu Status

Monitor status indication

- **EXE**: Executive monitor, **STB**: Standby monitor,
- Monitor 1 Monitor2
- Normal Warning Alarm Unplugged

Control status indication

- This symbol indicates that control is available on the PMDT. If the PMDT is a local, the keylock Switch is in **MAINT** or **LOCAL** position. If the PMDT is a remote, the keylock switch is in **REM**.
- This symbol indicates that control is not available on the PMDT. If the PMDT is a local, the keylock switch is in **REM** position. If the PMDT is a remote, the keylock switch is in **LOCAL** or **MAINT** position.

Login status

- As shown in Figure 4-41, the name and access privilege level of the user currently logged in are indicated.
- User Level 0 User Level 1 User Level 2 User Level 3

4.4.4.3. Menu and Toolbar Icons

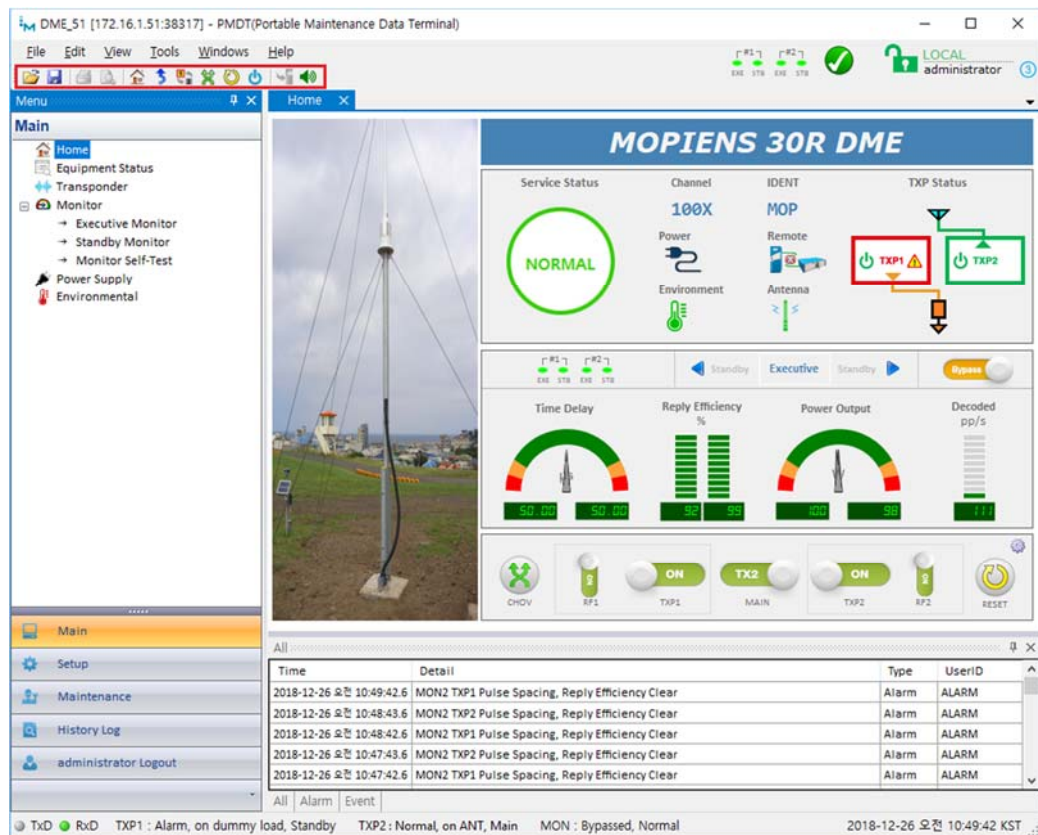


Figure 4-42. Menu Icon

Load Profile

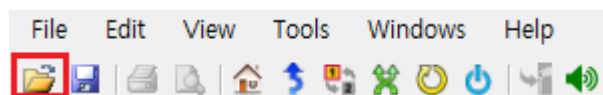


Figure 4-43. Menu Icon Load

- Loads the system configuration file in XML format into the equipment.
- The settings will be applied on execution of the command but it will not be automatically preserved after the power is turned off, unless the profile setting is saved to the non-volatile flash memory on the equipment.

Save Profile

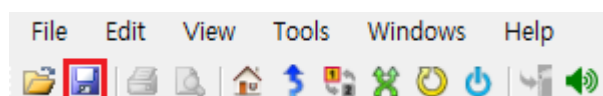


Figure 4-44. Menu Icon Save

- Saves the system configuration into a computer file in XML format on the PC.
- All the set items currently in effect are saved as an XML file, which can be loaded later

back to the equipment using **Load Configuration** command. The file also can be open and edited using a suitable text editor or xml editor.

Print

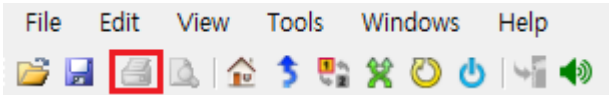


Figure 4-45. Menu Icon Print

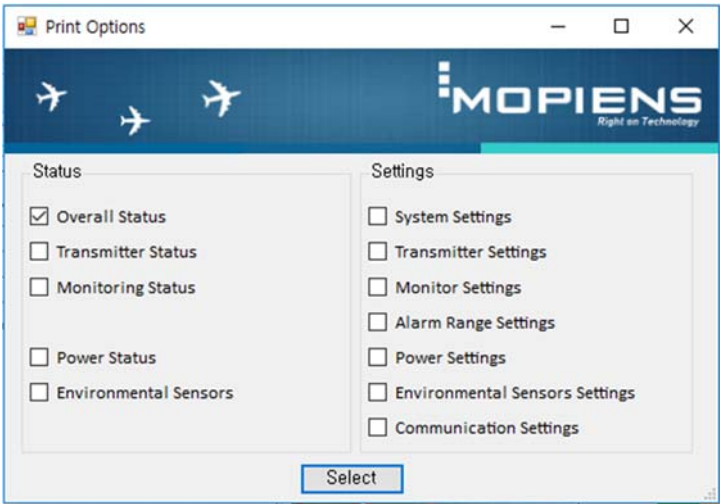


Figure 4-46. Print Options

- Prints output the current status and setting information of the equipment.
- Only checked items are printed out.

Print Preview

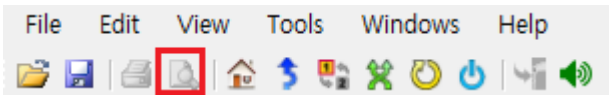


Figure 4-47. Print Preview

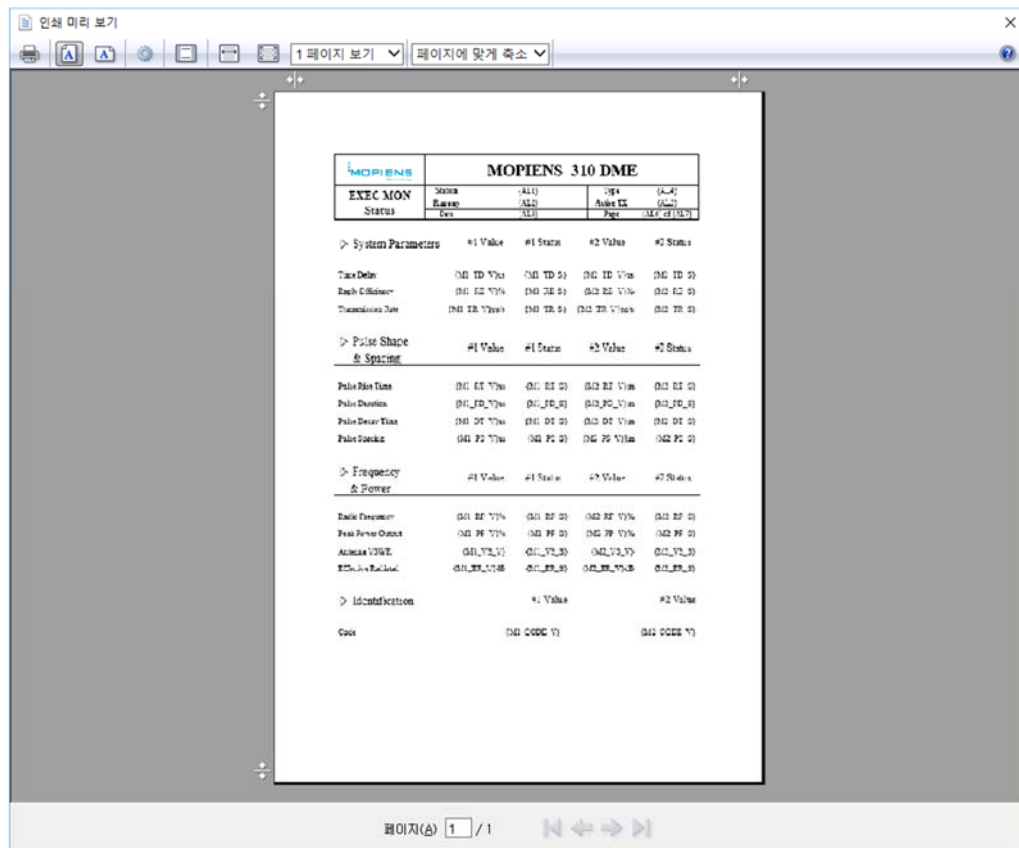


Figure 4-48. Preview

- The print preview function allows you to check what you want to print.
- Same as the Print menu, only checked item will appear on the preview.

Home

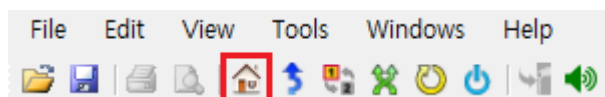


Figure 4-49. Menu Icon Home

- Clicking on this icon opens the Home screen as shown in Figure 4-50 or bring it back to top if it is already open.

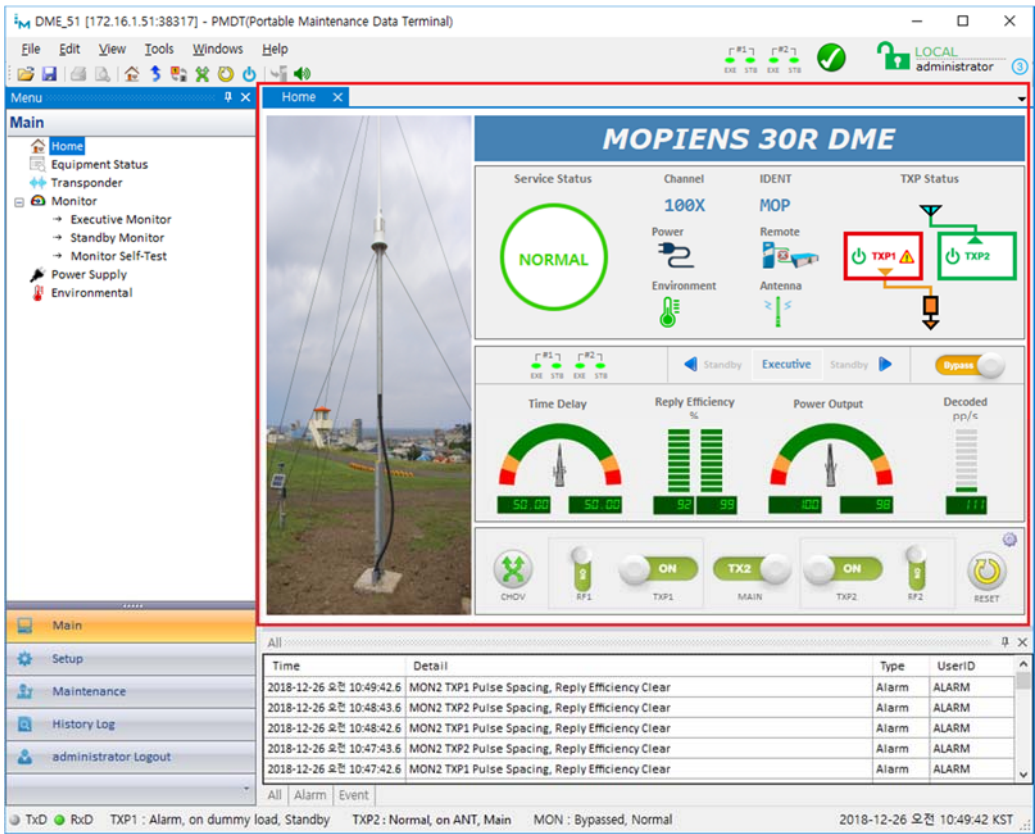


Figure 4-50. Home Screen

Monitor Bypass

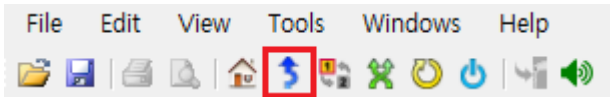


Figure 4-51. Menu Monitor Bypass

- Clicking on this icon will pop up the **Monitor Bypass** dialog box as shown in Figure 4-52.
- Pressing the **Monitor1** / **Monitor2** slide switch button will toggle between ‘**Bypass**’ and ‘**Auto**’ for each monitor.
- When Bypass is set, the executive action such as automatic transmitter changeover or shutdown will not be taken place even if there is a primary alarm being persisted for extended period of time.

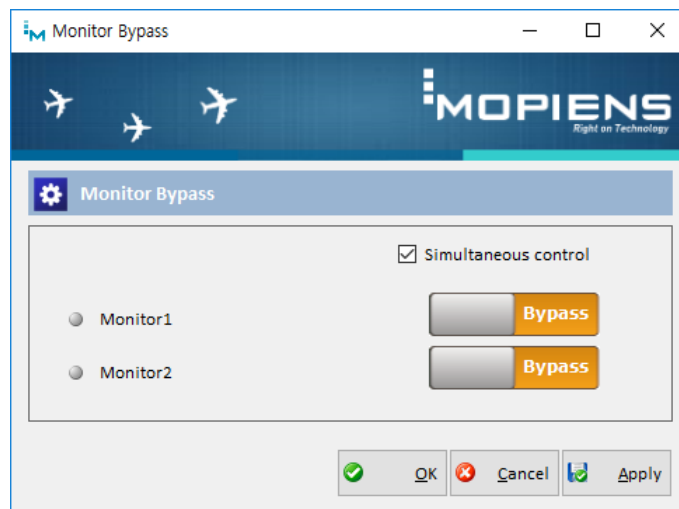


Figure 4-52. Monitor Bypass dialog box

Main TX

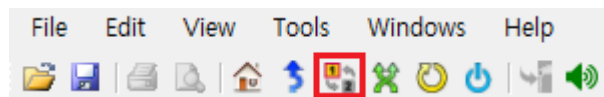


Figure 4-53. Menu Main TX

- Clicking on this icon will pop up the **Main Select** dialog box.
- Pressing the **Main Transponder** slide switch button on the dialog box toggles the transponder to be selected as main. The main transponder is the one that is to be connected to the antenna when the system powers up or reset.

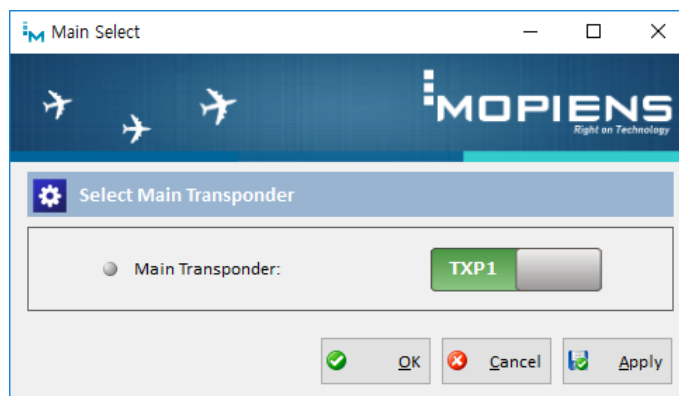


Figure 4-54. Main Select dialog box

Changeover

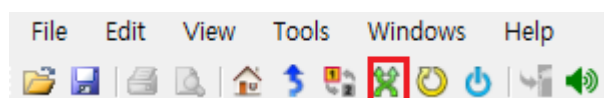


Figure 4-55. Menu Changeover

- Clicking on this icon will pop up the changeover dialog box as shown Figure 4-56.
- Pressing the **Changeover** button on the dialog box will make the transponder currently on the antenna to be transferred to the dummy load and the other transponder on the antenna.

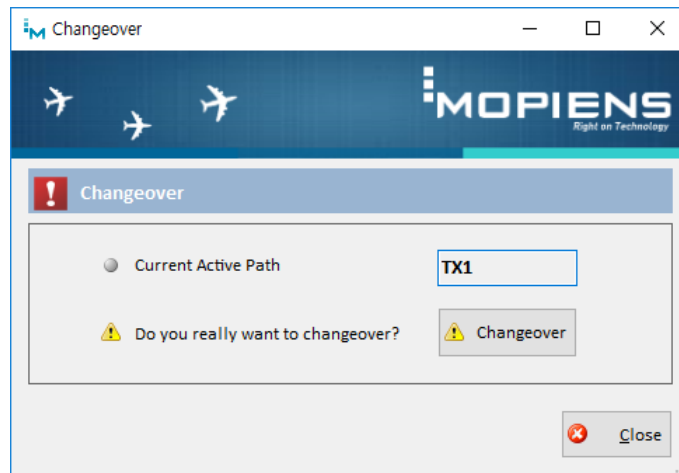


Figure 4-56. Changeover dialog box

Reset

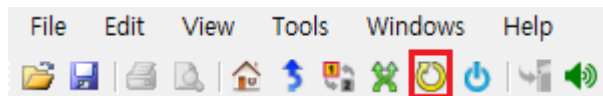


Figure 4-57. Menu Reset

- Clicking on this icon will pop up the Reset dialog box as shown in Figure 4-58.
- Pressing the Reset button on the dialog box will clear all the shutdown or changeover flags and switch on the main transponder and put it on the antenna. Also if configured as hot standby mode, it will switch on the standby transponder and put it on the dummy load.

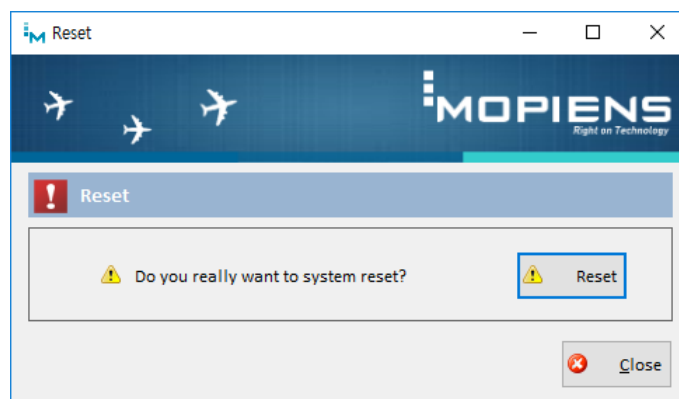


Figure 4-58. Reset dialog box

Power & Signal On / Off

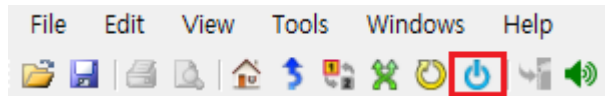


Figure 4-59. Menu Power On Off

- Clicking on this icon will pop up the **Power On/Off** dialog box as shown in Figure 4-60.
- Turning transponder OFF will cut off the DC power supply to the transponder.
- Turning RF Off will not cut off the DC power supply to the transponder but temporarily disable the RF power amplifier.

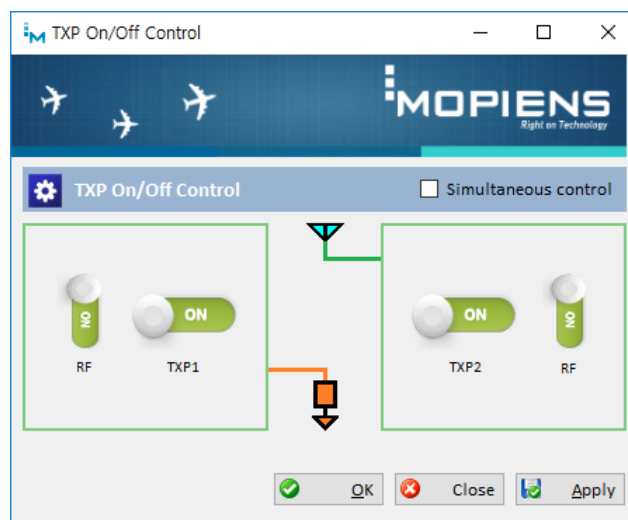


Figure 4-60. Power On/Off dialog box

Profile Save

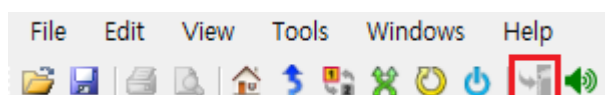


Figure 4-61. Menu Profile Save

- Clicking on this icon saves the current running profile into the non-volatile flash memory in the equipment.

Sound Control

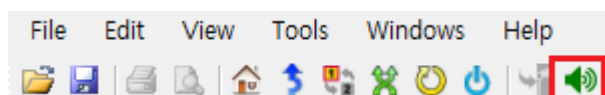


Figure 4-62. Menu Sound

- Clicking on this icon toggles the audible alert between on and off

4.4.4.4. Status Bar



Figure 4-63. Status Overview

- **TxD**: Indicates data is being transmitted from PMDT to the equipment
- **RxD**: Indicates that data is being received from the equipment by PMDT.
- **TXP1, TXP2**: Indicates each transponder status, i.e., Main or Standby, On Antenna or On Dummy load
- **EXEC MON**: Indicates status of the Executive Monitor, i.e. Bypass or Auto, Normal or Alarm or Warning.
- **Date / Time**: Shows current date and time of the PMDT computer.

4.4.4.5. Output Pane

Control		
Time	Detail	UserID
2018-12-14 오후 2:42:38.4	User Logged Out : administrator	administrator
2018-12-14 오후 12:50:57.8	User Logged Out : administrator	administrator
2018-12-14 오후 12:27:02.8	User Logged Out : administrator	administrator
2018-12-14 오후 2:36:25.9	User Logged Out :	
2018-12-14 오후 2:59:56.7	User Logged In : administrator(Level3)	administrator

Figure 4-64. Output Pane

- **All**: Shows all the event history record received by PMDT.
- **Alarm**: Shows only the alarm history record received by PMDT
- **Control**: Shows only the command history record received by PMD.
- **Event**: Shows only the event history record received by PMDT.

4.4.5. MAIN GROUP SCREENS

4.4.5.1. Home Screen

4.4.5.1.1. Introduction

The home screen is in the Main group. The home screen provides quick and convenient interface to most frequently accessed status data and control functions.

The home screen is divided into three sections in vertical.

- The upper section illustrates concise overall system status.
- The middle section shows monitor readings in both analog gauge and digital display.
- The lower section provides quick buttons to selected system control functions.

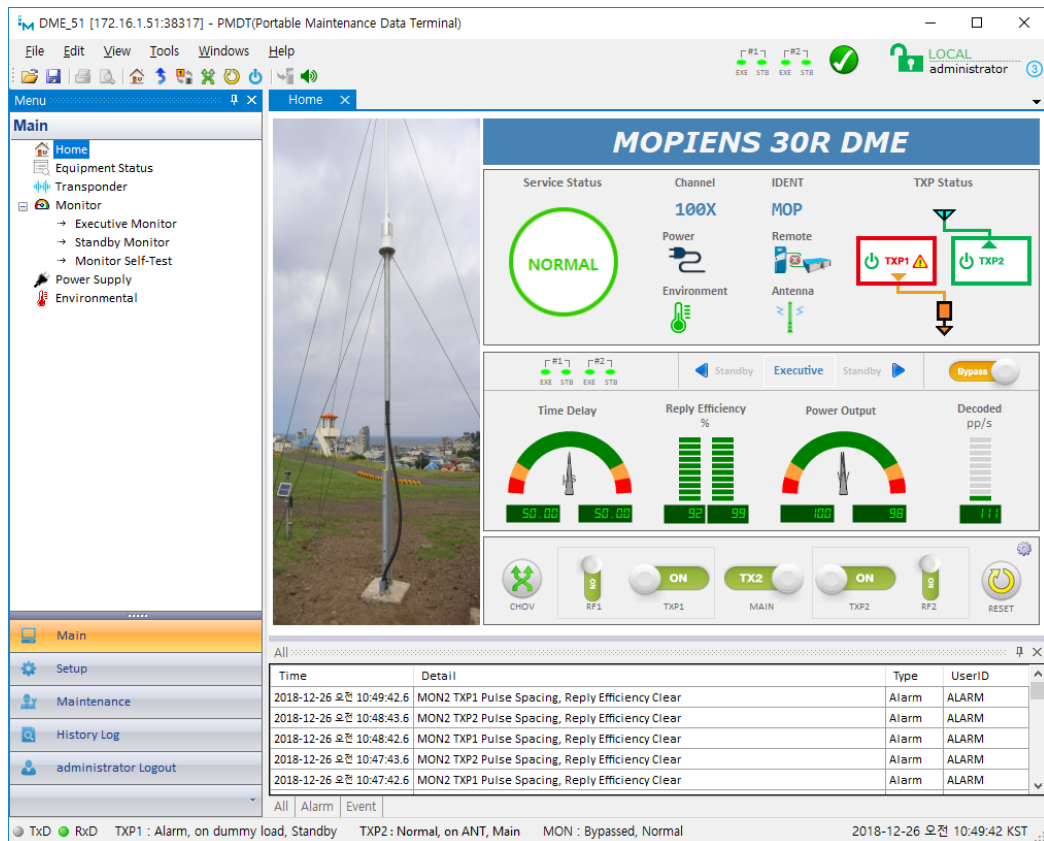


Figure 4-65. Main Home

4.4.5.1.2. System Status Section

Displays the overall status of the DME including:

- Service Status determined based on the monitors
- Channel number
- IDENT code
- Power supply status – on AC main or on battery
- Environmental monitor status
- Data link status to Remote Control Unit
- Antenna Status
- Transponder Status

Service Status

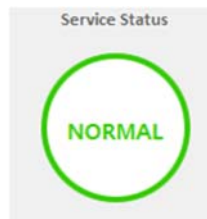


Figure 4-66. Service Status

-  This indicates that all the monitor values of the transmitted signal are normal.
-  This indicates that the monitor value of the transmitted signal is in pre-alarm range or that one or more maintenance alerts are reported.
-  This indicates that the monitor value of the transmitted signal is in alarm state.
-  This indicates that the equipment is shutdown.

Channel

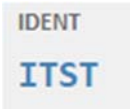


- This shows the channel number in operation of the DME transponder.

Power

-  This indicates that equipment is running on the AC main power.
-  This indicates that the equipment is running on the backup battery.

IDENT



- This shows the Ident Code of the station in operation.

Remote



- This indicates that the data link with the RCU is normal.



- This indicates that the data link with RCU is broken or malfunctioning.

Antenna



- This indicates that the antenna status is normal.



- This indicates that the antenna status is faulty.

TXP Status

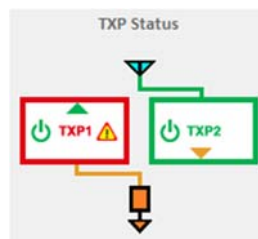


Figure 4-67. TXP Status



- This indicates that the transponder is On Antenna.



- This indicates that the transponder is on Dummy Load.



- This indicates that the station is turned off as it is deselected by Interlock.

- 

This indicates that the transponder is Normal.
- 

This indicates that the monitor is giving Warning on the transponder or the transponder has one or more maintenance alert.
- 

This indicates that the monitor is giving Alarm on transponder.
- 

This indicates that transponder is off or shutdown by control for maintenance reasons.
- 

This indicates that the transponder is faulty as it has been shutdown automatically by persistent alarm condition.

Monitor Readings

This section displays the major monitor parameters reported by MON 1 and MON 2. For dual systems, each monitor has executive channel and standby channel, but only one channel is displayed at a time. The monitor channel to be displayed can be selected by manually clicking on the left or right arrow in blue or can be configured to run automatic slideshow.



Figure 4-68. Monitor Readings

The monitor can be bypassed by clicking on the Bypass button. The bypass button applies to the all monitor channels together.

The button toggles between Bypass and Auto:

-  **Bypass** – In this state, monitor executive action is disabled. Therefore no Automatic Changeover or Shutdown will take place even if alarm condition persists.
-  **Auto** – In this state, executive action is enabled. Automatic Changeover or Shutdown will take place when alarm condition persists over preset limit.

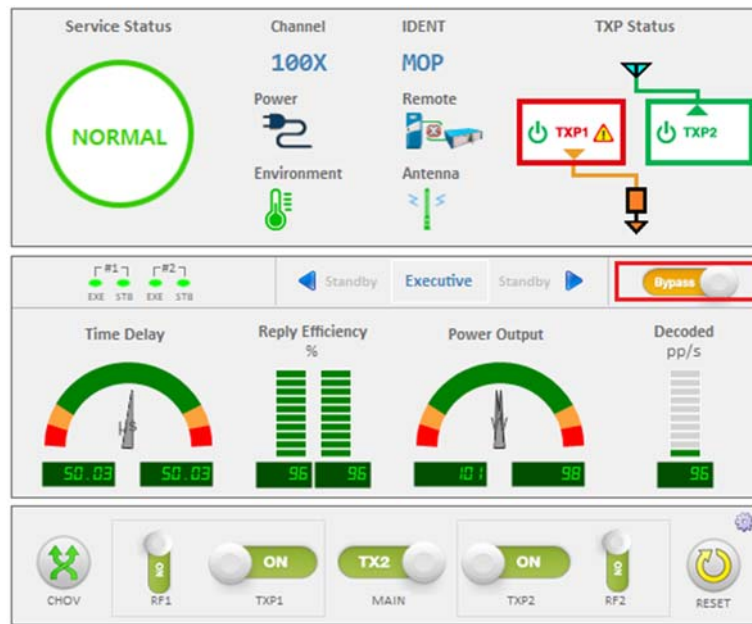


Figure 4-69. Bypass button

4.4.5.1.3. System Control Section

Change Over button

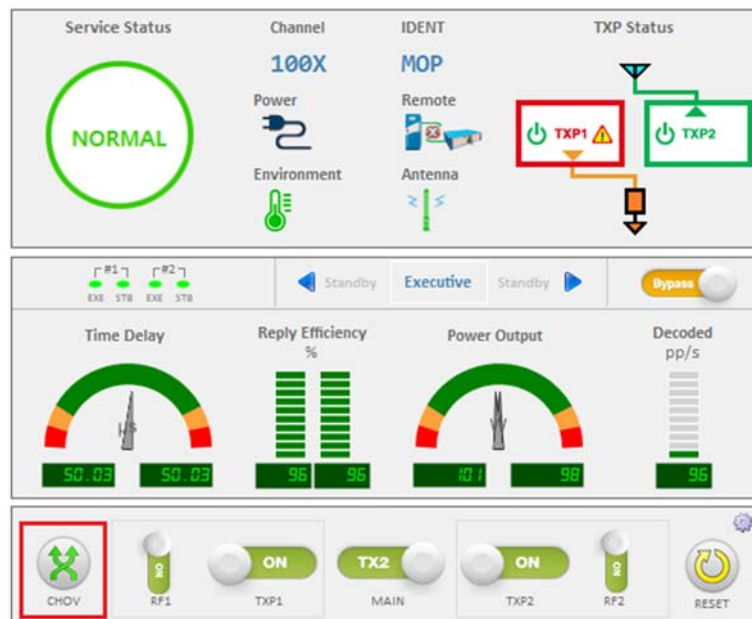


Figure 4-70. Changeover button

- Transfers the transponder currently on the antenna to dummy load and the other transponder on the antenna.

TX1 On/Off Control

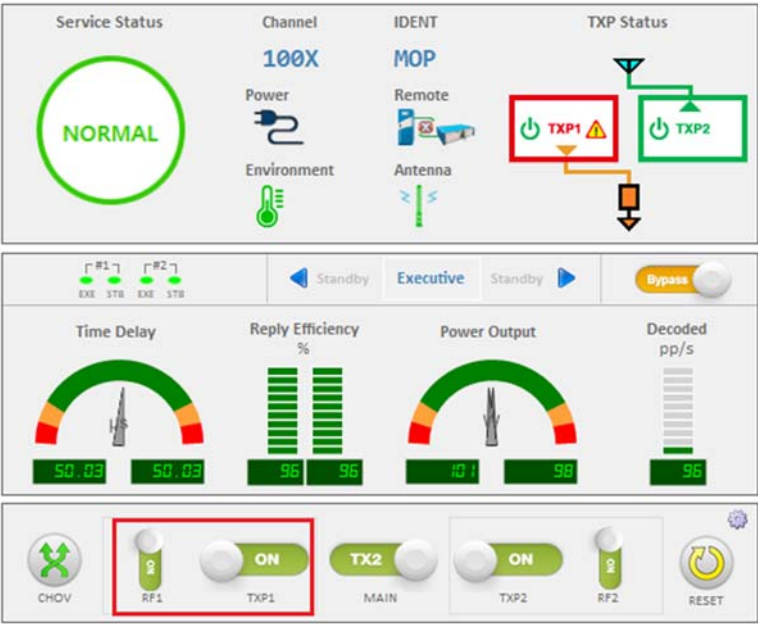


Figure 4-71. TX1 On

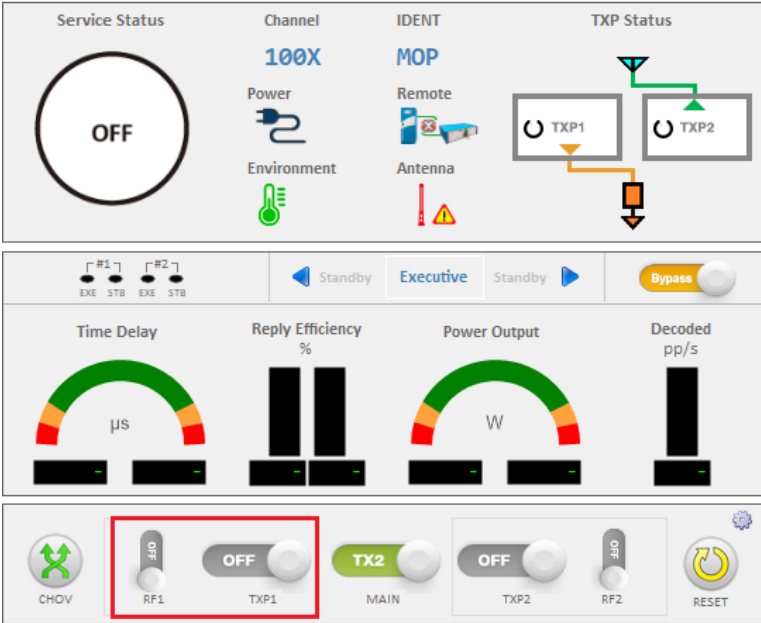


Figure 4-72. TXP1 Off

- TXP1 button toggles the power supply of the transponder #1.
- RF1 button toggles the RF output of the transponder #1 on or off.

TX2 On/Off Control

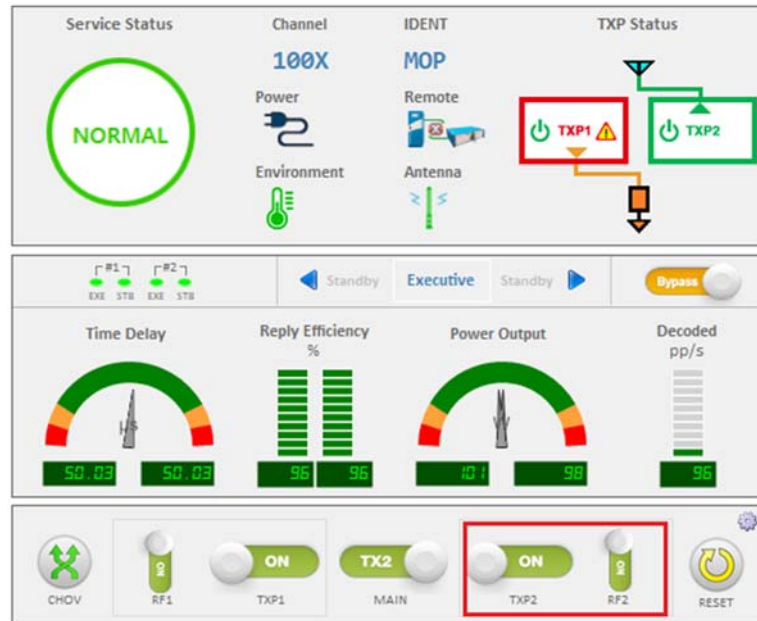


Figure 4-73. TX2 On

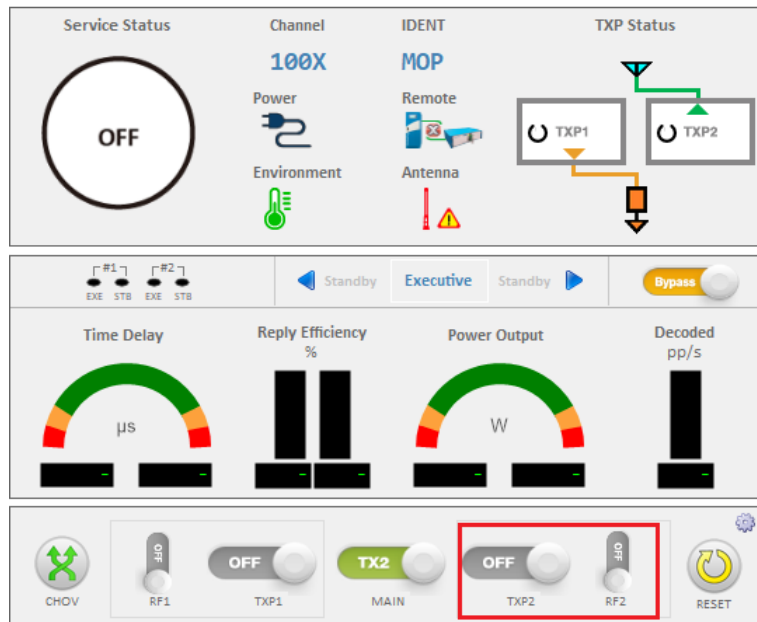


Figure 4-74. TXP2 Off

- TXP2 button toggles the power supply of the transponder #2 ON or OFF.
- RF2 button toggles the RF output of the transponder #2 ON or OFF.

Main TXP Select button

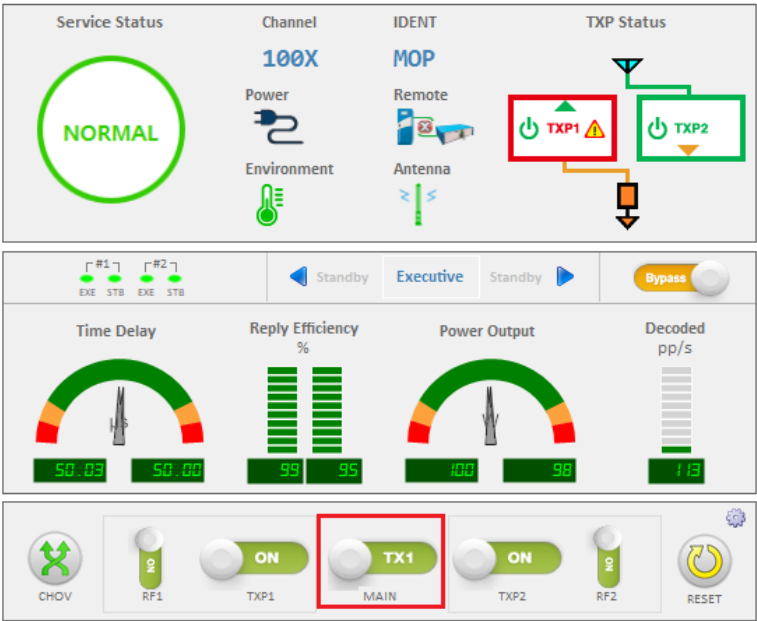


Figure 4-75. Main TXP1

- Pressing this button selects transponder #1 or #2 as the Main transponder.
- The main transponder is the one that is to be on air, i.e. connected to the antenna by default when the system powers up or reset.

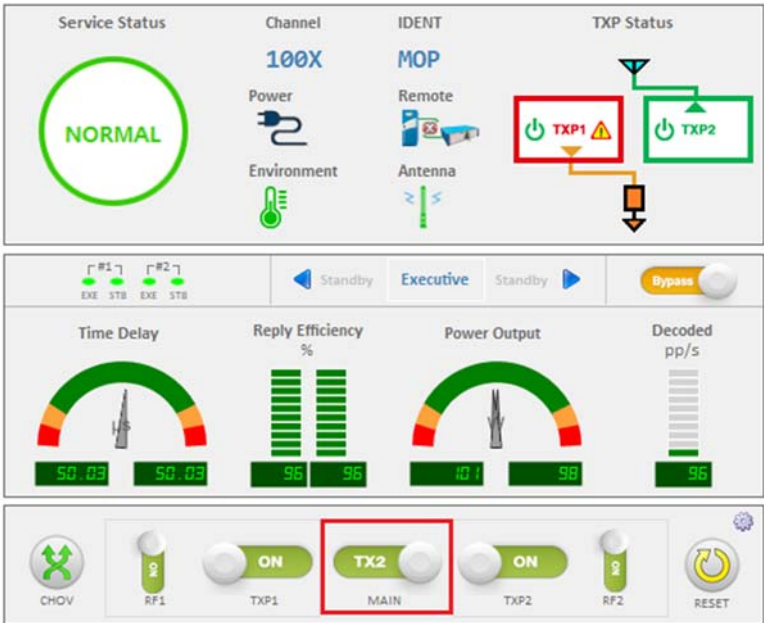


Figure 4-76. Main TXP2

Reset button

- Pressing the Reset button on the dialog box will clear all the shutdown or changeover flags and switch on the main transponder and put it on the antenna. Also if configured as hot standby mode, it will switch on the standby transponder and put it on the dummy load.

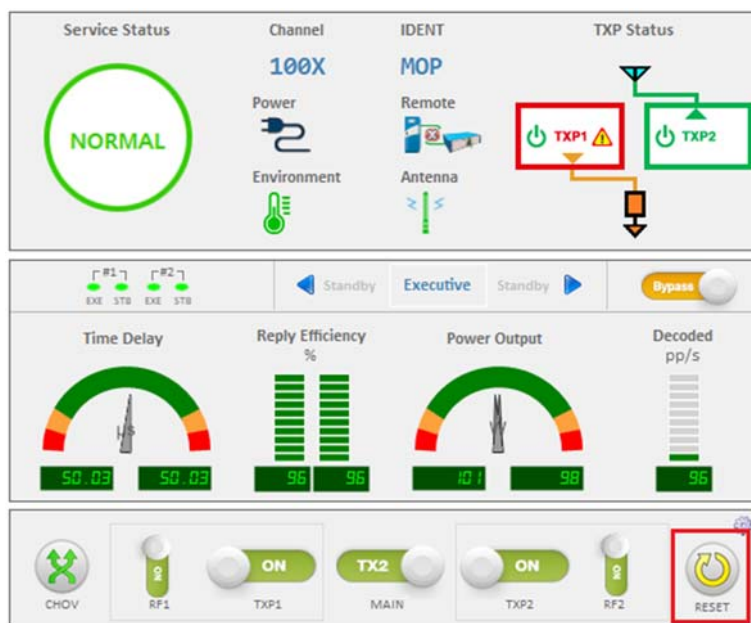


Figure 4-77. Reset button

4.4.5.2. Equipment Status Screen

Clicking on the Equipment Status from the navigation pane will display detailed status information for each part of the equipment. Also when further detailed data is available, it can be accessed by clicking on the item.

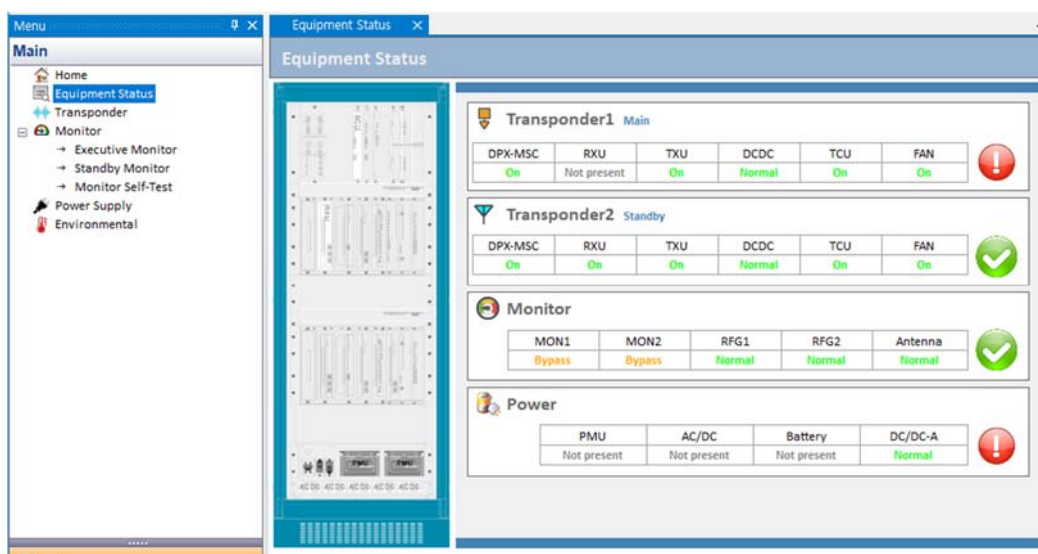


Figure 4-78. Equipment Status

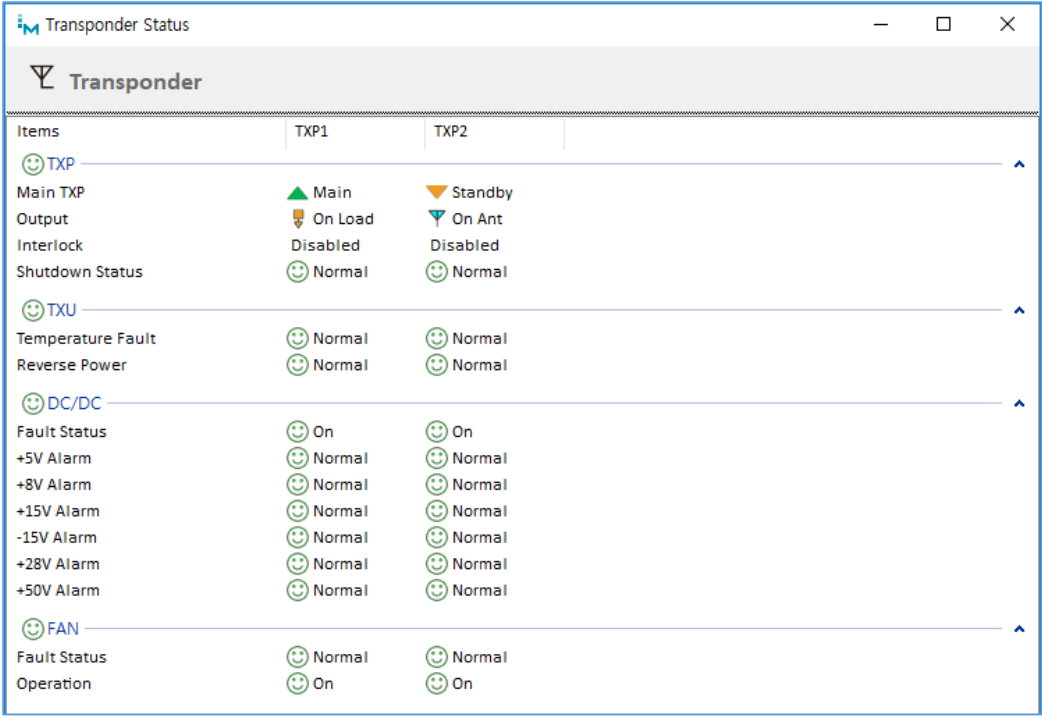
The information displayed on the Equipment Status screen is summarized in Table 4-7.

Table 4-7. Equipment Status

Category	Item	Description
Transponder 1 Transponder 2	DPX-MSC	Normal: No Alarm No Present: DPX Unplugged N/A: TCU Unplugged
	RXU	Normal: No Alarm No Present: RXU Unplugged N/A: TCU Unplugged
	TXU	Normal: No Alarm Fault: PLL Lock Fail, Temperature Fail No Present: TXU Unplugged N/A: TCU Unplugged
	DCDC	Normal: No Alarm Fault: DC/DC Alarm No Present: DC/DC Unplugged
	TCU	Normal: No Alarm Fault: TCU Hardware Fault Off: TX Power Off No Present: TCU Unplugged
	FAN	On: Fan On Off: Fan Off Fault: Hardware Fault N/A: TCU Unplugged
Monitor	MON1 MON2	Bypassed: MON Bypassed Normal: MON Auto, No Alarm Warning: MON Auto, Parameter Warning Alarm: MON Auto, Parameter Alarm No Present: MON Unplugged
	RFG1 RFG2	Normal: No Alarm Fault: MON Auto, Parameter Alarm No Present: RFG Unplugged N/A: MON Unplugged
	Antenna	Normal: No Alarm Fault: Antenna Fault
Power	AC/DC	Normal: No Alarm Fault: AC/DC Fault N/A: PMU Unplugged

	PMU1	Normal: No Alarm
	PMU2	Fault: Hardware Fault N/A: PMU Unplugged
	BAT1	Normal: No Alarm
	BAT2	Warning: Battery Voltage, Temperature Warning Alarm: Battery Voltage, Temperature Alarm No Battery: Battery Unplugged N/A: PMU Unplugged
	DC/DC-A	Normal: No Alarm Warning: DC/DC-A Temperature Warning Alarm: DC/DC-A Voltage Alarm, DC/DC-A Unplugged

Transponder Detail



Items	TXP1	TXP2
TXP		
Main TXP	▲ Main	▼ Standby
Output	🔥 On Load	🔥 On Ant
Interlock	Disabled	Disabled
Shutdown Status	😊 Normal	😊 Normal
TXU		
Temperature Fault	😊 Normal	😊 Normal
Reverse Power	😊 Normal	😊 Normal
DC/DC		
Fault Status	😊 On	😊 On
+5V Alarm	😊 Normal	😊 Normal
+8V Alarm	😊 Normal	😊 Normal
+15V Alarm	😊 Normal	😊 Normal
-15V Alarm	😊 Normal	😊 Normal
+28V Alarm	😊 Normal	😊 Normal
+50V Alarm	😊 Normal	😊 Normal
FAN		
Fault Status	😊 Normal	😊 Normal
Operation	😊 On	😊 On

Figure 4-79. Transponder Detail

Click the **Transponder** on the **Equipment Status** screen to go to the Transponder Status Detail screen. The information displayed on the Transponder Detail screen is summarized in Table 4-8.

Table 4-8. Transponder Status Detail

Category	Item	Description
TXP	Main TX	Main: Main Transmitter Standby : Standby Transmitter
	Output	On ANT

		On Load
	Interlock	Disable: Not use interlock Enable: Use Interlock
	Shutdown Status	Normal: No Alarm Off: Shutdown
TXU	Temperature Fault	Normal: No Alarm Fault: Temperature Alarm Unplugged: TXU Unplugged Unknown: TCU Unplugged
	Reverse Power	Normal: No Alarm Fault: Reverse Power Alarm Unplugged: TXU Unplugged Unknown: TCU Unplugged
DC/DC	Fault Status	Normal: No Alarm Warning: DC/DC Voltage Reading Warning Alarm: DC/DC Voltage Reading Alarm Off: DC/DC Off Unplugged: DC/DC Unplugged Unknown: TCU Unplugged
	+5V Alarm	Normal: No Alarm Warning: DC/DC 5V Voltage Reading Warning Alarm: DC/DC 5V Voltage Reading Alarm Off: DC/DC Off Unplugged: DC/DC Unplugged Unknown: TCU Unplugged
	+8V Alarm	Normal: No Alarm Warning: DC/DC 8V Voltage Reading Warning Alarm: DC/DC 8V Voltage Reading Alarm Off: DC/DC Off Unplugged: DC/DC Unplugged Unknown: TCU Unplugged
	+15V Alarm	Normal: No Alarm Warning: DC/DC 15V Voltage Reading Warning Alarm: DC/DC 15V Voltage Reading Alarm Off: DC/DC Off Unplugged: DC/DC Unplugged Unknown: TCU Unplugged

	-15V Alarm	Normal: No Alarm Warning: DC/DC -15V Voltage Reading Warning Alarm: DC/DC -15V Voltage Reading Alarm Off: DC/DC Off Unplugged: DC/DC Unplugged Unknown: TCU Unplugged
	+28V Alarm	Normal: No Alarm Warning: DC/DC 28V Voltage Reading Warning Alarm: DC/DC 28V Voltage Reading Alarm Off: DC/DC Off Unplugged: DC/DC Unplugged Unknown: TCU Unplugged
	+50V Alarm	Normal: No Alarm Warning: DC/DC 50V Voltage Reading Warning Alarm: DC/DC 50V Voltage Reading Alarm Off: DC/DC Off Unplugged: DC/DC Unplugged Unknown: TCU Unplugged
FAN	Fault Status	Normal: No Alarm Alarm: FAN Alarm Off: FAN Off Unplugged: FAN Unplugged Unknown: TCU Unplugged
	Operation	On FAN Off FAN

Monitor Detail

Click the **Monitor** on the Equipment Status screen to go to the Monitor Status Detail screen.

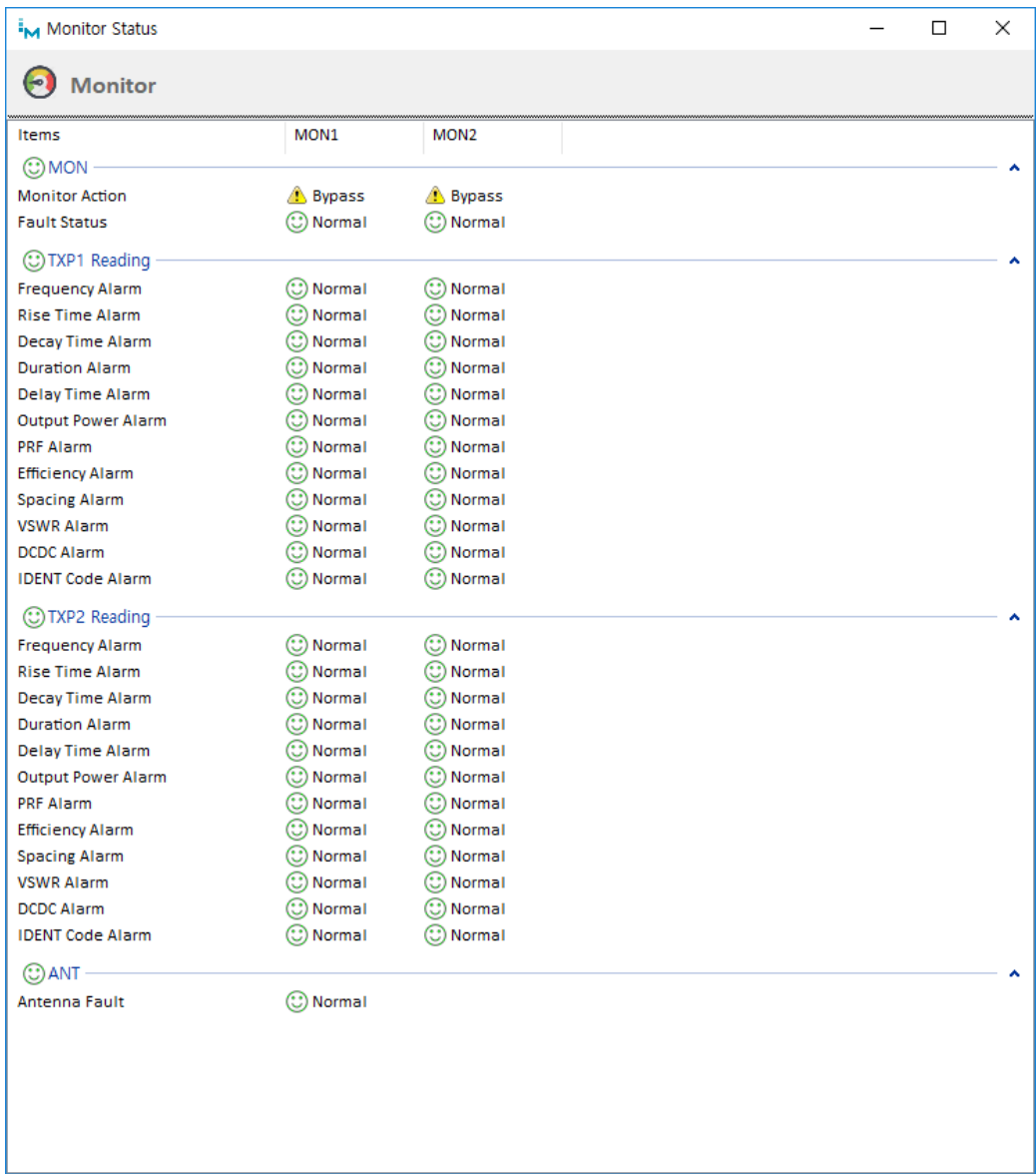


Figure 4-80. Monitor Status screen

The information displayed on the **Monitor Detail** screen is summarized in Table 4-9

Table 4-9. Monitor Status Detail

Category	Item	Description
Monitor	Monitor Action	Bypass: MON Bypass Mode Auto: MON Auto Mode No Present: MON Unplugged
	Fault Status	Normal: No Alarm Fault: Hardware Fault No Present: MON Unplugged
TXP1	Frequency Alarm	Normal: No Alarm

TXP2		Warning: Frequency Warning Alarm: Frequency Alarm No Present: MON Unplugged
	Rise Time Alarm	Normal: No Alarm Warning: Rise Time Warning Alarm: Rise Time Alarm No Present: MON Unplugged
	Decay Time Alarm	Normal: No Alarm Warning: Decay Time Warning Alarm: Decay Time Alarm No Present: MON Unplugged
	Duration Alarm	Normal: No Alarm Warning: Pulse Duration Warning Alarm: Pulse Duration Alarm No Present: MON Unplugged
	Delay Time Alarm	Normal: No Alarm Warning: Delay Time Warning Alarm: Delay Time Alarm No Present: MON Unplugged
	Output Power Alarm	Normal: No Alarm Warning: Output Power Warning Alarm: Output Power Alarm No Present: MON Unplugged
	PRF Alarm	Normal: No Alarm Warning: PRF Warning Alarm: PRF Alarm No Present: MON Unplugged
	Efficiency Alarm	Normal: No Alarm Warning: Efficiency Warning Alarm: Efficiency Alarm No Present: MON Unplugged
	Spacing Alarm	Normal: No Alarm Warning: Spacing Warning Alarm: Spacing Alarm No Present: MON Unplugged
	VSWR Alarm	Normal: No Alarm Warning: VSWR Warning

		Alarm: VSWR Alarm No Present: MON Unplugged
	DC/DC Alarm	Normal: No Alarm Warning: DC/DC Warning Alarm: DC/DC Alarm No Present: MON Unplugged
	IDENT Code Alarm	Normal: No Alarm Alarm: IDENT Code Alarm No Present: MON Unplugged
Antenna	Antenna Fault	Normal: No Alarm Fault: Antenna Fault

Power Detail

Click the **Power** on the **Equipment Status** screen to go to the **Power Status** screen.

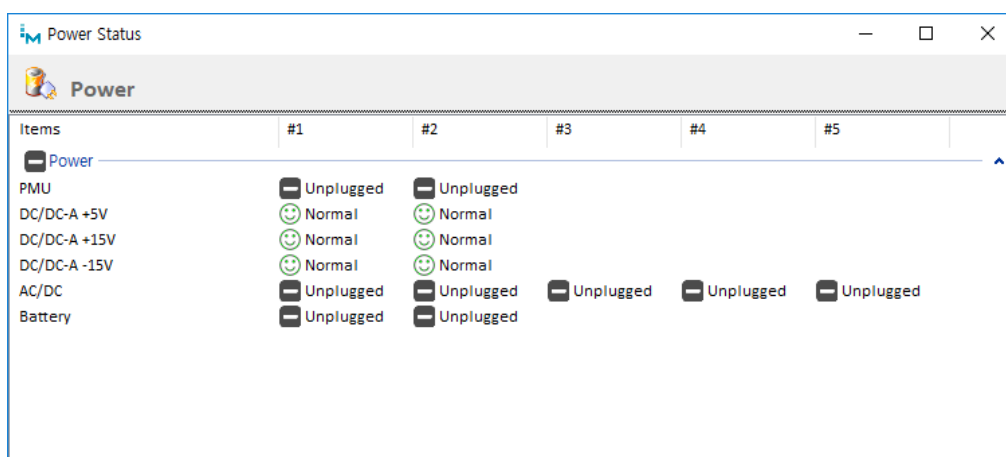


Figure 4-81. Power Status screen

The information displayed on the **Power Status** screen is summarized in Table 4-10.

Table 4-10. Power Status Detail

Category	Item	Description
Power Detail	PMU	Normal: No Alarm Fault: PMU Hardware Fault No Present: PMU Unplugged
	DC/DC-A +5V DC/DC-A +15V DC/DC-A -15V	Normal: No Alarm Alarm: Voltage/Current Alarm No Present: DC/DC-A Unplugged
	AC/DC	Normal: No Alarm Warning: CLR RF Level Warning Alarm: CLR RF Level Alarm No Present: MON Unplugged
	Battery	Normal: No Alarm Warning: CLR SDM Warning Alarm: CLR SDM Alarm

	No Battery: Battery Unplugged
--	-------------------------------

4.4.5.3. Transponder Status Screen

Transponder Status screen displays the transponder parameters reported by the TCU.

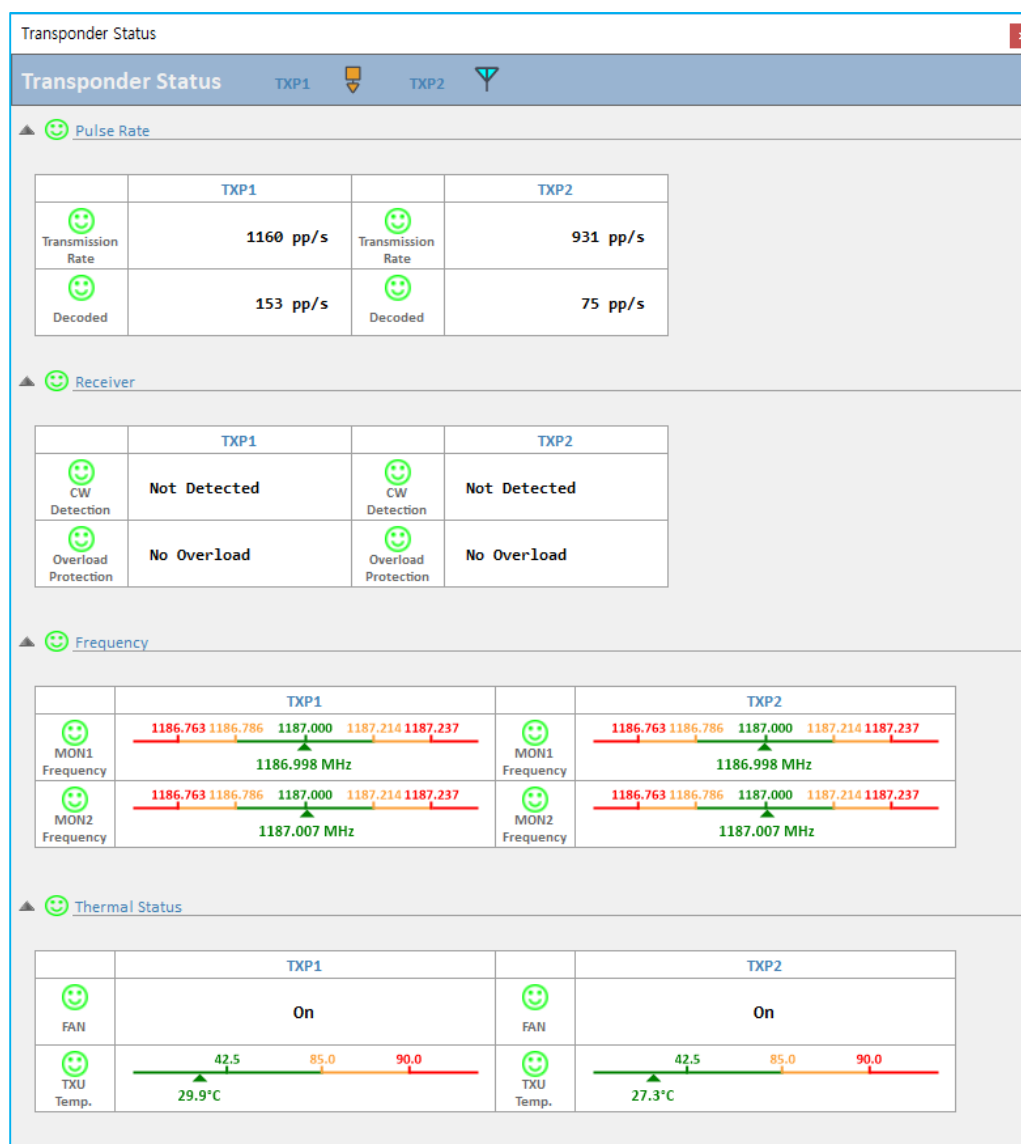


Figure 4-82. Transponder Status Screen

Different icon is displayed according to the state of the measured value as follows:

😊 Normal ⚠ Warning ⚠ Alarm 🚫 Unplugged

The information displayed on the **Transponder Status** screen is summarized in

Table 4-11.

Table 4-11. Transponder Readings

Category	Item	Description	Resolution	Unit
Pulse Rate	Transmission Rate	Transmission pulse pair per second (Reply + Squitter + IDENT)	1	pp/s
	Decoded	Decoded pulse pairs per second	1	pp/s
Receiver	CW Detection	- When Normal, i.e. CW is Not Detected: "Not Detected" message is displayed. - When CW is detected, the level of detected CW interference is displayed.	0.1	dBm
	Overload Sensitivity Level	- When Normal, i.e. the transponder is not overloaded, "No Overload" message is displayed. - When the transponder is overloaded, the current receiver sensitivity level is displayed.	0.1	dBm
Frequency	Frequency	Frequency reading	0.0001	MHz
Thermal Status	FAN	On/Off	-	-
	TXU Temp	TXU Temperature	0.1	°C

4.4.5.4. Monitor Screens

Executive Monitor Screen

Executive Monitor screen displays the executive monitor parameters reported by the monitors.

Executive Monitor screen can be accessed by selecting **Executive Monitor** from the navigation pane.



Figure 4-83. Executive Monitor

Different icon is displayed according to the state of the measured value as follows:

 Normal  Warning  Alarm  Unplugged

The information displayed on the screen is summarized in Table 4-12.

Table 4-12. Executive Monitor

Category	Item	Description	Resolution	Unit
System Parameter	Time Delay	Time Delay reading	0.01	μs
	Reply Efficiency	Reply Efficiency reading	1	%
	Transmission Rate	Transmission Rate reading	1	pp/s
Pulse Shape and Spacing	Pulse Rise Time	Pulse Rise Time reading	0.01	μs
	Pulse Duration	Pulse Duration reading	0.01	μs
	Pulse Decay Time	Pulse Decay Time reading	0.01	μs
	Pulse Spacing	Pulse Spacing reading	0.01	μs
Frequency and Power	Radio Frequency	Reply Frequency reading	0.01	MHz
	Peak Power Output	Output Power reading	1	W
	Antenna VSWR	Antenna VSWR reading	0.01	
	Effective Radiated	Effective Radiated Power	0.01	dB
Identification	Code	IDENT Code		

Executive Monitor Masking

In the Executive Monitor screen, when ERP Alarm occurs, deactivate the rest of the items except the Effective Radiated item.



Figure 4-84. Execution monitor when ERP alarm occurs

When the ERP alarm occurs, the signal level input to MON is out of the range that can be analyzed, and the reliability of the result value is lowered, so the remaining items except for the Effective Radiated item are marked as inactive (not valid).

Standby Monitor Screen

Standby Monitor screen displays the standby monitor parameters reported by the monitors.

Standby Monitor screen can be accessed by selecting **Standby Monitor** from the navigation pane.

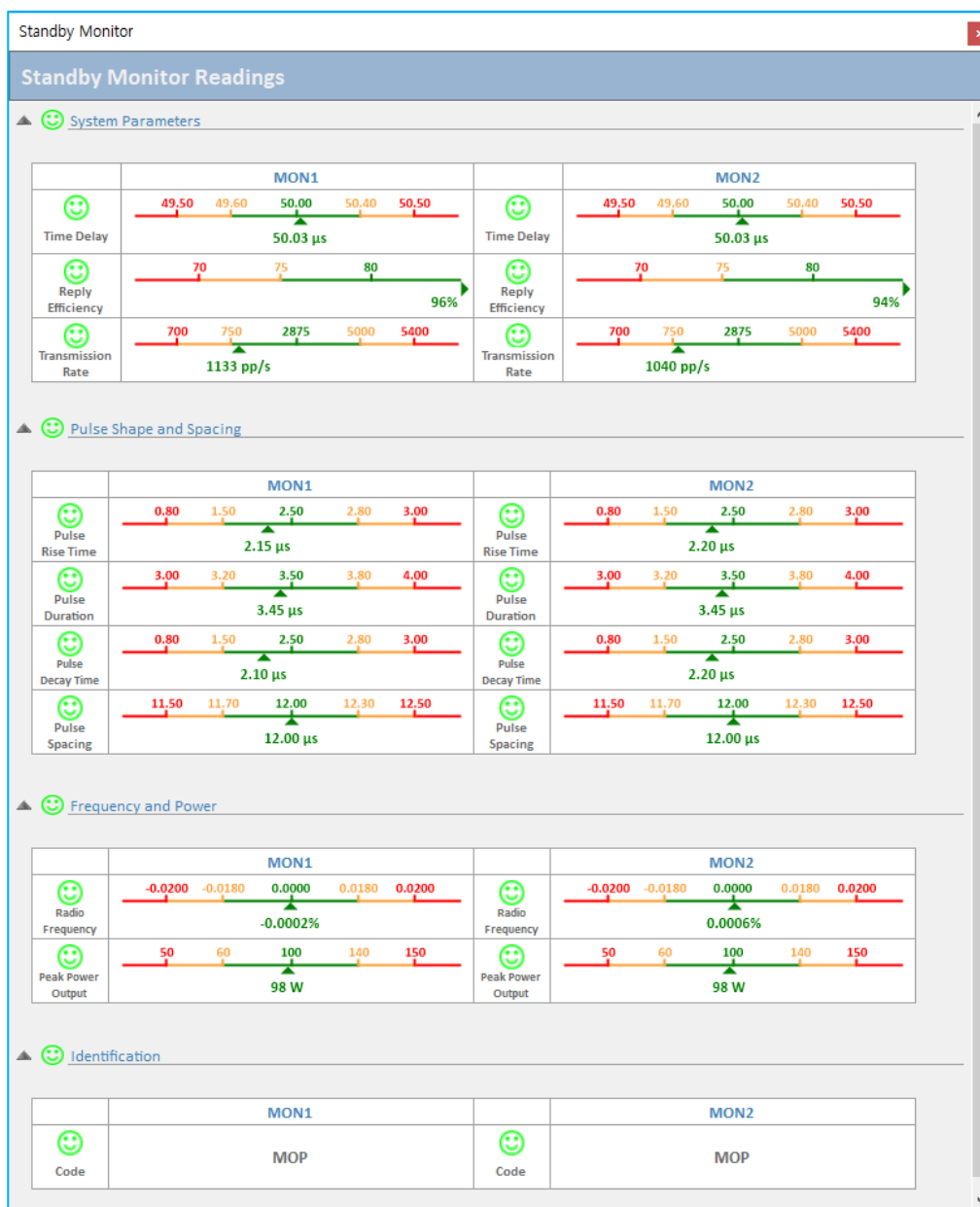


Figure 4-85. Standby Monitor Reading

Different icon is displayed according to the state of the measured value as follows:

😊 Normal ⚠ Warning 🚨 Alarm 🚪 Unplugged

The information displayed on the screen is summarized in Table 4-13.

Table 4-13. Standby Monitor Reading

Category	Item	Description	Resolution	Unit
System Parameter	Time Delay	Time Delay reading	0.01	μs
	Reply Efficiency	Reply Efficiency reading	1	%
	Transmission Rate	Transmission Rate reading	1	pp/s
Pulse Shape and Spacing	Pulse Rise Time	Pulse Rise Time reading	0.01	μs
	Pulse Duration	Pulse Duration reading	0.01	μs
	Pulse Decay Time	Pulse Decay Time reading	0.01	μs
	Pulse Spacing	Pulse Spacing reading	0.01	μs
Frequency and Power	Radio Frequency	Reply Frequency reading	0.0001	MHz
	Peak Power Output	Output Power reading	1	W
	Antenna VSWR	Antenna VSWR reading	0.01	
	Effective Radiated	Effective Radiated Level	0.1	
Identification	Code	IDENT Code		char

Monitor Self-Test Screen

Monitor Self-Test screen displays the status of the monitor self-test.

Monitor Self-Test screen can be accessed by selecting **Monitor Self-Test** from the navigation pane.

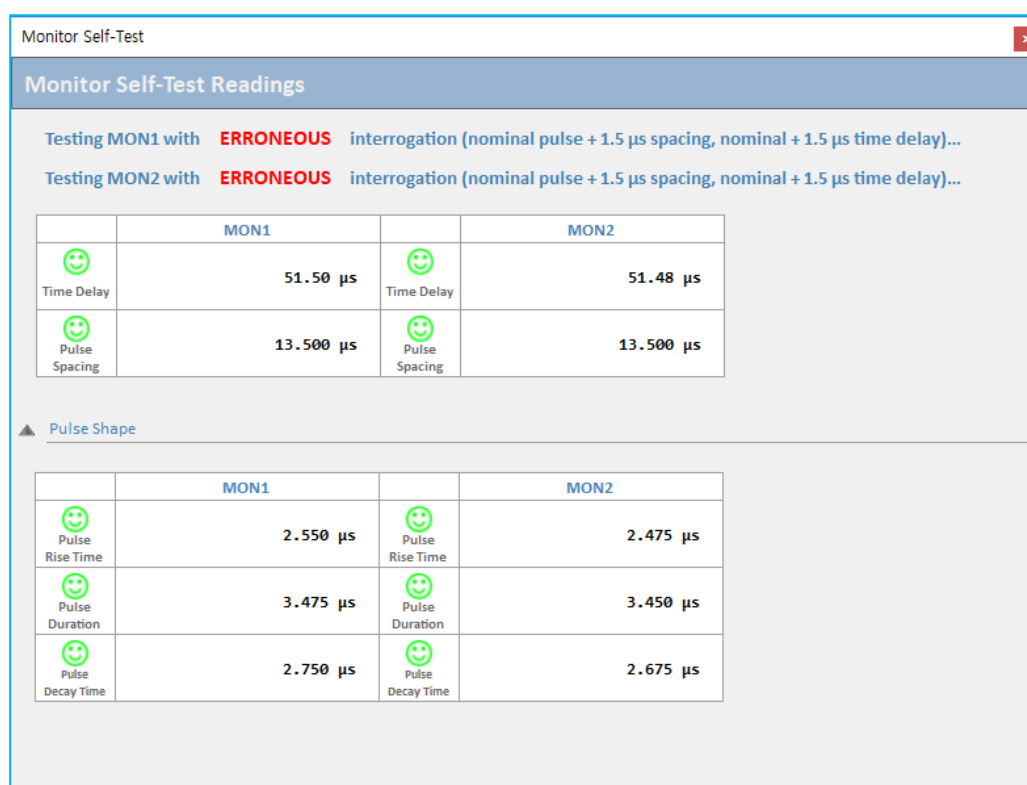


Figure 4-86. Monitor Self-Test

Different icon is displayed according to the state of the measured value as follows:

Normal Warning Alarm Unplugged

The information displayed on the screen is summarized in Table 4-14.

Table 4-14. Monitor Self-Test

Category	Item	Description	Resolution	Unit
Self-Test	Time Delay	Time Delay reading	0.01	μs
	Pulse Spacing	Pulse Spacing reading	0.01	μs
	Normal	Test result of normal interrogation.		
	Erroneous	Test result of intentional erroneous interrogations which are: • with time delay increased by +1.5 μs from the nominal value • with pulse spacing increased by +1.5 μs from the nominal value		
Pulse Shape and Spacing	Pulse Rise Time	Pulse Rise Time reading	0.01	μs
	Pulse Duration	Pulse Duration reading	0.01	μs
	Pulse Decay Time	Pulse Decay Time reading	0.01	μs

4.4.5.5. Power Supply Screen

Power Supply screen displays the power supply related parameters including:

- AC/DC parameters collected by PMU
- DC/DC parameters collected by each TCU
- DC/DC-A parameters collected by SCU
- Battery parameters collected by PMU

Power Supply screen can be accessed by selecting **Power Supply** from the navigation pane.

Power Supply					
Power Supply					
DC/DC					
Items	#1 Voltage (V)	#1 Current (A)	#2 Voltage (V)	#2 Current (A)	
DC/DC-A					
5V	5.07	0.47	5.07	0.52	
15V	15.05	0.43	15.07	0.18	
-15V	-15.05	0.03	-14.97	0.00	
DC/DC					
5V	4.95	1.40	5.05	1.77	
8V	8.05	0.74	8.02	0.71	
15V	15.12	0.72	15.10	0.17	
-15V	-14.95	0.20	-14.95	0.47	
28V	28.18	1.82	28.03	1.85	
50V	50.23	0.07	50.33	0.05	
AC/DC					
Items	Voltage (V)	Current (A)	Temperature (°C)	Status	
AC/DC-A	-	-	-	-	
AC/DC-B	-	-	-	-	
AC/DC-C	-	-	-	-	
AC/DC-D	-	-	-	-	
AC/DC-E	-	-	-	-	
Battery					
Items	Voltage (V)	Current (A)	Temperature (°C)	Status	
Battery #1	-	-	-		
Battery #2	-	-	-		

Figure 4-87. Power Supply

The information displayed on the screen is summarized in Table 4-15.

Table 4-15. Power Supply

Category	Item	Description	Resolution	Unit
DC/DC-A	5V Voltage	DC/DC-A 5V voltage reading	0.01	V
	5V Current	DC/DC-A 5V current reading	0.01	A
	15V Voltage	DC/DC-A 15V voltage reading	0.01	V
	15V Current	DC/DC-A 15V current reading	0.01	A
	-15V Voltage	DC/DC-A -15V voltage reading	0.01	V
	-15V Current	DC/DC-A -15V current reading	0.01	A
DC/DC	5V Voltage	DC/DC 5V voltage reading	0.01	V
	5V Current	DC/DC 5V current reading	0.01	A
	8V Voltage	DC/DC 8V voltage reading	0.01	V
	8V Current	DC/DC 8V current reading	0.01	A

	15V Voltage	DC/DC 15V voltage reading	0.01	V
	15V Current	DC/DC 15V current reading	0.01	A
	-15V Voltage	DC/DC -15V voltage reading	0.01	V
	-15V Current	DC/DC -15V current reading	0.01	A
	28V Voltage	DC/DC 28V voltage reading	0.01	V
	28V Current	DC/DC 28V current reading	0.01	A
	50V Voltage	DC/DC 50V voltage reading	0.01	V
	50V Current	DC/DC 50V current reading	0.01	A
AC/DC	Voltage	AC/DC voltage reading	0.1	V
	Current	AC/DC current reading	0.1	A
	Temp	AC/DC temperature reading	0.1	°C
	Status	Normal: No Alarm Alarm: AC/DC voltage/current Alarm N/A: AC/DC Unplugged		
Battery	Voltage	Battery voltage reading	0.1	V
	Current	Battery current reading	0.1	A
	Temp	Battery temperature reading	0.1	°C
	Status	Normal: No Alarm Alarm: Battery voltage/current Alarm N/A: Battery Unplugged		

4.4.5.6. Environmental Screen

Environmental screen displays the environmental monitor parameters collected by EMU. Environmental parameters include status of the environmental sensors that are smoke sensor, intrusion sensor and temperature sensor, which are supported by default, and also status of the expansion I/O. As the EMU is optional, the screen is displayed only when the EMU is installed and enabled.

Environmental screen can be accessed by selecting **Environmental** from the navigation pane.

The screenshot shows the 'Environmental Status' screen with a title bar and a close button. The main content area is divided into sections for different sensor types, each with a table of readings.

Environmental Status																	
Environmental Sensor																	
Status	Smoke	Intrusion	Temperature														
	-	-	-														
Analog Input Sensor																	
Readings	#1	#2	#3	#4	#5	#6	#7	#8	Remark								
	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0 ~ 5V								
Digital Input Sensor																	
Readings	#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12	#13	#14	#15	#16	Remark
	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0 or 1
Expansion Digital Input Sensor																	
Readings	#1	#2	#3	#4	#5	#6	#7	#8	Remark								
	-	-	-	-	-	-	-	0	0 or 1								

Figure 4-88. Environmental Status

The information displayed on the screen is summarized in Table 4-16.

Table 4-16. Environmental Status

Category	Item	Description	Resolution	Unit
Sensor Status	Smoke	Status of smoke sensor		
	Intrusion	Status of intrusion sensor		
	Temperature	Temperature sensor reading	0.1	°C
Analog Input	Analog Input # 1 through # 8	Analog Input channel voltage readings	0.1	V
Digital Input	Digital Input # 1 through # 16	Digital Input channel readings	0 or 1	int
Expansion Digital Input	Expansion Digital Input # 1 through # 8	Expansion Digital Input channel readings	0 or 1	int

4.4.6. SETUP GROUP SCREENS

4.4.6.1. Station Setup Screen

Station Setup screen as shown in Figure 4-89 is to set up those system parameters which are specific to the equipment station but common for both transponders.

The parameters that can be configured on the screen are summarized in Table 4-17.

Station Setup
Apply Save

Items	Settings	Units	Helper
General			
Station Name	MOPIENS		
Runway Designator	30R		
Equipment Type	DME, LPDE	read-only	
Basic			
Channel	100X		Helper
Power Output	100.0	W	
Delay Offset	0.00	μs	
Auto Delay Calibration	Never		
Sensitivity Level	-91.0	dBm	
Pulse Rate			
Minimum Pulse Rate	800	pp/s	
Echo Suppression			
SDES	On		
SDES Duration	6.0	μs	
LDES	Off		
LDES Duration	200.0	μs	
LDES Threshold Level	-50.0	dBm	
Dead Time	60.0	μs	
IDENT			
IDENT Code	MOP		
IDENT Keyer	Independent		
IDENT Sync	Code		
IDENT Sound	On Antenna		
Equalizing Pulse	Off		
Equipment Configuration			
Enable Interlock	No		
Standby Mode	Hot Standby		
On Boot			
Bypass Monitor	Yes		
Transmitter Output On	Yes		

Figure 4-89. Station Setup

Table 4-17. Station Setup

Category	Item	Description	Min	Max	Resolution	Unit
General	Station Name	Informative text of the station name	0	31	Enter text	
	Runway Designator	Informative text of the runway designator	0	3	Enter text	
	Equipment Type	This is non-alterable equipment type information. • LP: Low Power • HP: High Power • SE: Single Equipment • DE: Dual Equipment			Information Only	
Basic	Channel	DME channel number assigned to the station	1X 1Y	126X 126Y	Select from list.	
	Power Output	Peak power output • for 310 DME • for 320 DME	0 0	125 1250	0.1 1	W
	Time Delay Offset	Offset for Reply Delay	-15	35	0.1	μs

	Auto Delay Calibration	Use of Auto Delay Calibration function	- Always - Never		Choose option	
	Sensitivity Level	Receiver Sensitivity	-95	-60	0.1	dBm
Pulse Rate	Minimum Pulse Rate	Minimum Pulse Rate (Squitter pulses)	0	2700	1	pp/s
Echo Suppression	SDES	Use of Short Distance Echo Suppression function	- On - Off		Choose option	
	SDES Duration	Duration of SDES	0	34	0.1	μs
	LDES	Use of Long Distance Echo Suppression function	- On - Off		Choose option	
	LDES Duration	Duration of LDES	0	350	0.1	μs
	LDES Threshold Level	LDES Threshold Level	-110	9	0.1	dBm
	Dead Time	Dead Time	0	150	0.1	μs
IDENT	IDENT Code	IDENT Code	-	4	Enter text	
	IDENT Keyer	IDENT Keyer mode	- None - Independent - Master - Slave - Continuous		Select from list	
	IDENT Sync	IDENT Synchronization mode	- Code - Pulse		Select from list	
	IDENT Sound	Source of IDENT Keying sound playback	- MON1 - MON2 - STB MON - TX1 - TX2 - On Antenna - Off		Select from list	
	Equalizer Pulse	Use of Equalizing Pulse	- On - Off		Select from list	
Equipment Configuration	Enable Interlock	Use of Interlock	- Yes - No		Select from list	
	Standby Mode	Operation of Standby Transmitter	- Hot Standby - Cold Standby		Select from list	
On Boot	Bypass Monitor	Whether or not to bypass on boot up	- Yes - No		Select from list	
	Transmitter Output	Whether or not to turn the transmitter RF output ON	- Yes - No		Select from list	

4.4.6.2. Transponder Screens

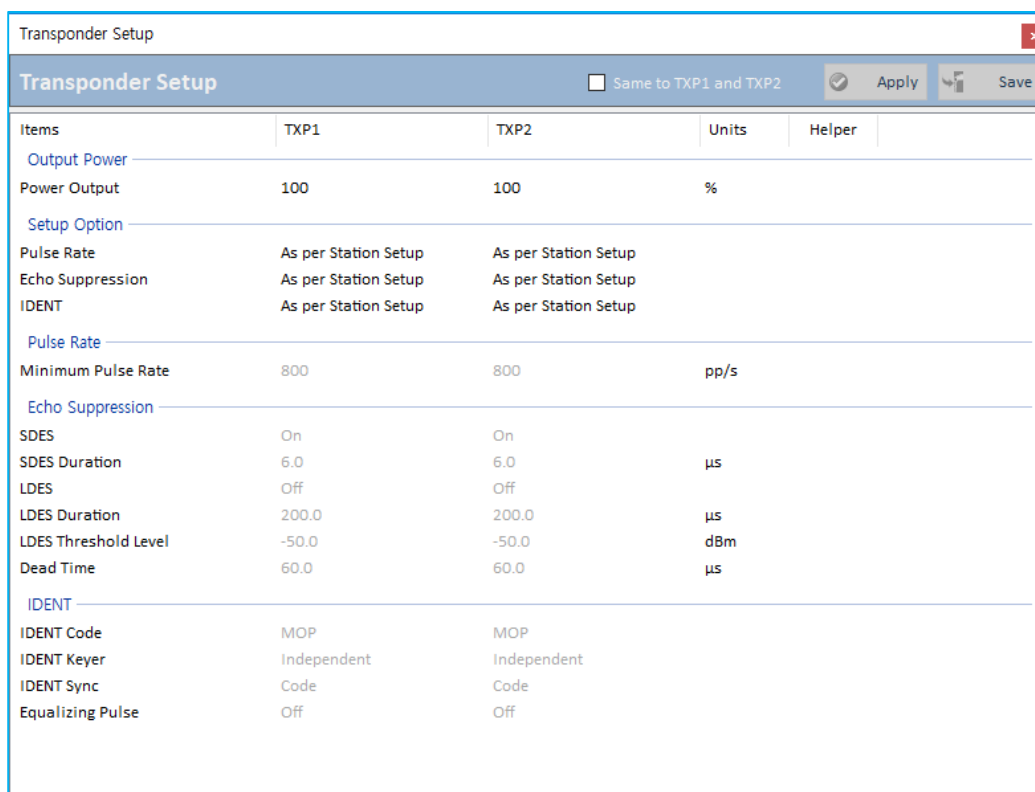
Transponder Setup Screen

Transponder Setup screen as shown in Figure 4-90 is to set up those parameters which are specific to each transponder.

Note that the some of the station parameters can be also configured in this screen for each individual transponder. If the value setup in the station setup is desired to be used for both transponders, the setup option “**As per Station Setup**” should be selected and the value set up in the Station Setup is automatically applied to both transponders and the corresponding parameter in the Transponder Setup screen will be disabled.

And it is also possible to globally apply the same setup values to both transponders, in which case the checkbox option “**Same to TXP1 and TXP2**” should be checked.

The parameters that can be configured on the screen are summarized in Table 4-18.



The screenshot shows the 'Transponder Setup' window. At the top, there's a title bar 'Transponder Setup' with a close button. Below it, a header bar contains the title 'Transponder Setup', a checkbox 'Same to TXP1 and TXP2', and buttons for 'Apply' and 'Save'. The main area is a table with columns: Items, TXP1, TXP2, Units, and Helper. The table is divided into sections by blue lines and headers: 'Output Power', 'Setup Option', 'Pulse Rate', 'Echo Suppression', and 'IDENT'.

Items	TXP1	TXP2	Units	Helper
Output Power				
Power Output	100	100	%	
Setup Option				
Pulse Rate	As per Station Setup	As per Station Setup		
Echo Suppression	As per Station Setup	As per Station Setup		
IDENT	As per Station Setup	As per Station Setup		
Pulse Rate				
Minimum Pulse Rate	800	800	pp/s	
Echo Suppression				
SDES	On	On		
SDES Duration	6.0	6.0	μs	
LDES	Off	Off		
LDES Duration	200.0	200.0	μs	
LDES Threshold Level	-50.0	-50.0	dBm	
Dead Time	60.0	60.0	μs	
IDENT				
IDENT Code	MOP	MOP		
IDENT Keyer	Independent	Independent		
IDENT Sync	Code	Code		
Equalizing Pulse	Off	Off		

Figure 4-90. Transponder Setup

The parameters that can be configured on the screen are summarized in Table 4-18.

Table 4-18. Transponder Setup

Category	Item	Description	Min	Max	Resolution	Unit
Output Power	Power Output	Power output per cent with reference to the power output in watts configured in the station setup.	0	100	1	%
Pulse Rate	Minimum Pulse Rate	Minimum Pulse Rate (Squitter pulses)	0	2700	1	pp/s
Echo Suppression	SDES	Use of Short Distance Echo Suppression	• On • Off		Choose option	
	SDES Duration	Duration of SDES	0	34	0.1	μs
	LDES	Use of Long Distance Echo Suppression	- On - Off			
	LDES Duration	Duration of LDES	0	350	0.1	μs
	LDES Threshold Level	LDES Threshold Level	-110	9	0.1	dBm
	Dead Time	Dead Time	0	150	1	μs
IDENT	Code	IDENT Code	-	4	Char	
	Keyer	IDENT Keyer mode	• None • Independent • Master • Slave • Continuous		Select from list	
	Sync	IDENT Synchronization mode	• Code • Pulse		Select from list	
	Equalizer Pulse	Use of Equalizing Pulse	• On • Off		Select from list	

Thermal Control Screen

Thermal Control screen is to set up operation of cooling fans and overheat protection. It can be accessed by selecting **Thermal Control Setup** from the navigation pane.

The screenshot shows a software window titled "Thermal Control". Inside, there's a header bar with "Thermal Control" and a checkbox "Same to TXP1 and TXP2" which is checked. To the right of the header are "Apply" and "Save" buttons. Below the header is a table with columns: Items, TX1, TX2, Units, and Helper. The table is divided into two sections: "Cooling Fans" and "Overheat Protection".

Items	TX1	TX2	Units	Helper
Cooling Fans				
Operation Mode	On	On		
Operation Temperature	65.0	65.0	°C	
Stop Temperature	67.0	67.0	°C	
Overheat Protection				
TXU Shutdown at	90.0	90.0	°C	
TXU Restart at	85.0	85.0	°C	

Figure 4-91. Thermal Control

The parameters that can be configured in the screen is summarized in Table 4-19

Table 4-19. Thermal Control

Category	Item	Description	Min	Max	Resolution	Unit
Cooling Fans	Operation Mode	FAN Operation Mode	• Auto • On • Off		Select from list	
	Operation Temperature	The temperature at which the fan start to run when the operation mode is Auto.	-100	200	0.1	°C
	Stop Temperature	The temperature at which the fan stop to run when the operation mode is Auto.	-100	200	0.1	°C
Overheat Protection	TXU Shutdown at	The temperature at which TXU shuts down automatically to protect.	-100	200	0.1	°C
	TXU Restart at	The temperature at which TXU restarts automatically.	-100	200	0.1	°C

4.4.6.3. Monitor Setup Screens

Executive Monitor Setup Screen

Executive Monitor Setup screen is to set up the parameters relating executive monitor. This screen can be accessed by selecting **Executive Monitor Setup** from the navigation pane.

Executive Monitor Setup

Executive Monitor Setup

ApplySave

Items	Settings	Units
Executive MON		
Monitor Voting Logic	OR config	
MON Action Delay	6.4	sec
Post Changeover Alarm Delay	1.00	sec
Identification Fault Condition	15.00	sec
Self Test Alarm Delay	4.00	sec
Power On Delay	3.00	sec
Primary Alarm		
IDENT Code Alarm	Primary	
ERP Alarm	Secondary	
DCDC Alarm	Secondary	
Self Test Alarm	Primary	

Figure 4-92. Executive Monitor Setup

The parameters that can be configured in the Executive Monitor Setup is summarized in Table 4-20.

Table 4-20. Executive Monitor Setup

Category	Item	Description	Min	Max	Resolution	Unit
Executive Monitor	MON Voting Logic	Monitor voting logic	• AND • OR		Choose option	
	MON Action Delay	Delay until taking monitor action	0.2	51	0.2	sec
	Post Changeover Alarm Delay	Hold-off time for alarms right after Changeover	0	100	0.1	sec
	Identification fault Condition	Threshold time for detecting absence of IDENT	0	100	0.1	sec
	Self-Test Alarm Delay	Hold-off time for Self-Test alarm	0	100	0.1	sec
	Power On Delay	Hold-off time for alarms right after power on	0	100	0.1	sec
Primary Alarm	Ident Code Alarm	Set Ident Code Alarm as Primary or Secondary	• Primary • Secondary		Select from list	
	ERP Alarm	Set ERP Alarm as Primary or Secondary	• Primary • Secondary		Select from list	
	DCDC Alarm	DCDC Alarm as Primary or Secondary	• Primary • Secondary		Select from list	
	Self-Test Alarm	Set Self-Test Alarm as Primary or Secondary	• Primary • Secondary		Select from list	

Monitor Limit Setup Screen

Monitor Limit Setup screen is to set up monitor alarm / warning limits and to set the monitor action attributes: alarm delay and classification of primary or secondary.

The Monitor Limit Setup can be accessed by selecting **Monitor Limit Setup** from the navigation pane.

For each monitor parameter, alarm limits and warning limits can be individually configured but the warning limits cannot be wider than the alarm limits. The default range for the warning limits can be defined globally

Primary alarms are executive and they will trigger a monitor action when the alarm condition persists over the duration of Monitor Action Delay which is configured in the Executive Monitor Alarm setup. Secondary alarm are non-executive and will not trigger a monitor action.

Monitor Limit Setup

☒ Same limits to both TXP1 and TXP2 Apply Save

Items	Lower Alarm	Lower Warning	Nominal	Upper Warning	Upper Alarm	Units	Helper	Primary Secondary	Alarm Delay	Unit
Time Delay	49.50	49.60	50.00	50.40	50.50	μs	Helper	Primary	1.00	sec
Pulse Spacing	11.50	11.70	12.00	12.30	12.50	μs	Helper	Primary	1.00	sec
Pulse Duration	3.00	3.20	3.50	3.80	4.00	μs	Helper	Primary	3.00	sec
Pulse Decay Time	0.80	1.50	2.00	2.80	3.00	μs	Helper	Primary	3.00	sec
Pulse Rise Time	0.80	1.50	2.00	2.80	3.00	μs	Helper	Primary	3.00	sec
Transmission Rate	700	750	N/A	5000	5400	pp/s		Primary	3.00	sec
Reply Efficiency	70	75	N/A	N/A	N/A	%		Primary	3.00	sec
Frequency	1186.763	1186.786	1187.000	1187.214	1187.237	MHz	Helper	Primary	3.00	sec
Power Output	50	60	70	140	150	W	Helper	Primary	3.00	sec
VSWR	N/A	N/A	N/A	2.00	3.00			Primary	3.00	sec

Figure 4-93. Executive Monitor Limit Setup

The parameters that can be configured in the Executive Monitor Limit Setup screen is summarized in Table 4-21.

Table 4-21. Executive Monitor Limit Setup

Category	Item	Description	Min	Max	Resolution	Unit
Monitor Limit Setup	Time Delay	Limits for Time Delay alarm/warning	0	125	0.01	μs
	Pulse Spacing	Limits for Pulse Spacing alarm / warning	0	125	0.01	μs
	Pulse Duration	Limits for Pulse Duration alarm / warning	0	125	0.01	μs
	Pulse Decay Time	Limits for Pulse Decay Time alarm / warning	0	125	0.01	μs
	Pulse Rise Time	Limits for Pulse Rise Time alarm / warning	0	125	0.01	μs
	Transmission Rate	Limits for Transmission Rate alarm / warning	0	7000	1	pp/s
	Reply Efficiency	Limits for Replay Efficiency alarm / warning	0	100	1	%
	Frequency	Limits for Frequency alarm / warning	0	2000	0.001	MHz
	Power Output	Limits for Power Output alarm / warning	0	2500	1	W
	VSWR	Limits for VSWR alarm / warning	1	15	0.01	dB
	ERP	Limits for Effective Radiated Power alarm / warning	-10	20	0.01	dB
Alarm Delay	Alarm Delay	Delay time until reporting alarm	0	30	1	Sec

4.4.6.4. System Setup Screen

System Setup screen is to set up parameters specific to the system configuration. It can be accessed by selecting **System Setup** from the navigation pane.

The screenshot shows the 'System Setup' window with a title bar and a close button. Below the title bar is a header with 'System Setup' and buttons for 'Apply' and 'Save'. The main content area is divided into sections by blue lines. The sections are: 'Login Options' with settings for 'Allow simultaneous login with same account' (Yes), 'Allow guest access without login/password' (Yes), and 'Automatic logout time' (60 minute(s)); 'IDENT keying' with 'When monitor bypassed' (No Change); 'Can modify system settings' with 'only when monitor bypassed' (Yes) and 'only at local' (No); and 'Automatic Shutdown Option' with 'Shut down on fault with RCU' (No), 'Shut down on fault with LMI' (No), 'Shut down on fault with CSP' (No), and 'Delay to shut down on fault' (3.0 sec).

Items	Settings	Units	Helper
Login Options			
Allow simultaneous login with same account	Yes		
Allow guest access without login/password	Yes		
Automatic logout time	60	minute(s)	
IDENT keying			
When monitor bypassed	No Change		
Can modify system settings			
only when monitor bypassed	Yes		
only at local	No		
Automatic Shutdown Option			
Shut down on fault with RCU	No		
Shut down on fault with LMI	No		
Shut down on fault with CSP	No		
Delay to shut down on fault	3.0	sec	

Figure 4-94. System Setup

The parameters that can be configured in the System Setup is summarized in Table 4-22.

Table 4-22. System Setup

Category	Item	Description	Min	Max	Resolution	Unit
Login Option	Allow Simultaneous Login with same account	• Yes • No			Choose option	
	Allow guest access without login/password	• Yes • No			Choose option	
	Automatic Logout Time		0	1440	1	Minute
Can modify system settings	Only when monitor bypassed	• Yes • No			Choose option	
	Only at local	• Yes • No			Choose option	
Automatic Shutdown Option	Shutdown on fault with RCU	• Yes • No			Choose option	
	Shutdown on fault with LMI	• Yes • No			Choose option	
	Shutdown on fault with CSP	• Yes • No			Choose option	
	Delay to shutdown on fault		0	30	0.1	sec

4.4.6.5. Environmental Setup Screen

Environmental Setup screen is to set up parameters relating environmental monitor, EMU. It can be accessed by selecting **Environmental Setup** from the navigation pane. As the EMU is optional, the menu item **Environmental Setup** is available only if the EMU board is present and otherwise not shown.

The screenshot shows the 'Environmental Setup' window with a title bar and 'Apply' and 'Save' buttons. The window is divided into several sections, each with a collapse/expand icon (triangle) and a section title.

- Environmental Sensor**: A table with columns 'Smoke', 'Intrusion', and 'Temper...'. The 'Enable' row has three checkboxes, all of which are unchecked.
- Analog Input**: A table with columns 'Input Port' and '#1' through '#8'. The 'Enable' row has eight checkboxes, all of which are checked.
- Analog Output**: A table with columns 'Output Port' and '#1' through '#8'. The 'Enable' row has eight checkboxes, all of which are unchecked. Below this is a 'Default Value' row with eight spinners, all set to '0'.
- Digital Input**: A table with columns 'Input Port' and '#1' through '#16'. The 'Enable' row has sixteen checkboxes, all of which are unchecked.
- Expansion Digital Input**: A table with columns 'Input Port' and '#1' through '#8'. The 'Enable' row has eight checkboxes, with the last one (port #8) checked.
- Digital Output**: A table with columns 'Output Port' and '#1' through '#16'. The 'Enable' row has sixteen checkboxes, all of which are unchecked. Below this is a 'Default High' row with sixteen checkboxes, with ports #1, #2, #4, #5, #6, #8, #9, #11, #12, #14, #15, and #16 checked.

Figure 4-95. Environmental Setup

The Input item is activated when the corresponding check box in the screen is checked.

The Output item is activated by checking the corresponding check box in the screen. Output value can be set once it is activated.

4.4.6.6. Communication Setup Screen

The screenshot shows the 'Communication Setup' window with the following settings:

Items	Settings	Units	Helper
Connection Limit (Max Total Connection = 16)			
Remote Connection Limit	8		
Local Connection Limit (include USB)	1		
PMDT RS-232			
Baudrate	115200	bps	
SCU1 Remote			
Line Type	RS232		
Baudrate	115200	bps	
Flow Control	None		
SCU2 Remote			
Line Type	RS232		
Baudrate	57600	bps	
Flow Control	None		
RCU			
Line Type	Ethernet		
Local PMDT on IP network			
IP Address Starting	172.16.1.5		
IP Address Ending	172.16.1.253		

Figure 4-96. Communication Setup

The parameters that can be configured in the Communication Setup screen is summarized in Table 4-23.

Table 4-23. Communication Setup

Category	Item	Description	Min	Max	Resolution	Unit
Connection Limit	Remote connection limit	Maximum number of simultaneous remote PMDT connections.	0	16	1	EA
	Local connection limit	Maximum number of simultaneous local PMDT connections.	0	16	1	EA
PMDT RS-232	Buadrate	Baud rate of the Local PMDT	2400	230400	Select from list	bps
SCU1 Remote	Type	Data link type for the remote port of SCU #1.	• RS-232 • Leased Line • Dialup		Select from list	
	Baudrate	Baud rate of the remote port of the SCU #1	2400	230400	Select from list	bps
	Flow Control	Use of Hardware Flow Control	• Yes • No		Choose option	

SCU2 Remote	Type	Data link type for the remote port of SCU #2.	• RS-232 • Leased Line • Dialup		Select from list	
	Baudrate	Baud rate of the remote port of SCU #2	2400	230400	Select from list	bps
	Flow Control	Use of Hardware Flow Control	• Yes • No		Choose option	
RCU	Line Type	Type of data link with RCU	• Ethernet • Modem • RS-232		Select from list	
Local PMDT On IP Network	IP Address Start	Start of the IP address range allocated for local PMDT	IP Address in dotted decimal form		Enter valid IP address	
	IP Address End	End of the IP address range allocated for local PMDT	IP Address in dotted decimal form		Enter valid IP address	

4.4.6.7. Miscellaneous Setup Screen

Miscellaneous Setup screen is to configure miscellaneous parameters which are not related to anywhere else.

Items	PMU1	PMU2	Units
Voltage Alert			
Warning Limit	22.00	22.00	V
Alarm Limit	21.00	21.00	V
Temperature Alert			
Warning Limit	40	40	°C
Alarm Limit	50	50	°C
Battery Limit			
Charging Current	10	10	A
Cut-off Voltage	20.00	20.00	V

Figure 4-97. Battery Setup

The parameters that can be configured in the Miscellaneous Setup screen is summarized in Table 4-24.

Table 4-24. Battery Setup

Category	Item	Description	Min	Max	Resolution	Unit
Voltage Alert	Warning Limit	Limits for Battery Voltage Warning	0	24	0.01	V
	Alarm Limit	Limits for Battery	0	24	0.01	V

		Voltage Alarm				
Temperature Alert	Warning Limit	Limits for Battery Temperature Warning	-128	127	1	°C
	Alarm Limit	Limits for Battery Temperature Alarm	-128	127	1	°C
Battery Limit	Charging Current	Limits for Battery Charging Current	10	30	0.01	V
	Cut-off Voltage	Limits for Cut-off Voltage	19	24	0.01	V

4.4.7. MAINTENANCE GROUP SCREENS

4.4.7.1. Calibration Screen

The 310/320 DME calibration consists of 10 steps as shown below

- 1) Transmitter Drive Level Optimization
- 2) Transmitter Power Output Calibration
- 3) Transponder Processing Delay Compensation
- 4) Auto Time Delay Calibration
- 5) Monitor Interrogation Level Calibration
- 6) Time Delay Monitor Reading Calibration
- 7) ERP Monitor Reading Calibration
- 8) VSWR Monitor Reading Calibration
- 9) Power Output Monitor Reading Calibration
- 10) Receiver Threshold Calibration

Calibration Test Setup

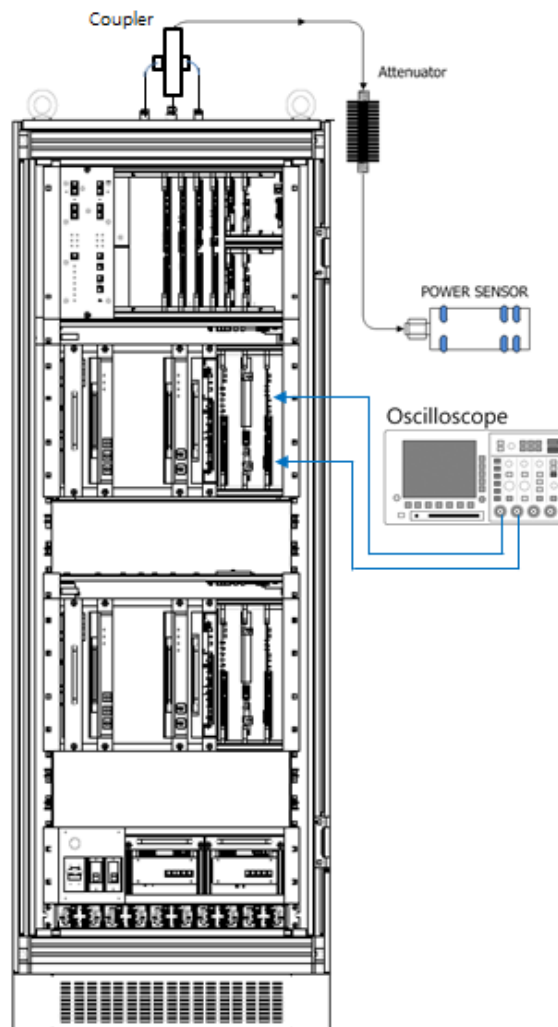


Figure 4-98. Calibration Test Setup

Necessary Test Equipment

- Wideband peak power sensor, R&S ERP-Z81 or equivalent
- RF power attenuator : 50 dB, 50 Ω , 250W
- Coupler : 30 dB
- Test Cable : N(M) to N(M) 50 Ω Coaxial Cable with known loss at the test frequency

Calibration Procedure

The calibration is carried out as following steps.

- a. Select **Maintenance** → **Calibration** and click on **Start** button.

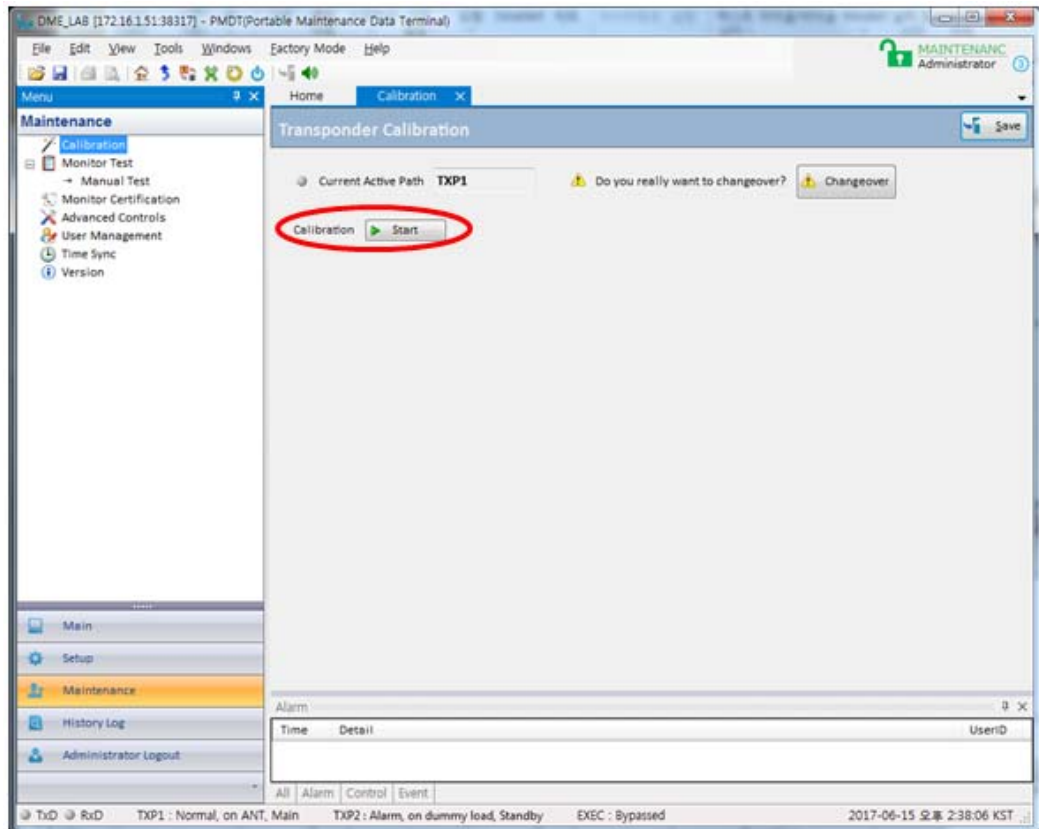


Figure 4-99. Calibration Screen

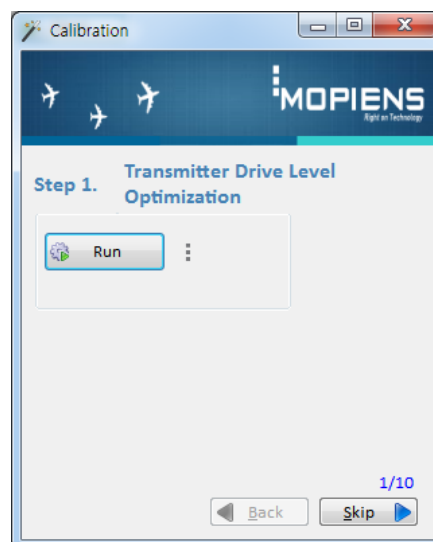


Figure 4-100. Step 1 Transmitter Drive Level Optimization

- b. Then the Calibration dialog box will pop up as shown in Figure 4-100. The step 1 is

Transmitter Drive Level Optimization. Click on **Run** button on the dialog box and wait until a green tick mark is displayed, which indicates step #1 has been completed successfully. Proceed to next step.

- c. Step 2 will show *Transmitter Power Output Calibration* dialog box as shown in Figure 4-101. Verify that the peak power output is close as the setting (usually 100 W for 310 DME, 1000 W for 320 DME) using peak power meter. If the reading is within $\pm 5\%$, it is not necessary to run this step, click on **Skip** button. Otherwise it is necessary to run Step 2.

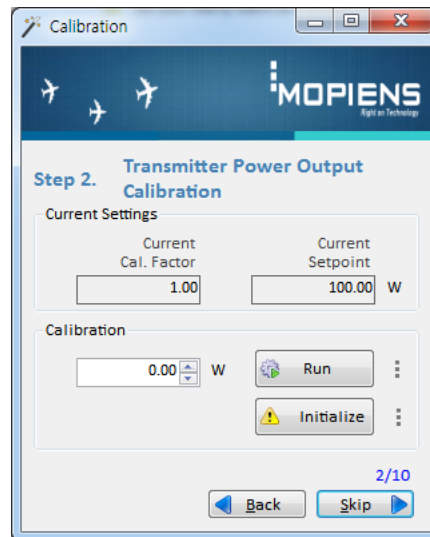


Figure 4-101. Step 2. Transmitter Power Output Calibration

- d. Measure the peak power output using a precision peak power meter. Refer to Table 4-25. Enter the measured value into the edit box in W and click on **Run** button. When it is done a green tick mark will be displayed. Then proceed to next step.

Table 4-25. Step 2. Transmitter Power Output Calibration

Item	Description
Current Setpoint	Shows the setting in Station Setup → Basic → Power Output
Current Cal. Factor	Current calibration factor, which is an internal scale factor
Calibration	Enter the real measurement into the edit box in W and click on Run button to calculate the new calibration factor. If required, the calibration factor can be initialized to the default value by clicking on the Initialize button.

- e. Step 3 will show *Transponder Processing Delay Compensation* dialog box as shown Figure 4-102. Check the reply time delay using accurate oscilloscope as shown in Figure 4-103. If the time delay greatly differs from the nominal value (i.e., 50 μ s for X channel and 56 μ s for Y channel), it is necessary to compensate it. The compensation can be done either automatically or manually. The step 3 is for manual compensation only and usually it can be skipped for auto compensation. If manual compensation is to be performed, get the time delay measurement accurately and enter to the edit box in μ s and click on **Run** button.

*** Note that the Step 3 should be executed only if Step 1 or Step 2 has been already performed.

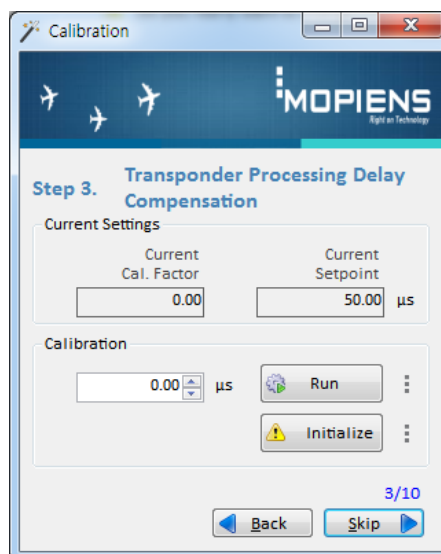


Figure 4-102. Step 3. Transponder Processing Delay Compensation

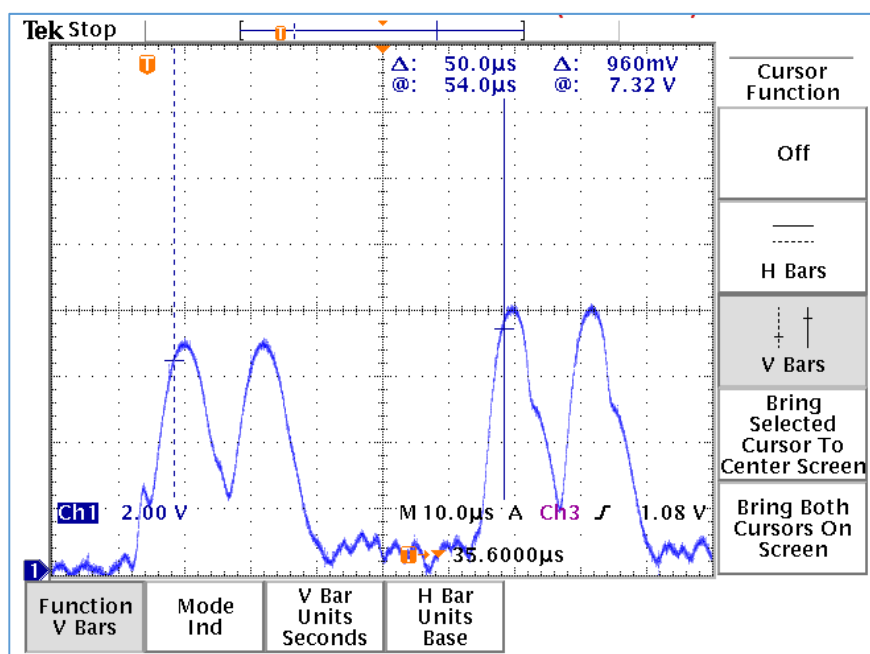


Figure 4-103. Time delay measurement using oscilloscope

Table 4-26. Step 3. Transponder Processing Delay Compensation

NAME	Description
Current Setpoint	Shows the setting in Station Setup → Basic → Delay Offset imposed on the nominal time delay (50 µs for X channel, 56 µs for Y channel). e.g. X channel with Delay Offset = 0.5 µs → Current Setpoint will be 50.5 µs
Current Cal. Factor	Current calibration factor, which is an internal compensation offset.
Calibration	Enter the compensation value calculated from the time delay measurement into the edit box in µs and click on Run button to apply the new compensation value. e.g. X channel, Current Setpoint = 50.5 µs

	If measured time delay is 50.6 μs, enter the error, i.e. the difference, 0.1 μs into the edit box for compensation If required, the calibration factor can be initialized to the default value by clicking on the Initialize button.
--	---

- f. Step 4 will show *Auto Time Delay Calibration* dialog box as shown in Figure 4-104. This is to compensate the internal delay automatically and necessary to run if the Step 3 is skipped. Just click on the **Run** button and wait until a green tick mark is displayed. Then proceed to the next step.

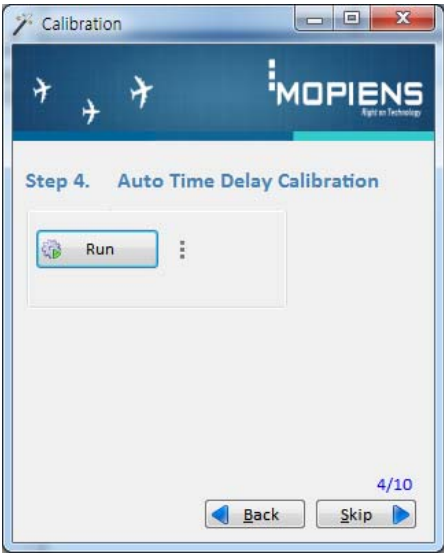


Figure 4-104. Step 4. Auto Time Delay Calibration

- g. Step 5 will show *Monitor Interrogation Level Calibration* dialog box as shown in Figure 4-105. Just click on **Run** buttons for MON1 and MON2 consecutively. When completed, green tick marks will be displayed. Then proceed to the next step.



Figure 4-105. Step 5. Monitor Interrogation Level Calibration

- h. Step 6 will show *Time Delay Monitor Reading Calibration* dialog box as shown in Figure 4-106. Just click on **Run** buttons for MON1 and MON2 consecutively. When completed, green tick marks will be displayed. Then proceed to the next step.

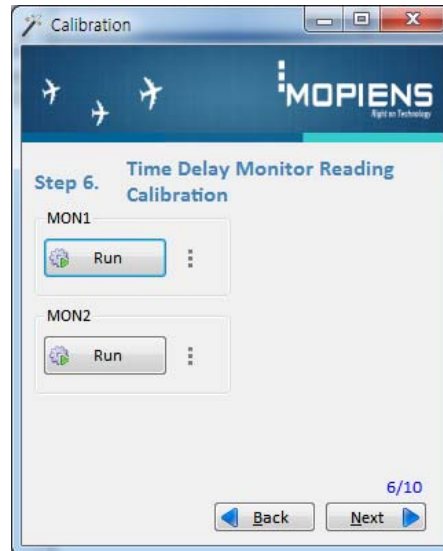


Figure 4-106. Step 6. Time Delay Monitor Reading Calibration

- i. Step 7 will show ERP Monitor Reading Calibration dialog box as shown in Figure 4-107. Make sure proper antenna with two monitor ports connected to the equipment and click on **Run** buttons for MON1 and MON2 consecutively. When completed, green tick marks will be displayed. Then proceed to the next step.

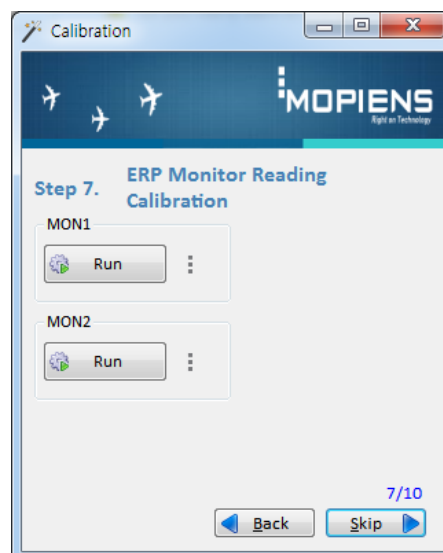


Figure 4-107. Step 7. ERP Monitor Reading Calibration

- j. Step 8 will show *VSWR Monitor Reading Calibration* dialog box as shown in Figure 4-108. This is to calibrate the VSWR monitor and ideally it is required to be performed under three conditions: on Load (terminated with the nominal impedance 50 Ω , on Open and on Short. But for this VSWR monitor calibration, the 'on Short' condition is omitted. Terminate the RF output port of the DME equipment with a proper 50 Ω load and Click on **Run** button for both MON1 and MON2 one by one.

- k. Disconnect the load and leave the output port open. Then clock on **Run** button for both MON1 and MON2 one by one. When all completed, green tick marks will be displayed. Connect the antenna feeder cable back to the RF output port of the DME equipment and proceed to the next step.
- *** Note that, an extra checkbox option to switch off RF output is available at the bottom of the dialog box, this process can be performed without the RF output power. However this is not guaranteed to always work.

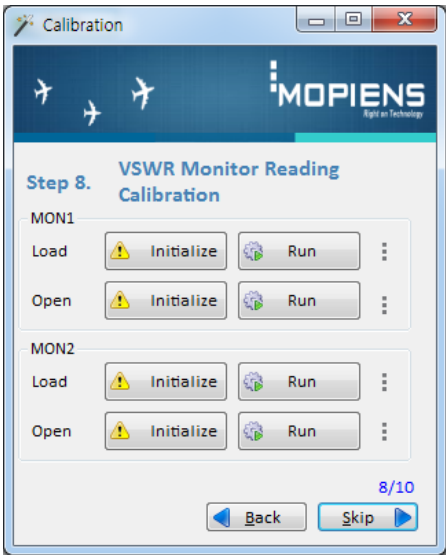


Figure 4-108. Step 8. VSWR Monitor Reading Calibration

- l. Step 9 will show Power Output Monitor Reading Calibration dialog box as shown in Figure 4-109. Refer to Table 4-27. Check the real power output using accurate peak power meter and compare with the current readings on the MON1 and MON2. If the indication on MON1 or MON2 is not accurate, enter the real power output into the edit box in W and clock on Run button. Do the same for both MON1 and MON2. When completed, green tick marks will be displayed. Now proceed to the next step.

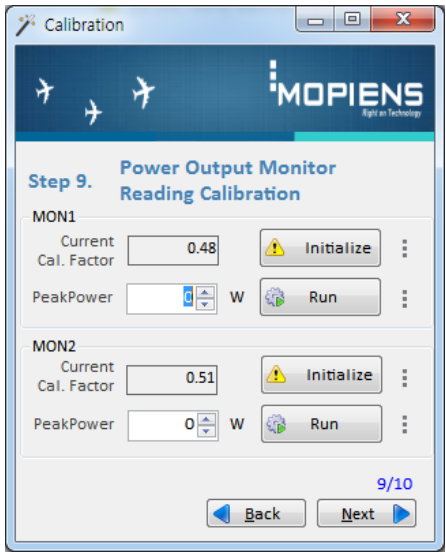


Figure 4-109. Step 9. Power Output Monitor Reading Calibration

Table 4-27. Step 9. Power Output Monitor Reading Calibration

NAME	Description
Current Cal. Factor	Current calibration factor, which is an internal scale factor.
Peak Power	Enter real value in W measured with a peak power meter.

- m. Step 10 will show *Receiver Threshold Calibration* dialog box as shown in Figure 4-110. Choose MON1 and click on Run button. When completed, a green tick mark will be displayed. Choose MON2 and carry out the same. Clicking on **Next** button will close the calibration dialog box.



Figure 4-110. Step 10. Receiver Threshold Calibration

Table 4-28. Step 10. Receiver Threshold Calibration

NAME	Description
Current Setpoint	Receiver Sensitivity Level, which is configured in Station Setup

- n. Now that one transponder is fully calibrated, change to the other transponder by clicking on Changeover button as shown in Figure 4-111 and carry out the complete steps as before.

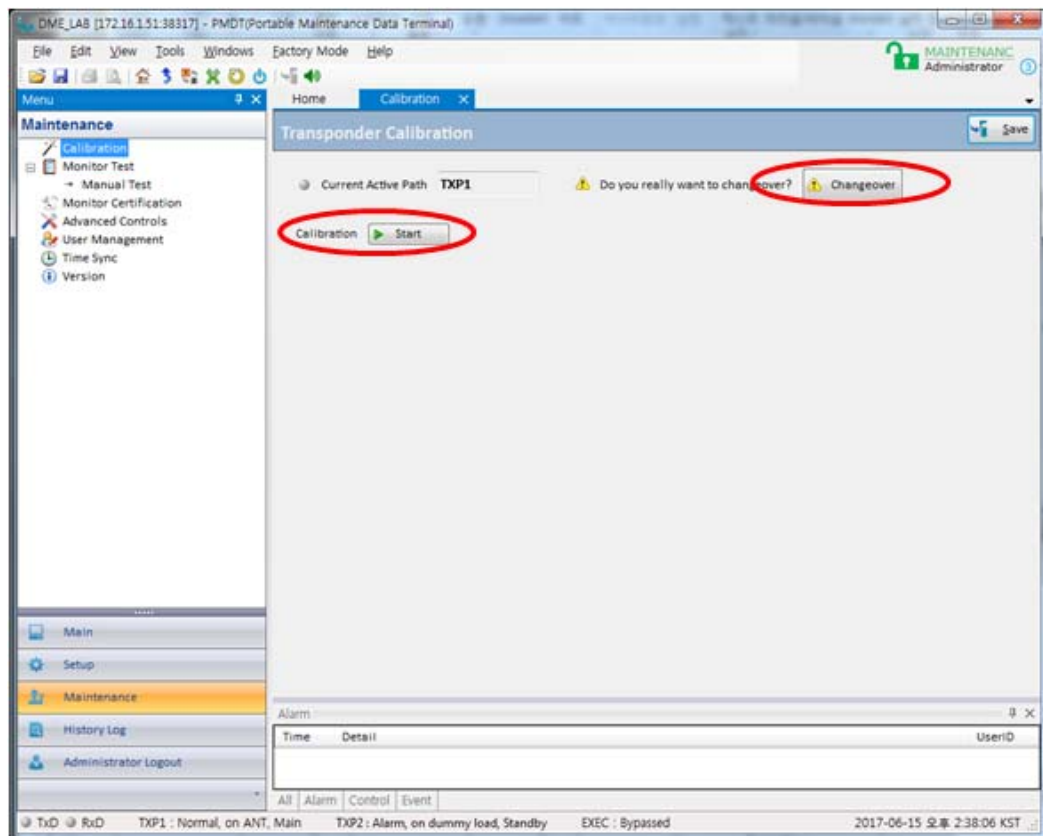


Figure 4-111. Calibration

4.4.7.2. Monitor Test Screen

Manual Test

310/320 DME has a manual test mode in which the test conditions can be manually set. The test parameters that can be set in the Manual Test screen include:

- Manual check TXP: The target transponder to test
- Interrogation Signal Level Peak power level of the test interrogation pulse pairs
- Check Number Number of test interrogation pulse pairs to inject
- Frequency Offset RF frequency offset of the interrogation signal with respect to the mid frequency of selected channel
- Check Number Sets of test to run

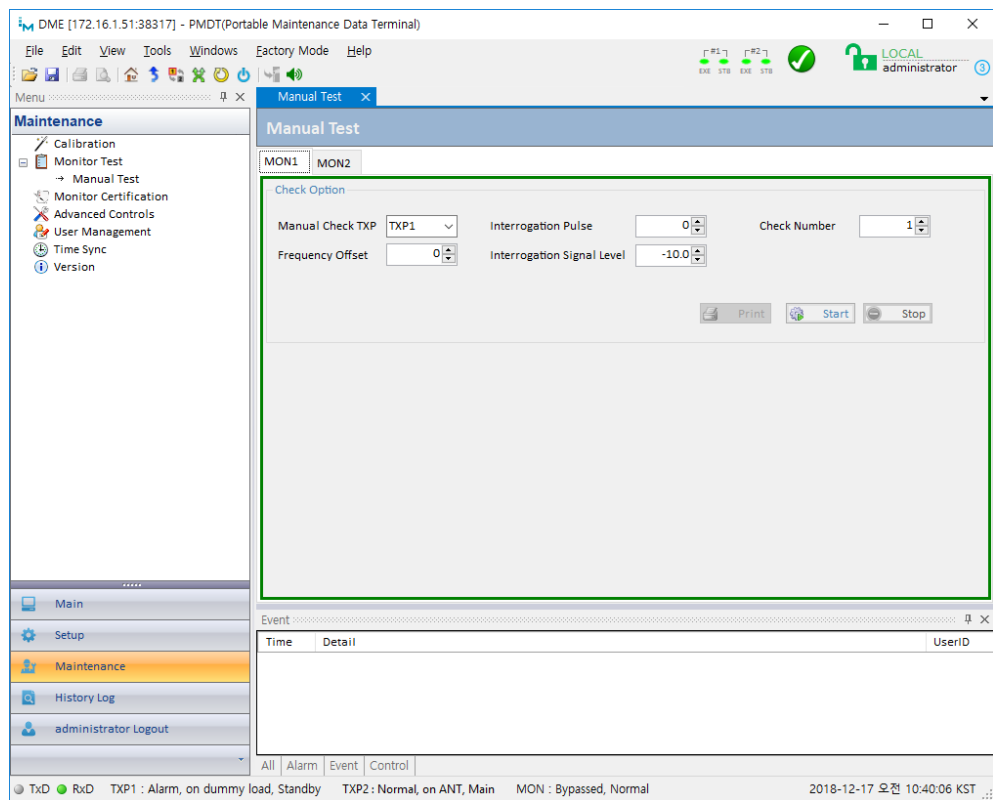


Figure 4-112. Manual Test screen

Select the tab of the MON unit to be tested and choose the target TXP. Enter the desired value for each edit box for the parameter under desired test conditions, and click on **Start** button and the result will be displayed when specified sets of test have been completed.

4.4.7.3. Monitor Certification Screen

As illustrated in Figure 4-113, 310/320 DME has a Monitor Certification screen to check if the monitor action of the equipment is correct. It can be accessed through **Maintenance → Monitor Certification** from the navigation pane. Parameters to be checked for the monitor certification include:

- Time Delay
- Pulse Spacing
- Reply Efficiency
- Peak Power Output
- Transmission Rate
- Identification Code

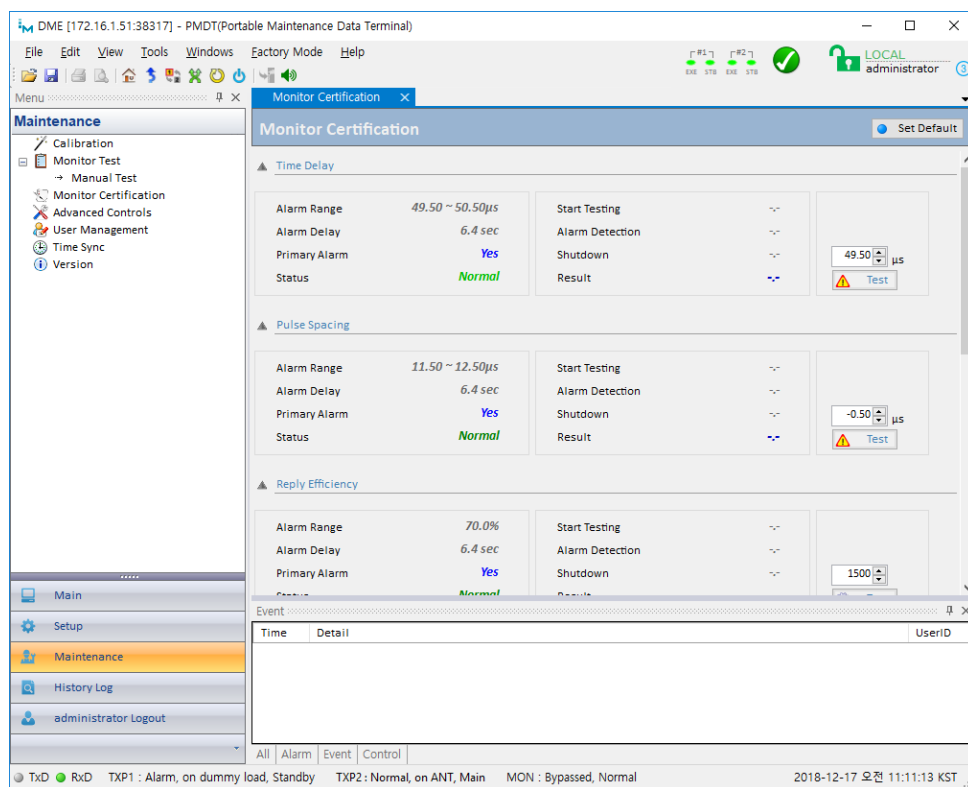


Figure 4-113. Monitor Certification

Tests and results can be checked for each item.

Alarm Range and Delay for each item, Primary Alarm and Monitor status values are displayed. For the test, the **Test** button is activated so that the test can be performed only when the **Primary Alarm** is set to "Yes". If you specify a test value (you must set it outside the alarm range) and click the **Test** button, the time of Start Testing, Alarm Detection, Shutdown, Result is displayed. Result is normal when it is included in error range (10%) of Alarm Delay (Fault Condition + Alarm Delay in case of IDENT) time.

4.4.7.4. Advanced Controls Screen

310/320 DME provides a number of special control functions for equipment maintenance. This function is initialized to the operational default state at the start-up of the equipment. The special function can be accessed through **Maintenance** → **Advanced Control** from the navigation pane.

TXP Advanced Control

There are four control items in the TXP tab of the Advanced Control screen. Figure 4-114 shows default state for each control item.

- Squitter Pulse
 - Stops the TXP Squitter function.
- IDENT Keying
 - Changes the IDENT Keying function of TXP to OFF or Continuous.
- RF Loopback (TX to RX)
 - Enable the RF loopback function so that the reply pulse pairs can be observed together with the interrogation pulse pairs at the same time on the T/P Log

Video Wide on the front panel of the RXU.

- Spacing
 - Intentionally adjusts the pulse spacing of the pulse pairs transmitted by the transponder.

The screenshot shows the TXP Advanced Control interface. It features two tabs at the top: TXP1 and TXP2. Below the tabs are four expandable sections, each with a green arrow icon and a title. The first section, 'Squitter Pulse', has radio buttons for 'On' (selected) and 'Off', and an 'Apply' button. The second section, 'IDENT Keying', has radio buttons for 'On' (selected), 'Off', and 'Continuous', and an 'Apply' button. The third section, 'RF Loopback (TX to RX)', has radio buttons for 'On' and 'Off' (selected), and an 'Apply' button. The fourth section, 'Spacing', has a 'Spacing Offset' input field set to '0' with a unit of 'μs', and an 'Apply' button.

Figure 4-114. TXP Advanced Control

Monitor Interrogation

The MON Tab on the Advanced Control screen provides the ability to manually set and test the monitor interrogation signals.

Figure 4-115 shows the screen with Manual Trigger Mode selected.

The screenshot shows the MON Advanced Control interface. It features two tabs at the top: MON1 and MON2. Below the tabs is an expandable section titled 'Monitor Interrogation'. Inside this section, there are two main areas. The left area has 'Trigger Mode' with radio buttons for 'Automatic' and 'Manual' (selected), and 'TX' with radio buttons for 'Active' (selected) and 'Standby'. The right area, titled 'ManualOption', contains five parameters: 'Interrogation Signal Level' (-50.0 dBm), 'Interrogation Pulse Rate' (100 per Sec), 'Average Count' (50), 'Frequency Offset' (0.0000 MHz), and 'Spacing' (12.000 μs). Each parameter has a corresponding input field or dropdown menu. At the bottom right of the section is an 'Apply' button.

Figure 4-115. MON Advanced Control with 'Manual Trigger Mode' selected

When *Manual Trigger mode* is selected, the following parameters can be manually set:

- **Interrogation Signal Level** Peak power level of the test interrogation pulse pairs
- **Interrogation Pulse Rate** Pulse pair rate of the test interrogation signals
- **Frequency Offset** RF frequency offset of the interrogation pulse pair
- **Spacing** Pulse pair spacing of the test interrogation pulse pairs
- **Average Count** Number of average to take on the result

The monitor result of the changed setting can be checked from the screens on **Main → Monitor → Executive Monitor / Standby Monitor** from the PMDT.

4.4.7.5. User Management

310/320 DME provides PMDT user account management. It can be accessed by selecting **Maintenance → User Management** from the navigation pane.

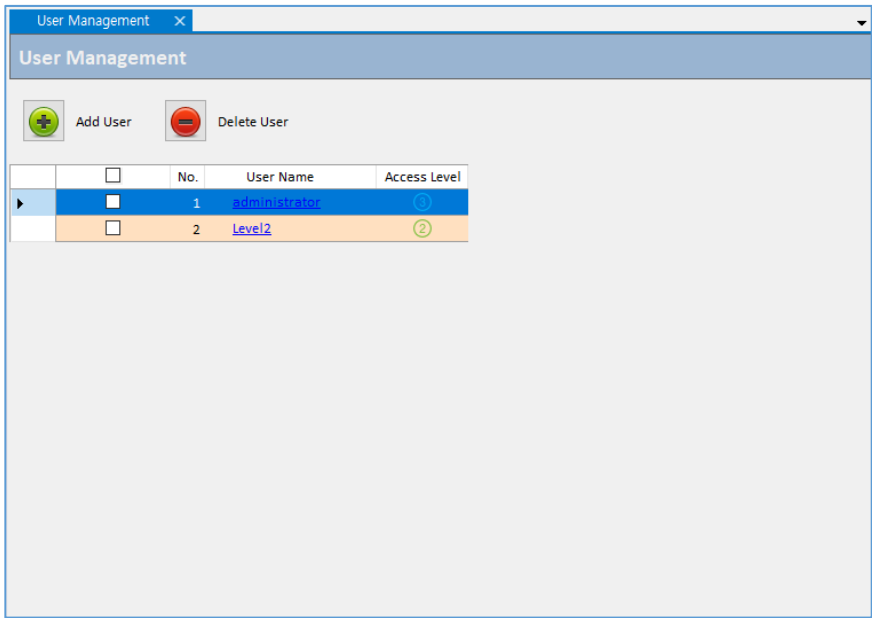


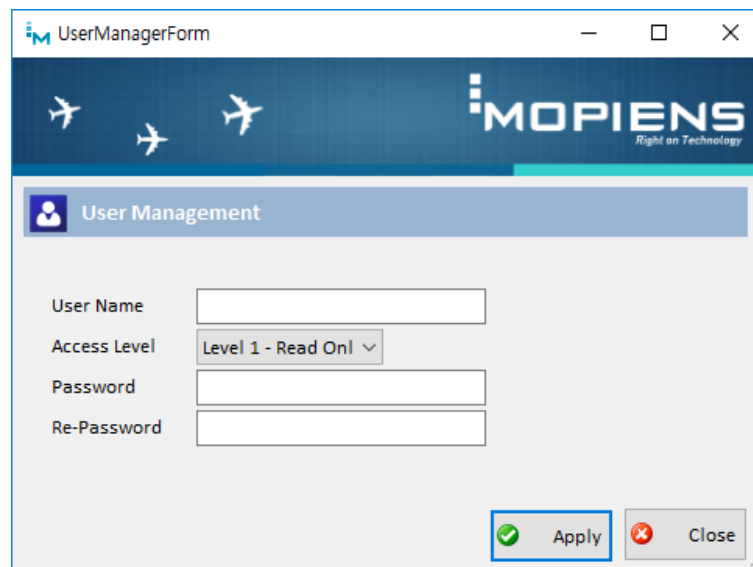
Figure 4-116. User Management Screen

Figure 4-116 shows the User Management screen.

The contents displayed on this screen are as follows.

-  : Add a new user
-  : Delete the selected user
- No. : Serial number for the user
- User Name : User name, i.e., login account
- Access Level : Access level of the user

If you click the Add User button, you will see the screen to input user information as shown in Figure 4-117 and you can add user by entering user name, level and password.



The screenshot shows a window titled "UserManagerForm" with a standard Windows title bar (minimize, maximize, close buttons). The window has a dark blue header bar with the MOPIENS logo and three airplane icons. Below the header is a light blue bar with a user icon and the text "User Management". The main area is light gray and contains four input fields: "User Name" (text box), "Access Level" (dropdown menu showing "Level 1 - Read Onl"), "Password" (text box), and "Re-Password" (text box). At the bottom right are two buttons: "Apply" (with a green checkmark icon) and "Close" (with a red X icon).

Figure 4-117. Add User Screen

The user level is divided into three levels: 1, 2, and 3.

The privileges for each level are as follows:

- Level 1: Read-only account. Only logged in without control
- Level 2: Limited rights account. Basic setting is possible, but maintenance functions such as calibration are limited.
- Level 3: Full control

4.4.7.6. Time Sync

310/320 DME provides a time synchronization function to compensate for equipment visual errors as shown in Figure 4-118 and Figure 4-119. There are two methods of time synchronization: Manual or Automatic. It can be selected from the Time Sync screen, which is available on the PMDT by selecting **Maintenance→Time Sync** from the navigation pane.

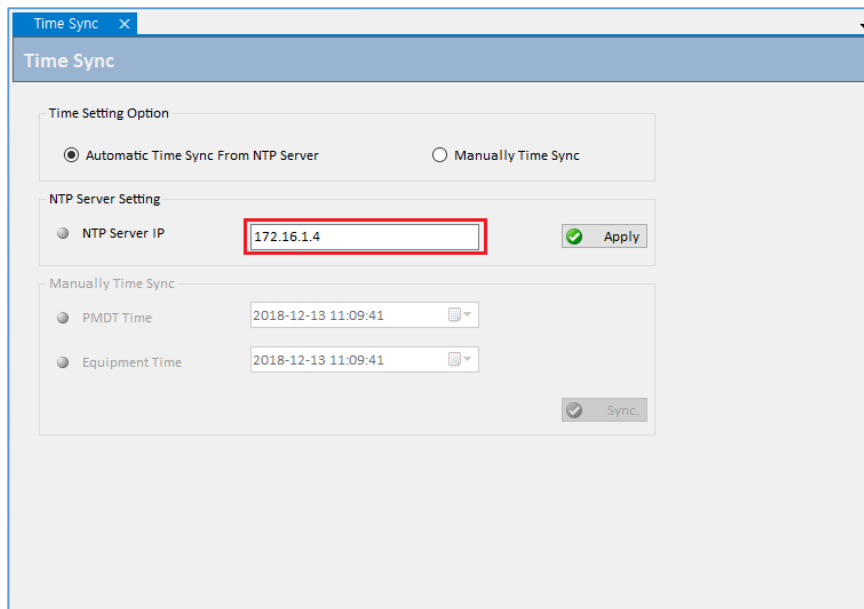
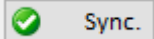
The screenshot shows the 'Time Sync' window with a title bar. Inside, there's a 'Time Setting Option' section with two radio buttons: 'Automatic Time Sync From NTP Server' (selected) and 'Manually Time Sync'. Below this is the 'NTP Server Setting' section with a radio button for 'NTP Server IP' and a text box containing '172.16.1.4', which is highlighted with a red rectangle. An 'Apply' button with a green checkmark is next to the text box. At the bottom is the 'Manually Time Sync' section with two radio buttons for 'PMDT Time' and 'Equipment Time', each followed by a time selection dropdown showing '2018-12-13 11:09:41'. A 'Sync.' button with a green checkmark is at the bottom right of this section.

Figure 4-118. Time Sync using NTP Server

Details are as follows.

- Automatic Time Synchronization from NTP Server
 - Select **Automatic Time Sync from NTP Server** option in **Time Setting Option**.
 - Enter the IP address of the NTP Server in the edit box and click on **Apply** button. The LMI of the equipment will perform automatic time synchronization with the NTP server periodically, once a week.
- Manual Time sync with PMDT PC
 - Select **Manually Time Sync** option in **Time Setting Option**.
 - After checking the equipment time and PMDT time, click on the button .

The synchronized time can be checked at the bottom right of the PMDT screen as shown in Figure 4-119.

Manually Time Sync

☐ PMDT Time: 2018-12-13 11:13:08
 ☐ Equipment Time: 2018-12-13 11:13:08

Event		
Time	Detail	UserID

All | Alarm | Control | Event

TxD RxT TX1 : Normal, on dummy load, Standby TX2 : Normal, on ANT, Main EXEC MON : Bypassed, Warning

2018-12-13 오전 11:13:08 KST

Figure 4-119. Equipment Time

4.4.7.7. Version

The version screen shows Reference Designation Part Number, Hardware Revision, SW Version and Uptime of each unit of the equipment as Figure 4-120. The version screen can be accessed by selecting In the **Maintenance** → **Version** from the navigation pane.

Version Information X							
Version							
Sub	Unit	Ref Des.	P/N	H/W Rev.	S/W Ver.	Up Time	
AUX	CSP	1A1A1	0330-7560	B			
	SCU	1A1A2	0130-7520	Unplugged	-	00:00:00	
	SCU	1A1A3	0130-7520	B	1.0.28 #03973	01:37:36	
	EMU	1A1A4	0130-7540	B	1.0.5 #03889	00:42:34	
	DC/DC-A	1A1A5	0130-7720	B			
	DC/DC-A	1A1A6	0130-7720	B			
	MODEM	1A1A7	0130-7420	B			
	MODEM	1A1A8	0130-7420	B			
TXP1	DPX-MSC	1A2A1	0330-7120	O			
	RXU	1A2A2	0330-7180	Unplugged			
	TXU	1A2A4	0330-7140	C			
	DC/DC	1A2A5	0330-7710	B			
	TCU	1A2A6	0330-7570	B	1.0.4 #03984	01:38:09	
	FPGA				#183		
	RFG	1A2A7	0330-7910	B			
	MON	1A2A8	0330-7550	C	1.0.16 #03994	01:38:10	
	FPGA				#208		
TXP2	FAN	1A2A9	0130-7910	A			
	DPX-MSC	1A3A1	0330-7120	O			
	RXU	1A3A2	0330-7180	C			
	TXU	1A3A4	0330-7140	C			
	DC/DC	1A3A5	0330-7710	B			
	TCU	1A3A6	0330-7570	C	1.0.4 #03984	01:38:08	
	FPGA				#183		
	RFG	1A3A7	0130-7910	A			
	MON	1A3A8	0330-7550	B	1.0.16 #03914	01:38:10	
	FPGA				#208		
	FAN	1A3A9	0130-7910	A			

Figure 4-120. Version Screen

- **Sub:** Subrack
- **Unit:** Name of the Unit
- **Ref Des.:** Reference Designation
- **P/N:** Part Number
- **H/W Rev.:** Hardware Revision Code
- **S/W Version:** Software Version Number
- **Up time:** Elapsed time since the boot up of each unit

Firmware/FPGA Upgrade

The 310/320 DME provides software (firmware) and FPGA download functionality via PMDT. When you click the button  on the right of the selected unit from the screen where Version is displayed, a dialog box for firmware update will pop up.

4.4.8. HISTORY LOG GROUP SCREENS

4.4.8.1. Introduction

The 310/320 DME has a history log function that allows you to check the history of device events. The history log is divided into the following two types according to the storage location.

- LMI History Log
 - Logs are stored in the LMI on the front door of the equipment.
 - Events are recorded as long as the equipment is on.
 - The records can be downloaded to PMDT and examined.
- PMDT History Log
 - Logs are stored in the computer where the PMDT is running on.
 - Events are recorded only while the PMDT is connected to the equipment.
 - The records can be examined directly using the PMDT.

The items stored in the history log are roughly divided into three categories.

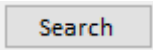
- Control: User control such as setting change, calibration
- Alarm: Equipment status such as monitor alarm
- Event: Equipment events such as automatic changeover or shutdown

4.4.8.2. PMDT History Log Screen

PMDT History Log can be accessed by selecting **History Log → PMDT History Log** from the navigation pane.

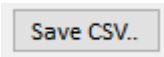
The screenshot shows a web-based application window titled "PMDT History Log". The window has a header bar with the title. Below the header, there is a search section. On the left, under "Search Date/Time", there are two date pickers: "From:" and "To:", both set to "2018년 12월 17일 월요일". To the right of these are two time pickers: "오전 12:00:00" and "오후 12:21:23". On the right side of the search section, there is a "Search Class" section with three checkboxes: "Alarm" (checked), "Control" (checked), and "Event" (checked). Next to these is a "Search" button and a "Save CSV..." button. The main body of the window is a large, empty gray rectangle.

Figure 4-121. PMDT History Log Screen

Select the time and class you want to search for and click the button  to search the device for the history log and display it on the screen.

PMDT History Log				
Search Date/Time From: 2018년 12월 17일 월요일 오전 12:00:00 To: 2018년 12월 17일 월요일 오후 12:16:53 Search Class ☒ Alarm ☒ Control ☒ Event Search Save CSV..				
ID	Time	Detail	Type	UserID
10288	2018-12-17 오후 12:10:40.4	TX1 TXU Digital Atten : 3.75 -> 0 (Profile : Active)	Control	administrato
10289	2018-12-17 오후 12:10:40.4	TX1 Pedestal PulseLevel : 523 -> 0 (Profile : Active)	Control	administrato
10290	2018-12-17 오후 12:10:40.4	TX1 Gaussian Pulse Level : 3217 -> 0 (Profile : Active)	Control	administrato
10291	2018-12-17 오후 12:10:40.4	TX1 Receiver Threshold Level : 823 -> 905 (Profile : Active)	Control	administrato
10292	2018-12-17 오후 12:10:40.4	TX1 LDES Duration : 200 -> 100 (Profile : Active)	Control	administrato
10293	2018-12-17 오후 12:10:40.4	TX1 SDES Duration : 3.5 -> 3.5 (Profile : Active)	Control	administrato
10294	2018-12-17 오후 12:10:40.4	TX1 Minimum Pulse Rate : 800 -> 800 (Profile : Active)	Control	administrato
10295	2018-12-17 오후 12:10:40.4	TX1 Delay Offset : 0 -> 0 (Profile : Active)	Control	administrato
10296	2018-12-17 오후 12:10:40.4	TX1 Dead Time : 60 -> 56 (Profile : Active)	Control	administrato
10297	2018-12-17 오후 12:10:40.4	TX1 Half Point Offset : 218 -> 218 (Profile : Active)	Control	administrato
10298	2018-12-17 오후 12:10:40.5	TX1 LDES Threshold Level : -50 -> -50 (Profile : Active)	Control	administrato
10301	2018-12-17 오후 12:10:40.5	TX1 Rectangular Pulse Level : 0 -> 4092 (Profile : Active)	Control	administrato
10302	2018-12-17 오후 12:10:40.5	TX1 Pedestal PulseLevel : 0 -> 247 (Profile : Active)	Control	administrato
10303	2018-12-17 오후 12:10:40.5	TX1 Gaussian Pulse Level : 0 -> 2860 (Profile : Active)	Control	administrato
10304	2018-12-17 오후 12:10:40.5	TX1 Drive Amp Output Level : 2802 -> 2676 (Profile : Active)	Control	administrato
10305	2018-12-17 오후 12:10:40.5	TX1 Output Level : 1441 -> 1423 (Profile : Active)	Control	administrato
10306	2018-12-17 오후 12:10:40.5	TX1 TXU Analog Atten : 0 -> 0 (Profile : Active)	Control	administrato
10307	2018-12-17 오후 12:10:40.5	TX1 TXU Digital Atten : 0 -> 4.75 (Profile : Active)	Control	administrato
10308	2018-12-17 오후 12:10:40.5	TX1 Gaussian Shape Bank : 3.65 -> 3.85 (Profile : Active)	Control	administrato
10309	2018-12-17 오후 12:10:40.5	TX1 Pedestal Output Level : 2546 -> 2730 (Profile : Active)	Control	administrato
10310	2018-12-17 오후 12:10:40.5	TX1 Pedestal 2nd Pulse Level Offset : 0 -> 0 (Profile : Active)	Control	administrato
10312	2018-12-17 오후 12:10:40.5	TX1 HPA Temp Sleep : 90 -> 90 (Profile : Active)	Control	administrato

Figure 4-122. PMDT History Log

The search results can be saved as a CSV file by clicking on the  button.

4.4.8.3. LMI History Log Screen

The LMI History Log can be accessed by selecting **History Log** → **LMI History Log** from the navigation pane.

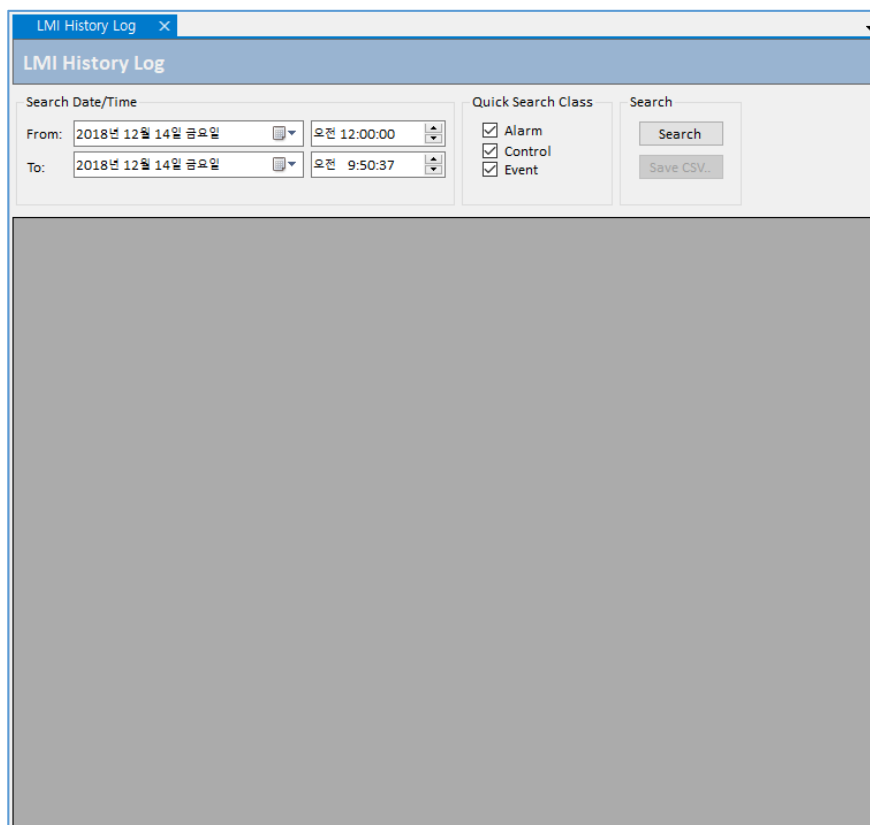
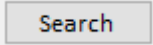
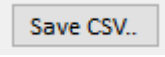


Figure 4-123. LMI History Log Screen

Select the time and class you want to search for and click the button  to search the device for the history log and display it on the screen.

The search results can be saved as a CSV file by clicking on the  button.

4.5. LMI

4.5.1. INTRODUCTION

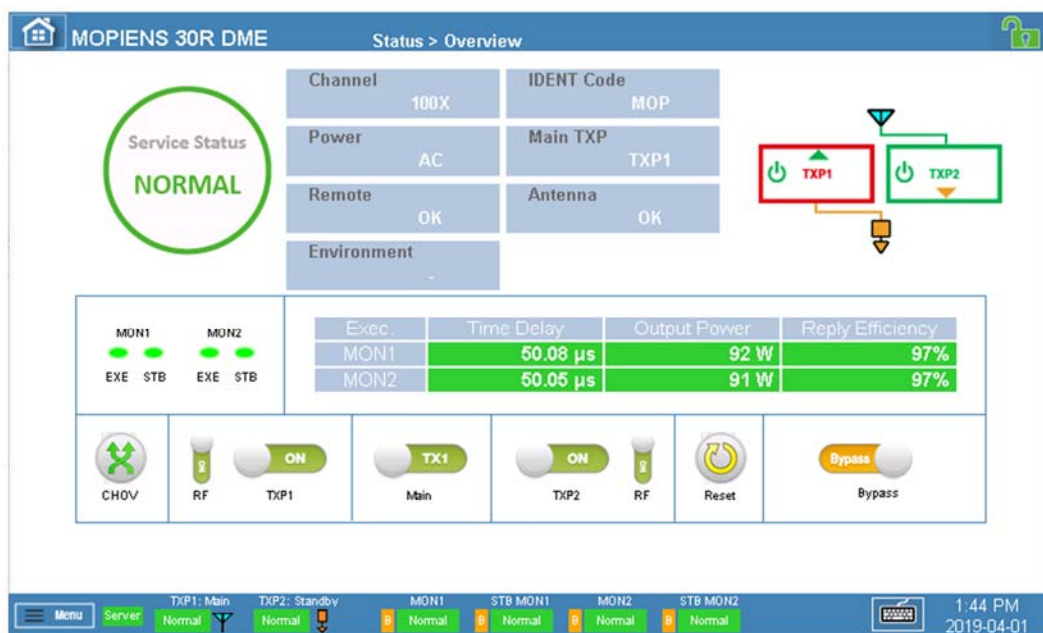


Figure 4-124. LMI Screen Layout

4.5.1.1. Title Bar



Figure 4-125. Title Bar

The title bar has the following elements:

-  Home button:
Clicking on this button bring the 'Home' screen (Main → Overview) to top.
- Informative text indicating 'Site Name' is displayed next to the home button.
- Informative text indicating 'Runway Designator' is displayed next to the 'Site Name'.
- Navigation path of the Current Page is displayed in the middle of the bar.
- Local/Remote mode of operation is displayed on the right by an icon:
 - o  This indicates that the local control is enabled and remote control is disabled. Remote status display is still available. In this state, it is allowed to control the equipment from LMI or CSP. This is when the keylock witch is in **LOCAL** or **MAINT** position.
 - o 

This indicates that the local control is disabled and remote control is enabled. In this state, it is not allowed to control the equipment from LMI or CSP. This is when the keylock switch is in **REM** position

4.5.1.2. Status Bar



Figure 4-126. Status Bar

-  Touching on this icon will pop up the Menu Control Panel mainly for navigating through the pages. Refer to 4.5.1.3.
-  The green icon indicates the server process is running OK and data exchange with SCU is normal. When the server process stopped or no communication with SCU, it turns to red.
-  These two blocks indicate status of transponders. When the transponder is on antenna, the antenna symbol  is displayed. When the transponder is on dummy load, the load symbol  is displayed. Main or Standby transponder indication attached to the transponder number.
-  The four blocks indicate status of monitors. When the monitor is bypassed, the bypass symbol  is displayed on the left of the monitor status box. The color of monitor status box is green when the monitor gives Normal, amber when the monitor gives Warning, or red when the monitor give Alarm indication.
-  Touching on this icon will pop up a soft keyboard to enter text.
-  Date and Time of the LMI is displayed. Clicking on this area will pop up a dialog box for configuring the date, time and time zone setting.

4.5.1.3. Menu Control Panel

The menu screen appears when touching on the area  at the bottom left of LMI.

The menu screen is divided into **Group, Select and Control** blocks and a number of icons that can be touched to select are under each block. Brief description on each item is summarized in Table 4-29. Details of each screen page is described in the next sections.

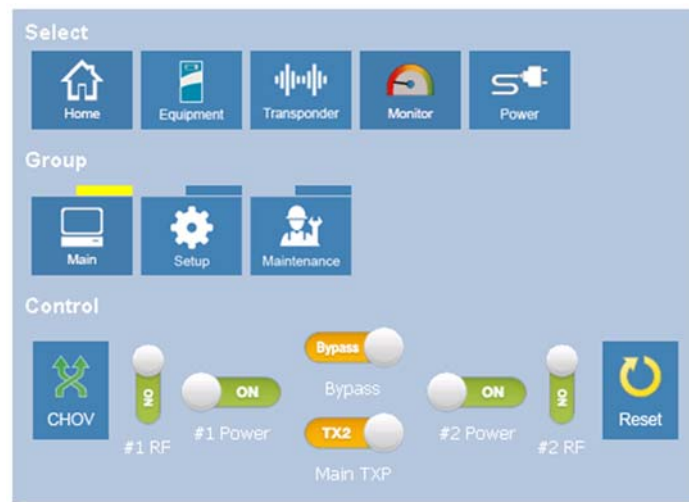


Figure 4-127. Menu Control Panel

Screen navigation

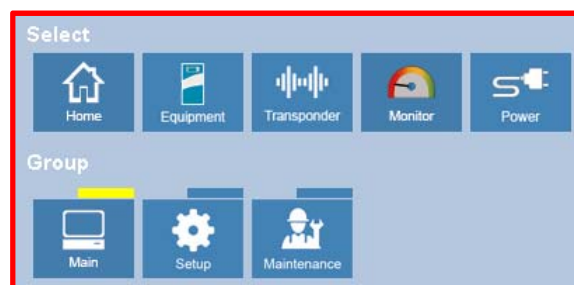


Figure 4-128. Menu Groups and Items

Control

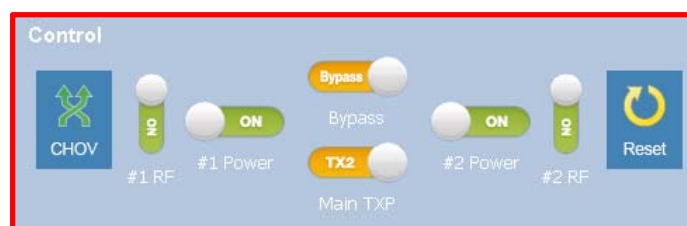


Figure 4-129. Controls on Menu Window

Table 4-29. LMI Menu

Group	Select	Description
 Main	 Home	This icon is to navigate the Home page where status of the system in overall is displayed in concise form and basic controls such as on/off, changeover, reset, etc. are available.
	 Equipment	This icon is to navigate to the Equipment page where status of the equipment is displayed.
	 Transponder	This icon is to navigate to the Transponder page where status of the transponder is displayed.
	 Monitor	This icon is to navigate to the Monitor page where status of the monitors is displayed.
	 Power	This icon is to navigate to the Power page where status of the power supply is displayed.
 Setup	 Station Setup	This icon is to navigate to the Station Setup page where those parameters specific to the station can be examined and configured.
	 TXP Setup	This icon is to navigate to the Transponder Setup page where those parameters specific to transponders can be examined and configured.
	 MON Setup	This icon is to navigate to the Monitor Setup where those parameters specific to monitors can be examined and configured.
	 System Setup	This icon is to navigate to the System Setup page where those parameters which are specific to the equipment station but common for both transponders can be examined and configured.
	 Comm. Setup	This icon is to navigate to the Communication Setup where the data link related parameters can be examined and configured.
	 Misc. Setup	This icon is to navigate to the Miscellaneous Setup where other miscellaneous parameters can be examined and configured.
 Maintenance	 Adv. Control	This icon is to navigate to the Advanced Control page where advanced control function is available.

		This icon is to navigate to the Version page where hardware revision and software version information are available.
--	---	---

4.5.1.4. Using Controls

The basic control such as transponder on/off, main/standby changeover, reset, monitor bypass, etc. are available on Home screen as well as on the popup dialog box *Menu Control Panel*.

If an icon is touched, the control is immediately executed without any confirmation.

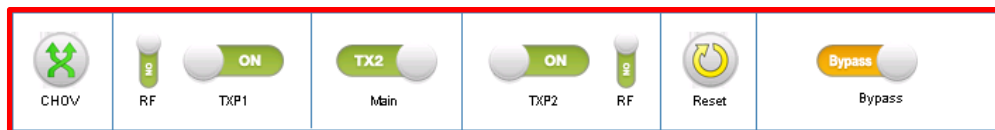


Figure 4-130. Using Controls on Home screen

Changeover



This is to transfer the transponder currently on antenna onto dummy load and the other transponder onto antenna.

Main/Standby



This is to select one transponder as Main and the other transponder as standby. Touching on the switch icon toggles between TX1 and TX2. The transponder currently selected becomes the main transponder.

Power On/Off



This is to turn on or off the transponder. Touching on the switch icon toggles between on and off. Turning off the transponder will cut off the DC power supply to the transponder.

RF On/Off



This is to turn the RF output of the transponder on or off. Turning the RF output off will not cut off the DC power supply to the transponder but temporarily disable the RF power amplifiers.

MON Bypass/Auto



This is to switch the executive action of the monitor. Touching on the switch icon toggles between Bypass and AUTO. When it is in AUTO, monitor executive action will take place when such condition is encountered. When it is Bypass, monitor is bypassed and no executive action will take place even if such condition is persistent.

System Reset



This is to reset the equipment. Reset will clear the internal flag indicating fault state, if it is set; put the main transponder on antenna and the standby transponder on dummy load; and turn on the transponders.

4.5.2. STATUS SCREENS

4.5.2.1. Status Overview Screen

This is the Home screen and displays the overall status of the system. Touching on the **Home** icon on the upper left will show up the Status Overview screen.

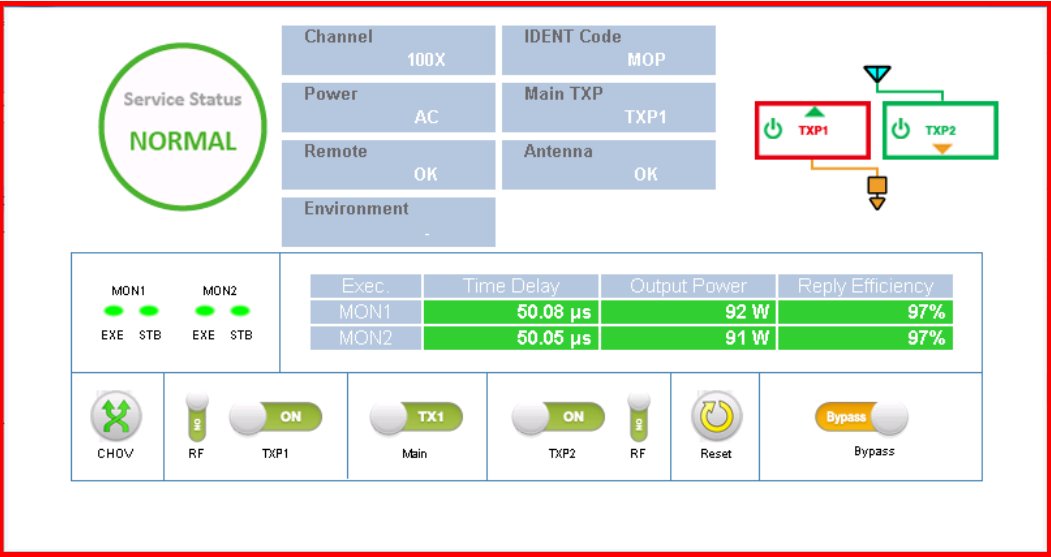


Figure 4-131. Status Overview Screen

Service Status

The service status icon shows status of the signal being transmitted:

-  This indicates that the signal being transmitted is normal.
-  This indicates that the signal being transmitted is in warning, i.e. pre-alarm condition.
-  This indicates that the signal being transmitted is in alarm condition.
-  This indicates that the transponder TX1 and TX2 are turned off and no service available.

Channel

Shows the DME channel number in operation.

Power

Shows status of main power supply

-  This indicates that the equipment is running on the AC main.
-  This indicates that the equipment is running on the backup battery.

Environment

Shows status of environmental monitoring.

-  When the EMU is not installed or no supported sensors are installed, no status information is displayed.
-  This indicates that the environmental monitoring is normal.
-  This indicates that environmental monitoring is giving an alarm.

IDENT Code

Shows the IDENT Code assigned to the DME station in operation.

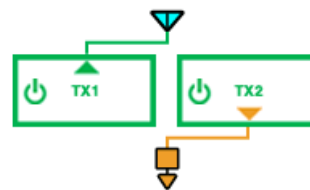
Main TX

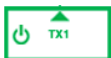
Shows the transponder currently selected as Main.

Remote

Shows the status of the data link with RCU

TX Status



-  This indicates that the transponder is On Antenna.
-  This indicates that the transponder is On Dummy Load.
-  The green upward triangle on the top indicates that the transponder is currently selected as Main. The green color indicates the transponder is normal. The switch symbol on the left indicates that the transponder is turned on.
-  The amber downward triangle on the bottom indicates that the transponder is currently selected as Standby. The green color indicates the transponder is normal. The switch symbol on the left indicates that the transponder is turned on.
-   The color of the box and the text inside indicates operational status of the transponder: If it is in red, it indicates the transponder is on alarm. If it is green, it indicates the transponder is normal.
-  The diagonal cross indicates the transponder has been shutdown automatically due to persistent alarm condition.

Monitor Overview



This indicates status monitors. The legends are:

- **EXE:** Executive, **STB:** Standby
- Normal Warning Alarm Unplugged

Monitor Readings

Displays the major parameters reported by the executive monitors.

Exec.	Time Delay	Output Power	Reply Efficiency
MON1	50.03 μ s	96 W	98%
MON2	50.03 μ s	97 W	97%

Controls

When the local control is enabled as indicated by the icon, basic controls are available on the screen.



- CHOV: Changeover
- RF: RF On/Off
- TXP1/2: TXP1/2 On/Off
- Main: Select Main TX
- Reset: Reset the equipment
- Bypass: Monitor Bypass/Auto

4.5.2.2. Equipment Status Screen



Displays detailed information for each unit of the equipment. The icon at the end of each item indicates the overall status of the group. Touch the icon to go to the detail page of the group.

The indication by the icons are:

- Normal
- Alert
- Off or not available

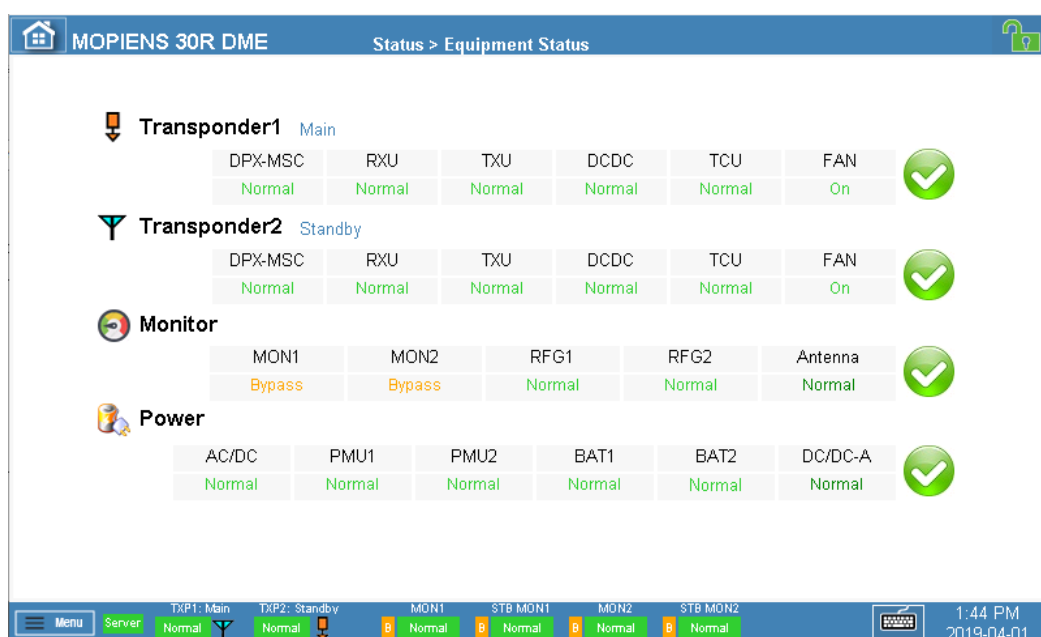


Figure 4-132. Equipment Status Screen

The information displayed on the Equipment Status screen is summarized in Table 4-30.

Table 4-30. Equipment Status

Category	Item	Description
Transponder 1 Transponder 2	DPX-MSC	Normal: No Alarm No Present: DPX Unplugged N/A: TCU Unplugged
	RXU	Normal: No Alarm No Present: RXU Unplugged N/A: TCU Unplugged
	TXU	Normal: No Alarm Fault: PLL Lock Fail, Temperature Fail No Present: TXU Unplugged N/A: TCU Unplugged
	DCDC	Normal: No Alarm Fault: DC/DC Alarm No Present: DC/DC Unplugged
	TCU	Normal: No Alarm Fault: TCU Hardware Fault Off: TX Power Off No Present: TCU Unplugged
	FAN	On: Fan On Off: Fan Off Fault: Hardware Fault N/A: TCU Unplugged

Monitor	MON1 MON2	Bypassed: MON Bypassed Normal: MON Auto, No Alarm Warning: MON Auto, Parameter Warning Alarm: MON Auto, Parameter Alarm No Present: MON Unplugged
	RFG1 RFG2	Normal: No Alarm Fault: MON Auto, Parameter Alarm No Present: RFG Unplugged N/A: MON Unplugged
	Antenna	Normal: No Alarm Fault: Antenna Fault
Power	AC/DC	Normal: No Alarm Fault: AC/DC Fault N/A: PMU Unplugged
	PMU1 PMU2	Normal: No Alarm Fault: Hardware Fault N/A: PMU Unplugged
	BAT1 BAT2	Normal: No Alarm Warning: Battery voltage, temperature Warning Alarm: Battery voltage, temperature Alarm No Battery: Battery Unplugged N/A: PMU Unplugged
	DC/DC-A	Normal: No Alarm Warning: DC/DC-A voltage Warning Alarm: DC/DC-A voltage Alarm, DC/DC-A Unplugged

Transponder Detail

Transponder Detail screen displays status information of TXP1, TXP2. The information displayed on this screen is summarized in Table 4-31.

TX Detail					
TXP	TXP1	TXP2	DCDC	TXP1	TXP2
Main TXP	Standby	Main	Fault Status	Normal	Normal
Output	On Ant	On Load	+5V Alarm	Normal	Normal
Interlock	Disable	Disable	+8V Alarm	Normal	Normal
Shutdown Status	Normal	Normal	+15V Alarm	Normal	Normal
TXU	Temperature Fault	Normal	-15V Alarm	Normal	Normal
			+28V Alarm	Normal	Normal
			+50V Alarm	Normal	Normal
	Reverse Power	Normal			
	PLL Status	Normal			
Close					

Figure 4-133. TXP Detail

Table 4-31 . TXP Detail

Category	Item	Description
TXP	Main TX	Main: Main Transmitter Standby : Standby Transmitter
	Output	On ANT On Load
	Interlock	Disable: Not use interlock Enable: Use Interlock
	Shutdown Status	Normal: No Alarm Off: Shutdown
TXU	Temperature Fault	Normal: No Alarm Fault: Temperature Alarm Unplugged: TXU Unplugged Unknown: TCU Unplugged
	Reverse Power	Normal: No Alarm Fault: Reverse Power Alarm Unplugged: TXU Unplugged Unknown: TCU Unplugged
	PLL Status	Normal: No Alarm Fault: PLL Lock Fail Unplugged: TXU Unplugged Unknown: TCU Unplugged
DC/DC	Fault Status	Normal: No Alarm Warning: DC/DC voltage reading Warning Alarm: DC/DC voltage reading Alarm Off: DC/DC Off Unplugged: DC/DC Unplugged Unknown: TCU Unplugged

	+5V Alarm	Normal: No Alarm Warning: DC/DC 5V voltage reading Warning Alarm: DC/DC 5V voltage reading Alarm Off: DC/DC Off Unplugged: DC/DC Unplugged Unknown: TCU Unplugged
	+8V Alarm	Normal: No Alarm Warning: DC/DC 8V voltage reading Warning Alarm: DC/DC 8V voltage reading Alarm Off: DC/DC Off Unplugged: DC/DC Unplugged Unknown: TCU Unplugged
	+15V Alarm	Normal: No Alarm Warning: DC/DC 15V voltage reading Warning Alarm: DC/DC 15V voltage reading Alarm Off: DC/DC Off Unplugged: DC/DC Unplugged Unknown: TCU Unplugged
	-15V Alarm	Normal: No Alarm Warning: DC/DC -15V voltage reading Warning Alarm: DC/DC -15V voltage reading Alarm Off: DC/DC Off Unplugged: DC/DC Unplugged Unknown: TCU Unplugged
	+28V Alarm	Normal: No Alarm Warning: DC/DC 28V voltage reading Warning Alarm: DC/DC 28V voltage reading Alarm Off: DC/DC Off Unplugged: DC/DC Unplugged Unknown: TCU Unplugged
	+50V Alarm	Normal: No Alarm Warning: DC/DC 50V voltage reading Warning Alarm: DC/DC 50V voltage reading Alarm Off: DC/DC Off Unplugged: DC/DC Unplugged Unknown: TCU Unplugged

Monitor Detail

The Monitor Detail screen displays status information of MON1 and MON2. Touching the arrows at both ends will move to the reading screen for the other transponder.

Monitor			
		MON1	MON2
Monitor	Monitor Action	Bypass	Bypass
	Fault Status	Normal	Normal
TXP1	Frequency Alarm	Normal	Normal
	Rise Time Alarm	Normal	Normal
	Decay Time Alarm	Normal	Normal
	Duration Alarm	Normal	Normal
	Delay Time Alarm	Normal	Normal
	Output Power Alarm	Normal	Normal
	PRF Alarm	Normal	Normal
	Efficiency Alarm	Normal	Normal
	Spacing Alarm	Normal	Normal
	VSWR Alarm	Normal	Normal
	DCDC Alarm	Normal	Normal
	IDENT Code Alarm	Normal	Normal
	ERP Alarm	Normal	Normal
Close			

Figure 4-134. Monitor Detail

The information displayed on this screen is summarized in Table 4-32.

Table 4-32. Monitor Detail

Category	Item	Description
Monitor	Monitor Action	If the monitor is in bypass state, Bypass in amber is displayed. If the monitor is not bypass state, Auto in green is displayed. If the monitor is unplugged Not Present is displayed.
	Fault Status	If the parameter is normal, Normal is displayed in green. If there is hardware fault, Fault is displayed in red. If monitor is unplugged, Not Present is displayed.
TXP1 TXP2	Frequency Alarm	If the parameter is normal, Normal in green is displayed. If the parameter is in warning condition, Warning in amber is displayed. If the parameter is in alarm condition, Alarm in red is displayed. If the monitor is not present, “ MON Unplugged ” is displayed
	Rise Time Alarm	
	Decay Time Alarm	
	Duration Alarm	
	Delay Time Alarm	
	Output Power Alarm	
	PRF Alarm	
	Efficiency Alarm	
	Spacing Alarm	
	VSWR Alarm	
	DC/DC Alarm	
	IDENT Code Alarm	
	ERP Alarm	

Power Detail

The Power Status screen displays PMU, AC/DC, DC/DC-A and battery status information.

Power Detail						
	#1	#2	#3	#4	#5	
PMU	Normal	Normal				
Battery	Charging	Charging				
AC/DC	Normal	Normal	Normal	Normal	Normal	

DC/DC-A #1	Voltage	Current	DC/DC-A #2	Voltage	Current
+5V	Normal	Normal	+5V	Normal	Normal
+15V	Normal	Normal	+15V	Normal	Normal
-15V	Normal	Normal	-15V	Normal	Normal

Close

Figure 4-135. Power Status

The information displayed on the screen is summarized in Table 4-33.

Table 4-33. Power Status

Category	Item	Description
Power Detail	PMU	Normal: No Alarm Fault: PMU Hardware Fault N/A: PMU Unplugged
	Battery	Full: Full Charging Charging: Charging Discharging: Discharging No Battery: Battery Unplugged
	AC/DC	Normal: No Alarm Fault: Hardware Fault No AC Input: AC/DC Input Fail N/A: AC/DC Unplugged
	DC/DC-A +5V DC/DC-A +15V DC/DC-A -15V	Normal: No Alarm Warning: Voltage/Current Alarm Alarm: Voltage/Current Alarm N/A: DC/DC-A Unplugged

4.5.2.3. Transponder Status Screen

 **Menu** → **Main** → **Transponder**

This screen displays the readings of the transmitter reported by the TCU. The icon  indicates that the transponder is on Antenna and the icon  indicates that the transponder is on Dummy Load.

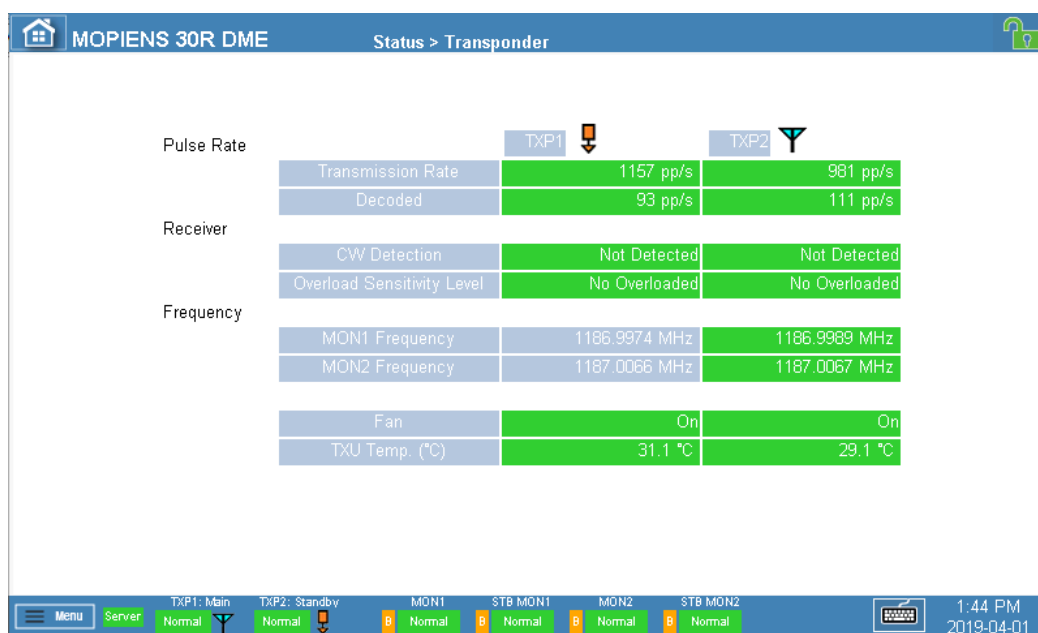
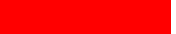


Figure 4-136. Transmitter Reading

The color of the text box changes according to the state of the measured value.

Normal  Warning  Alarm 

The items displayed on this screen are summarized in Table 4-34. If TCU is not installed, N/A is displayed.

Table 4-34. Transmitter Reading

Category	Item	Description	Resolution	Unit
Pulse Rate	Transmission Rate	Transmission pulse pair rate of Reply + Squitter + IDENT	1	pp/s
	Decoded	Number of decoded interrogation pulse pair per second	1	pp/s
Receiver	CW Detection	If CW interference is detected, the level of the CW interference signal is displayed. If not, Not Detected is displayed	0.1	dBm
	Overload Sensitivity Level	If the transponder is overloaded, the receiver sensitivity level is displayed. If not, No Overload is displayed	0.1	dBm
Frequency	Frequency	Frequency reading	0.0001	MHz
Thermal Status	FAN	On/Off status	-	-
	TXU Temp	TXU Temperature	0.1	°C

4.5.2.4. Monitor Reading Screen

 **Menu** → **Main** → **Monitor**

Display reading measured by Monitor. Touch the arrows on both ends of the screen to display the standby monitor screen.

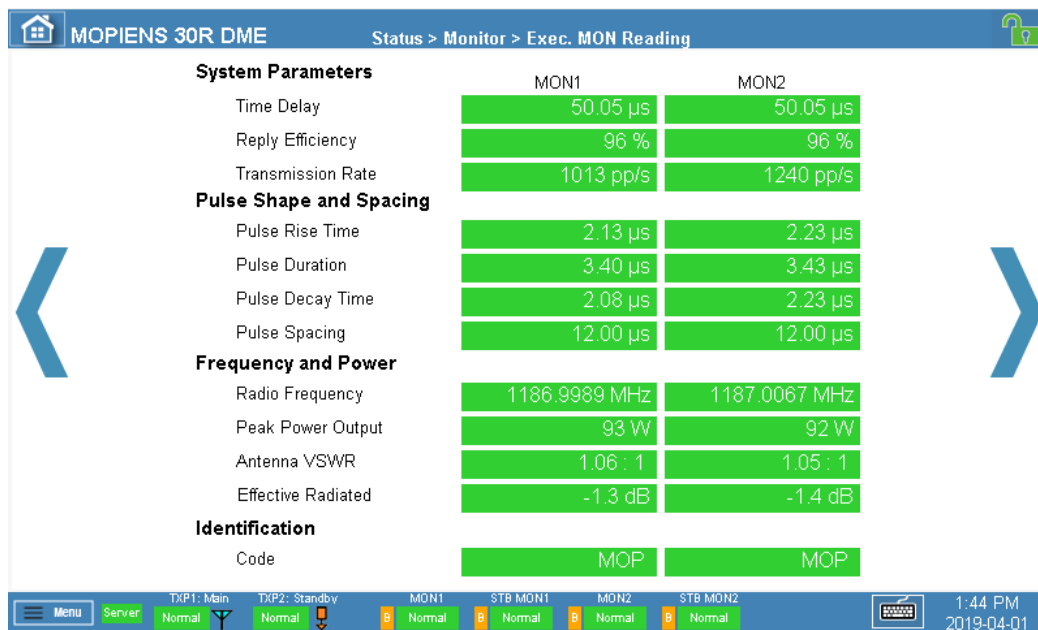


Figure 4-137. Executive Monitor Reading

The color of the text box changes according to the state of the measured value.

Normal Warning Alarm

The items displayed on this screen are summarized in Table 4-35. If MON is not installed, N/A is displayed.

Table 4-35. Executive Monitor Reading

Category	Item	Description	Resolution	Unit
System Parameter	Time Delay	Time Delay reading	0.01	μ s
	Reply Efficiency	Reply Efficiency reading	1	%
	Transmission Rate	Transmission Rate reading	1	pp/s
Pulse Shape and Spacing	Pulse Rise Time	Pulse Rise Time reading	0.01	μ s
	Pulse Duration	Pulse Duration reading	0.01	μ s
	Pulse Decay Time	Pulse Decay Time reading	0.01	μ s
	Pulse Spacing	Pulse Spacing reading	0.01	μ s
Frequency and Power	Radio Frequency	Reply Frequency reading	0.0001	MHz
	Peak Power Output	Output Power reading	1	W
	Antenna VSWR	Antenna VSWR reading	0.01	
	Effective Radiated	Effective Radiated Level	0.1	
Identification	Code	IDENT Code		

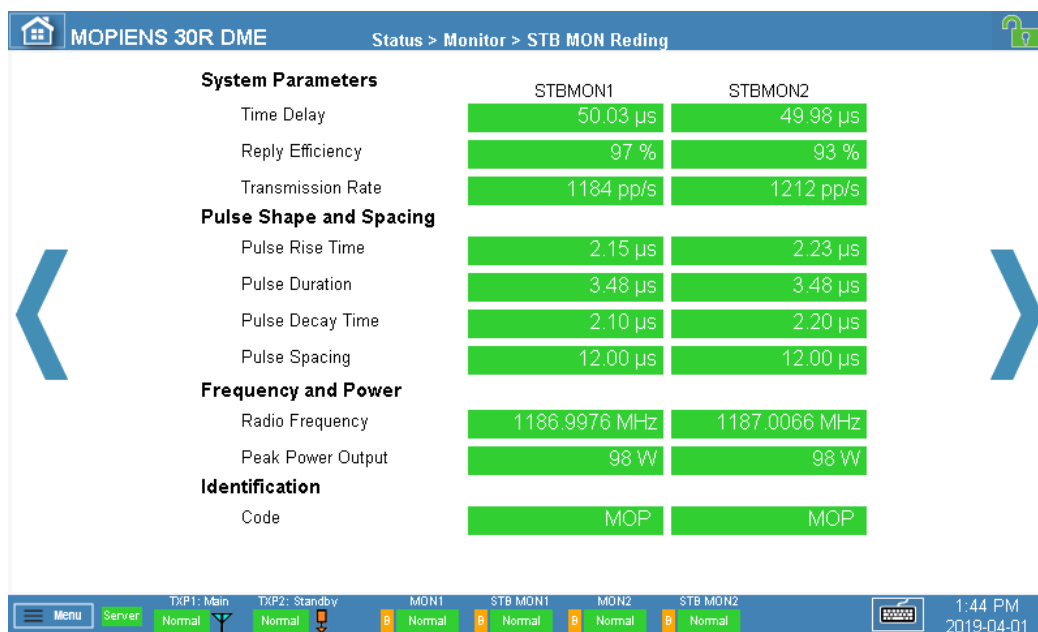


Figure 4-138. Standby Monitor Reading

The color of the text box changes according to the state of the measured value.

Normal Warning Alarm

The items displayed on this screen are summarized in Table 4-36. If MON is not installed, N/A is displayed.

Table 4-36. Standby Monitor Reading

Category	Item	Description	Resolution	Unit
System Parameter	Time Delay	Time Delay reading	0.01	μ s
	Reply Efficiency	Reply Efficiency reading	1	%
	Transmission Rate	Transmission Rate reading	1	pp/s
Pulse Shape and Spacing	Pulse Rise Time	Pulse Rise Time reading	0.01	μ s
	Pulse Duration	Pulse Duration reading	0.01	μ s
	Pulse Decay Time	Pulse Decay Time reading	0.01	μ s
	Pulse Spacing	Pulse Spacing reading	0.01	μ s
Frequency and Power	Radio Frequency	Reply Frequency reading	0.0001	MHz
	Peak Power Output	Output Power reading	1	W
Identification	Code	IDENT Code		

4.5.2.5. Power Supply Screen

 **Menu** → **Main** → **Power Supply**

AC/DC, DC/DC-A and Battery readings are displayed. The information displayed on the screen is summarized in Table 4-37.

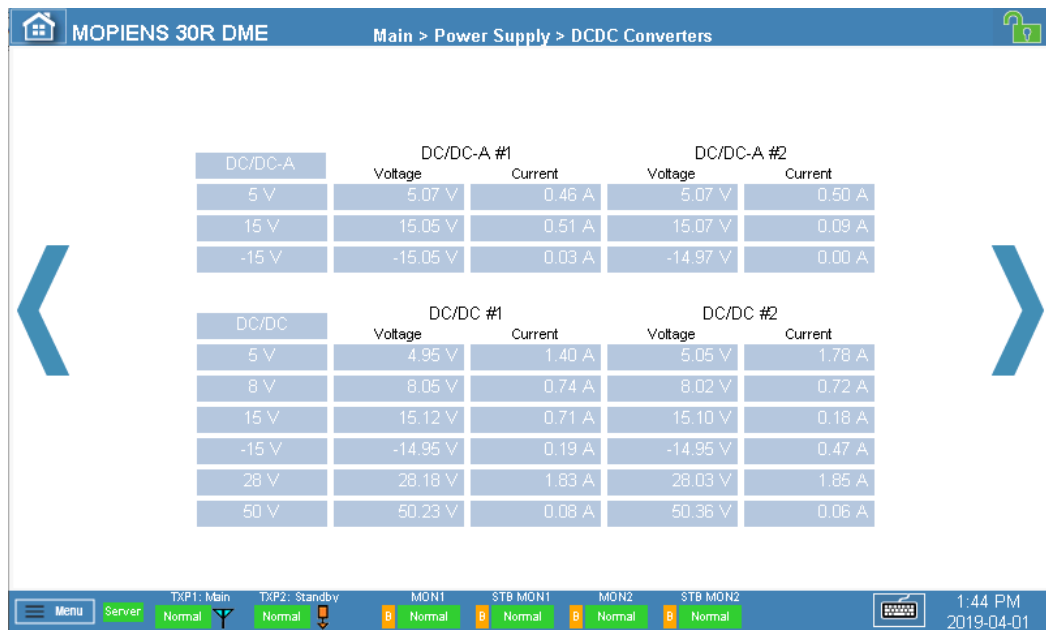


Figure 4-139. Power Supply Reading

Table 4-37. Power Supply Reading

Category	Item	Description	Resolution	Unit
DC/DC-A	5V Voltage	DC/DC-A 5V voltage reading	0.01	V
	5V Current	DC/DC-A 5V current reading	0.01	A
	15V Voltage	DC/DC-A 15V voltage reading	0.01	V
	15V Current	DC/DC-A 15V current reading	0.01	A
	-15V Voltage	DC/DC-A -15V voltage reading	0.01	V
	-15V Current	DC/DC-A -15V current reading	0.01	A
DC/DC	5V Voltage	DC/DC 5V voltage reading	0.01	V
	5V Current	DC/DC 5V current reading	0.01	A
	8V Voltage	DC/DC 8V voltage reading	0.01	V
	8V Current	DC/DC 8V current reading	0.01	A
	15V Voltage	DC/DC 15V voltage reading	0.01	V
	15V Current	DC/DC 15V current reading	0.01	A
	-15V Voltage	DC/DC -15V voltage reading	0.01	V
	-15V Current	DC/DC -15V current reading	0.01	A
	28V Voltage	DC/DC 28V voltage reading	0.01	V
	28V Current	DC/DC 28V current reading	0.01	A
AC/DC	Voltage	AC/DC voltage reading	0.1	V
	Current	AC/DC current reading	0.1	A
	Temp	AC/DC temperature reading	0.1	°C
	Status	Normal: No Alarm Alarm: AC/DC voltage/current Alarm N/A: AC/DC Unplugged		
Battery	Voltage	Battery voltage reading	0.1	V

	Current	Battery current reading	0.1	A
	Temp	Battery temperature reading	0.1	°C
	Status	Normal: No Alarm Alarm: Battery voltage/current Alarm N/A: Battery Unplugged		

4.5.3. SETUP SCREENS

4.5.3.1. Station Setup

 → Setup → Station Setup

Station Setup screen enables the user to examine and set up the parameters specific to the station but common to both transponders. Touching on a box where current setting is displayed will pop up a setup dialog. The parameters that can be accessed from the Station Setup is as summarized in Table 4-17

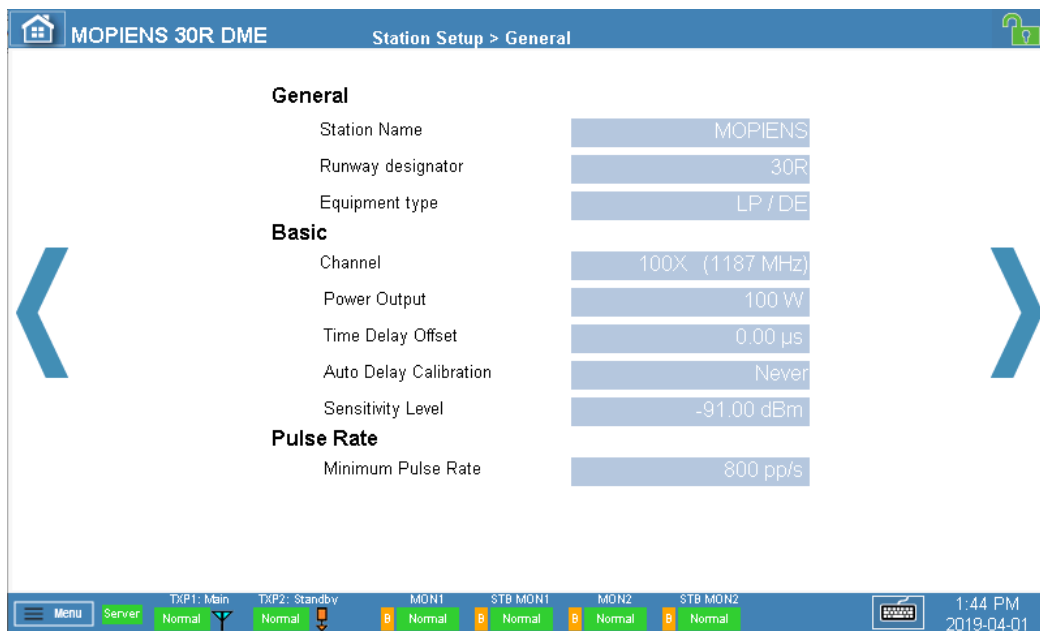


Figure 4-140. Station Setup

4.5.3.2. TXP Setup

 → Setup → TXP Setup

Transponder Setup is the screen for detailed setting of transponder. Touching on a box where current setting is displayed will pop up a setup dialog. When the option box **As Per Station Setup** is checked, the settings configured in the station setup are applied to the transponder setup. The items that can be set in Transponder Setup are as summarized in Table 4-18

MOPIENS 30R DME Transponder Setup > Output Power

Output Power

	TXP1	TXP2
Power Output	100%	100%

Pulse Rate

	TXP1	TXP2
Minimum Pulse Rate	☒ As Per Station Setup 800 pp/s	☒ As Per Station Setup 800 pp/s

IDENT

	TXP1	TXP2
Code	☒ As Per Station Setup MOP	☒ As Per Station Setup MOP
Keyer	Independent	Independent
Sync	Code	Code
Equalizing Pulse	Off	Off

Menu Server TXP1: Main TXP2: Standby MON1 STB MON1 MON2 STB MON2 1:44 PM 2019-04-01

Figure 4-141. Transponder Setup(1/3)

MOPIENS 30R DME Transponder Setup > Echo Suppression

Echo Suppression

	TXP1	TXP2
SDES	☒ As Per Station Setup On	☒ As Per Station Setup On
SDES Duration	6.0 μs	6.0 μs
LDES	Off	Off
LDES Duration	200.0 μs	200.0 μs
LDES Threshold Level	-50.0 dBm	-50.0 dBm
Dead Time	60 μs	60 μs

Menu Server TXP1: Main TXP2: Standby MON1 STB MON1 MON2 STB MON2 1:44 PM 2019-04-01

Figure 4-142. Transponder Setup(2/3)

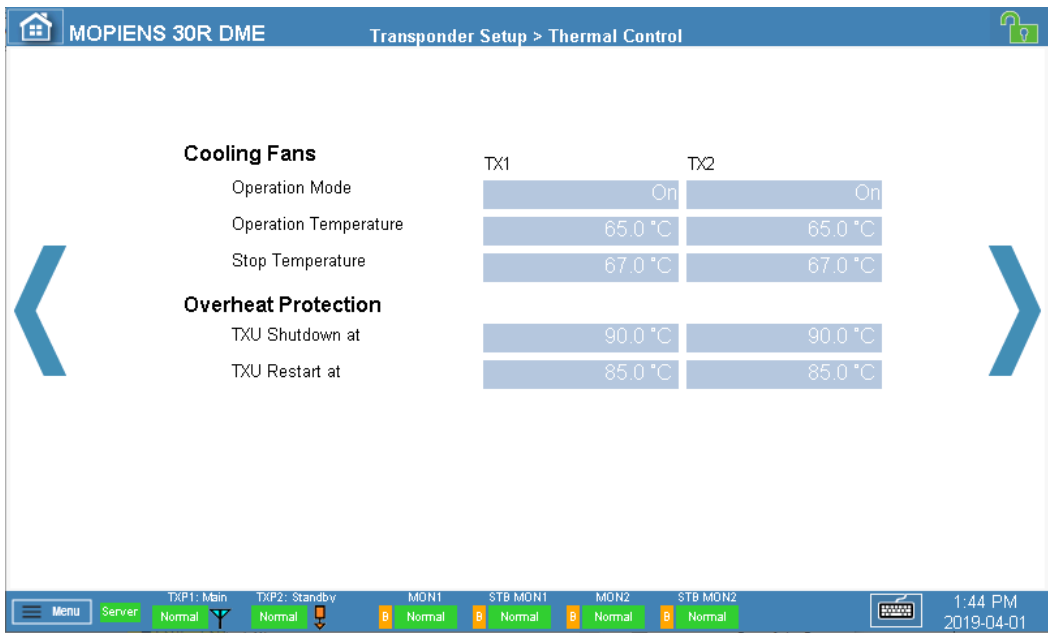


Figure 4-143. Transponder Setup(3/3)

4.5.3.3. MON Setup

Menu → Setup → MON Setup

Monitor Setup is the screen for detailed setting of the monitors. Touching on a box where current setting is displayed will pop up a setup dialog. The items that can be set in Transponder Setup are as summarized in Table 4-20 and Table 4-21.

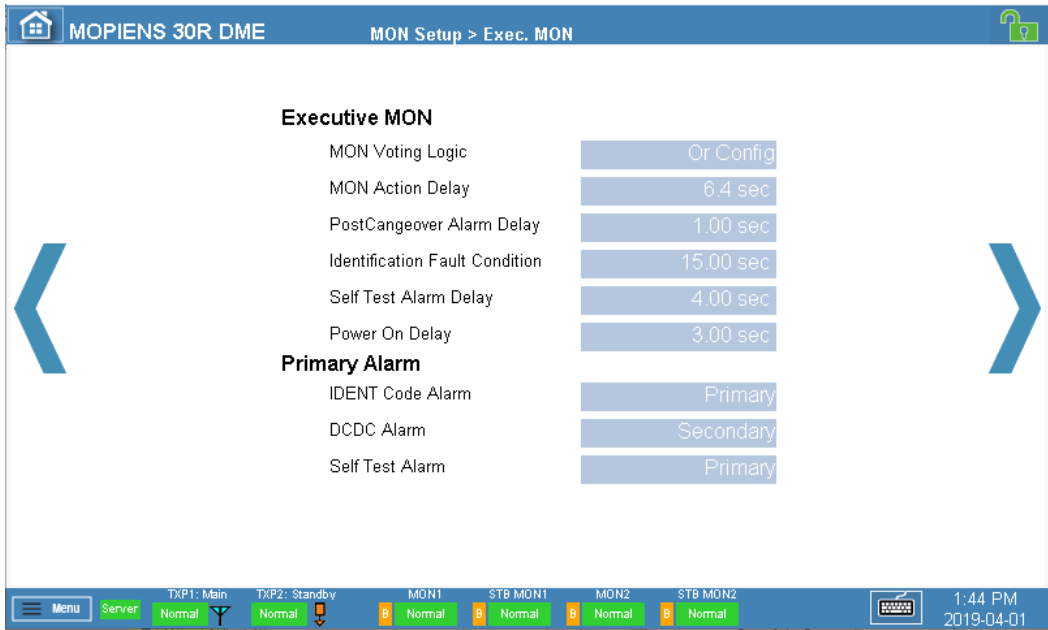


Figure 4-144. Monitor Setup(1/2)



Figure 4-145. Monitor Setup(2/2)

4.5.3.4. System Setup

Menu → **Setup** → **System Setup**

Monitor Setup is the screen for detailed setting of the monitors. Touching on a box where current setting is displayed will pop up a setup dialog. The items that can be set in Transponder Setup are as summarized in Table 4-22.

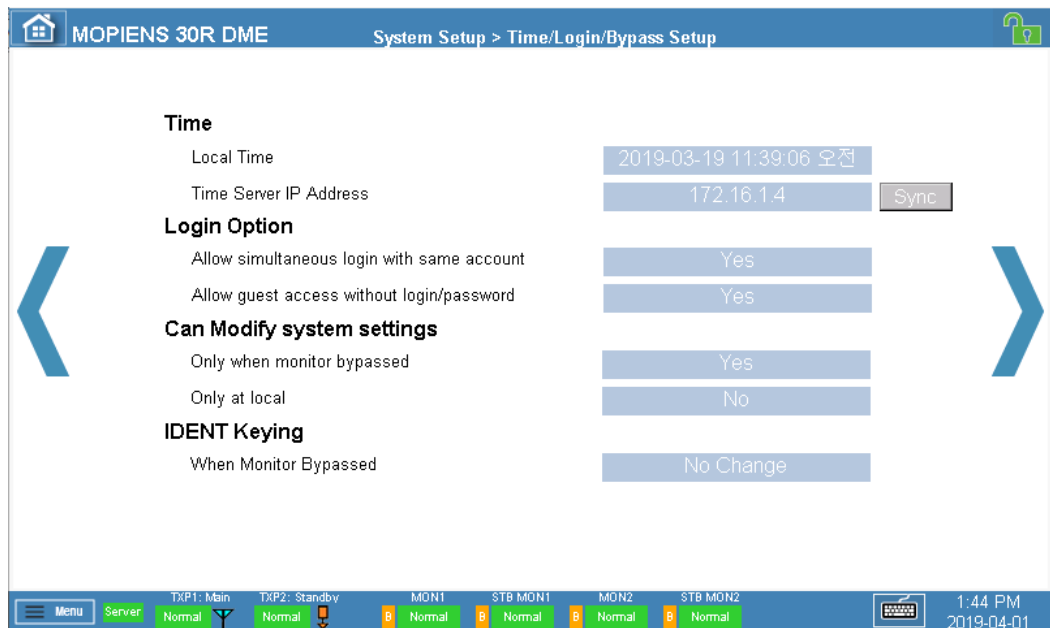


Figure 4-146. System Setup

4.5.3.5. Comm. Setup

 → Setup → Communication Setup

Communication Setup is the screen for detailed setting of the interface for data link with remote control / monitoring. Touching on a box where current setting is displayed will pop up a setup dialog. The items that can be set in Communications Setup are as summarized in Table 4-23.

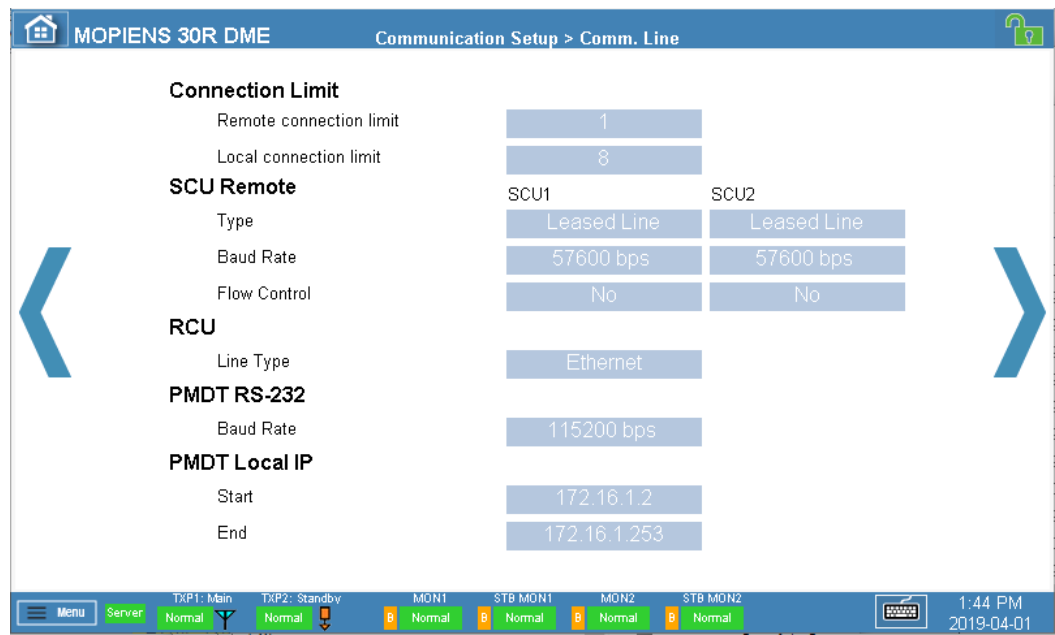


Figure 4-147. Communication Setup

4.5.3.6. Date/Time Setup



Figure 4-148. Date/Time Display

The date and time and time zone can be set on the Date/Time Properties dialog box which can be popped up by touching on the date / time display area at the bottom right of the screen.

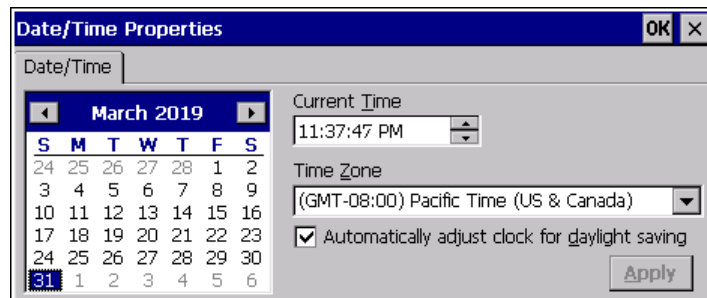


Figure 4-149. Date/Time Setup

After setting the date and time and time zone, press the **Apply** button to save the changes. When completed, press the **OK** button to close the dialog box.

4.5.4. MAINTENANCE SCREENS

4.5.4.1. Advanced Control

 **Menu** → **Maintenance** → **Advanced Control**

Advanced Control screen is to enable users to test and control the equipment in special operating conditions temporarily. When the equipment is restarted, the control settings on this screen are initialized to the default normal operating condition. The items that can be accessed in the screen is as described in Section 4.4.7.4.

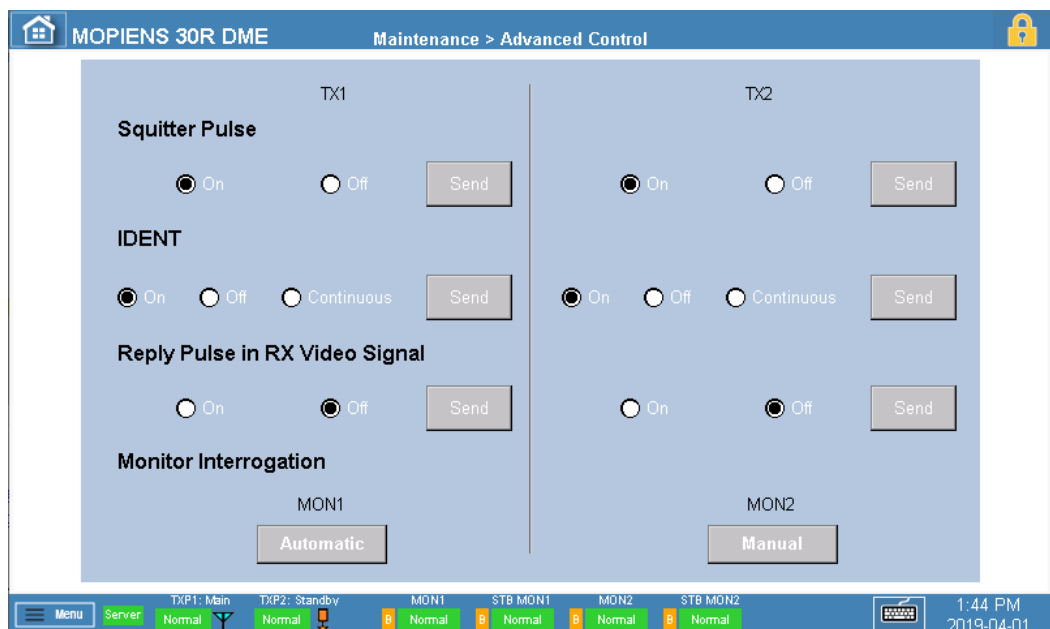


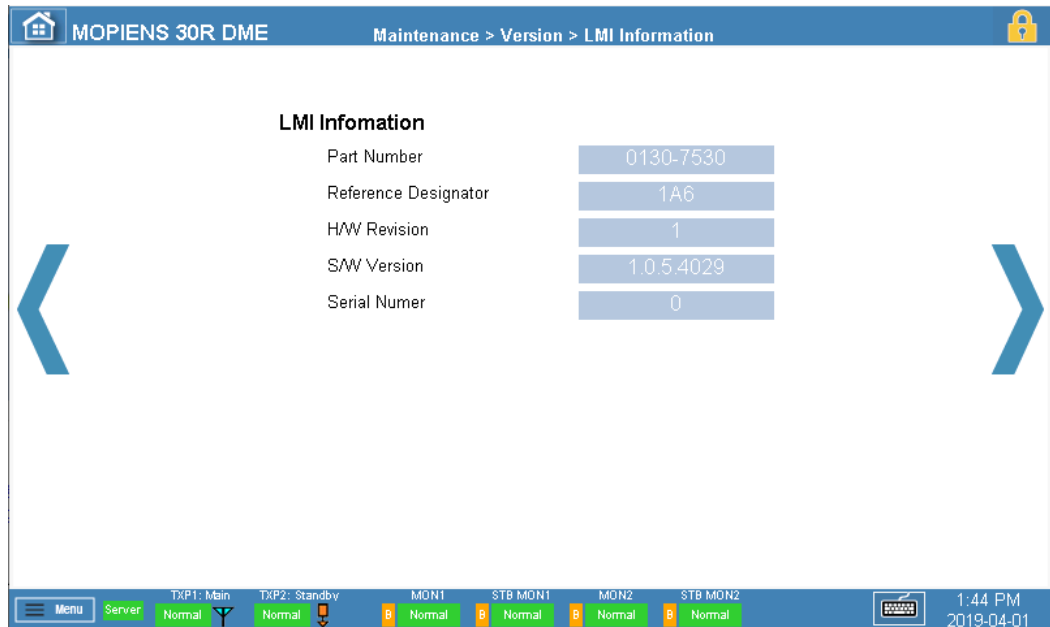
Figure 4-150. Advanced Control

4.5.4.2. Version Screen

LMI Information

 → Maintenance → Version → LMI Information

This page shows version and logistics information on the LMI.



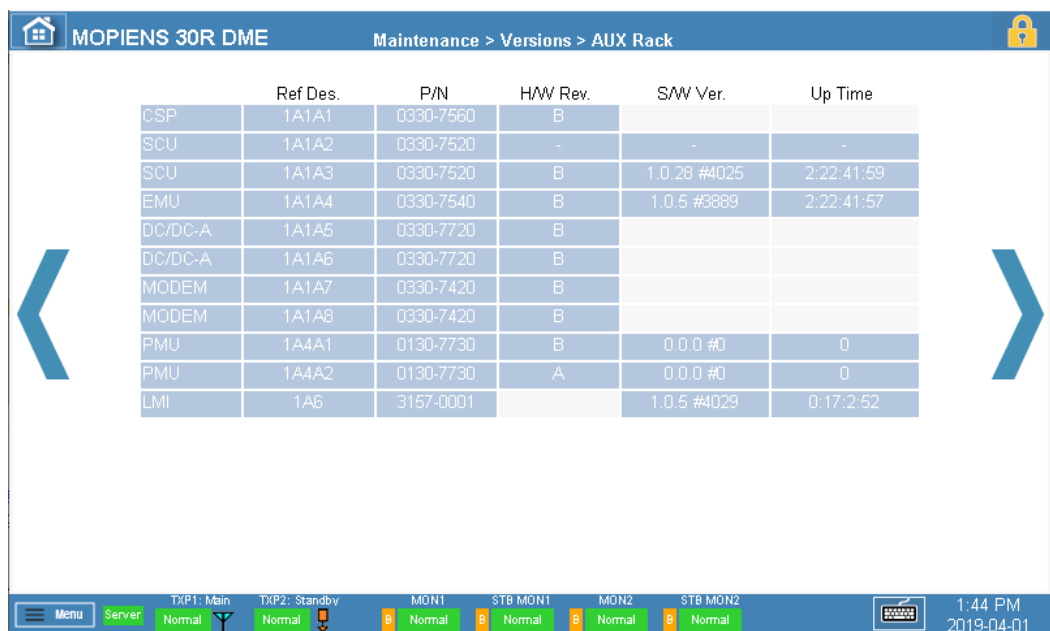
LMI Information	
Part Number	0130-7530
Reference Designator	1A6
HW Revision	1
SW Version	1.0.5.4029
Serial Number	0

Figure 4-151. LMI Information

Aux Rack Versions

 → Maintenance → Version → Aux Rack

This page shows versions and logistics information of the units on the Aux subrack.



	Ref Des.	P/N	HW Rev.	SW Ver.	Up Time
CSP	1A1A1	0330-7560	B	-	-
SCU	1A1A2	0330-7520	-	-	-
SCU	1A1A3	0330-7520	B	1.0.28 #4025	2:22:41:59
EMU	1A1A4	0330-7540	B	1.0.5 #3889	2:22:41:57
DC/DC-A	1A1A5	0330-7720	B	-	-
DC/DC-A	1A1A6	0330-7720	B	-	-
MODEM	1A1A7	0330-7420	B	-	-
MODEM	1A1A8	0330-7420	B	-	-
PMU	1A4A1	0130-7730	B	0.0.0 #0	0
PMU	1A4A2	0130-7730	A	0.0.0 #0	0
LMI	1A6	3157-0001	-	1.0.5 #4029	0:17:2:52

Figure 4-152. Aux Rack Unit Version

TXP1 Versions

 **Menu** → Maintenance → Version → TXP1

This page shows versions and logistics information of the units on the TXP1 subrack.

MOPIENS 30R DME Maintenance > Version > TXP1					
	Ref Des.	P/N	H/W Rev.	S/W(FPGA) Ver.	Up Time
DPX-MS	1A2A1	0330-7120	0		
RXU	1A2A2	0330-7180	-		
TXU	1A2A4	0330-7140	C		
DC/DC	1A2A5	0330-7110	B		
TCU	1A2A6	0330-7570	B	1.0.4 #4018(#183)	1:15:53:15
RFG	1A2A7	0330-7190	B		
MON	1A2A8	0330-7550	C	1.0.17 #4037(#)	1:22:9:20
FAN	1A2A9	0330-7910	A		

 **Menu**
Server
TXP1: Main Normal
TXP2: Standby Normal
MON1 Normal
STB MON1 Normal
MON2 Normal
STB MON2 Normal
1:44 PM 2019-04-01

Figure 4-153. TXP1 Unit Version

TXP2 Versions

 **Menu** → Maintenance → Version → TXP2

This page shows versions and logistics information of the units on the TXP2 subrack.

MOPIENS 30R DME Maintenance > Version > TXP2					
	Ref Des.	P/N	H/W Rev.	S/W Ver. (FPGA)	Up Time
DPX-MS	1A3A1	0330-7120	0		
RXU	1A3A2	0330-7180	C		
TXU	1A3A4	0330-7140	C		
DC/DC	1A3A5	0330-7110	B		
TCU	1A3A6	0330-7570	C	1.0.4 #4018(#183)	1:22:10:44
RFG	1A3A7	0330-7190	A		
MON	1A3A8	0330-7550	B	1.0.17 #4037(#212)	1:22:10:47
FAN	1A3A9	0330-7910	A		

 **Menu**
Server
TXP1: Main Normal
TXP2: Standby Normal
MON1 Normal
STB MON1 Normal
MON2 Normal
STB MON2 Normal
1:44 PM 2019-04-01

Figure 4-154. TXP2 Unit Version

SECTION 5

STANDARDS AND TOLERANCES

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5. STANDARDS AND TOLERANCES

5.1. INTRODUCTION

This section lists and describes the MOPIENS 310/320 DME standards and tolerances.

5.2. 300 DME STANDARDS AND TOLERANCES

Table 5-1 provides the system standards and tolerances for the performance checks and maintenance procedures of 300 DME. It is advisable that these parameters be checked at the recommended intervals and maintenance procedures are performed on schedule for efficient performance of the equipment.

Table 5-1. Transponder Standards and Tolerances

Parameter	Standard	Tolerance Limit
TRANSMITTER		
Rated Power Output	(310 DME) 100 W	-3 dB
	(320 DME) 1000 W	-3 dB
Frequency	Assigned frequency	±0.001 %
Pulse Spacing	(X channel) 12 µs	±0.1 µs
	(Y channel) 30 µs	±0.1 µs
Pulse Shape		
· Pulse duration	3.5 µs	± 0.5 µs
· Pulse rise time	2.5 µs	± 0.5 µs
· Pulse decay time	2.5 µs	-1 µs / +0.5 µs
Pulse Rate	> 700 pp/s	
Time Delay	(X channel) 50 µs	±0.2 µs
	(Y channel) 56 µs	±0.2 µs
Identification	1350 pp/s	± 10 pp/s
RECEIVER		
Frequency	Assigned frequency	±0.001 %
Sensitivity	(310 DME) -81 dBm	±2 dB
	(320 DME) -91 dBm	±2 dB
Sensitivity at		
· ±100 kHz	No change	< 3 dB
· ±900 kHz	No reply	Reply efficiency < 5%
SDES duration	(X channel) 6 µs	
	(Y channel) 24 µs	

Table 5-2. Transponder Standards and Tolerances (continued)

Parameter	Standard	Tolerance Limit
DECODER		
X channel; when spacing is	Reply efficiency:	Reply efficiency:
· 10.0 μ s:	< 10%	< 10%
· 11.0 μ s:	100%	> 80%
· 12.0 μ s:	100%	> 80%
· 13.0 μ s:	100%	> 80%
· 14.0 μ s	< 10%	< 10%
Y channel; when spacing is	Reply efficiency:	Reply efficiency:
· 34.0 μ s:	< 10%	< 10%
· 35.0 μ s:	100%	> 80%
· 36.0 μ s:	100%	> 80%
· 37.0 μ s:	100%	> 80%
· 38.0 μ s:	< 10%	< 10%

Table 5-3. Monitor Standards and Tolerances

Parameter	Standard	Tolerance Limit
Test Interrogation Signal		
Frequency	Assigned frequency	$\pm 0.001\%$
Pulse Spacing	(X channel) 12 μs	$\pm 0.1\ \mu\text{s}$
	(Y channel) 36 μs	$\pm 0.1\ \mu\text{s}$
Pulse Shape		
· Pulse duration	3.5 μs	$\pm 0.5\ \mu\text{s}$
· Pulse rise time	2.5 μs	$\pm 0.5\ \mu\text{s}$
· Pulse decay time	2.5 μs	-1 μs / +0.5 μs
Identification Fault Condition	40 seconds	
Monitor Action Delay	4 seconds	≤ 10 seconds

Table 5-4. Monitor Alarm Limits

Parameter	Nominal Value	Lower Limit	Upper Limit
Time Delay	(X channel) 50.0 μs	49.6 μs	50.4 μs
	(Y channel) 56.0 μs	55.6 μs	56.4 μs
Pulse Spacing	(X channel) 12.0 μs	11.6 μs	12.4 μs
	(Y channel) 30.0 μs	29.6 μs	30.4 μs
Pulse Rise Time	2.5 μs	2.0 μs	3.0 μs
Pulse Decay Time	2.5 μs	2.0 μs	3.0 μs
Pulse Duration	3.5 μs	3.0 μs	4.0 μs
Peak Power Output	100 W	50 W	
	1000 W	500 W	
Transmission Rate	N/A	700 pp/s	5400 pp/s
Reply Efficiency	> 70%	70%	100%
Frequency	Assigned frequency	-0.001%	0.001%
VSWR	N/A	N/A	3 : 1
ERP	0 dB	-3 dB	3 dB

Table 5-5. Power Supply Standards and Tolerances

Parameter	Standard	Tolerance Limit
AC Line Voltage	110 V / 220 V	85 V to 265 V
Frequency	50 Hz / 60 Hz	47 Hz to 63 Hz
DC Bus Voltages	5 V	4.75 V to 5.25 V
	8 V	7.60 V to 8.40 V
	15 V	14.25 V to 15.75 V
	-15 V	-15.75 V to -14.25 V
	28 V	26.60 V to 29.40 V
	50 V	47.50 V to 52.50 V
Backup Battery Voltage when fully charged	27 V	> 26 V

SECTION 6

PERIODIC MAINTENANCE

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6. PERIODIC MAINTENANCE

6.1. INTRODUCTION

This section describes the required performance checks that must be conducted in order to ensure safe operation of MOPIENS 300 DME equipment. The supervising personnel are responsible for scheduling these performance checks based on the recommended intervals. The intervals given for the performance checks in Table 6-1, however, are not intended to replace, substitute for or supersede their official regulations or directives which should be consulted for final authority.

The supervisor shall notify the responsible authorities, such as air traffic control tower for example, of possible alarm indication or interruption of DME navigation service during the maintenance work.

6.2. SAFETY PRECAUTIONS

The purpose of safety precautions is to protect persons and property, and therefore all attentions must be given to them.

If improperly handled, the equipment may be damaged resulting in reduced performance and lifetime since MOPIENS 300 DME equipment contains modules and parts equipped with ESD-sensitive semiconductor components. When replacing modules and parts containing ESD-sensitive components, special precautionary measures should be taken during removal, transport and installation in order to prevent damage to the components. For protection against ESD, the use of appropriate kits is recommended, such as portable static-dissipative field kit with dissipative mat, wrist band, and ground wire or equivalent protection connected to same potential as the equipment.

Work should not be carried out outside the shelter or on the antenna system during a storm due to the danger of injury by lightning.

6.3. TEST EQUIPMENT REQUIREMENT

The test equipment inherent errors should be at least five times smaller than the specified tolerances. Given below is the recommended list of test equipment, or equivalent, to make the measurements:

Table 6-1. Test Equipment Required

Equipment	Make	Model
USB Power Sensor	Rohde & Schwarz	NRP-Z81
Frequency Counter	Keysight	53131A with option 015
Digital Multi-Meter	FLUKE	187
Digital Oscilloscope	TEKTRONIX	TDS1012B
RF Directional Coupler	Any	30 dB, 960 MHz – 1215 MHz

Table 6-2. Optional Test Equipment for Depot-level Maintenance

(*For information only*)

Equipment	Make	Model
Avionic Signal Generator	Rohde & Schwarz	SMA100A with option K26
High Power Attenuator	Any	30 dB, fixed
Network Analyzer	Keysight	8714ES
USB Power Sensor Adaptor	Rohde & Schwarz	NRP-Z4
Spectrum Analyzer	Advantest	R3267

6.4. MAINTENANCE SCHEDULE

Table 6-3. Maintenance Schedule

Performance Check		Reference Paragraph		
		ICAO Doc 8071	Standards and Tolerances	Maintenance Procedure
MONTHLY	Operation Status Check	n/a	n/a	6.5.1
	Equipment Cleaning	n/a	n/a	6.6.1
	Reply Pulse Shape	3.2.6	Table 5-1	6.5.4
	Reply Pulse Pair Spacing	3.2.7	Table 5-1	6.5.4
	RF Power	3.2.8	Table 5-1	6.5.3
	Transmitter Pulse Rate	3.2.10	Table 5-1	6.5.6
	Receiver Sensitivity	3.2.12	Table 5-1	6.5.7
	Time Delay	3.2.16	Table 5-1	6.5.2
	Power Supply	n/a	Table 5-4	6.5.9
ANNUALY	RF Frequency	3.2.4 3.2.11	Table 5-1	6.5.10
	Monitor Alarm	3.2.18	Table 5-3	6.5.8
	Monitor Action Delay	n/a	Table 5-2	6.5.11
	Verification of BITE	n/a	n/a	6.5.12
	Antenna and Equipment Ground Resistance	n/a	n/a	6.5.13
	Damage Check	n/a	n/a	6.6.2

6.5. ON-SITE PERFORMANCE CHECKS

6.5.1. OPERATION STATUS CHECK

Refer to Section 5 Operations for details.

6.5.2. REPLY DELAY CHECK

6.5.2.1. Standards and Tolerances

	Test point	Standard	Tolerance
System Time Delay	LOG VIDEO	X CH: 50 μ sec Y CH: 56 μ sec	$\pm 0.5 \mu$ sec

6.5.2.2. Test Setup

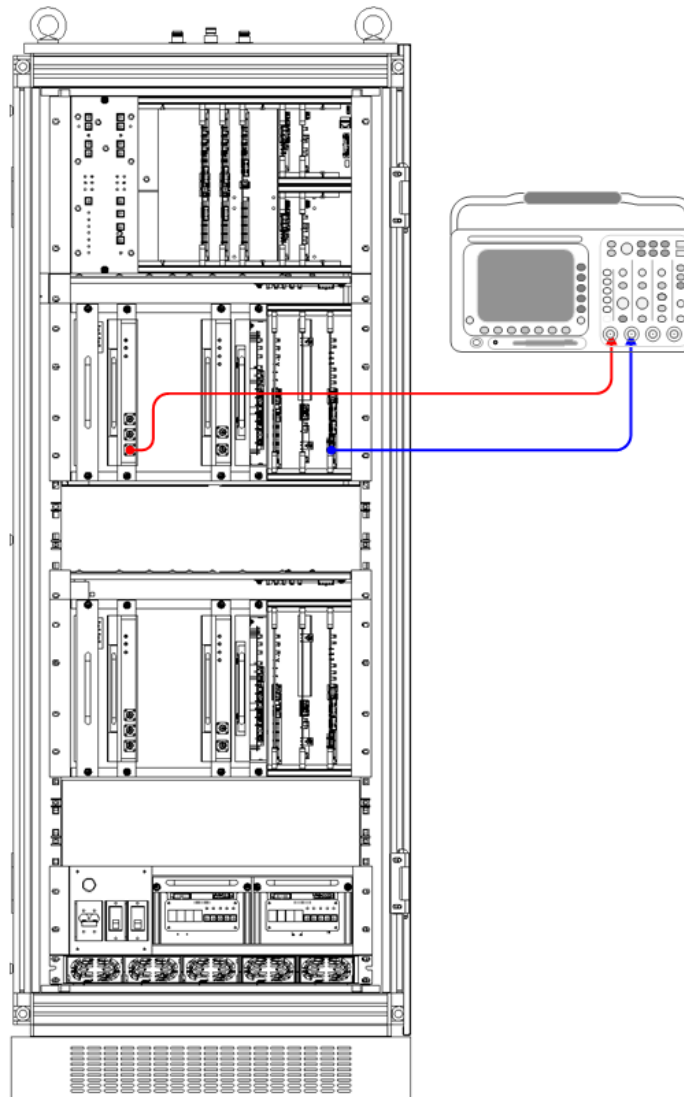


Figure 6-1. Test Setup for System Delay Measurement

6.5.2.3. Test Equipment

- Oscilloscope, Tektronix TDS 3012 or equivalent
- Oscilloscope Probe, 10X, 100 MHz or equivalent

- Test Cable, BNC(M) to BNC(M) 50 Ω Coaxial cable

6.5.2.4. Procedure

- Connect one end of the test cable to **LOG VIDEO** port on the front panel of RXU
- Connect the other end of the test cable to CH1 input of the oscilloscope.
- Connect the grabber of the oscilloscope probe to the **Interrogation Trigger** test point on the MON unit as shown in Figure 6-2.

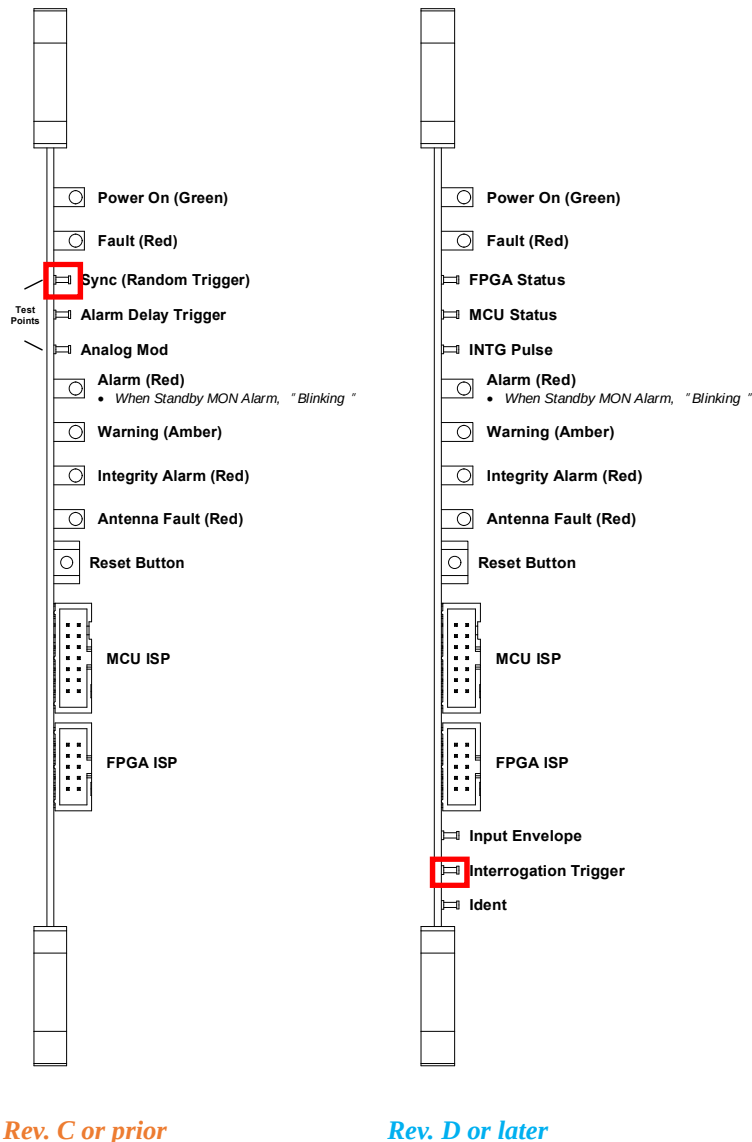


Figure 6-2. Test point for Interrogation Trigger on MON unit

- Connect the oscilloscope probe to the CH2 input of the oscilloscope.
- Start PMDT and log in.
- Refer to Go to **Maintenance** → **Advanced Control** → **RF Loopback (TX to RX)** and select the option **On**. Then click on the **Apply** button next to it.

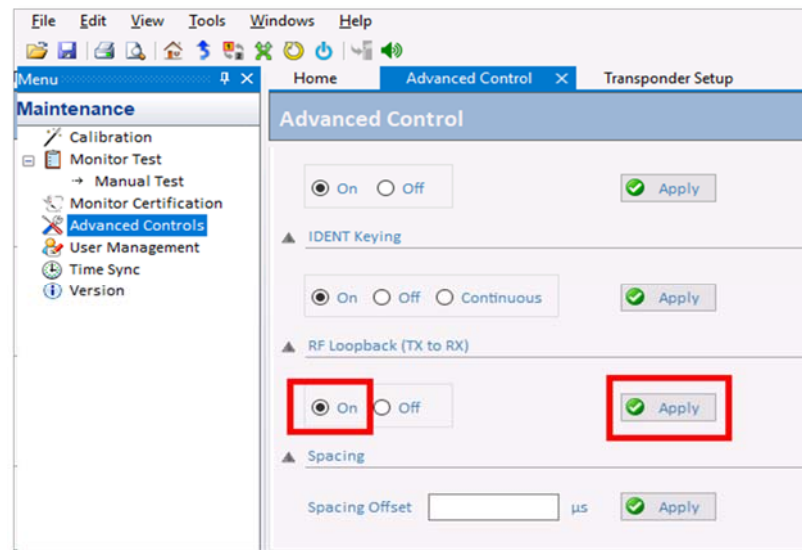


Figure 6-3. RF Loopback (TX to RX) option

- g. Set the oscilloscope as follows and adjust the trigger level accordingly:
 - o horizontal scale: 10 μ s/div
 - o vertical scale: 2 V/div
 - o trigger source: CH2
 - o trigger mode: Single
- h. Press the SINGLE SEQ button on the oscilloscope and check to see if the interrogation and the reply pulses are acquired together adequately as shown in Figure 6-4. If not, repeat pressing the SINGLE SEQ button until it is captured as the figure.
- i. Find peak points on the leading pulses of two pulse pairs on the oscilloscope screen. Find two points which are 0.7 V down from the peak on the rising edge as shown in Figure 6-4. Measure the time difference between the two points.
- j. Refer to Go to **Maintenance** $\rightarrow$ **Advanced Control** $\rightarrow$ **RF Loopback (TX to RX)** and select the option **Off**. Then click on the **Apply** button next to it.

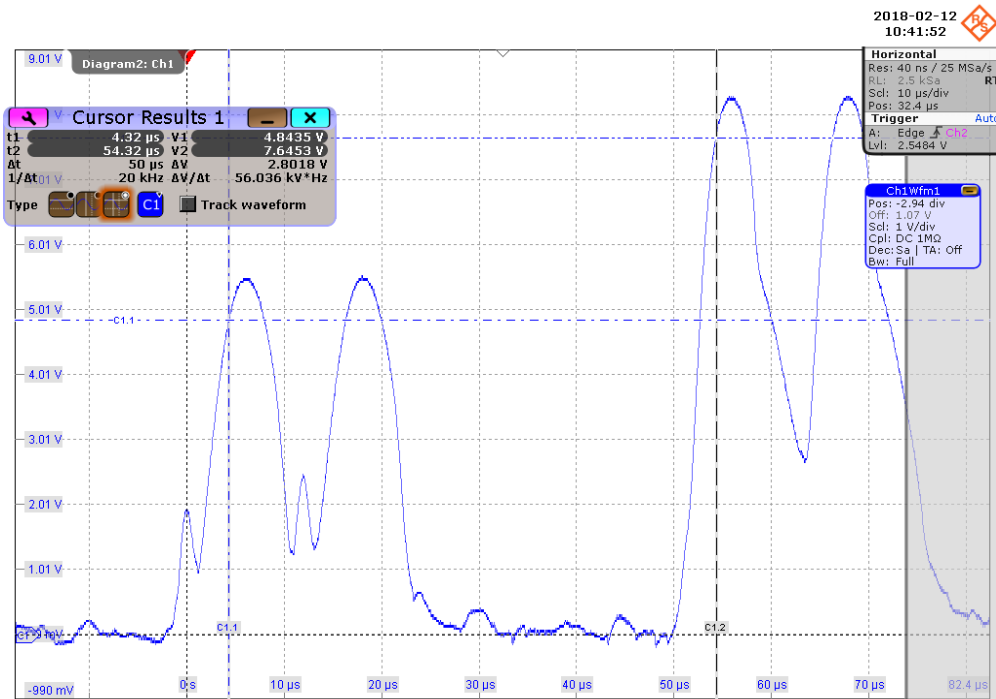


Figure 6-4. Waveforms on oscilloscope

6.5.3. RF POWER OUTPUT CHECK

6.5.3.1. Standards and Tolerances

	Test point	Standard	Tolerance
Transmitter peak power output	ANT (J2)	Assigned power	± 1 dB

6.5.3.2. Test Setup

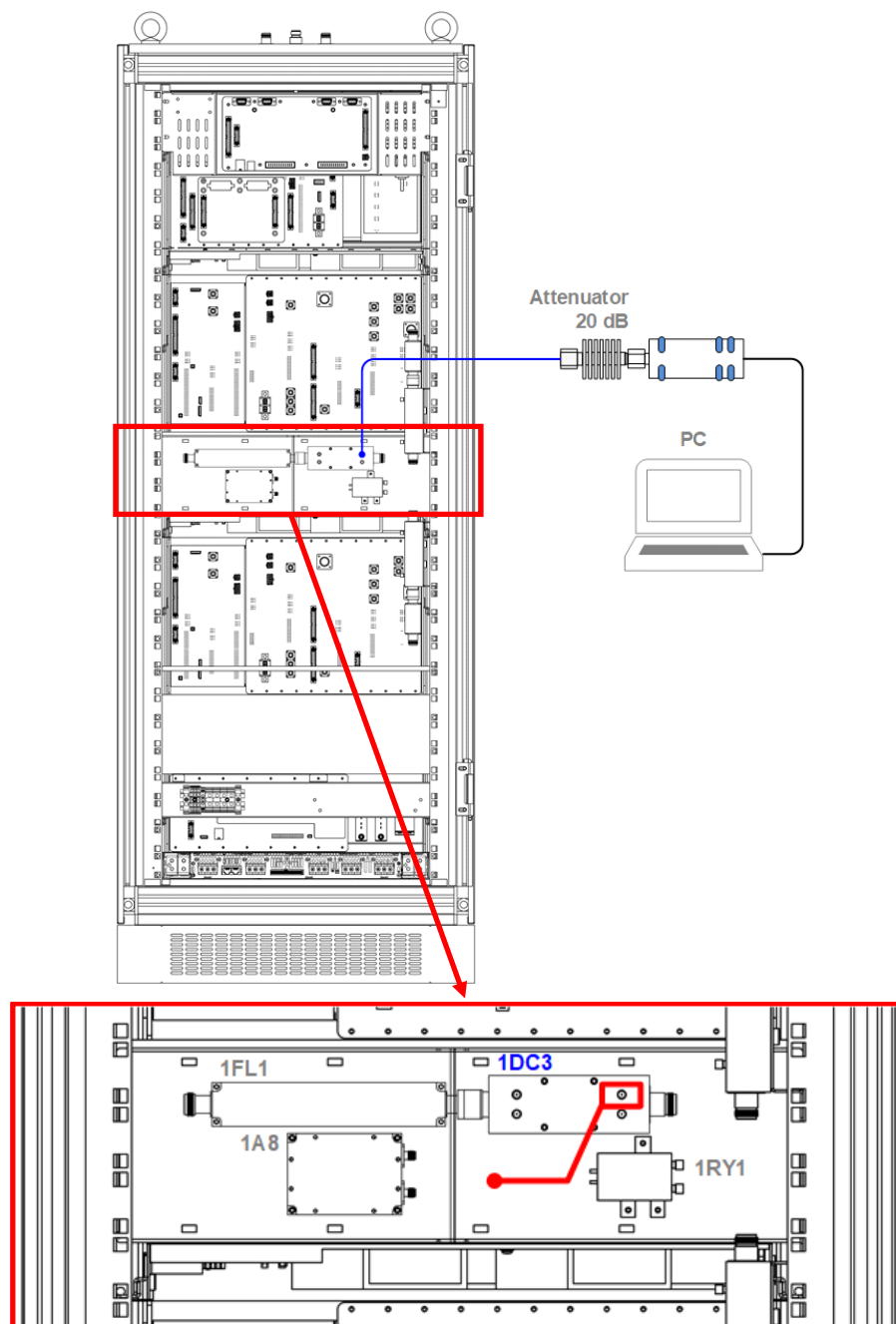


Figure 6-5. Test Setup for Power Output Measurement (s/n 00022 or prior)

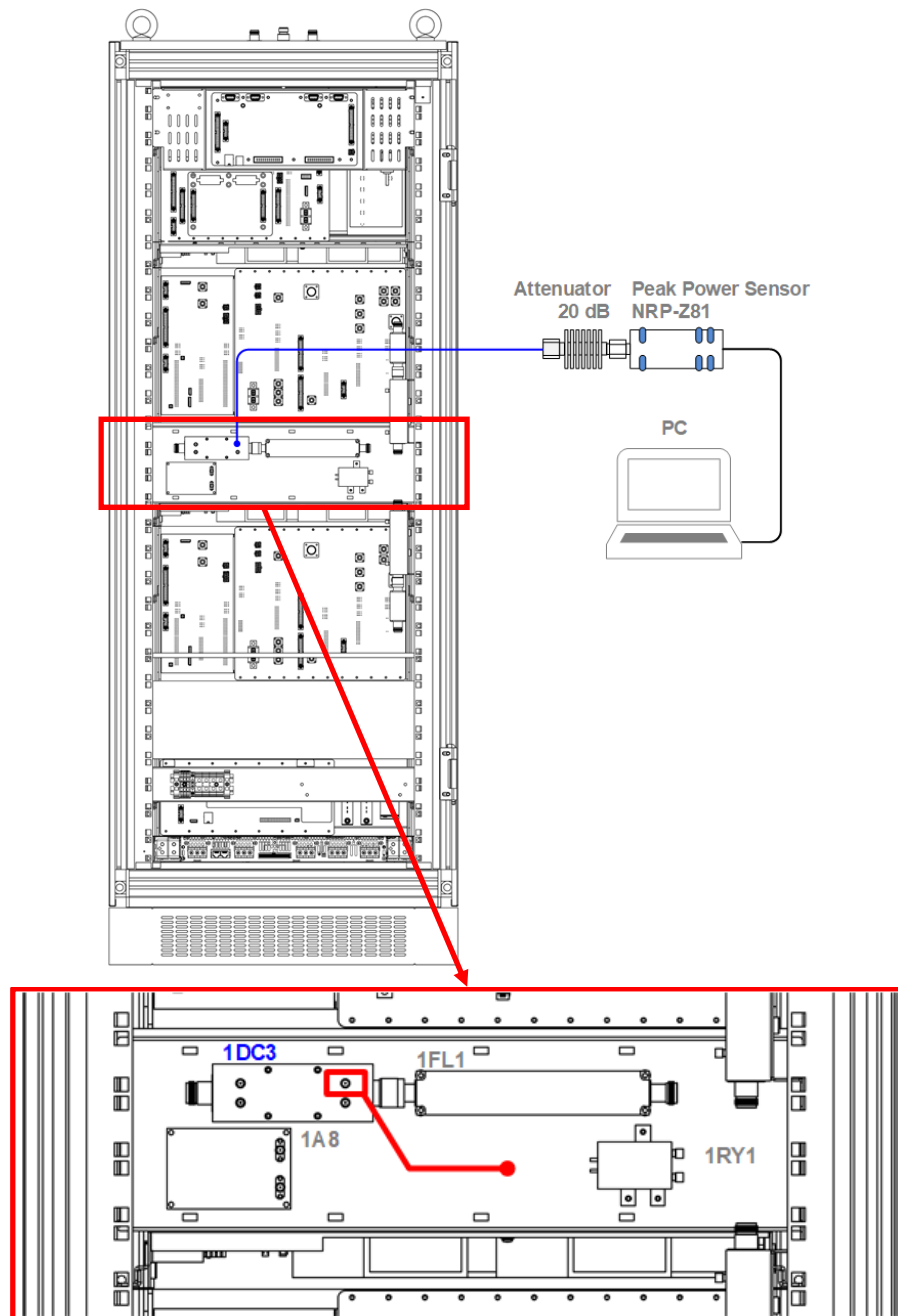


Figure 6-6. Test Setup for Power Output Measurement (s/n: 00023 or later)

6.5.3.3. Test Equipment

- Wideband peak power sensor, R&S ERP-Z81 or equivalent
- PC with R&S® Power Viewer Plus software installed.
- RF power attenuator: 20 dB, 50Ω, 5 W continuous.
- Test Cable: N(M) to N(M) 50 Ω Coaxial Cable with known loss at the test frequency

6.5.3.4. Procedure

- a. Open the rear door of the equipment cabinet.
- b. Disconnect the SMA termination (dummy load) from the directional coupler, 1DC3 as shown in Figure 6-5 (cabinet s/n: 00022 or prior) or Figure 6-6 (cabinet s/n: 00023 or later).
- c. Connect one end of the test cable to the coupled port on the directional coupler.
- d. Connect the other end of the test cable to input port of the power attenuator.
- e. Connect the peak power sensor to the output port of the power attenuator.
- f. Run the R&S® Power Viewer Plus Software on the PC which the power sensor is connected to.
- g. Consulting the R&S® Power Viewer Plus Software manual, set the offset value, 30 dB (or the actual couple ratio, of the directional coupler, measured using an accurate a network analyzer) + 20 dB (or the actual attenuation ratio, of the power attenuator, measured using an accurate network analyzer) + cable loss to the power sensor
- h. Read and note the peak power output

6.5.4. REPLY PULSE CHARACTERISTICS CHECK

6.5.4.1. Standards and Tolerances

	Test point	Standard	Tolerance
Transmitter Reply Pulse Rising Time	TXU (J2) for Low power DME HPA (J7) for High power DME	2.0 μ s	< 3.0 μ s
Transmitter Reply Pulse Duration	TXU (J2) for Low power DME HPA (J7) for High power DME	3.5 μ s	$\pm$ 0.5 μ s
Transmitter Reply Pulse Decay Time	TXU (J2) for Low power DME HPA (J7) for High power DME	2.5 μ s	< 3.5 μ s

6.5.4.2. Test Setup

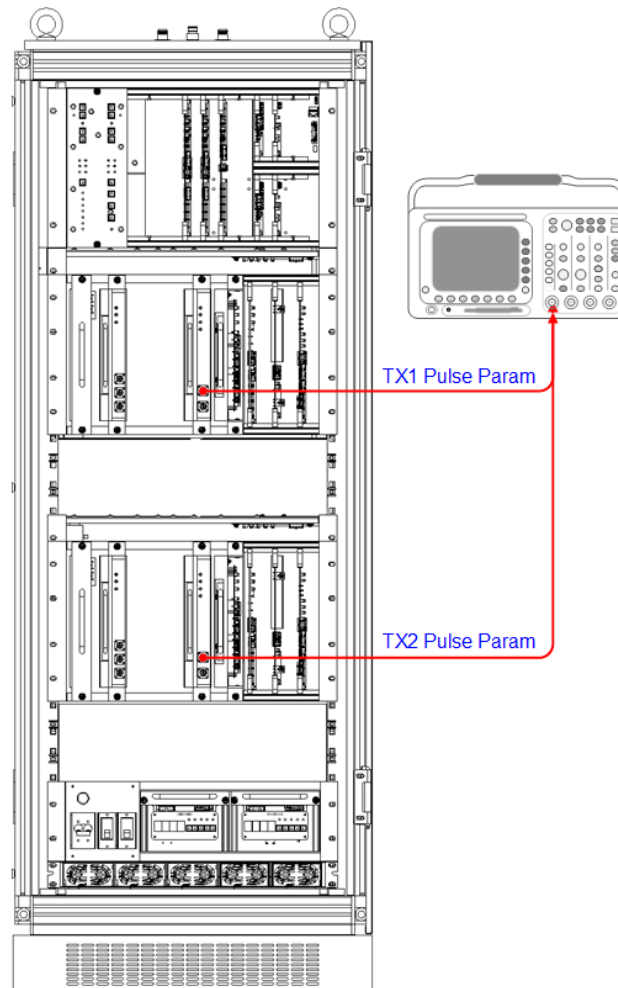


Figure 6-6. Reply Pulse Characteristics Test Setup for 310 DME

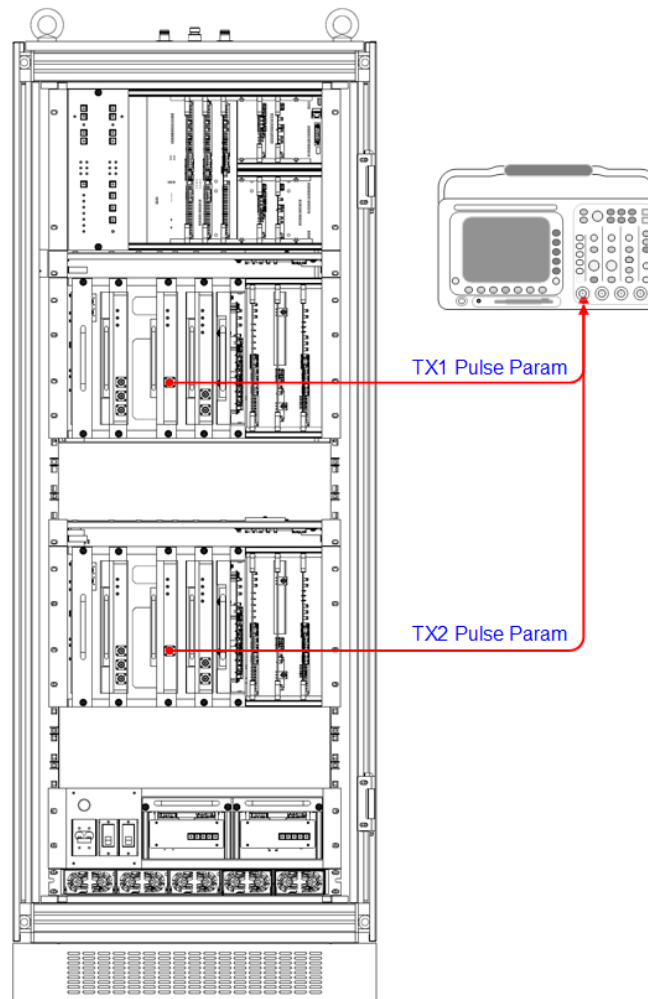


Figure 6-7. Reply Pulse Characteristics Test Setup for 320DME

6.5.4.3. Test Equipment

- Oscilloscope, Tektronix TDS 3012 or equivalent
- Oscilloscope Probe, 10X, 100 MHz or equivalent
- Test Cable, BNC(M) to BNC(M) 50 Ω Coaxial cable

6.5.4.4. Procedure for Low Power DME

- a. Connect one end of the test cable to **Detected VIDEO** port on the front panel of TXU
- b. Connect the other end of the test cable to CH1 input of the oscilloscope.
- c. Set the oscilloscope as follows and adjust the trigger level accordingly:
 - horizontal scale: 1 μ s/div;
 - vertical scale: adjust to fit the full height of the grid;
 - trigger source: CH1
- d. Using oscilloscope, measure the pulse rise time, pulse duration and pulse decay time:

- For the pulse rise time, measure the time from the 10% voltage point to 90% voltage point on the rising edge.
- For the pulse rise time, measure the time from the 50% voltage point on the rising edge to 50% voltage point on the falling edge.
- For the pulse decay time, measure the time from the 90% voltage point to 10% voltage point on the falling edge.

6.5.4.5. Procedure for High Power DME

- a. Connect one end of the test cable to **Detected VIDEO** port on the front panel of HPA
- b. Connect the other end of the test cable to CH1 input of the oscilloscope.
- c. Set the oscilloscope as follows and adjust the trigger level accordingly:
 - horizontal scale: 1 μ s/div;
 - vertical scale: adjust to fit the full height of the grid;
 - trigger source: CH1
- d. Using oscilloscope, measure the pulse rise time, pulse duration and pulse decay time:
 - For the pulse rise time, measure the time from the 10% voltage point to 90% voltage point on the rising edge.
 - For the pulse rise time, measure the time from the 50% voltage point on the rising edge to 50% voltage point on the falling edge.
 - For the pulse decay time, measure the time from the 90% voltage point to 10% voltage point on the falling edge.

6.5.5. PULSE PAIR SPACING CHECK

6.5.5.1. Standards and Tolerances

	Test point	Standard	Tolerance
Transmitter Reply Pulse Pair Spacing	TXU (J2) for Low power DME HPA (J7) for High power DME	(X Ch.) 12 μ s (Y Ch.) 30 μ s	< 0.10 μ s

6.5.5.2. Test Setup

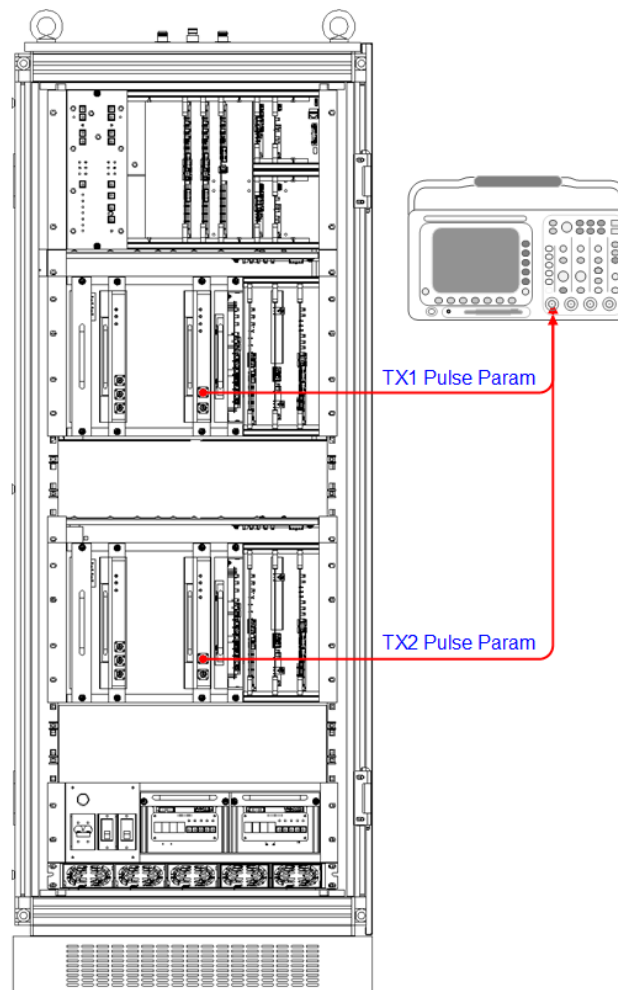


Figure 6-6. Reply Pulse Pair Spacing Test Setup for 310 DME

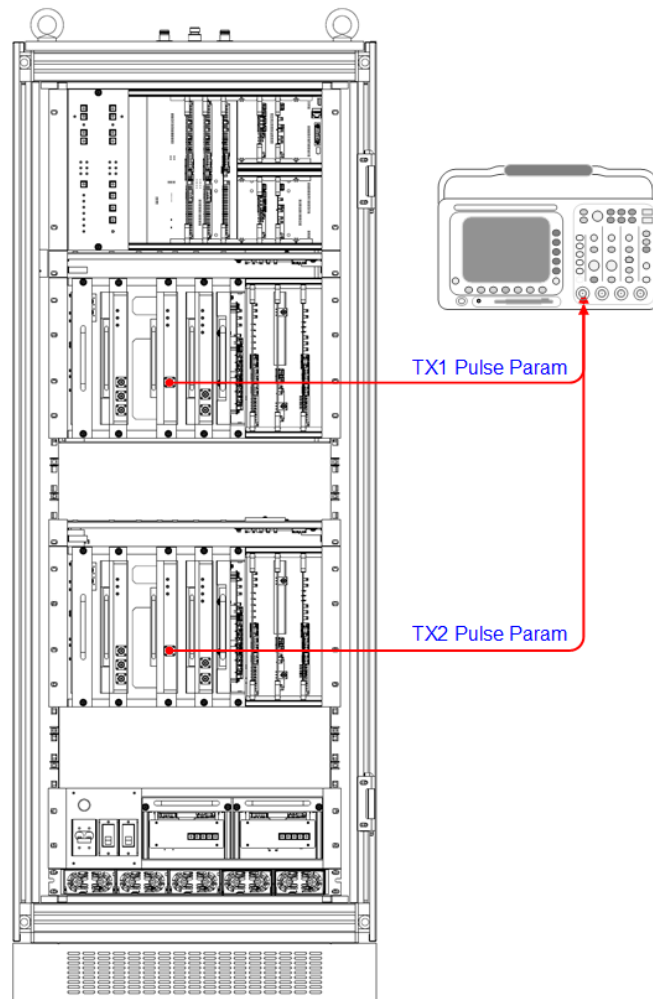


Figure 6-8. Reply Pulse Pair Spacing Test Setup for 320DME

6.5.5.3. Test Equipment

- Oscilloscope, Tektronix TDS 3012 or equivalent
- Oscilloscope Probe, 10X, 100 MHz or equivalent
- Test Cable, BNC(M) to BNC(M) 50 Ω Coaxial cable

6.5.5.4. Procedure for Low Power DME

- a. Connect one end of the test cable to **Detected VIDEO** port on the front panel of TXU
- b. Connect the other end of the test cable to CH1 input of the oscilloscope.
- c. Set the oscilloscope as follows and adjust the trigger level accordingly:
 - horizontal scale: 2 μ s/div;
 - vertical scale: adjust to fit the full height of the grid;
 - trigger source: CH1
- d. Adjust the horizontal position so that one full pulse pair waveform fits in.

- e. Using oscilloscope, measure the time between the 50% voltage point on the rising edge of the first pulse and the 50% voltage point on the rising edge of the second pulse.

6.5.5.5. Procedure for High Power DME

- a. Connect one end of the test cable to **Detected VIDEO** port on the front panel of HPA
- b. Connect the other end of the test cable to CH1 input of the oscilloscope.
- c. Set the oscilloscope as follows and adjust the trigger level accordingly:
 - horizontal scale: 2 μ s/div;
 - vertical scale: adjust to fit the full height of the grid;
 - trigger source: CH1
- d. Adjust the horizontal position so that one full pulse pair waveform fits in.
- e. Using oscilloscope, measure the time between the 50% voltage point on the rising edge of the first pulse and the 50% voltage point on the rising edge of the second pulse.

6.5.6. OUTPUT PULSE COUNT CHECK

6.5.6.1. Standards and Tolerances

	Test point	Standard	Tolerance
Transmitter Reply Pulse Rate	TXU (J2)	700 pp/s	> 700 pp/s

6.5.6.2. Test Setup

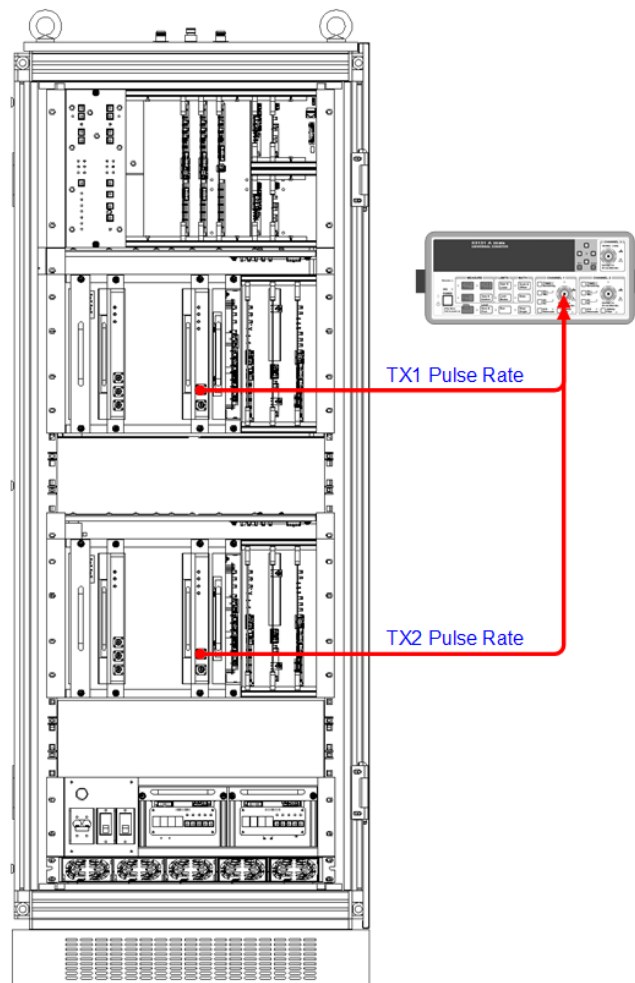


Figure 6-8. Pulse Rate Test Setup

6.5.6.3. Test Equipment

- Frequency Counter: 1.5 GHz minimum, Keysight 53131A with option 015 or equivalent
- Test Cable: BNC(M) to BNC(M) 50 Ω Coaxial Cable

6.5.6.4. Procedure

- Connect one end of the test cable to **Detected VIDEO** port on the front panel of TXU as shown in Figure 6-208.

- b. Connect the other end of the test cable to the appropriate input port of the frequency counter.
- c. Read and note the reading.
- d. Calculate the pulse pair transmission rate by dividing the reading by two.

6.5.7. RECEIVER SENSITIVITY CHECK

6.5.7.1. Standards and Tolerances

	Test point	Baseline	Tolerance
Receiver Sensitivity Level	PMDT	-81dBm for Low Power DME -91dBm for High Power DME	< -81dBm for 310 DME < -91dBm for 320 DME

6.5.7.2. Procedure for Low Power DME

- Start PMDT and log in
- Bypass the monitors
- Select **Main** → **Advanced Control** screen.
- Turn off the squitter pulse and IDENT Keying.

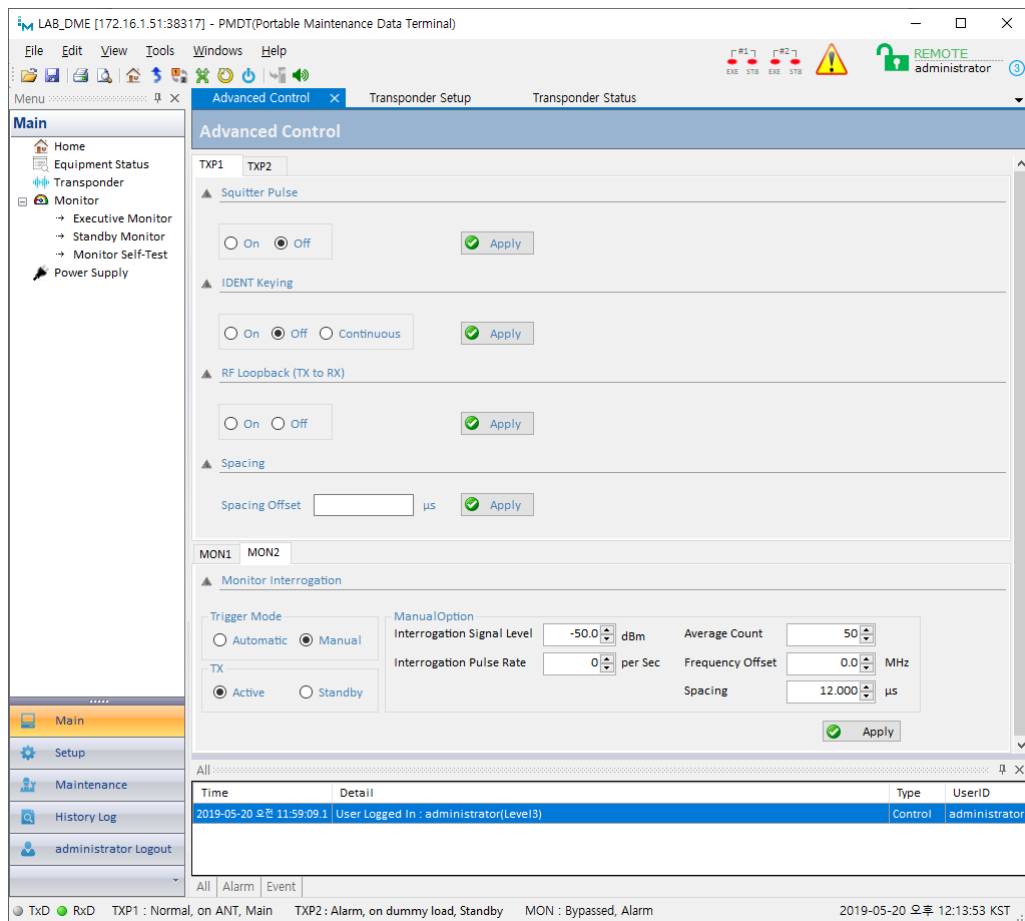


Figure 6-9. Advanced Control screen on PMDT

- For MON2, Set the **Trigger Mode** to **Manual** and the **Interrogation Pulse Rate** to 0 pp/s and click on the **Apply** button.

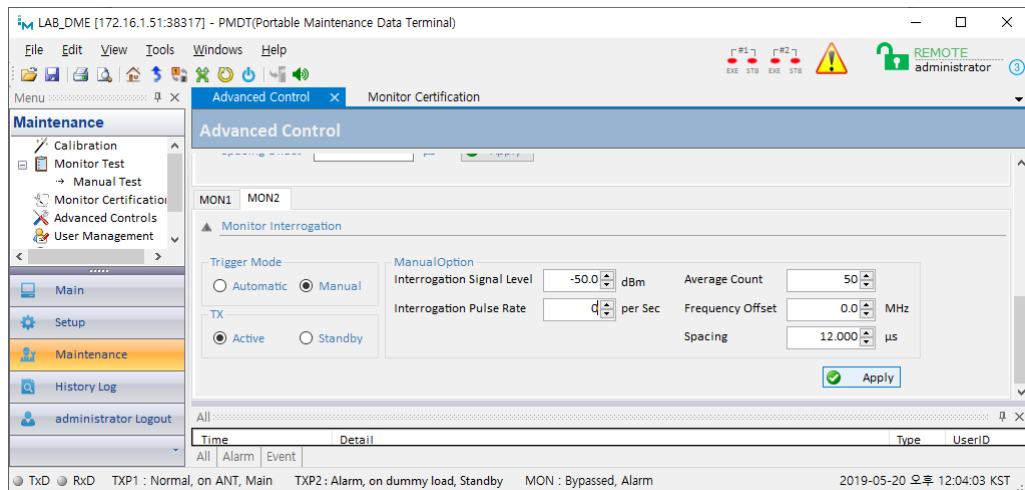


Figure 6-10. Monitor Interrogation Section in the Advanced Control

- f. For MON2, Set the **Trigger Mode** to **Manual**, the **Interrogation Pulse Rate** to 100 pp/s, and the **Interrogation Pulse Level** to -81dBm and click on the **Apply** button. For 320 DME, Set the **Interrogation Pulse Level** to -91dBm.
- g. Open **Transponder Status** screen.
- h. Verify that the Transmission Rate is 70 pp/s or more.
- i. Changeover to the standby transponder.
- j. Repeat the procedure above for the standby transponder.

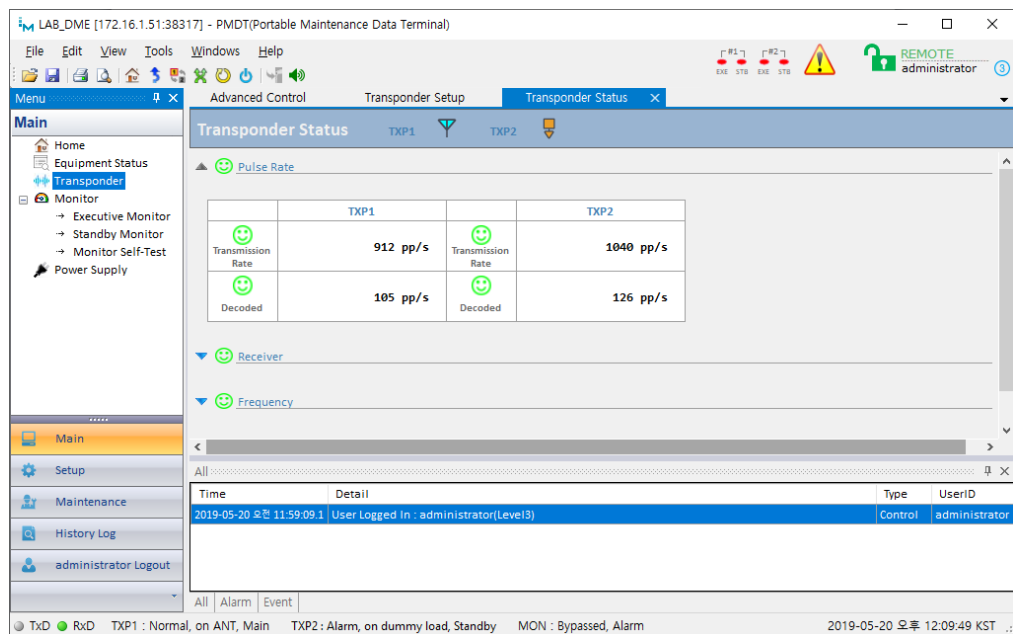


Figure 6-11. Transponder Status screen on PMDT

6.5.8. MONITOR ALARM POINTS CHECK

6.5.8.1. Standards and Tolerances

	Test point	Standard	Tolerance
Monitor Alarm Points	LED indicators on MON and CSP	- The red ALARM LED illuminates when alarm condition. - The amber WARNING LED illuminates when warning condition.	-

6.5.8.2. Procedure for Low Power DME

- Start PMDT and log in
- Bypass monitors.
- Go to **Main** → **Setup** → **Monitor Limit Setup** and set the alarm/warning ranges for **Pulse Spacing** alarm as shown in Figure 6-19

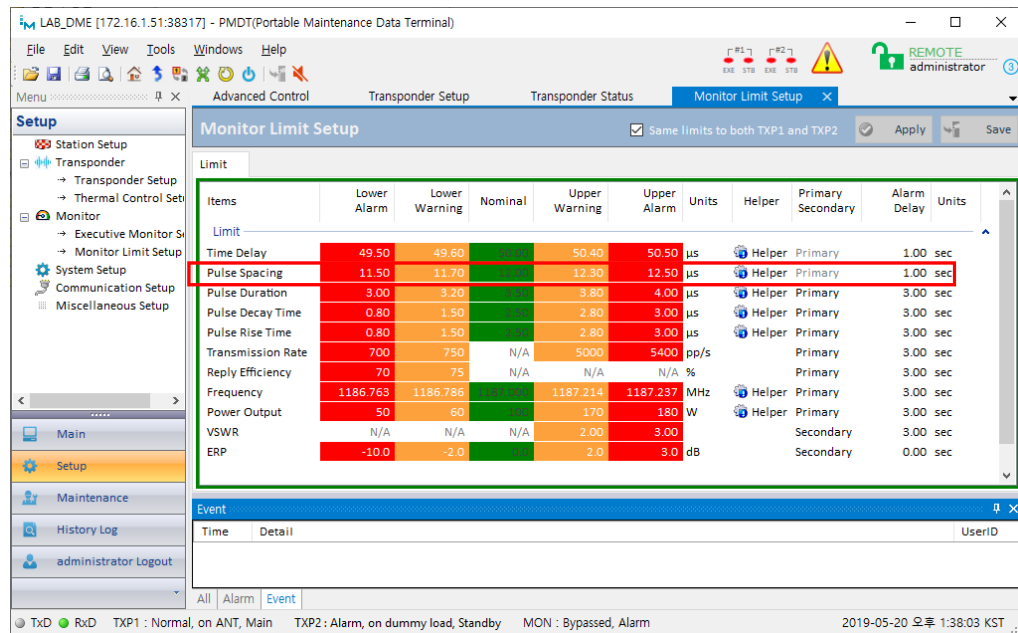


Figure 6-12. Advanced Control screen on PMDT

- Put the transponder 1 on antenna.
- Open **Advanced Control** screen
- Set **Spacing Offset** for Transponder1 to 0.6 μs and click on **Apply** button.
- Verify that the red LED indicator **ALARM** on MON1 and MON2 illuminate.
- If the value for **Pulse Spacing** displayed on the **Executive Monitor** is less than 12.5 μs, increase the **Spacing Offset** of Transponder1 until it reaches 12.5 μs.

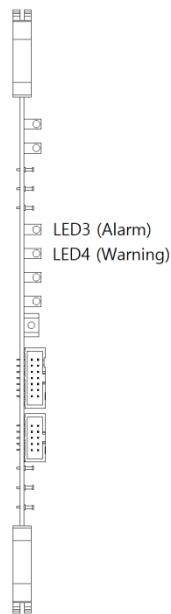


Figure 6-13. Alarm / Warning Indicators on MON

- i. Verify that monitor executive ALARM is indicated on the front of the monitor and on CSP, LMI or PMDT.
- j. Set **Spacing Offset** for Transponder1 to 0.4 μ s and click on **Apply** button.
- k. Adjust the **Spacing Offset** for transponder 1 until the **Pulse Spacing** displayed on the Executive Monitor screen is between 12.3 μ s and 12.5 μ s.

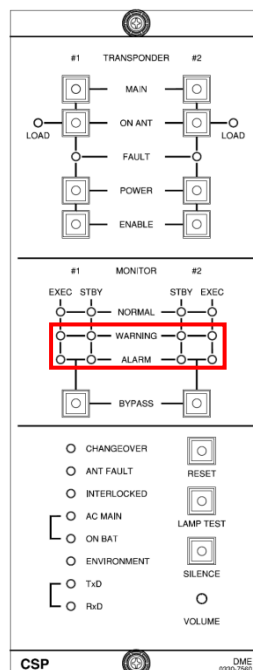


Figure 6-14. Monitor ALARM /WARNING indicators on CSP

- l. Verify that monitor executive WARNING is indicated on the front of the monitor and on CSP, LMI or PMDT.

6.5.9. POWER SUPPLY CHECK

6.5.9.1. Standards and Tolerances

- DC/DC-A

	Test point	Standard	Tolerance
DC/DC-A Output Voltages	+ 5 V	+ 5 V	± 0.25 V
	+ 15 V	+ 15 V	± 0.75 V
	- 15 V	- 15 V	± 0.75 V

- DC/DC

	Test point	Standard	Tolerance
DC/DC Output Voltages	+ 5 V	+ 5 V	± 0.25 V
	+ 15 V	+ 15 V	± 0.75 V
	- 15 V	- 15 V	± 0.75 V
	+ 8 V	+ 8 V	± 0.4 V
	+ 50 V	+ 50 V	± 2.5 V

- AC/DC

	Test point	Standard	Tolerance
AC/DC Output Voltage	AC/DC VOLT on PMU	+ 28 V	± 1.4 V

6.5.9.2. Test Setup

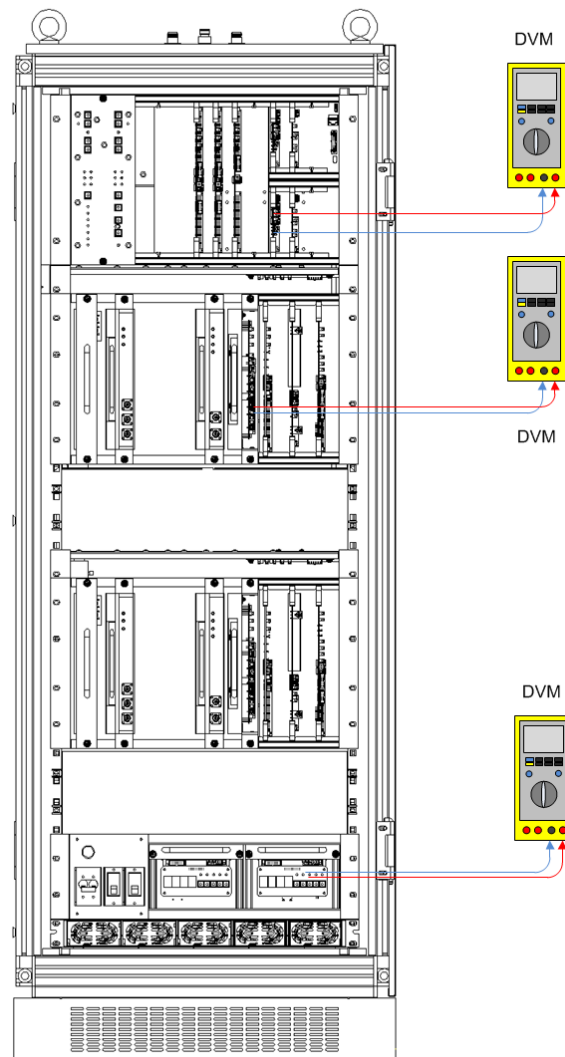


Figure 6-15. Test Setup for power supply measurement

6.5.9.3. Test Equipment

- Digital Multimeter (DMM) or Voltmeter (DVM)
- Test leads

6.5.9.4. Procedure

AC/DC Output Voltages

- Insert the negative tip (black) of the test leads into the tip jack, **GND** on the front panel of the PMU as shown in Figure 6-16.
- Insert positive tip (red) of the test leads into the tip jack, **ACDC VOLT** on the front panel of the PMU as shown in Figure 6-16.
- Read and note the measured value.

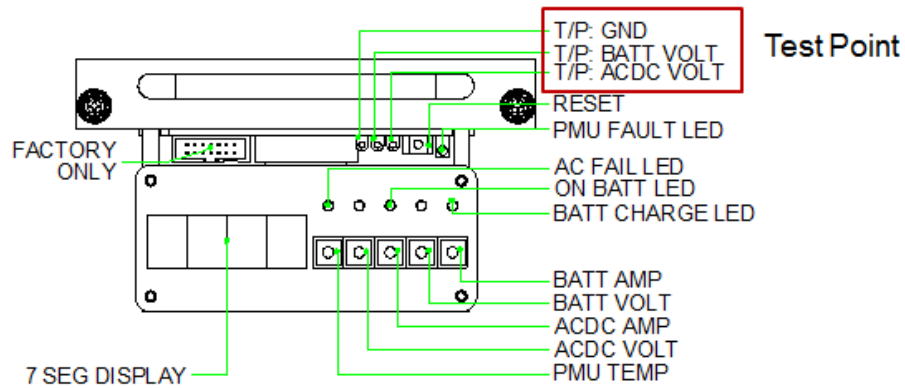


Figure 6-16. Test points on the front panel of PMU

DC/DC-A Voltages

- Insert the negative tip (black) of the test leads into the tip jack, **GND** on the front of the DC/DC-A unit as shown in Figure 6-17.
- Insert positive tip (red) of the test leads into the tip jack, **+5V** on the front of the DC/DC-A Figure 6-17.
- Read and note the measured value.
- Repeat above steps for the test points, **+15V** and **-15V**.

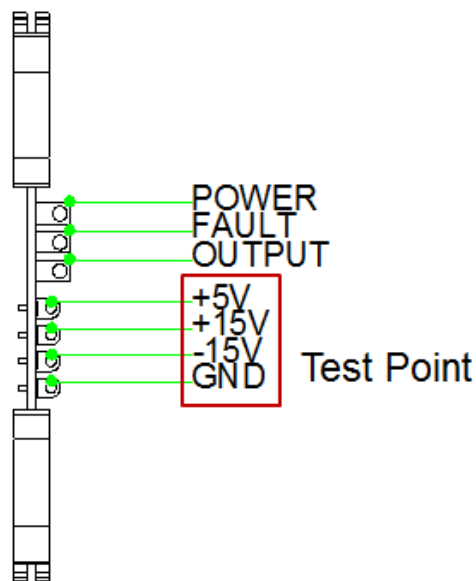


Figure 6-17. Test Points on DC/DC-A

DC/DC #1, #2 Voltages

- Insert the negative tip (black) of the test leads into the tip jack, **GND** on the front of the DC/DC unit as shown in Figure 6-18.
- Insert positive tip (red) of the test leads into the tip jack, **+5V** on the front of the DC/DC as shown in Figure 6-18.
- Read and note the measured value.

- d. Repeat above steps for the test points, +8V, +15V, -15V, +28V, +50V.

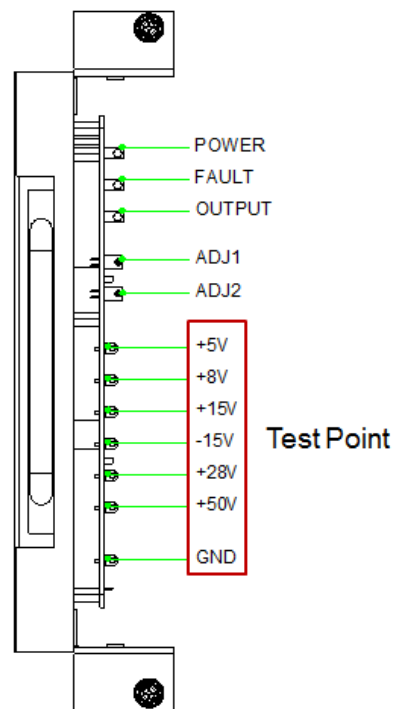


Figure 6-18. Test points on DC/DC

6.5.10. TRANSMITTER FREQUENCY CHECK

6.5.10.1. Standards and Tolerances

	Test point	Standard	Tolerance
Transmitter Frequency	TX FREQ	Assigned frequency	$\pm 0.001\%$

6.5.10.2. Test Setup

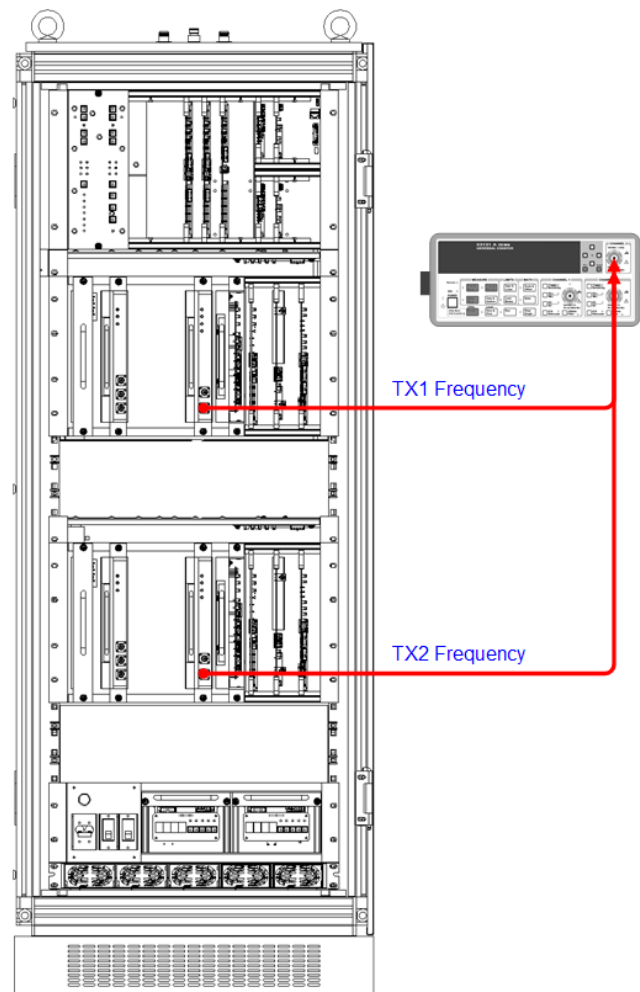


Figure 6-19. Test Setup for Transmit Frequency Measurement

6.5.10.3. Test Equipment

- Frequency Counter: 1.5 GHz minimum, Keysight 53131A with option 015 or equivalent
- Test Cable: BNC(M) to BNC(M) 50 Ω Coaxial Cable

6.5.10.4. Procedure

- Connect one end of the test cable to **TX FREQ** port on the front panel of TXU as shown in Figure 6-20.

- b. Connect the other end of the test cable to the appropriate input port of the frequency counter.
- c. Read and note the reading

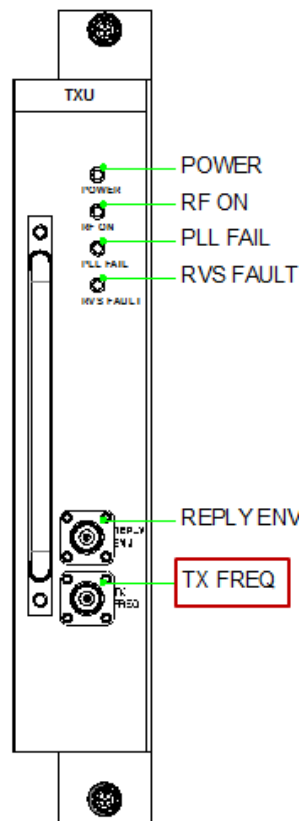


Figure 6-20. Test Point: TX FREQ on front panel of TXU

6.5.11. MONITOR ALARM DELAY CHECK

6.5.11.1. Standards and Tolerances

	Test point	Standard	Tolerance
Alarm Delay time	PMDT	4.0 sec	±0.4 sec

6.5.11.2. Procedure

- Start PMDT and log in
- Bypass the monitors.
- Open Executive Monitor Setup screen
- Set the **MON Action Delay** of the **Executive MON** to 4.0 sec.

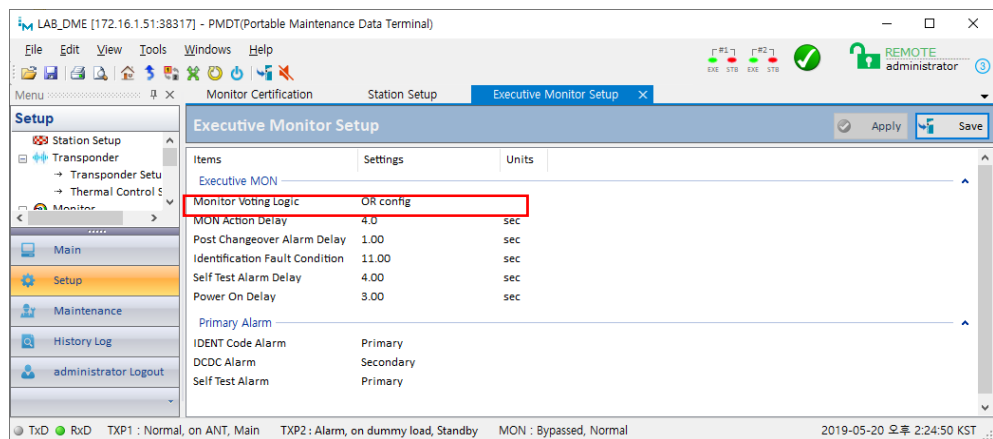


Figure 6-11.Executive Monitor Setup on PMDT

- Open Monitor Certification screen

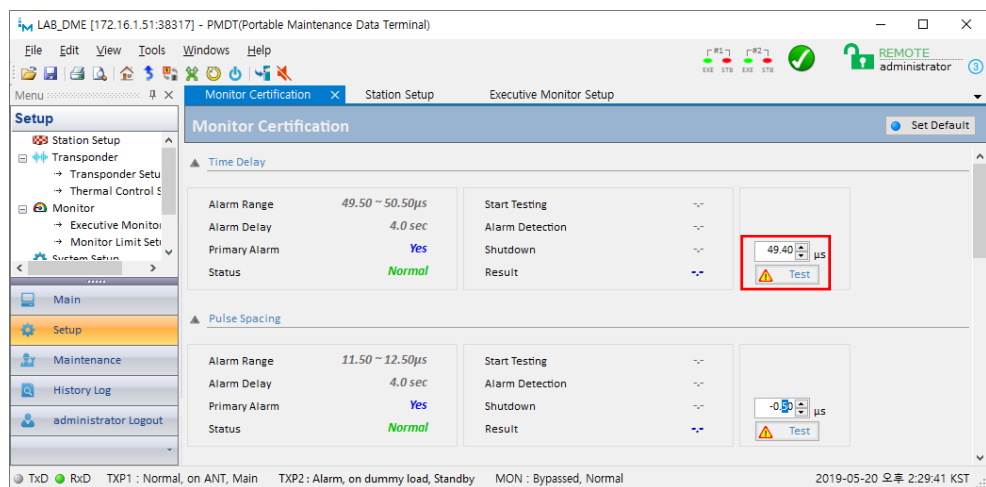


Figure 6-11.Monitor Certification screen on PMDT

- Set the Time Delay to 49.4 μs

- g. Click on the **Test** button in **Time Delay** group and verify that the **Result** displayed is 4.0 $\pm$ 0.4 sec.

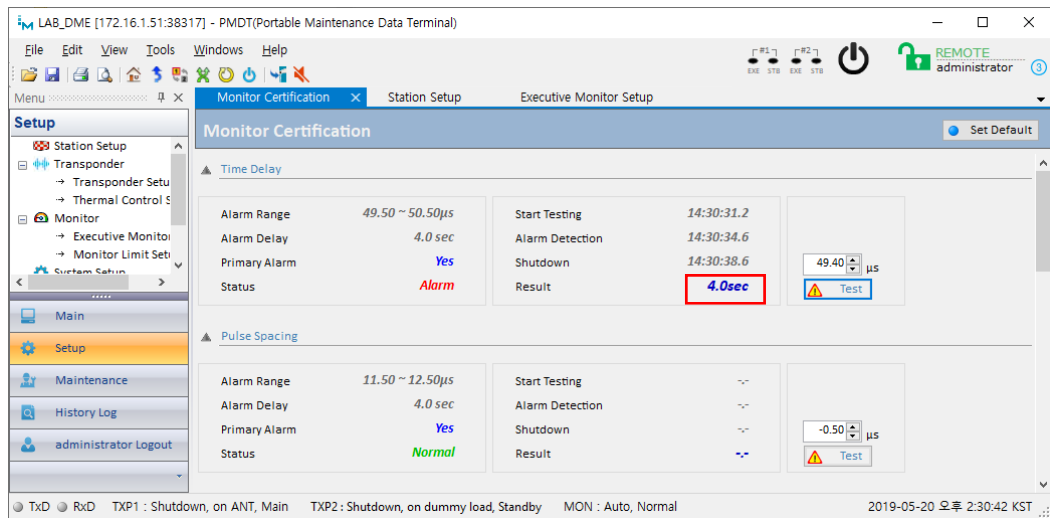


Figure 6-21. Monitor Certification Test screen on PMDT

6.5.12. BITE CERTIFICATION CHECK

6.5.12.1. Standards and Tolerances

	Test point	Standard	Tolerance
BITE Certification	PMDT	MON status indicated correctly	-

6.5.12.2. Procedure

- Start PMDT and log in
- Bypass the monitors.
- Open Monitor Self-Test Readings screen.

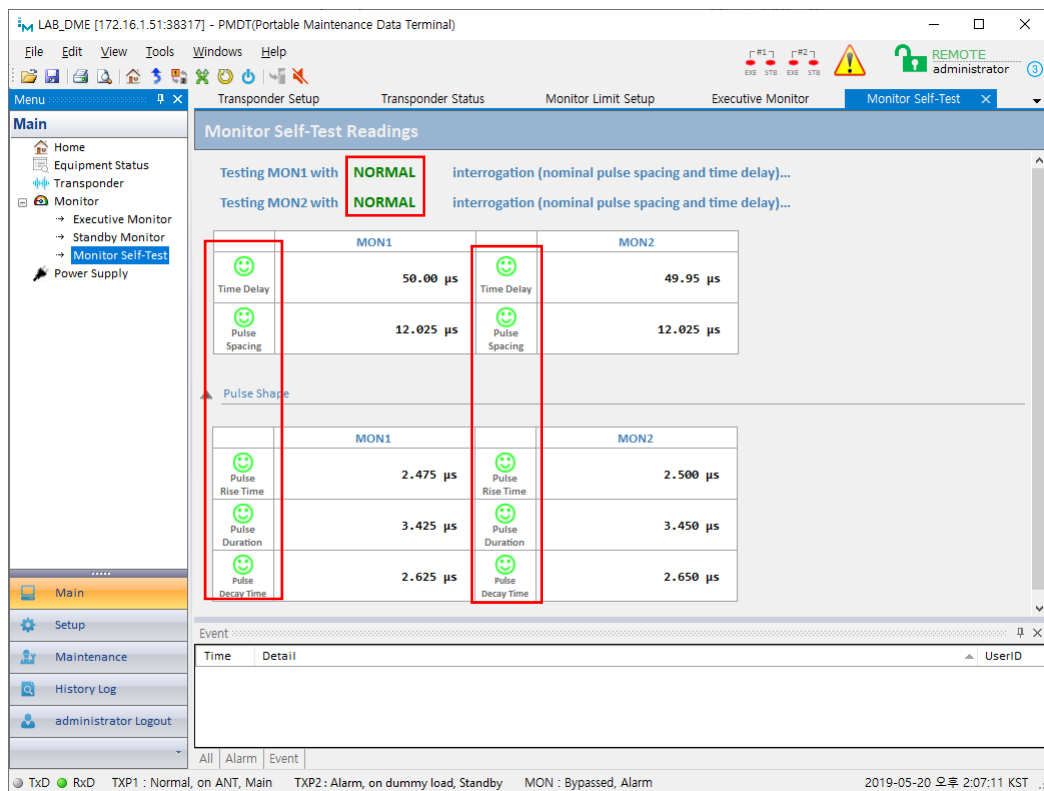


Figure 6-22. Monitor Self-Test Readings for NORMAL interrogations

- Verify that the result for the NORMAL test interrogation signal for MON1 and MON2 is all green.

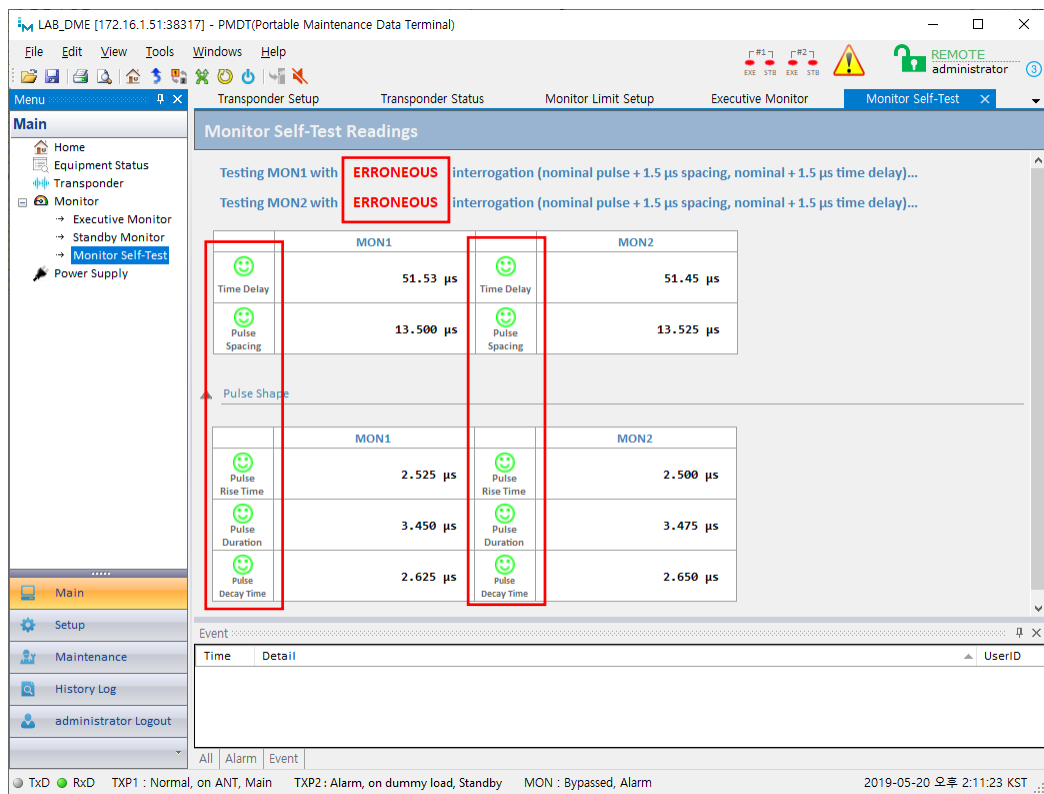


Figure 6-23. Self-Test Readings for ERRONEOUS interrogations

- e. Verify that the result for the ERRORNEOUS test interrogation signal for MON1 and MON2 is all green.

6.5.13. ANTENNA AND EQUIPMENT GROUND RESISTANCE CHECK

6.5.13.1. Standards and Tolerances

	Test point	Standard	Tolerance
Ground Resistance	Ground point	less than 10Ω	-

6.5.13.2. Test Equipment

- Suitable Ground Resistance Tester (a.k.a. Earth Tester)

6.5.13.3. Procedure

- Set up the Ground Resistance Tester, consulting with the instruction manual. Typical three-electrode type earth tester is installed as shown in Figure 6-24.
- Measure and record the ground resistance.

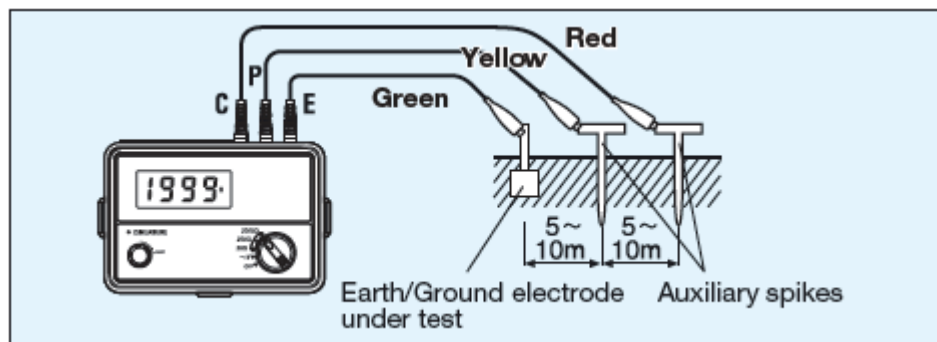


Figure 6-24. Earth Tester Setup

6.6. OTHER ON-SITE MAINTENANCE

The other maintenance tasks include periodic cleaning and inspection of the shelter and equipment.

It also includes inspection of batteries for corrosion or any kind of deterioration. Periodic checks and tightening of RF and battery connections should also be done to ensure continuous and efficient operation of the equipment.

Check air filters in shelter vent and air conditioning unit. Clean or replace any part if required.

Check the installed environmental sensors to ensure normal functioning.

Perform a complete visual inspection of the equipment, antenna array and distribution network assembly.

6.6.1. EQUIPMENT CLEANING

Tools required:

- Cleaning aids: brushes and dusters with antistatic material
- Vacuum cleaner
- Others: suitable LCD cleaner liquid or isopropyl alcohol (IPA) and clean water and as required

Notify the air traffic control authorities of possible interruption of DME navigation service

Check the shelter temperature sensor if available and verify the normal operation of air conditioner, heater and fan.

Clean dust from cabinet using air or soft brush. If required, turn off switches on DC/DC and AC/DC converters.

Check the air filters of ventilator(s) or air conditioner. If accumulation of dirt is detected, the air filters should be replaced by new ones to prevent clog.

In case the new filter is out of stock, on an exceptional basis, the dirt may be banged out of the old filter for a limited reuse of it.

In the shelter, do not use a broom but use a vacuum cleaner to avoid transferring dirt into the equipment cabinet and its associated parts.

Remove layers of dirt on the LMI panel with a cloth moistened a little with a suitable LCD cleaner liquid. If the LCD cleaner liquid is not available, a mixture of isopropyl alcohol and clean water with a ratio of 1:1 can be used instead.

WARNING

Isopropyl Alcohol (IPA) is a flammable liquid.

If a visual check detects dust on units or PCB's, they should be dusted using a soft brush with the aid of a vacuum cleaner.

On completion of cleaning work, turn on the equipment and verify normal operation.

6.6.2. DAMAGE CHECK FOR EQUIPMENT, SHELTER AND ANTENNA

- Check tightness of all internal and external RF cable connections
- Check antenna distribution, connectors and supporting pedestals
- Remove the dust, corrosion and foreign dirt, if any
- Check the mounting solidity and tightness of all mechanical parts (nuts & bolts)
- Check for sign of corrosion of antenna elements including any sign of salinity
- Check for signs of cracks or breakage of coaxial feed cables
- Check antenna obstruction lights

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SECTION 7

CORRECTIVE MAINTENANCE

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7. CORRECTIVE MAINTENANCE

This section includes information for locating and troubleshooting faults at the site for normal system operation.

7.1. FAULT ISOLATION

Identify the symptom and locate the fault

- List the symptoms monitored and described by the technicians.
- Check the status of operation with PMDT.
- Locate the faulty module or unit.
- Perform a careful visual inspection of the suspected module or unit.
- Check for signs of excessive heat
- Check that all ICs are firmly seated in their sockets and that the ICs have no bent pins
- Check that PCB edge connectors are clean and seated fully
- Check the ribbon-cable connectors (if any)
- Check the remote control line
- Check power supply which is frequently a very common source of faults.
- If the fault is suspected outside the equipment cabinet or shelter, check coaxial feed cables for antenna distribution.
- Identify and use the proper test equipment.
- Check signals of the suspected module or unit using the test equipment.

7.2. TROUBLESHOOTING

7.2.1. INTRODUCTION

This section provides general information for the site specialists to restore a faulty system to a ready-to-operate condition with the least possible delay.

7.3. LRU REPLACEMENT PROCEDURES

7.3.1. INTRODUCTION

This section describes the procedures to replace the line replaceable unit (LRU).

The MOPIENS 300 DME equipment is designed based on a maintenance concept that simplifies on-site maintenance, maximizing availability of equipment and minimizing life-cycle costs. On-site repair is reduced to the use of the equipment's built-in capabilities to isolate equipment faults to the LRU level. The faulty LRU can be replaced by on-site maintenance personnel, restoring the equipment to service quickly. The faulty LRU can then be sent to the manufacturer's service center or an authorized depot facility for repair.

7.3.2. CSP **1A1A1**

7.3.2.1. Typical Failure Symptoms

- LED indicators on the front panel are not lit at all.
- LED indicators for specific function is not illuminated when it is supposed to be so.
- Push buttons for specific function do not work as it is supposed to do so.
- No audible sound alert is heard even if there is a persistent alarm.
- The volume control on the front face does not work.

7.3.2.2. Necessary Tools

None.

7.3.2.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Open the front door of the equipment cabinet.
- c. Disconnect the cable 1W64 from the back of the CSP board.
- d. Unscrew the two captive screws on top and bottom and remove the CSP assembly.
- e. Connect the 1W64 to the replacement CSP.
- f. Holding the CSP in position, fasten the two mounting screws on the top and bottom.
- g. Return the keylock switch to the original position.

7.3.2.4. Checkup after Replacement

- Press the **LAMP TEST** button and check if all the LED indicators are lit up and continuous beep sound is coming out of the loud speaker on the CSP circuit board.
- Press the **BYPASS** button and check to see if the indicator on the button is illuminated and the monitor bypass indication is observed on the LMI or PMDT.

7.3.3. SCU 1A1A3, 1A1A4

7.3.3.1. Typical Failure Symptoms

- The Fault indicator on the SCU is illuminated.
- The SCU that used be the slave has been changed over to the master automatically.
- DC/DC power supplies has been switched off and attempts to switch them on keep failing.
- Connection with LMI, RCU or PMDT is not established.

7.3.3.2. Necessary Tools

None.

7.3.3.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. If the SCU to replace is the master, press the reset button on the slave SCU, so that the slave takes over before unplugging of the SCU.
- c. Open the front door of the equipment cabinet.
- d. Pull up the upper ejector lever and pull down the lower lever together at the same time.
- e. Holding the ejector levers, carefully take the unit out of the rack.
- f. Insert the replacement unit into the rack carefully.
- g. Push down the upper ejector lever and push up the lower lever together at the same time until the unit is firmly mated to the backplane.
- h. Wait approximately 30 seconds or so until the configuration data is copied into the new SCU.
- i. Press the reset switch on the master SCU so that the newly inserted SCU becomes the master.
- j. Turn the key-lock switch to **LOCAL** position.

7.3.3.4. Checkup after Replacement

- a. Press the Bypass button on the CSP and check to see if it works.
- b. Touch on the Bypass button on the LMI and check to see if it works.
- c. Verify that the readings and settings on the PMDT are as expected.
- d. Click on the Bypass button on PMDT and check to see if it works.
- e. Using PMDT or LMI, check the hardware revision code and the software version number and verify that the displayed data is correct.

7.3.4. EMU **1A1A4**

7.3.4.1. Typical Failure Symptoms

- Environmental alarms are not activated even when such conditions are actually encountered and sensors are functioning.
- Expansion input/output do not work.

7.3.4.2. Necessary Tools

None.

7.3.4.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Open the front door of the equipment cabinet.
- c. Pull up the upper ejector lever and pull down the lower lever together at the same time.
- d. Holding the ejector levers, carefully take the unit out of the rack.
- e. Insert the replacement unit into the rack carefully.
- f. Pull down the upper ejector lever and pull up the lower lever together at the same time until the unit is firmly mated to the backplane.

7.3.4.4. Checkup after Replacement

- a. Connect and login to PMDT.
- b. Go to **Main → Environmental** and check if the sensors status is correct.
- c. If intrusion sensor is installed, make it trip by simulating intrusion. Check to see if **ENVIRONMENTAL** alarm indicator is activated on CSP, LMI or PMDT.
- d. Make it normal and check the CSP, LMI or PMDT to see if the **ENVIRONMENTAL** alarm is deactivated.
- e. If other sensors are installed, verify them as well to see if they work correctly.

7.3.5. DC/DC-A *1A1A5, 1A1A6*

7.3.5.1. Typical Failure Symptoms

- The green POWER indicator LED on the front of the unit does not illuminate.
- Out of tolerance voltages are measured at the test points on the front of the unit.

7.3.5.2. Necessary Tools

- Digital Multimeter

7.3.5.3. Replacement Procedure

- a. Open the front door of the equipment cabinet.
- b. Pull up the upper ejector lever and pull down the lower lever together at the same time.
- c. Holding the ejector levers, carefully take the unit out of the rack.
- d. Insert the replacement unit into the rack carefully.
- e. Push down the upper ejector lever and push up the lower lever together at the same time until the unit is firmly mated to the backplane.

7.3.5.4. Checkup after Replacement

- a. Verify that the POWER indicator LED on the front illuminates.
- b. Measure the voltages on the test points on the front and verify that all the voltages are within tolerance.
- c. Temporarily remove the other DC/DC-A unit so that the replaced DC/DC-A unit runs alone. Check to see if that all the units in the AUX subrack are powered up and work as normal.

7.3.6. MODEM 1A1A7, 1A1A8

7.3.6.1. Typical Failure Symptoms

- The RCU linked using the modem do not show the system status and all the LED indicators on the front face are flashing.
- The PMDT linked using the modem cannot establish a connection to the equipment.

7.3.6.2. Necessary Tools

None.

7.3.6.3. Replacement Procedure

- a. Open the front door of the equipment cabinet.
- b. Pull up the upper ejector lever and pull down the lower lever together at the same time.
- c. Holding the ejector levers, carefully take the unit out of the rack.
- d. Insert the replacement unit into the rack carefully.
- e. Push down the upper ejector lever and push up the lower lever together at the same time until the unit is firmly mated to the backplane.

7.3.6.4. Checkup after replacement

- a. Verify that the RCU linked using the modem displays correct system status.
- b. Verify that the control buttons work correctly as expected.

7.3.7. DPX-MSC 1A2A1, 1A3A1

7.3.7.1. Typical Failure Symptoms

- The receiver sensitivity is degraded abruptly, which may result in:
 - o Reply Efficiency alarm
 - o Time Delay alarm

7.3.7.2. Necessary Tools

None.

7.3.7.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. If the transponder of the unit to replace is on antenna, put it on dummy load using CSP, LMI or PMDT.
- c. Turn off the transponder using CSP, LMI or PMDT.
- d. Open the front door of the equipment cabinet.
- e. Fully unfasten the upper and lower captive screws on the front panel of the unit to replace.
- f. Holding the carrying handle, carefully take the unit out of the rack.
- g. Insert the replacement unit into the rack carefully.
- h. Push it a little bit hard until the unit is firmly mated to the backplane until the unit is firmly mated to the backplane.
- i. Fully fasten the upper and lower captive screws on the front panel of the replacement unit.
- j. Turn on the transponder and enable the RF output back using CSP, LMI or PMDT.

7.3.7.4. Adjustment after replacement

- a. Turn the keylock switch to **MAINT** position.
- b. Put the transponder with the replaced unit on antenna.
- c. Connect PMDT to the equipment.
- d. Go to **Maintenance → Calibration → Calibration**.
- e. Skip *Step1. Transponder Drive Level Optimization*.
- f. Skip *Step2. Transponder Power Output Calibration*.
- g. Skip *Step3. Transponder Processing Delay Compensation*.
- h. Skip *Step4. Auto Time Delay Calibration*.
- i. Run *Step5. Monitor Interrogation Level Calibration* for both MON1 and MON2.
- j. Run *Step6. Time Delay Monitor Reading Calibration* for MON1 and MON2.
- k. Skip *Step7. ERP Monitor Reading Calibration*.
- l. Skip *Step8. VSWR Monitor Reading Calibration*.
- m. Run *Step9. Power Output Monitor Reading* for MON1 and MON2.
- n. Run *Step10. Receiver Threshold Calibration* for MON1

- o. Close the Calibration dialog box.
- p. Click on **Save to Flash** button.
- q. Turn the keylock switch to **REM** position.

7.3.8. RXU 1A2A2, 1A3A2

7.3.8.1. Typical Failure Symptoms

- The receiver sensitivity is degraded abruptly, which may result in:
 - o Reply Efficiency alarm
 - o Time Delay alarm
- The POWER indicator LED on the front panel of RXU does not illuminate.

7.3.8.2. Necessary Tools

None.

7.3.8.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. If the transponder of the unit to replace is on antenna, put it on dummy load using CSP, LMI or PMDT.
- c. Turn off the transponder using CSP, LMI or PMDT.
- d. Open the front door of the equipment cabinet.
- e. Verify that the **POWER** indicator on the front panel of the unit does not illuminate.
- f. Fully unfasten the upper and lower captive screws on the front panel of the unit to replace.
- g. Holding the carrying handle, carefully take the unit out of the rack.
- h. Insert the replacement unit into the rack carefully.
- i. Push it a little bit hard until the unit is firmly mated to the backplane.
- j. Fully fasten the upper and lower captive screws on the front panel of the replacement unit.
- k. Turn on the transponder and enable the RF output back using CSP, LMI or PMDT.

7.3.8.4. Adjustment after Replacement

- a. Turn the keylock switch to **MAINT** position.
- b. Put the transponder with the replaced unit on antenna.
- c. Connect PMDT to the equipment.
- d. Go to **Maintenance → Calibration** and click on the **Start** button.
- e. Skip *Step1. Transponder Drive Level Optimization*.
- f. Skip *Step2. Transponder Power Output Calibration*.
- g. Skip *Step3. Transponder Processing Delay Compensation*.
- h. Skip *Step4. Auto Time Delay Calibration*.
- i. Run *Step5. Monitor Interrogation Level Calibration* for both MON1 and MON2.
- j. Run *Step6. Time Delay Monitor Reading Calibration* for MON1 and MON2.
- k. Skip *Step7. ERP Monitor Reading Calibration*.
- l. Skip *Step8. VSWR Monitor Reading Calibration*.

- m. Run *Step9. Power Output Monitor Reading* for MON1 and MON2.
- n. Run *Step10. Receiver Threshold Calibration* for MON1
- o. Close the Calibration dialog box.
- p. Click on **Save to Flash** button.
- q. Turn the keylock switch to **REM** position.

7.3.9. HPA 1A2A3, 1A3A3

7.3.9.1. Typical Failure Symptoms

- No power output or much lower power output than normal, which may result in:
 - o Power Output Alarm
 - o Reply Efficiency Alarm
 - o Time Delay Alarm
 - o All alarms in System Parameters
 - o All alarms in Pulse Shape and Spacing
 - o All alarms in Frequency and Power except Radio Frequency Alarm
 - o Identification Code alarm
- The green indicator LED on the HPA front panel, **RF** does not blinking.
- The red indicator LED on the HPA front panel, **FAULT** illuminates.
- The green indicator LED on the HPA front panel **POWER** does not illuminates.

7.3.9.2. Necessary Tools

None.

7.3.9.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. If the transponder of the unit to replace is on antenna, put it on dummy load using CSP, LMI or PMDT.
- c. Turn off the transponder using CSP, LMI or PMDT.
- d. Open the front door of the equipment cabinet.
- e. Verify that the **POWER** indicator on the front panel of the unit does not illuminate.
- f. Fully unfasten the upper and lower captive screws on the front panel of the unit to replace.
- g. Holding the carrying handle, carefully take the unit out of the rack.
- h. Insert the replacement unit into the rack carefully.
- i. Push it a little bit hard until the unit is firmly mated to the backplane.
- j. Fully fasten the upper and lower captive screws on the front panel of the replacement unit.
- k. Turn on the transponder and enable the RF output back using CSP, LMI or PMDT.

7.3.9.4. Adjustment after Replacement

- a. Turn the keylock switch to **MAINT** position.
- b. Turn the RF output of the transponder off (disabled) using CSP, LMI or PMDT.
- c. Disconnect the RF cable from the DME transponder RF port, 1J1.
- d. Connect suitable RF peak power meter to 1J1.
- e. Turn the RF output of the transponder on (enabled) using CSP, LMI or PMDT.

- f. Read and record the peak power output on the peak power meter
- g. Turn the RF output of the transponder off (disabled) using CSP, LMI or PMDT.
- h. Disconnect the peak power meter from the connector 1J1 on the equipment.
- i. Reconnect the RF cable to the DME transponder RF port, 1J1.
- j. Turn the RF output on the transponder on (enabled) using CSP, LMI or PMDT.
- k. Connect PMDT to the equipment.
- l. Put the transponder with the replaced unit on antenna.
- m. Go to **Maintenance** → **Calibration** and click on the **Start** button.
- n. Run Step1. *Transponder Drive Level Optimization*
- o. In Step2. *Transponder Power Output Calibration*, Enter the measured value of the peak power and click on the **Run** button.
- p. Skip Step3. *Transponder Processing Delay Compensation*.
- q. Run Step4. *Auto Time Delay Calibration*.
- r. Close the Calibration dialog box.
- s. Click on **Save to Flash** button.
- t. Turn the keylock switch to **REM** position.

7.3.10. TXU *1A2A4, 1A3A4*

7.3.10.1. Typical Failure Symptoms

- No power output or much lower power output than normal, which may result in:
 - o Power Output Alarm
 - o Reply Efficiency Alarm
 - o Time Delay Alarm
 - o All alarms in System Parameters
 - o All alarms in Pulse Shape and Spacing
 - o All alarms in Frequency and Power except Radio Frequency Alarm
 - o Identification Code alarm
- The green indicator LED on the TXU front panel, **RF** does not blinking.
- The red indicator LED on the TXU front panel, **FAULT** illuminates.
- The green indicator LED on the TXU front panel **POWER** does not illuminates

7.3.10.2. Necessary Tools

None.

7.3.10.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. If the transponder of the unit to replace is on antenna, put it on dummy load using CSP, LMI or PMDT.
- c. Turn off the transponder using CSP, LMI or PMDT
- d. Open the front door of the equipment cabinet.
- e. Verify that the **POWER** indicator on the front panel of the unit does not illuminate.
- f. Fully unfasten the upper and lower captive screws on the front panel of the unit to replace.
- g. Holding the carrying handle, carefully take the unit out of the rack.
- h. Insert the replacement unit into the rack carefully.
- i. Push it a little bit hard until the unit is firmly mated to the backplane.
- j. Fully fasten the upper and lower captive screws on the front panel of the replacement unit.
- k. Turn on the transponder and enable the RF output back using CSP, LMI or PMDT.

7.3.10.4. Adjustment after Replacement

- a. Turn the keylock switch to **MAINT** position.
- b. Turn the RF output of the transponder off (disabled) using CSP, LMI or PMDT.
- c. Disconnect the RF cable from the DME transponder RF port, 1J1.
- d. Connect suitable RF peak power meter to 1J1.
- e. Turn the RF output of the transponder on (enabled) using CSP, LMI or PMDT.

- f. Read and record the peak power output on the peak power meter
- g. Turn the RF output of the transponder off (disabled) using CSP, LMI or PMDT.
- h. Disconnect the peak power meter from the connector 1J1 on the equipment.
- i. Reconnect the RF cable to the DME transponder RF port, 1J1.
- j. Turn the RF output on the transponder on (enabled) using CSP, LMI or PMDT.
- k. Connect PMDT to the equipment.
- l. Put the transponder with the replaced unit on antenna.
- m. Go to **Maintenance** → **Calibration** and click on the **Start** button.
- n. Run Step1. *Transponder Drive Level Optimization*
- o. In Step2. *Transponder Power Output Calibration*, Enter the measured value of the peak power and click on the **Run** button.
- p. Skip Step3. *Transponder Processing Delay Compensation*.
- q. Run Step4. *Auto Time Delay Calibration*.
- r. Close the Calibration dialog box.
- s. Click on **Save to Flash** button.
- t. Turn the keylock switch to **REM** position.

7.3.11. DC/DC 1A1A2A5, 1A1A3A5

7.3.11.1. Typical Failure Symptoms

- TCU, TXU, RXU and HPA are not powered up.
- The green POWER indicator LED on the front of the unit does not illuminate.
- Out of tolerance voltages are measured at the test points on the front of the unit.

7.3.11.2. Necessary Test Instrument

DVM

7.3.11.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. If the transponder of the unit to replace is on antenna, put it on dummy load using CSP, LMI or PMDT.
- c. Turn off the transponder using CSP, LMI or PMDT.
- d. Open the front door of the equipment cabinet.
- e. Verify that the **POWER** indicator on the front panel of the unit does not illuminate.
- f. Fully unfasten the upper and lower captive screws on the front panel of the unit to replace.
- g. Holding the carrying handle, carefully take the unit out of the rack.
- h. Insert the replacement unit into the rack carefully.
- i. Push it a little bit hard until the unit is firmly mated to the backplane.
- j. Fully fasten the upper and lower captive screws on the front panel of the replacement unit.
- k. Turn on the transponder and enable the RF output back using CSP, LMI or PMDT.

7.3.11.4. Checkup after Replacement

- a. Verify that the POWER indicator LED on the front illuminates.
- b. Measure the voltages on the test points on the front and verify that all the voltages are within tolerance.
- c. Check to see if that all the units in the transponder subrack are powered up and work as normal.

7.3.12. TCU **1A2A6, 1A3A6**

7.3.12.1. Typical Failure Symptoms

- Even if there is valid interrogation pulse received, reply pulse pair is not produced at all.
- All the executive monitor alarms are activated.
- Transponder status data is not displayed on PMDT or LMI.
- The green **POWER** LED indicator on the front of the unit does not illuminate.
- The red **FAULT** LED indicator on the front of the unit illuminates.

7.3.12.2. Necessary Tools

None.

7.3.12.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. If the transponder of the unit to replace is on antenna, put it on dummy load using CSP, LMI or PMDT.
- c. Turn off the transponder using CSP, LMI or PMDT.
- d. Open the front door of the equipment cabinet.
- e. Verify that the **POWER** indicator on the front panel of the unit does not illuminate.
- f. Pull up the upper ejector lever and pull down the lower lever together at the same time.
- g. Holding the ejector levers, carefully take the unit out of the rack.
- h. Insert the replacement unit into the rack carefully.
- i. Push down the upper ejector lever and push up the lower lever together at the same time until the unit is firmly mated to the backplane.
- j. Turn on the transponder and enable the RF output back using CSP, LMI or PMDT.

7.3.12.4. Checkup after Replacement

- a. Using PMDT, Go to **Main → Executive Monitor** and verify that all the monitor parameters are as expected.
- b. If standby transponder is available, Go to **Main → Standby Monitor** and verify that all the monitor parameters are as expected.

7.3.13. RFG *1A2A7, 1A3A7*

7.3.13.1. Typical Failure Symptoms

- No test interrogation signal is produced.
- All the monitor alarms except the *Frequency and Power alarm* and *ERP alarm* are activated.
- The green **POWER** LED indicator on the unit does not illuminate.
- The green RF LED indicator on the front of the unit does not blink.
- The red FAULT LED indicator on the front of the unit illuminates.

7.3.13.2. Necessary Tools

None.

7.3.13.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Open the front door of the equipment cabinet.
- c. Fully unfasten the upper and lower captive screws on the front panel of the unit to replace.
- d. Holding the carrying handle, carefully take the unit out of the rack.
- e. Insert the replacement unit into the rack carefully.
- f. Push it a little bit hard until the unit is firmly mated to the backplane.
- g. Fully fasten the upper and lower captive screws on the front panel of the replacement unit.

7.3.13.4. Adjustment after Replacement

- a. Turn the keylock switch to **MAINT** position.
- b. Connect PMDT to the equipment.
- c. Go to **Maintenance → Calibration** and click on **Start** button.
- d. Skip Step1. *Transponder Drive Level Optimization*.
- e. Skip Step2. *Transponder Power Output Calibration*.
- f. Skip Step3. *Transponder Processing Delay Compensation*.
- g. Skip Step4. *Auto Time Delay Calibration*.
- h. Run Step5. *Monitor Interrogation Level Calibration* for the monitor of the replaced RFG.
- i. Run Step6. *Time Delay Monitor Reading Calibration* for the monitor of the replaced RFG.
- j. Skip Step7. *ERP Monitor Reading Calibration*.
- k. Skip Step8. *VSWR Monitor Reading Calibration*.
- l. Run Step9. *Power Output Monitor Reading* for the monitor of the replaced RFG.
- m. Close the Calibration dialog box.
- n. Click on **Save to Flash** button.
- o. Turn the keylock switch to **REM** position.

7.3.14. MON *1A2A8, 1A3A8*

7.3.14.1. Typical Failure Symptoms

- All the monitor status is not displayed or displayed with garbage data on LMI and PMDT.
- All the monitor parameters are blank or not refreshed for extended period of time.
- Monitor self-test failure alarm is reported.
- Monitor certification test fails.
- Normal service is available with the transponder, if the monitor is bypassed.

7.3.14.2. Necessary Tools

None.

7.3.14.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Open the front door of the equipment cabinet.
- c. Pull up the upper ejector lever and pull down the lower lever together at the same time.
- d. Holding the ejector levers, carefully take the unit out of the rack.
- e. Insert the replacement unit into the rack carefully.
- f. Push down the upper ejector lever and push up the lower lever together at the same time until the unit is firmly mated to the backplane.

7.3.14.4. Adjustment after Replacement

- a. Turn the keylock switch to **MAINT** position.
- b. Connect PMDT to the equipment.
- c. Go to **Maintenance** → **Calibration** and click on **Start** button.
- d. Skip Step1. *Transponder Drive Level Optimization*.
- e. Skip Step2. *Transponder Power Output Calibration*.
- f. Skip Step3. *Transponder Processing Delay Compensation*.
- g. Skip Step4. *Auto Time Delay Calibration*.
- h. Run Step5. *Monitor Interrogation Level Calibration* for the monitor of the replaced RFG.
- i. Run Step6. *Time Delay Monitor Reading Calibration* for the monitor of the replaced RFG.
- j. Skip Step7. *ERP Monitor Reading Calibration*.
- k. Run Step8. *VSWR Monitor Reading Calibration*.
- l. Run Step9. *Power Output Monitor Reading* for the monitor of the replaced RFG.
- m. Close the Calibration dialog box.
- n. Click on **Save to Flash** button.
- o. Turn the keylock switch to **REM** position.

7.3.14.5. Checkup after Replacement

- a. Using PMDT, Go to **Main → Executive Monitor** and verify that all the monitor parameters are as expected.
- b. If standby transponder is available, Go to **Main → Standby Monitor** and verify that all the monitor parameters are as expected.
- c. Using PMDT, Go to **Main → Monitor-Self-Test** and verify that monitor self-test result is as expected and all green.

7.3.15. FAN *1A2A9, 1A3A9*

7.3.15.1. Typical Failure Symptoms

- FAN on/off control from LMI or PMDT does not work
- FAN on/off control works but the fan does not run.
- The green POWER LED indicator on the front of the unit does not illuminate.
- The one or more red FAULT LED indicators on the front of the unit illuminates.

7.3.15.2. Necessary Tools

- Phillips Screwdriver (+)

7.3.15.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Open the front door of the equipment cabinet.
- c. Disconnect the cable 1A2W1 or 1A3W1 on the front of the unit.
- d. Unscrew the two mounting bolts on the left and right side of the unit.
- e. Keeping the cable out of the way, take the unit out of the rack.
- f. Insert the replacement fan unit into the rack.
- g. Put the two mounting screws in place and fasten them.
- h. Reconnect the cable 1A2W1 or 1A3W1 into the connector on the front of the unit.

7.3.15.4. Checkup after Replacement

- a. Verify that the green POWER LED indicator on the front illuminates.
- b. Verify that none of the red FAULT LED indicators on the front of the unit illuminates.
- c. Using PMDT or LMI, verify that the manual activation of FAN on/off control works as expected.

7.3.16. PMU *1A4A1, 1A4A2*

7.3.16.1. Typical Failure Symptoms

- The seven-segment LED indicators on the faceplate of the PMU does not illuminate or displays garbage data.
- AC/DC status and backup battery status on the LMI or PMDT is not displayed or incorrect.
- The battery voltage is lower than nominal and is not charged, despite AC main power is available.

7.3.16.2. Necessary Tools

None.

7.3.16.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Open the front door of the equipment cabinet.
- c. Turn off the **BATT 1** or **BATT 2** circuit breaker switch of the PMU to replace.
- d. Fully unfasten the left and right captive screws on the front panel of the unit to replace.
- e. Holding the carrying handle, carefully take the unit out of the rack.
- f. Insert the replacement unit into the rack carefully.
- g. Push it a little bit hard until the unit is firmly mated to the backplane.
- h. Fully fasten the left and right captive screws on the front panel of the replacement unit.
- i. Turn on the **BATT 1** or **BATT 2** circuit breaker switch back.

7.3.16.4. Checkup after Replacement

- a. Verify that the seven-segment LED indicators are turned on.
- b. Press the buttons on the faceplate of the unit one by one and verify that correct data corresponding the button pressed is displayed.
- c. Turn off the AC main power switch (circuit breaker) and check to see if battery backup is working as expected. Also battery discharge current is correctly displayed on the 7-segment LED indicator.
- d. Turn on the AC main power switch back and check to see if the battery is being charged and the charging current is correctly displayed on the 7-segment LED indicator.

7.3.17. AC/DC *1A5A2, 1A5A4*

7.3.17.1. Typical Failure Symptoms

- Even if the main AC power supply input is available and the AC power switch is on, the green **AC** indicator LED does not illuminate.
- The red **DC fault** LED indicator illuminates.
- The status of the AC/DC is not displayed or incorrect.

7.3.17.2. Necessary Tools

None.

7.3.17.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Open the front door of the equipment cabinet.
- c. Push up and hold the release knob and pull out the handle of the unit to replace.
- d. Holding the handle, carefully take the unit out of the rack.
- e. Insert the replacement unit into the rack.
- f. Push it a little bit hard until the unit is firmly mated to the backplane.

7.3.17.4. Checkup after Replacement

- a. Verify that the green LED indicator **AC** on the front of the unit illuminates.
- b. Verify that the red LED indicator **DC** fault on the front of the unit does not illuminate.
- c. Verify that AC/DC status information is correct on LMI or PMDT.

7.4. OFF-SITE MAINTENANCE OF LRU's

Most of the modules equipped in this equipment are LRU's and not user-serviceable. The module and unit repair is to be performed off site; i.e. at the factory or authorized repair shop. Defective modules are, therefore, to be returned to the factory or repair shop.

Customers are requested to contact an authorized representative of MOPIENS in advance of sending modules to repair and get appropriate RMA (Return Merchant Authorization) number for reference.

Modules to be shipped to the factory or repair shop for service must always be identified by module number, serial number and the RMA number.

Special care must be given to the packing of module to prevent any possible damage during its transportation to the factory or back to the site.

7.5. SPECIAL REPLACEMENT PROCEDURES

7.5.1. INTRODUCTION

This section describes the procedures to replace the units and modules that require special attention. These units and modules are typically mounted on the side walls at rear side of the electronics cabinet. Those units and modules are designed to last for the whole lifecycle of the electronics cabinet without replacement or special maintenance.

Nonetheless, if it ever come to replace any of those units and modules, the procedures described in this section should be carried out with more care than normal maintenance.

7.5.2. ANTENNA 2

7.5.2.1. Typical Failure Symptoms

- Antenna **VSWR** alarm is activated
- **Effective Radiated Power** alarm is activated.

7.5.2.2. Necessary Tools

- Torque wrench for type N connectors.
- Adjustable wrench.

7.5.2.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Open the front door of the equipment cabinet.
- c. Turn off system completely - both **BATT** and **AC MAIN** switches.
- d. Disconnect the RF feeder cables from the antenna.
- e. Replace the antenna
- f. Reconnect the RF feeder cables to the antenna.
- g. Turn on the system back.

7.5.2.4. Adjustment after Replacement

- a. Turn the keylock switch to **MAINT** position.
- b. Connect PMDT to the equipment.
- c. Go to **Maintenance → Calibration** and click on the **Start** button.
- d. Skip Step1. *Transponder Drive Level Optimization*
- e. Skip Step2. *Transponder Power Output Calibration*
- f. Skip Step3. *Transponder Processing Delay Compensation.*
- g. Run Step4. *Auto Time Delay Calibration.*
- h. Run Step5. *Monitor Interrogation Level Calibration.*
- i. Run Step6. *Time Delay Monitor Reading Calibration.*
- j. Run Step7. *ERP Monitor Reading Calibration.*
- k. Close the Calibration dialog box.
- l. Click on **Save to Flash** button.
- m. Turn the keylock switch to **REM** position.

7.5.3. LMI: IEC-1000 **1A6A1**

7.5.3.1. Typical Failure Symptoms

- The LMI does not turn on at all.
- The operating system or application software fails to boot up:
 - o The application screen does not come up.
 - o The screen freezes showing the boot logo.
 - o The LMI keeps restarting without a warning.
- The touchscreen interface does not work.
- Ethernet connection to the equipment keeps dropping or takes extremely long or unstable.

7.5.3.2. Necessary Tools

- Screwdriver PH #2 (+)

7.5.3.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Open the front door of the equipment cabinet.
- c. Fully unfasten the two captive screws on the protective cover of the LMI assembly, which is mounted on the front door of the equipment cabinet. Open the cover and leave it hanging down.
- d. Disconnect the cables from the IEC-1000: 1A6W1, 1A6W2, 1A6W4 1W1 and 1W2.
- e. Remove the mounting screws on the IEC-1000.
- f. Carefully remove the IEC-1000.
- g. Put and hold the replacement IEC-1000 in place.
- h. Put the mounting screws back and fasten them all firmly.
- i. Connect the cables back: 1A6W1, 1A6W2, 1A6W4 1W1 and 1W2.
- j. Close the protective cover and fasten the two captive screws.

7.5.3.4. Checkup after Replacement

- a. Verify that the operating system and the application software boots up normally.
- b. Verify that the touch screen interface is working by trying to control something simple, e.g. put monitors in bypass.
- c. Verify that PMDT can connect to the equipment via the Ethernet interface.
- d. Verify that RCU with Ethernet interface can connect to the equipment.

7.5.4. LMI: LMI-IF **1A6A2**

7.5.4.1. Typical Failure Symptoms

- The four indicators on the front door of the equipment: NORMAL, WARNING, ALARM and MAINT, do not illuminate at all, or illuminate all the time or dim.

7.5.4.2. Necessary Tools

- Screwdriver PH #2 (+)

7.5.4.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Open the front door of the equipment cabinet.
- c. Fully unfasten the two captive screws on the protective cover of the LMI assembly, which is mounted on the front door of the equipment cabinet. Open the cover and leave it hanging down.
- d. Disconnect the cables from the LMI-IF board: 1A6W3, 1W3 and 1W4.
- e. Remove the mounting screws on the LMI-IF board.
- f. Carefully remove the LMI-IF board.
- g. Put and hold the replacement LMI-IF board in place.
- h. Put the mounting screws back and fasten them all firmly.
- i. Connect the cables back: 1A6W3, 1W3 and 1W4.
- j. Close the protective cover and fasten the two captive screws.

7.5.4.4. Checkup after Replacement

- a. Verify that the one of the status indicators **.NORMAL**, **WARNING** or **ALARM** illuminates.
- b. Turn the keylock switch to **MAIN** position, and the **MAINT** indicator illuminates.
- c. Create a warning/alarm condition temporarily, e.g. by turning off the transponder on antenna, and verify that the **WARNING** or **ALARM** indicator illuminate.
- d. Clear the warning/alarm condition that has been created temporarily.
- e. Turn the keylock switch back to **REM** position.

7.5.5. IFB 1A7

7.5.5.1. Typical Failure Symptoms

- PMDT or RCU via the RS-232C serial data link does not work and gets disconnected.
- PMDT or RCU via the modem line does not work and gets disconnected.
- PMDT or RCU via the Ethernet interface does not work and gets disconnected.
- PMDT via the USB interface does not work and gets disconnected.
- The environmental sensors are not read correctly.
- The IDENT synchronization with the collocated equipment does not work.

7.5.5.2. Necessary Tools

- Screwdriver PH #2 (+)

7.5.5.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Open the rear door of the equipment cabinet.
- c. Disconnect all the external interface cables.
- d. Disconnect the cables from IFB: 1W5, 1W6, 1W7, 1W8.
- e. Remove the plastic label plate on the IFB board by unscrewing the two mounting bolts.
- f. Remove the mounting screws from the IFB board.
- g. Take the IFB board out carefully.
- h. Put and hold the replacement IFB board in place.
- i. Put the mounting screws back and fasten them all firmly.
- j. Install the plastic label plate back.
- k. Connect the cables: 1W5, 1W6, 1W7, 1W8 back.
- l. Connect the external interface cables back.

7.5.5.4. Checkup after Replacement

- a. Verify that the PMDT or RCU via RS-232C serial data link works as normal.
- b. Verify that the PMDT or RCU via the modem line works as normal.
- c. Verify that the PMDT or RCU via the Ethernet interface works as normal.
- d. Verify that the PMDT via USB interface works as normal.
- e. Verify that the environmental sensors are read correctly.
- f. Verify that the IDENT synchronization with the collocated equipment works correctly.

7.5.6. VSWR MONITOR 1A8

7.5.6.1. Typical Failure Symptoms

- The VSWR monitor gives no or incorrect readings, which may result in a false alarm of antenna failure.

7.5.6.2. Necessary Tools

- Torque wrench for type SMA connectors
- Screwdriver PH #2 (+)

7.5.6.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Open the rear door of the equipment cabinet.
- c. Disconnect the cable connected to the J1 of the module, 1W119 (p/n: 0330-4118).
- d. Disconnect the cable connected to the J2 of the module, 1W118 (p/n: 0330-4117).
- e. Disconnect the cable connected to the PL1 of the module, 1W21 (p/n: 0330-4424).
- f. Disconnect the cable connected to the PL2 of the module, 1W23 (p/n: 0330-4429).
- g. Remove the four mounting screws from the module.
- h. Take the module out of the cabinet.
- i. Put and hold the replacement unit in place.
- j. Put the mounting screws back and fasten them all firmly.
- k. Connect the cables 1W23, 1W21, 1W118, 1W119 back.

7.5.6.4. Adjustment after Replacement

- a. Turn the keylock switch to **MAINT** position.
- b. Connect PMDT to the equipment.
- c. Go to **Maintenance** → **Calibration** and click on **Start** button.
- d. Skip Step1. *Transponder Drive Level Optimization*.
- e. Skip Step2. *Transponder Power Output Calibration*.
- f. Skip Step3. *Transponder Processing Delay Compensation*.
- g. Skip Step4. *Auto Time Delay Calibration*.
- h. Run Step5. *Monitor Interrogation Level Calibration* for the monitor of the replaced VSWR monitor.
- i. Run Step6. *Time Delay Monitor Reading Calibration* for the monitor of the replaced VSWR monitor.
- j. Skip Step7. *ERP Monitor Reading Calibration*.
- k. Run Step8. *VSWR Monitor Reading Calibration*.
- l. Close the Calibration dialog box.
- m. Click on **Save to Flash** button.

- n. Turn the keylock switch to **REM** position.

7.5.7. RF DETECTOR 1A9, 1A10

7.5.7.1. Typical Failure Symptoms

- Effective Radiated Power (ERP) measurement gives no or incorrect readings, which may result in a false alarm on ERP for the transponder.

7.5.7.2. Necessary Tools

- Torque wrench for type N connectors
- Screwdriver PH #2 (+)

7.5.7.3. Replacement Procedures

- Turn the keylock switch to **MAINT** position.
- Disconnect the two monitor cables connected to the antenna from the connectors, J1 and J3 on top of the equipment cabinet.
- Open the rear door of the equipment cabinet.
- Disconnect the cables from PL1 of the module, 1W20 (p/n: 0330-4423), 1W22 (p/n: 0330-4428).
- Remove the four mounting screws from the module.
- Take the module out of the cabinet.
- Put and hold the replacement unit in place.
- Put the mounting screws back and fasten them all firmly.
- Connect the cables 1W20, 1W22 back.
- Close the rear door of the equipment cabinet.
- Connect the two monitor cables connected to the antenna back to the connectors, J1 and J3 on top of the equipment cabinet.

7.5.7.4. Adjustment after Replacement

- Turn the keylock switch to **MAINT** position.
- Connect PMDT to the equipment.
- Go to **Maintenance** → **Calibration** and click on **Start** button.
- Skip Step1. *Transponder Drive Level Optimization*.
- Skip Step2. *Transponder Power Output Calibration*.
- Skip Step3. *Transponder Processing Delay Compensation*.
- Skip Step4. *Auto Time Delay Calibration*.
- Run Step5. *Monitor Interrogation Level Calibration* for the monitor of the replaced VSWR monitor.
- Run Step6. *Time Delay Monitor Reading Calibration* for the monitor of the replaced VSWR monitor.
- Run Step7. *ERP Monitor Reading Calibration*.
- Close the Calibration dialog box.

- l. Click on **Save to Flash** button.
- m. Turn the keylock switch to **REM** position.

7.5.8. LPF *1FL1*

7.5.8.1. Typical Failure Symptoms

- An ERP alarm and/or a false VSWR alarm is activated for both TX1 and TX2 while the antenna and RF feeder cables are normal.
- The receiver sensitivity for both transponders is degraded, which may result in alarms on most of the monitor parameters except for the radio frequency.
- Excessive RF harmonic components, which may result in interference to other device or equipment in the harmonic frequency range.

7.5.8.2. Necessary Tools

- Torque wrench for type N connectors
- Screwdriver PH #2 (+)

7.5.8.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Open the front door of the equipment cabinet.
- c. Turn the **BATT1** and **BATT2** switches off.
- d. Turn the **AC MAIN** switch off.
- e. Open the rear door of the equipment cabinet.
- f. Disconnect the cables from J1 of the module, 1W117 (p/n: 0330-4102).
- g. Remove the four mounting screws from the module.
- h. Unfasten the connector tied with the directional coupler 1DC3 (p/n: 3152-0102).
- i. Take the module out of the cabinet.
- j. Put and hold the replacement unit in place.
- k. Fasten the connector tied with the directional coupler 1DC3.
- l. Put the mounting screws back and fasten them all firmly.
- m. Switch on the **AC MAIN** switch back.
- n. Switch on the **BATT1** and **BATT2** switches back.

7.5.8.4. Check up after Replacement

- a. Using PMDT, Go to **Main → Executive Monitor** and verify that all the monitor parameters are as expected.
- b. If standby transponder is available, Go to **Main → Standby Monitor** and verify that all the monitor parameters are as expected.

7.5.9. DIRECTIONAL COUPLERS *1DC1, 1DC2, 1DC3*

7.5.9.1. Typical Failure Symptoms

When failure of 1DC1 or 1DC2:

- An ERP alarm and/or a false VSWR alarm is activated while the antenna and RF feeder cables are normal.
- The receiver sensitivity for the transponder which is TXP1 for 1DC1 or TXP2 for 1DC2 is degraded, which may result in alarms on most of the monitor parameters except for the radio frequency.

When failure of 1DC3:

- An ERP alarm and/or a false VSWR alarm is activated while the antenna and RF feeder cables are normal.
- Indication of antenna VSWR on LMI and PMDT is incorrect.

7.5.9.2. Necessary Tools

- Torque wrench for type N connectors
- Screwdriver PH #2 (+)

7.5.9.3. (1DC1) Replacement Procedure

- Turn the keylock switch to **MAINT** position.
- Turn the transponder #1 off.
- Open the rear door of the equipment cabinet.
- Disconnect the cable, 1W105 (p/n: 0330-4107) from J3 of the module.
- Disconnect the cable, 1W106 (p/n: 0330-4108) from J4 of the module.
- Disconnect the cable, 1W115, (p/n: 0330-4104) from J2 of the module.
- Remove the RF termination, 1RT2 (p/n: 3152-0013) from J5 of the module.
- Remove the RF termination, 1RT3 (p/n: 3152-0013) from J6 of the module.
- Remove the four mounting screws from the module.
- Fully unfasten the type N connector P1 of the module, which is directly connected to the Circulator, 1CIR1 (p/n: 3152-0078), and detach the module from the circulator.
- Attach the replacement unit to the circulator 1CIR1 by mating the connector P1 of the module.
- Put the mounting screws back and fasten them all firmly.
- Turn the transponder #1 on back.

7.5.9.4. (1DC2) Replacement Procedure

- Turn the keylock switch to **MAINT** position.
- Turn the transponder #2 off.
- Open the rear door of the equipment cabinet.
- Disconnect the cable, 1W121 (p/n: 0330-4110) from J3 of the module.

- e. Disconnect the cable, 1W116 (p/n: 0330-4109) from J4 of the module.
- f. Disconnect the cable, 1W115, (p/n: 0330-4105) from J2 of the module.
- g. Remove the RF termination, 1RT4 (p/n: 3152-0013) from J5 of the module.
- h. Remove the RF termination, 1RT5 (p/n: 3152-0013) from J6 of the module.
- i. Remove the four mounting screws from the module.
- j. Fully unfasten the type N connector P1 of the module, which is directly connected to the Circulator, 1CIR2 (p/n: 3152-0078), and detach the module from the circulator.
- k. Attach the replacement unit to the circulator 1CIR2 by mating the connector P1 of the module.
- l. Put the mounting screws back and fasten them all firmly.
- m. Turn the transponder #2 on back.

7.5.9.5. (1DC3) Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Open the front door of the equipment cabinet.
- c. Open the front door of the equipment cabinet.
- d. Turn the **BATT1** and **BATT2** switches off.
- e. Turn the **AC MAIN** switch off.
- f. Open the rear door of the equipment cabinet.
- g. Disconnect the cable 1W119 (p/n: 0330-4118) from J3 of the module
- h. Disconnect the cable 1W118 (p/n: 0330-4117) from J6 of the module.
- i. Disconnect the cable 1W101 (p/n: 0330-4101) form J2 of the module.
- j. Remove the RF termination 1RT106 (p/n: 3152-0013) from J4 of the module.
- k. Remove the RF termination 1RT107 (p/n: 3152-0013) from J5 of the module.
- l. Remove the four mounting screws from the module.
- m. Fully unfasten the type N connector P1 of the module, which is directly connected to the LPF 1FL1 (p/n: 3139-0079), and detach the module from the LPF.
- n. Attach the replacement unit to the circulator to the 1FL1 by mating the connector P1 of the module.
- o. Put the mounting screws back and fasten them all firmly.
- p. Turn the transponder #2 on back.

7.5.9.6. Adjustment after Replacement

- a. Turn the keylock switch to **MAINT** position.
- b. Connect PMDT to the equipment.
- c. Go to **Maintenance → Calibration** and click on **Start** button.
- d. Skip Step1. *Transponder Drive Level Optimization.*
- e. Skip Step2. *Transponder Power Output Calibration.*
- f. Skip Step3. *Transponder Processing Delay Compensation.*

- g. Skip Step4. *Auto Time Delay Calibration*.
- h. Run Step5. *Monitor Interrogation Level Calibration* for the monitor of the replaced VSWR monitor.
- i. Run Step6. *Time Delay Monitor Reading Calibration* for the monitor of the replaced VSWR monitor.
- j. Skip Step7. *ERP Monitor Reading Calibration*.
- k. Skip Step8. *VSWR Monitor Reading Calibration*.
- l. Run Step9. *Power Output Monitor Reading*.
- m. Run Step10. *Receiver Threshold Calibration*.
- n. Close the Calibration dialog box.
- o. Click on **Save to Flash** button.

7.5.10. COAXIAL RELAY **1RY1**

7.5.10.1. Typical Failure Symptoms

- Unable to do the transponder changeover between on antenna and on dummy load.
- Indication of which transponder is on antenna and which transponder is on dummy load on CSP, LMI and PMDT does not match with actual transponders.
- Although it appears as if the transponder has been changed on CSP, LMI and PMDT, no click sound comes out of the relay when changeover command is issued.

7.5.10.2. Necessary Tools

- Torque wrench for type SMA connectors
- Screwdriver PH #2 (+)

7.5.10.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Open the front door of the equipment cabinet.
- c. Turn the **BATT1** and **BATT2** switches off.
- d. Turn the **AC MAIN** switch off.
- e. Open the rear door of the equipment cabinet.
- f. Disconnect the cable 1W102 (p/n: 0330-4103) from J1 of the module.
- g. Disconnect the cable 1W115 (p/n: 0330-4104) from J2 of the module.
- h. Disconnect the cable 1W116 (p/n: 0330-4105) from J3 of the module.
- i. Disconnect the cable 1W117 (p/n: 0330-4102) from J4 of the module.
- j. Remove the three mounting screws from the module.
- k. Put and hold the replacement unit in place.
- l. Put the mounting screws back and fasten them all firmly.
- m. Turn the **AC MAIN** switch back on.
- n. Turn the **BATT1** and **BATT2** switches back on.
- o. Verify that the system status is stabilized as normal using CSP, LMI or PMDT.
- p. Verify that the changeover actually works and the relay click sound is heard.

7.5.11. CIRCULATOR *1CIR1, 1CIR2*

7.5.11.1. Typical Failure Symptoms

- An ERP alarm and/or a false VSWR alarm is activated while the antenna and RF feeder cables are normal.
- The receiver sensitivity for the transponder which is TXP1 for 1CIR1 or TXP2 for 1CIR2 is degraded, which may result in alarms on most of the monitor parameters except for the radio frequency.

7.5.11.2. Necessary Tools

- Torque wrench for type N connectors
- Screwdriver PH #2 (+)

7.5.11.3. (1CIR1) Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Turn the transponder #1 off.
- c. Open the rear door of the equipment cabinet.
- d. Disconnect the cable 1W103 (p/n: 0331-4101) or 1W128 (p/n: 0332-4101) from J1 of the module.
- e. Disconnect the cable 1W104 (p/n: 0331-4106) from J2 of the module
- f. Fully unfasten the type N connector P1 of the directional coupler 1DC1 (p/n: 3152-0102), which is directly attached to the J1 of the module, and detach the module from the directional coupler.
- g. Attach the replacement unit to the directional coupler 1DC1 by mating the connector P1 of the circulator to J1 of the module.
- h. Put the mounting screws back and fasten them all firmly.
- i. Turn the transponder #1 on back

7.5.11.4. (1CIR2) Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Turn the transponder #1 off.
- c. Open the rear door of the equipment cabinet.
- d. Disconnect the cable 1W123 (p/n: 0331-4101) for 310 DME or 1W130 (p/n: 0332-4101) for 320 DME from J1 of the module.
- e. Disconnect the cable 1W122 (p/n: 0331-4106) from J2 of the module
- f. Fully unfasten the type N connector P1 of the directional coupler 1DC2 (p/n: 3152-0102), which is directly attached to the J1 of the module, and detach the module from the directional coupler.
- g. Attach the replacement unit to the directional coupler 1DC2 by mating the connector P1 of the circulator to J1 of the module.
- h. Put the mounting screws back and fasten them all firmly.
- i. Turn the transponder #2 on back

7.5.11.5. Adjustment after Replacement

- j. Turn the keylock switch to **MAINT** position.
- k. Connect PMDT to the equipment.
- l. Go to **Maintenance** → **Calibration** and click on **Start** button.
- m. Skip Step1. *Transponder Drive Level Optimization*.
- n. Skip Step2. *Transponder Power Output Calibration*.
- o. Skip Step3. *Transponder Processing Delay Compensation*.
- p. Skip Step4. *Auto Time Delay Calibration*.
- q. Run Step5. *Monitor Interrogation Level Calibration* for the monitor of the replaced VSWR monitor.
- r. Run Step6. *Time Delay Monitor Reading Calibration* for the monitor of the replaced VSWR monitor.
- s. Skip Step7. *ERP Monitor Reading Calibration*.
- t. Skip Step8. *VSWR Monitor Reading Calibration*.
- u. Run Step9. *Power Output Monitor Reading*.
- v. Run Step10. *Receiver Threshold Calibration*.
- w. Close the Calibration dialog box.
- x. Click on **Save to Flash** button.

7.5.12. RF TERMINATION 1RT1

7.5.12.1. Typical Failure Symptoms

- The transponder on dummy load is not terminated with proper load impedance and it will trip the reverse power protection circuit of the TXU (for 310 DME) or HPA (for 320 DME).
- When the reverse power protection circuit is tripped, the red **FAULT** indicator on the front panel of TXU (for 310 DME) or HPA (for 320 DME) illuminates.

7.5.12.2. Necessary Tools

- Torque wrench for type N connectors
- Screwdriver PH #2 (+)

7.5.12.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Open the front door of the equipment cabinet.
- c. Open the front door of the equipment cabinet.
- d. Turn the **BATT1** and **BATT2** switches off.
- e. Turn the **AC MAIN** switch off.
- f. Open the rear door of the equipment cabinet.
- g. Disconnect the cable 1W102 (p/n: 0330-4103) from the J1 of the module.
- h. Remove the hex nut on the J1, while holding the module.
- i. Put and hold the replacement unit in place.
- j. Place and fasten the hex nut on the J1 firmly, while holding the module.
- k. Turn the **AC MAIN** switch back on.
- l. Turn the **BATT1** and **BATT2** switches back on.
- m. Verify that the red **FAULT** indicator on the front panel of the TXU (for 310 DME) or HPA (for 320 DME) does not illuminate.
- n. Verify that the system status is stabilized as normal using CSP, LMI or PMDT.

7.5.13. LIGHTNING ARRESTER *1AR1*

7.5.13.1. Typical Failure Symptoms

- An ERP alarm and/or a false VSWR alarm is activated for both TX1 and TX2 while the antenna and RF feeder cables are normal.
- The receiver sensitivity for both transponders is degraded, which may result in alarms on most of the monitor parameters except for the radio frequency.

7.5.13.2. Necessary Tools

- Torque wrench for type N connectors
- Screwdriver PH #2 (+)

7.5.13.3. Replacement Procedure

- a. Turn the keylock switch to **MAINT** position.
- b. Open the front door of the equipment cabinet.
- c. Open the front door of the equipment cabinet.
- d. Turn the **BATT1** and **BATT2** switches off.
- e. Turn the **AC MAIN** switch off.
- f. Disconnect the antenna feeder cable connected to J1 on the top plate of the equipment cabinet.
- g. Open the rear door of the equipment cabinet.
- h. Disconnect the cable 1W101 (p/n: 0330-4101) from the connector P2 of the module.
- i. Remove the hex nut on the J1 on the top plate of the equipment cabinet, while holding the module.
- j. Put and hold the replacement unit in place.
- k. Place and fasten the hex nut on the J1 firmly, while holding the module.
- l. Turn the **AC MAIN** switch back on.
- m. Turn the **BATT1** and **BATT2** switches back on.
- n. Verify that the system status is stabilized as normal using CSP, LMI or PMDT.

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SECTION 8

PARTS LIST

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8. PARTS LIST

8.1. INTRODUCTION

A list of the modules and units used in the 310/320 DME is shown in Table 8-1 for dual configuration.

Table 8-1. 310S DME Parts List

Ref. Des.	Item	P/N
1	Equipment Cabinet	0331-7001
1A1	AUX Rack	0330-7010
1A1A1	CSP	0330-7560
1A1A2	SCU	0130-7520
1A1A4	EMU	0130-7540
1A1A5	DC/DC-A	0130-7720
1A1A7	Modem	0130-7420
1A1A9	Backplane, BP-A	0330-3010
1A2	TX1 Rack	0330-7020
1A2A1	DPX-MSD	0330-7120
1A2A3	HPA	0332-7150
1A2A4	TXU	0330-7140
1A2A5	DC/DC	0330-7710
1A2A6	TCU	0330-7570
1A2A7	RFG	0330-7190
1A2A8	MON	0330-7550
1A2A9	FAN	0130-7910
1A2A10	Backplane, BP-BL	0330-3021
1A2A11	Backplane, BP-BR	0330-3022
1A4	PMU Rack	0130-7030
1A4A1	PMU	0130-7730
1A4A3	Backplane, BP-PM	0130-3030
1A5	AC/DC Rack	3151-0017
1A5A2	AC/DC	3151-0016
1A6	LMI	0130-7530
1A6A1	IEC-1000	3157-0001
1A6A2	LMI Interface Board	0130-7530
1A7	IFB	0130-3410

Ref. Des.	Item	P/N
1A8	VSWR Monitor	0130-7310
1A9	RF Detector	0330-7330
1FL1	Low Pass Filter	3139-0079
1DC1	Directional Coupler	3152-0102
1DC2	Directional Coupler	3152-0102
1CIR1	Circulator	3152-0078
1AR1	Lightning Arrester	0320-6001

Table 8-2. 320S DME Parts List

Ref. Des.	Item	P/N
1	Equipment Cabinet	0332-7001
1A1	AUX Rack	0330-7010
1A1A1	CSP	0330-7560
1A1A2	SCU	0130-7520
1A1A4	EMU	0130-7540
1A1A5	DC/DC-A	0130-7720
1A1A7	Modem	0130-7420
1A1A9	Backplane, BP-A	0330-3010
1A2	TX1 Rack	0330-7020
1A2A1	DPX-MSC	0330-7120
1A2A2	RXU	0330-7180
1A2A3	HPA	0332-7150
1A2A4	TXU	0330-7140
1A2A5	DC/DC	0330-7710
1A2A6	TCU	0330-7570
1A2A7	RFG	0330-7190
1A2A8	MON	0330-7550
1A2A9	FAN	0130-7910
1A2A10	Backplane, BP-BL	0330-3021
1A2A11	Backplane, BP-BR	0330-3022
1A4	PMU Rack	0130-7030
1A4A1	PMU	0130-7730
1A4A3	Backplane, BP-PM	0130-3030
1A5	AC/DC Rack	3151-0017
1A5A2	AC/DC	3151-0016
1A6	LMI	0130-7530
1A6A1	IEC-1000	3157-0001
1A6A2	LMI Interface Board	0130-7530
1A7	IFB	0130-3410
1A8	VSWR Monitor	0130-7310
1A9	RF Detector	0330-7330
1FL1	Low Pass Filter	3139-0079
1DC1	Directional Coupler	3152-0102

Ref. Des.	Item	P/N
1DC2	Directional Coupler	3152-0102
1CIR1	Circulator	3152-0078
1AR1	Lightning Arrester	0320-6001

Table 8-3. 320 DME Parts List

Ref. Des.	Item	P/N
1	Equipment Cabinet	0332-7002
1A1	AUX Rack	0330-7010
1A1A1	CSP	0330-7560
1A1A2	SCU	0130-7520
1A1A3	SCU	0130-7520
1A1A4	EMU	0130-7540
1A1A5	DC/DC-A	0130-7720
1A1A6	DC/DC-A	0130-7720
1A1A7	Modem	0130-7420
1A1A8	Modem	0130-7430
1A1A9	Backplane, BP-A	0330-3010
1A2	TX1 Rack	0330-7020
1A2A1	DPX-MSC	0330-7120
1A2A2	RXU	0330-7180
1A2A3	HPA	0332-7150
1A2A4	TXU	0330-7140
1A2A5	DC/DC	0330-7710
1A2A6	TCU	0330-7570
1A2A7	RFG	0330-7190
1A2A8	MON	0330-7550
1A2A9	FAN	0130-7910
1A2A10	Backplane, BP-BL	0330-3021
1A2A11	Backplane, BP-BR	0330-3022
1A3	TX2 Rack	0330-7020
1A3A1	DPX-MSC	0330-7120
1A3A2	RXU	0330-7180
1A3A3	HPA	0332-7150
1A3A4	TXU	0330-7140
1A3A5	DC/DC	0330-7710
1A3A6	TCU	0330-7570
1A3A7	RFG	0330-7190
1A3A8	MON	0330-7550
1A3A9	FAN	0130-7910

Ref. Des.	Item	P/N
1A3A10	Backplane, BP-BL	0330-3021
1A3A11	Backplane, BP-BR	0330-3022
1A4	PMU Rack	0130-7030
1A4A1	PMU	0130-7730
1A4A3	Backplane, BP-PM	0130-3030
1A5	AC/DC Rack	3151-0017
1A5A2	AC/DC	3151-0016
1A5A4	AC/DC	3151-0016
1A6	LMI	0130-7530
1A6A1	IEC-1000	3157-0001
1A6A2	LMI Interface Board	0130-7530
1A7	IFB	0130-3410
1A8	VSWR Monitor	0130-7310
1A9	RF Detector	0330-7330
1A10	RF Detector	0332-7330
1FL1	Low Pass Filter	3139-0079
1DC1	Directional Coupler	3152-0102
1DC2	Directional Coupler	3152-0102
1DC3	Directional Coupler	3152-0102
1RY1	Coaxial Relay	3152-0083
1RT1	RF Termination	3152-0101
1CIR1	Circulator	3152-0078
1CIR2	Circulator	3152-0013
1AR1	Lightning Arrester	0320-6001

Table 8-4. 310 DME Parts List

Ref. Des.	Item	P/N
1	Equipment Cabinet	0331-7002
1A1	AUX Rack	0330-7010
1A1A1	CSP	0330-7560
1A1A2	SCU	0130-7520
1A1A3	SCU	0130-7520
1A1A4	EMU	0130-7540
1A1A5	DC/DC-A	0130-7720
1A1A6	DC/DC-A	0130-7720
1A1A7	Modem	0130-7420
1A1A8	Modem	0130-7430
1A1A9	Backplane, BP-A	0330-3010
1A2	TX1 Rack	0330-7020
1A2A1	DPX-MSC	0330-7120
1A2A2	RXU	0330-7180
1A2A4	TXU	0330-7140
1A2A5	DC/DC	0330-7710
1A2A6	TCU	0330-7570
1A2A7	RFG	0330-7190
1A2A8	MON	0330-7550
1A2A9	FAN	0130-7910
1A2A10	Backplane, BP-BL	0330-3021
1A2A11	Backplane, BP-BR	0330-3022
1A3	TX2 Rack	0330-7020
1A3A1	DPX-MSC	0330-7120
1A3A2	RXU	0330-7180
1A3A3	HPA	0332-7150
1A3A4	TXU	0330-7140
1A3A5	DC/DC	0330-7710
1A3A6	TCU	0330-7570
1A3A7	RFG	0330-7190
1A3A8	MON	0330-7550
1A3A9	FAN	0130-7910
1A3A10	Backplane, BP-BL	0330-3021

Ref. Des.	Item	P/N
1A3A11	Backplane, BP-BR	0330-3022
1A4	PMU Rack	0130-7030
1A4A1	PMU	0130-7730
1A4A3	Backplane, BP-PM	0130-3030
1A5	AC/DC Rack	3151-0017
1A5A2	AC/DC	3151-0016
1A5A4	AC/DC	3151-0016
1A6	LMI	0130-7530
1A6A1	IEC-1000	3157-0001
1A6A2	LMI Interface Board	0130-7530
1A7	IFB	0130-3410
1A8	VSWR Monitor	0130-7310
1A9	RF Detector	0330-7330
1A10	RF Detector	0332-7330
1FL1	Low Pass Filter	3139-0079
1DC1	Directional Coupler	3152-0102
1DC2	Directional Coupler	3152-0102
1DC3	Directional Coupler	3152-0102
1RY1	Coaxial Relay	3152-0083
1RT1	RF Termination	3152-0101
1CIR1	Circulator	3152-0078
1CIR2	Circulator	3152-0013
1AR1	Lightning Arrester	0320-6001

SECTION 9

INSTALLATION

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9. INSTALLATION

9.1. INTRODUCTION

The site must be carefully selected and prepared for the installation of DME. Prior to the installation work, the equipment should be removed from shipping containers and unpacked for inspection as soon as possible to ensure that it is complete and intact in terms of numerical quantity and mechanical quality. Any discrepancy or damage shall be immediately reported. Each inspected item is to be repacked to prevent damage and kept in a dry storage until they are unpacked again as needed during the installation procedure.

Equipment grounding, main commercial power supplies, antenna mast and the input/output (I/O) connections shall be ready as appropriate. Depending on the beacon configuration selected and the options used, it may be necessary to connect the facility to the associated equipment or to other equipment.

9.2. SAFETY PRECAUTIONS

- To avoid any possible accidents, strict attention should be paid to safety regulations issued by the local authorities and the following safety rules should be observed:
- Keep the safety tools (given below) ready and wear them as considered appropriate.
- Goggles, safety gloves, sturdy shoes, safety helmets and etc.
- For emergency situation, everyone working on the installation site should be aware of:
 - escape routes (Escape routes should always be cleared)
 - the places where the First Aid Kit and Fire Extinguisher are kept in, and
 - Emergency contacts such as the nearest casualty doctor, eye specialist and fire station.
- Any pointed or sharp objects are to be removed from clothing pockets while working on the site.
- Remove chains and rings especially when working with electrical devices.
- Beware of electricity.
- Electrical devices and extension cables should be checked before you use them.
- Fuses are to be removed before working on the main power supply.
- Alcohol is strictly prohibited.
- Do not allow any type of alcohol to be carried into the installation site.
- Anyone under the influence of alcohol will not be tolerated on the installation site.
- Orderly arrangement and proper use of tools
- Sand off burr from chisels and punches.
- Test striking tools for tightness of fit.
- Always check ladders and planks carefully before use.
- Do not tread on protruding plank sections.
- Remove protruding nails, strips, etc. immediately.
- Never leave objects on scaffolding or ladders.

- Erect sturdy scaffolding and frames and always test them thoroughly before using them.
- When the shelter (equipment room) is unoccupied, it should be locked.

9.3. SITE INFORMATION

In general, it is recommended to select the highest ground with level terrain in the vicinity of proposed DME site. An ideal site is that all objects are cleared for a radius of at least 1000 meters, and no obstructions extend above the horizontal plane of the antenna within the service range of the facility. In reality, however, it is not possible to locate a site which fully satisfies such ideal requirements. Instead, every effort must be made to achieve the best site selection as far as possible.

The general location is initially determined by the type of facility; i.e. a terminal facility or an en route facility. The location of a terminal facility is constrained by the location of the airport and the orientation of the runway. The location of en route facility is determined largely by the location of the airways which they serve and, to a lesser extent, by the distance to other facilities serving the same airways. With respect to general requirements, a good site for ILS or VOR installation also satisfies the DME site requirements. Consequently the criteria for siting of ILS or VOR will be broadly applied to the site selection for collocation of DME with ILS or VOR. For reference, given below are some restrictions for siting of VOR:

- Obstructions within 300 meters of the antenna are to be removed.
- No groups of trees should be within a 300-meter radius or subtend a vertical angle of more than 2 degrees as measured from the antenna.
- Single trees, however, of moderate height (up to 9 meters) may be tolerated beyond 150 meters.
- At mountain top sites, no trees within 300 meters should be visible from the antenna array.
- Ordinary farm-type wire fences about 1.2 meters high are not permitted within 60 meters of the antenna. At mountain top sites, however, wire fences may be permitted within 60 meters of the antenna if they do not extend above the line of sight from the top of the antenna to the edge of the level area; i.e. they are within the shadow area of the ground counterpoise.
- Fences of the chain type (1.8 meters or more in height) are not permitted within 150 meters of the antenna.
- No wire fence should extend more than 0.5 degree above the horizontal beyond 150 meters of the antenna.
- The above three requirements may be relaxed for fences essentially radial to the antenna.
- Power and control line extensions should be installed underground for a minimum distance of 180 meters from the antenna.
- Beyond 180 meters, however, overhead power and control lines may be installed but should be essentially radial to the antenna for a minimum distance of 360 meters.
- No conductor should extend more than 0.5 degree above the horizontal plane containing the antennas, unless they are essentially radial (within ± 10 degrees) to the antenna array.
- At mountain top sites, the conductors will be permitted within 360 meters of the antenna, provided they do not extend above the conical surface formed by the top of the antenna and the edge of the leveled area.
- No structures should be permitted within 300 meters of the antenna, except for buildings such as the transmitter building at a mountain top site located on a slope below the ground level of the antenna so that they are not visible from the antenna.

- Partly or entirely metallic structures shall not subtend vertical angles of 1.2 degrees or more as measured from ground elevation at the antenna site.
- Wooden structures with negligible metallic content and little prospect of future metallic additions (such as roofs and wiring) may be tolerated if subtending vertical angles of less than 2.5 degrees.
- However, at airports, where a single hangar or line of hangars, administrative buildings, etc., may have considerable length, it is necessary to look upon such structures as producing interference in the same manner as power and telephone lines, and the criteria for power, control, and telephone lines will apply.

The selection of the DME location for combination with ILS facility also depends on nearby obstacles, clearance, and runway configuration.

The site selection guidelines above are only general recommendations. The final site decision should be made locally, prior to and during installation.

9.3.1. EQUIPMENT SITE LOCATION

The ground beacon may be installed in a control room or inside a shelter that complies with the environmental temperature, humidity, and pressure values listed in section 4. The equipment has the following overall dimensions:

- Height: 1600 mm
- Width: 580 mm
- Depth: 610 mm

The equipment requires the following clearances:

- If rear access is required, there must be at least 60 cm between the rear part of the equipment and the wall and any piece of equipment.
- A minimum of 60 cm between the top of the beacon and the ceiling of the control room or the shelter to leave space for the external connection cables and to allow access to the antenna connector and to the antenna probes connectors.
- A minimum of 1.2 meters between the front of the equipment and the wall and any other piece of equipment to allow the operator to open the front door.

The base must be able to support the total weight of the equipment--about 200 kg, including the optional modules, within the range of dynamic stress envisaged for the equipment.

The beacon does not normally need securing; however, if it is to be secured to the base, use four M12 bolts.

9.4. INSTALLATION PROCEDURES

9.4.1. RF CONNECTIONS

The RF input / output connectors are on the top plate of the equipment cabinet as shown in Figure 9-1.

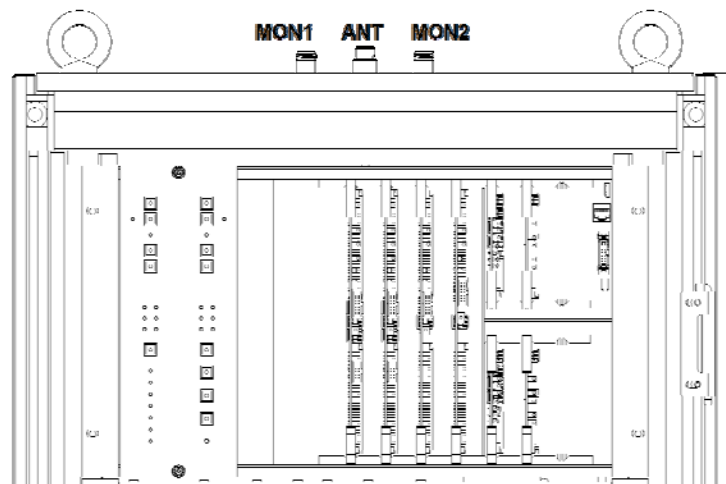


Figure 9-1. RF Connectors Locations

Run the RF cable and two monitor cables to the type N connector on top of the equipment cabinet. Typically 1/2" foam dielectric flexible corrugated cable is recommended for the RF cable and 1/4" foam dielectric flexible corrugated cable is recommended for the monitor cable.

Table 9-1. RF Connectors

Label	Type	Description	Cable
MON1	Type N	Monitor probe input #1	1/4" corrugated, flexible
ANT	Type N	RF input/output for antenna	1/2" corrugated, flexible
MON2	Type N	Monitor probe input #2	1/4" corrugated, flexible

9.4.2. AC / DC POWER CONNECTIONS

Suitable power sources as shown in Table 9-2 are required to operate the equipment.

Table 9-2. Power Source Requirements

Power Source	Requirements	
AC Input	Voltage	85 VAC – 265 VAC
	Frequency	47 Hz – 63 Hz
	Phase	Single
	Current	4A, max at 220 VAC 8A, max at 110 VAC
DC Input	Voltage	24 VDC nominal 22 V – 36 V
	Configuration	Two 24 V banks Two 12 V cells in series per each bank
	Battery Type	Maintenance-free Sealed Lead Acid Battery
	Backup Time	4+ hours with standard 65Ah batteries

Power sources are connected to the equipment via the terminal block at the back side of the cabinet as shown in Figure 9-2.

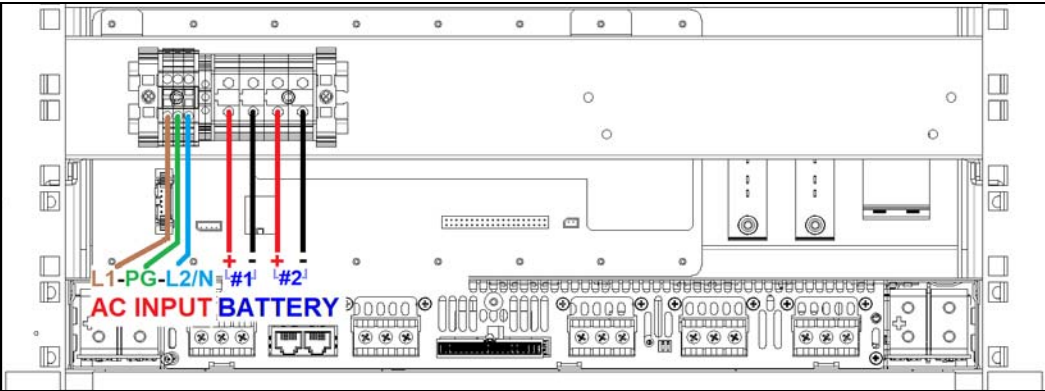


Figure 9-2. Power Input Terminal Block

Wire the AC power line and two backup batteries to the terminal block referring to Table 9-3.

Table 9-3. Power Input Wiring

Input	Terminal	Name	Wire Color
AC Line Input	L1	Line 1 (Live)	Brown
	PG	Protective Ground	Green
	L2/N	Line 2 / Neutral	Light Blue
Battery #1	+	Positive	Red
	–	Negative	Black
Battery #2	+	Positive	Red
	–	Negative	Black

9.4.3. I/O CONNECTIONS

Most of the I/O connections including data links are made on the IFB at the back side of the cabinet. The IFB contains connectors and terminal blocks for external connections and provides protection from surge or transient voltages.

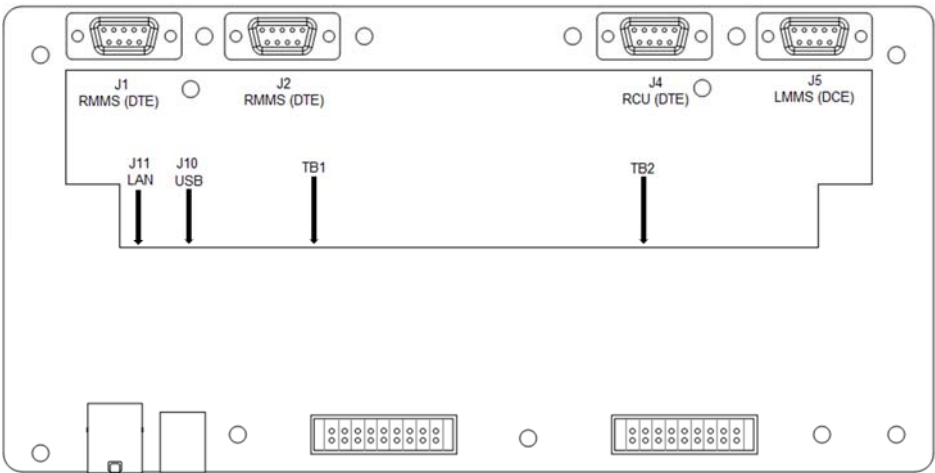


Figure 9-3. External Connectors on IFB

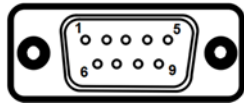
Connector	Pin	Name	Note
J1 (PL1) RMMS (DTE)  DB-9 Male	1	DCD	• A RS-232C, DTE, connector for RMMS (Remote PMDT) • Internally connected to SCU #1. • Up to 230 kbps, 8 bit, no parity, 1 stop bit • Externally connected to the PMDT PC via external DCE, i.e. modem. • Usually left unconnected when PMDT is connected over the Ethernet LAN.
	2	RxD	
	3	TxD	
	4	DTR	
	5	GND	
	6	DSR	
	7	RTS	
	8	CTS	
	9	RI	

Figure 9-4. J1 (PL1), RMMS

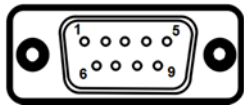
Connector	Pin	Name	Note
J2 (PL2) RMMS (DTE)  DB-9 Male	1	DCD	• A RS-232C, DTE, connector for RMMS (Remote PMDT) • Internally connected to SCU #2. • Up to 230 kbps, 8 bit, no parity, 1 stop bit • Externally connected to the PMDT PC via external DCE, i.e. modem. • Usually left unconnected when PMDT is connected over the Ethernet LAN.
	2	RxD	
	3	TxD	
	4	DTR	
	5	GND	
	6	DSR	
	7	RTS	
	8	CTS	
	9	RI	

Figure 9-5. J2 (PL2), RMMS

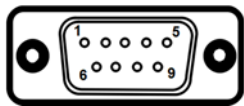
Connector	Pin	Name	Note
J4 (PL4) RCU (DTE)  DB-9 Male	1	-	• A RS-232C, DTE, connector for RCU • Internally connected to the master SCU. • 1200 bps, 8 bit, even parity, 1 stop bit • Externally connected to the RCU via external DCE, i.e. modem. • Usually left unconnected when RCU is connected over built-in modem or the Ethernet LAN.
	2	RxD	
	3	TxD	
	4	-	
	5	GND	
	6	-	
	7	-	
	8	-	
	9	-	

Figure 9-6. J4 (PL4), RCU

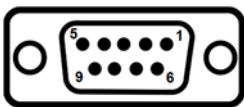
Connector	Pin	Name	Note
J5 (SK5) LMMS (DCE)  DB-9 Female	1	DCD	• A RS-232C, DCE, connector for LMMS (Local PMDT) • Internally connected to the master SCU. • Up to 230 kbps, 8 bit, no parity, 1 stop bit • Externally connected to the PMDT PC using a direct cable (a.k.a. null modem). • Usually left unconnected when PMDT is connected over the Ethernet LAN.
	2	RxD	
	3	TxD	
	4	DTR	
	5	GND	
	6	DSR	
	7	RTS	
	8	CTS	
	9	RI	

Figure 9-7. J5 (SK5), LMMS

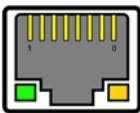
Connector	Pin	Name	Note
J11 (SK11) LAN  RJ-45 Female	1	TX+	• A100BASE-TX Ethernet LAN for PMDT or RCU • Internally connected to LMI
	2	TX–	
	3	RX+	
	4	tp1+	
	5	tp1–	
	6	RX–	
	7	tp2+	
	8	tp2–	

Figure 9-8. J11 (SK11), LMMS

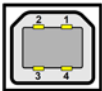
Connector	Pin	Name	Note
J10 (SK10) USB  USB type B Female	1	V _{BUS}	• A USB type B connector LMMS (Local PMDT) • Usually left unconnected when PMDT is connected over the Ethernet LAN or RS-232C.
	2	Data+	
	3	Data–	
	4	GND	

Figure 9-9. J10 (SK10), LMMS

Connector	Name	Pin	Pin	Name	
TB1 Detachable Terminal Block	(+)	1	10	(–)	Keyer Input from associated equipment
	(+)	2	11	(–)	Status Input from associated equipment
	(Tip)	3	12	(Ring)	Modem line for RMMS1
	(Tip)	4	13	(Ring)	Modem line for RMMS2
	(Tip)	5	14	(Ring)	Modem line for RCU1
	(Tip)	6	15	(Ring)	Modem line for RCU2
	(+)	7	16	(–)	Voice Input
	(+)	8	17	(–)	Intrusion Sensor Input
	NC	9	18	NC	Not connected

Figure 9-10. Terminal Block, TB1

Connector	Name	Pin	Pin	Name	
TB2 Detachable Terminal Block	+5V	1	10	GND	I ² C Temperature Sensor
	CL	2	11	DA	
	OS	3	12	NC	
	(+)	4	13	(–)	Smoke Sensor
	(+)	5	14	(–)	DC5V 1A, max
	(+)	6	15	(–)	DC12V 1A, max
	(+)	7	16	(–)	DC28V 0.5A, max
	(+)	8	17	(–)	Keyer Output to associated equipment
	(+)	9	18	(–)	Status Output to associated equipment

Figure 9-11. Terminal Block, TB2

9.5. INSPECTION

Make sure that all the units and assemblies are firmly assembled and mounted.

Turn on the AC main switch. Turn on the BATT switches.

Connect PMDT to the equipment.

Turn the keylock switch to **MAINT** position.

9.6. INITIAL STARTUP AND PRELIMINARY ADJUSTMENTS

Using CSP, LMI or PMDT, Turn off all the RF outputs.

Using PMDT, set up the STATION and TRANSPONDER settings properly according to the station data.

Station Setup to configure include:

- Basic – Channel: the channel number assigned to the station
- Basic – Power Output: 100 Watt or 1000 Watt
- Basic – Delay Offset, 0 μ s unless specified
- Basic – Auto Delay Calibration: default (Never)
- Basic – Sensitivity Level: -91 dBm for 320 DME or -81dBm for 310 DME
- Pulse Rate – Minimum Pulse Rate: 800 pp/s is recommended.
- Echo Suppression – SDES : On
- Echo Suppression – SDES Duration: 6.0 μ s for X channel, 30 μ s for Y channel
- Echo Suppression – LDES : Off
- Echo Suppression – LDES Duration : 200 μ s
- Echo Suppression – LDES Threshold Level : -80 dBm
- Echo Suppression – Dead Time : 60 μ s
- IDENT – IDENT Code: Up to four letters of alphabet character identification assigned to the station
- IDENT – IDENT Keyer: Slave if the DME is collocated with VOR or LOC.
- IDENT – IDENT Sync: Code or Pulse
- IDENT – IDENT Sound: any choice
- IDENT – Equalizing Pulse: Off
- Equipment Configuration – Enable Interlock: enable if interlock is used.
- Equipment Configuration – Standby Mode:
 - On Boot – Bypass Monitor: Yes, usually
 - On Boot – Transmitter Output On: Yes.

Transponder Setup to configure include:

- Output – Power Output: 100%
- Setup Option – Pulse Rate: As per Station Setup
- Setup Option – Echo suppression: As per Station Setup
- Setup Option – IDENT: As per Station Setup
- Thermal Control Setup:
 - o Cooling Fans – Operation Mode: On

- o Overheat Protection – TXU Shutdown at: 90 °C
- o Overheat Protection – TXU Restart at: 80 °C

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